

# Logic from Language

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## Chapter 0

# Truth, Consequence, and Inference

Logic is the study of *inference in language*. If someone says a sentence  $S$ , then they are usually committed to more than just  $S$ , and a hearer would typically *infer* the sentences that *follow* from  $S$ . Even more, if the speaker said a few sentences in a row,  $S_1, \dots, S_n$ , then there might well be sentences which follow from this list which do not follow from the individual sentences.

For example, if a speaker of English said *All of my friends can ride a bicycle* and *My friend Robin lives in Maine*, then the hearer should infer sentences like *Someone named Robin who lives in Maine can ride a bicycle*. This is a rather complicated example, more so than anything we'll do in this book. This is because the “distance” from the two statements to what we infer is large. A simpler example would be if the speaker said *All of my friends can ride a bicycle* and then *Sandy is my friend*, and then we inferred *Sandy can ride a bicycle*.

For a second example, consider a speaker of Nyanja<sup>1</sup> who says *osewerera mpira onse ndi amtali* and *anzake onse ndi osewera mpira*. Then it follows that *anzake ena ndiamtali*. We would like some account of this inference, some precise and clear theory (thus, a mathematical theory) which allows us to predict that the third sentence indeed follows from the first two. The way we do this is with a set of interlaced concepts that we are going to call *the framework of model-theoretic semantics and formal inferences*. You can find the basic ingredients of this framework in Figure 0.1. We'll be discussing each of these ingredients in this chapter, but doing so in a brief way. It is not really necessary that you learn about this framework now, and if you are impatient for “content” you might prefer to skip ahead to Section 0.3 or even to Chapter 1, where we see the framework in action (without much of a discussion of it.)

Another way to state things: a sentence follows from one or more other sentences if its meaning is somehow *implied* by the meaning of the assumptions. But what is this “meaning”? The overall concept of *meaning*, while fascinating and controversial, is not the topic of this book. We settle on the best-known approach, called *model-theoretic semantics*. We do this mainly because it leads to a set of mathematical definitions that we can get to quickly. We describe it next. And then we shall look at the second concept which we need, formal inference.

Up until now, we have been concerned with *intuitive inference*. This is the first concept in Figure 0.1. The second is *fragments of language*. What we mean here the very simple fragments of language studied either in formal language theory or (more rarely) in courses on natural language

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<sup>1</sup>This language is also called Chichewa, and it is spoken by around 12 million people in Malawi, Zambia, Mozambique, and Zimbabwe. The meanings of these sentences are not important here. Usually our examples will be from English, but there is a value in thinking about this matter of inference in a language which one does not understand.



**Pedagogic Note 0.1:** This chapter is very general background and could be skimmed or skipped.

The chapter has two purposes: it presents motivational examples on several central topics in the book, and it also presents some mathematics background. It has a very general discussion of model-theoretic semantics and also a very general discussion of formal proofs. One of the points of emphasis is that we do logic to model phenomena of interest such as inference and knowledge, and we approach the logic as we would any branch of applied mathematics; the goal is to make models which are precise and which make predictions which match our intuitions about the phenomena of interest.

The chapter also introduces preorders and equivalence relations for later use. Those introductions are little more than the definitions and examples. In terms of formal proofs, the goals are to illustrate what *formal* thinking is, and how it relates to model-theoretic semantics. We also have a toy soundness/completeness result dealing with the logic of identity statements between constants. Finally, we have a section on modal logic that is based on the interior and closure operations on sets of real numbers.

A course based on this book could start with Chapter 1 on syllogistic logic. It could also start with Chapter 5 on propositional logic; this would be a good choice for a course aimed at propositional modal logic.

In any case, an instructor would select sections of this chapter depending on the goals of a course and on the background of the students. (For example, a course which would not cover modal logic would probably omit the sections about knowledge and sets of real numbers.) The selected sections could be presented at the very start of the course. Alternatively, those sections could come whenever needed.

syntax. Notice that this point itself contains an idealization because we are approximating the natural notion of *grammaticality* with what formal language theory gives us.

Once we have a precise syntax for a fragment, the next step is to provide it with a precise semantics. The key move here is that there is no such thing as truth *simpliciter*, only truth *in a model*. We have a little to say about models in Section 0.1.1 just below. After we look at a model, we come to the the all-important definition of *logical consequence*.

## 0.1 Model-theoretic semantics

Probably the key point of logic is that there is a distinction between *syntax* and *semantics*. The idea is that syntax is the raw symbols. On the other hand, semantics indicates some “meaning.” For us, semantics will usually bring in abstract, mathematical objects. To see why this is so, let us look at a simple example of a *context* or *model* which serves as a source of meaning.

Let  $M = \{1, 2, 3, 4, 5, 6, 7, 8\}$ : the orange rectangle.

Let  $\llbracket \textit{runners} \rrbracket = \{1, 2, 3, 4, 5, 6\}$ , in the black oval.

Let  $\llbracket \textit{skiers} \rrbracket = \{1, 4\}$ , shown in the blue oval.

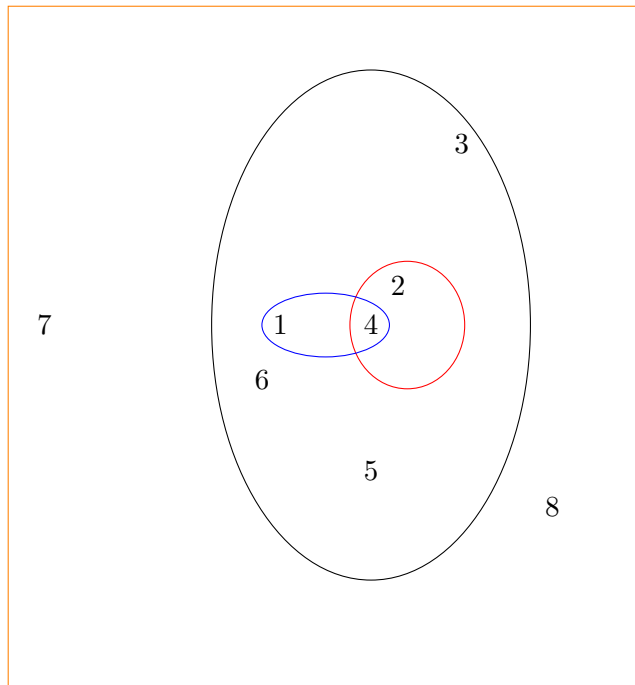
Let  $\llbracket \textit{skaters} \rrbracket = \{2, 4\}$ , in the red oval.

1. Intuitive inference (Section 0.1).
2. Precisely specified fragments of language.
3. Models, formal semantics of sentences from the previous point, and definitions of semantic consequence (Definition 0.3).
4. Logical rules as formal operations, leading to the concept of proof-theoretic consequence (Section 0.2).
5. The connection of semantic consequence, proof-theoretic consequence, and intuitive inference (Section ??).

Figure 0.1: The framework of model-theoretic semantics and formal inferences.

|                         |                       |
|-------------------------|-----------------------|
| All skaters are runners | is true in our model  |
| All runners are skiers  | is false in our model |
| All skaters are skiers  | is false in our model |
| All skaters are runners | is true in our model  |
| All runners are runners | is true in our model  |

The point is that the sentence All skaters are runners is not true or false *on its own*. It makes no sense to even ask: Is All skaters are runners true? The question can only be broached when supplies a missing context. This is what a semantics gives us. If a model is specified precisely, then the determination of truth or falsehood can be done. In fact, it can be done rather mindlessly.



### 0.1.1 Inference based on models

Sentences are not true or false on their own: they are true or false *in models*! But some sentences are true *no matter what the model says*. *All  $x$  are  $x$*  is one example.

We often want to make *temporary assumptions* about models.

Logical reasoning is reasoning that works for all models. Logical reasoning *cannot* tell you what is true and what is false. It can tell you that based on some temporary assumptions, some conclusions follow. In fact, this is what *logic wants to do*. Here is an example.

$$\begin{array}{l} \text{All sagatricians are maltnomans.} \\ \text{All sagatricians are aikims.} \\ \hline \text{All maltnomans are aikims.} \end{array}$$

If we have any model in the whole world where the two sentences above the line are true, then the sentence below the line is also true in that model.

We have analyzed our small piece of reasoning using models, and this opens the door for mathematics. To see why, we mention the central definitions of *semantic entailment* in Definition 0.3.

First, we remind you of some Greek letters which we use as notation all the time.

#### Notation 0.2: Greek letters for use throughout the book

| sentences | name | sets of sentences | name  |
|-----------|------|-------------------|-------|
| $\varphi$ | phi  | $\Gamma$          | gamma |
| $\psi$    | psi  | $\Delta$          | delta |
| $\chi$    | chi  |                   |       |

Please note that the meaning of the word *sentence* varies from chapter to chapter. In fact, in this chapter, it changes from section to section. But what we are aiming at here is the overall framework that we mentioned in Figure 0.1, not in any particular example of that framework.

#### Definition 0.3: semantic consequence

Each logical system has its own definition of when a model  $\mathcal{M}$  satisfies a sentence  $\varphi$ , written  $\mathcal{M} \models \varphi$ .

Let  $\Gamma$  be a set of sentences in some logical system, and  $\mathcal{M}$  a model.

We say that  $\mathcal{M} \models \Gamma$  iff  $\mathcal{M} \models \varphi$  for every  $\varphi \in \Gamma$ .

We say that  $\Gamma \models \varphi$  iff for all  $\mathcal{M}$ : if  $\mathcal{M} \models \Gamma$ , then also  $\mathcal{M} \models \varphi$ .

We read  $\Gamma \models \varphi$  in a number of ways:

Every model of  $\Gamma$  is a model of  $\varphi$   
 $\Gamma$  *logically implies*  $\varphi$   
 $\Gamma$  *semantically implies*  $\varphi$   
 $\varphi$  is a *semantic consequence* of  $\Gamma$ .

We write  $\Gamma \not\models \varphi$  if it is not the case that  $\Gamma \models \varphi$ . In plain terms, for *some* model  $\mathcal{M}$ ,  $\mathcal{M} \models \Gamma$  but  $\mathcal{M} \not\models \varphi$ .

## 0.2 Formal proofs and informal proofs

Our next concept is that of a *formal proof*.

A proof in mathematics is a piece of reasoning which convinces us that a given set of assumptions implies a given conclusion. Such a proof is almost always given in a natural language. It may use text, diagrams, citations of other results. It frequently skips steps. For all of these reasons, it is *informal*.

A formal proof is caricature of an informal proof. (That is, the informal proof is the real thing, and a formal proof is an approximation.) A formal proof makes use of sentences that are given in some precise way, usually this means that they are determined by a *formal grammar*. In this book, we are almost never going to need a sophisticated form of grammar, or even a very realistic one. That is, we are basically not going to study inference using sentences where a realistic grammar would help. And we are not going to be interested in ambiguous sentences: all of the grammars will be unambiguous. The formal proofs will use *logical rules*. Again, these will come in rather simple formats which can be specified in a completely precise way. Not only are the sentences in a formal grammar drawn from a very simple grammar, but the logical rules themselves will be taken in a very simple format. In fact, the grammar and the rules will be special cases of *syntax* as we mentioned on page 5.

A person should be able to check if a sentence is well-formed according to a grammar without understanding what it means. A person should also be able to check if a formal proof is correct according to its definition without understanding what any of the sentences mean and also without understanding what the rules are about, either. At first glance, this seems paradoxical, or even wrong-headed: we are supposed to be studying “meaning”, and the basic concepts which we use should be understandable without understanding what anything means(?!).

Now, what are mathematically precise concepts of *sentences* and *formal proofs*? For us, sentences will always be given by very simple inductive definitions. For a very simple example (actually used in Chapter 3), we could start with some set of *nouns*, say *dogs*, *cats*, *rabbits*. And then we might declare our sentences to be the expressions

$$\text{all } n_1 \ n_2 \quad \text{some } n_1 \ n_2 \quad \text{no } n_1 \ n_2$$

Or another example of a formal language is the set  $\mathcal{T}$  of terms of *boolean algebra*. We start with a collection  $p_0, p_0, \dots$ , of *atomic sentences*, and then we take the set  $\mathcal{T}$  to be the smallest set containing the atomic sentences, and also closed in the following ways: (i) **T** and **F** belong to  $\mathcal{T}$ ; (ii) if  $\varphi \in \mathcal{T}$ , then  $(\neg\varphi) \in \mathcal{T}$ ; (iii) if  $\varphi, \psi \in \mathcal{T}$ , then the following also belong to  $\mathcal{T}$ :  $(\varphi \wedge \psi)$ ,  $(\varphi \vee \psi)$ , and  $(\varphi \leftrightarrow \psi)$ . It is not important at this point what these mean. (Rather: it is important at this point that *these do not mean anything*, since we haven’t given any kind of semantics for any of this.). It’s only important to see what a precise collection of mathematical objects is.

A formal proof shares the same kinds of properties as our sentences. They can be constructed, taken apart, and manipulated without understanding what they are about. They can be fruitfully understood as trees with labels on the nodes, much like the elements of  $\mathcal{T}$  above. (But our formal proofs will eventually have more structure than this.) And again, they are caricatures of human proofs in much the same way as “sentences” in formal languages are caricatures of red-blooded sentences in real life.

### 0.3 Preorders

Our study of logic will involve a few kinds of structures from mathematics. These include *graphs* and *preorders*. At this point, we want to discuss preorders because they are related to the up/down arrows which we have already seen.

#### Definition 0.4: Preorders

A *preorder* is a pair  $\mathbb{P} = (P \leq)$ , where  $P$  is a set, and  $\leq$  is a relation on  $P$  which is both *reflexive* and *transitive*.

Reflexivity means that  $p \leq p$  for all  $p \in P$ .

Transitivity means that for all  $p, q$ , and  $r \in P$ , if  $p \leq q$  and  $q \leq r$ , then also  $p \leq r$ .

#### Example 0.5

Let  $\mathbb{N}$  be the natural numbers  $\{0, 1, 2, \dots\}$  with the usual ordering relation  $\leq$ . This is a preorder because the usual ordering is reflexive and transitive.

Similarly, the real numbers with its natural ordering gives a preorder that we call  $\mathbb{R}$ .

In both cases, the preorder is technically a pair, not a set. But following standard usage, we refer to the preorder by the set when the order is clear.

#### Example 0.6

Here is another family of natural examples of preorders. Let  $X$  be any set. Let  $\mathcal{P}(X)$  be the set of subsets of  $X$ , and let  $\leq$  be the *inclusion relation*  $\leq$  on  $\mathcal{P}(X)$ :  $A \subseteq B$  means that every element of  $A$  is also an element of  $B$ .

#### Example 0.7

Here are some sets and relations which are not preorders.

Let  $A = \{a, b, c\}$ , and let  $R = \{(a, b), (b, c)\}$ . Then  $(A, R)$  is not a preorder:  $R$  is neither reflexive nor transitive.

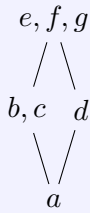
Let  $N$  be the set of natural numbers, and let  $E = \{(n, m) : n - m \leq 1 \in N\}$ . Then  $(N, E)$  is reflexive but not transitive, and so  $(N, E)$  is not a preorder.

Let  $R = \{(n, m) : n < m\}$ . Then  $R$  is transitive but not reflexive, and so  $(N, R)$  is not a preorder.

The reason we use the symbol  $R$  here instead of  $\leq$  as before is that whenever we use  $\leq$  in this book, we will be working with a preorder.

Here is another example of a preorder, a more artificial example, but one which allows us to introduce a point of notation and terminology. We take  $P$  to be the set  $\{a, b, \dots, g\}$ . Then we take the poset to be as shown below:

### A picture is worth a thousand words



$a \leq a, a \leq b, a \leq c, a \leq d, a \leq e, a \leq f,$   
 $a \leq g, b \leq b, b \leq c, b \leq e, b \leq e, b \leq f,$   
 $b \leq g, c \leq b, c \leq c, c \leq e, c \leq e, c \leq f,$   
 $c \leq g, d \leq d, d \leq e, d \leq f, d \leq g, e \leq e, e \leq f,$   
 $e \leq g, f \leq e, f \leq f, f \leq g, g \leq e, g \leq f, g \leq g$

On the left we have a picture, and on the right the same picture written out as a big set of pairs. You can see why most people prefer the picture.

Either way, we have a preorder  $\mathbb{P} = (P, \leq)$ . Note that  $b \leq c \leq b$ . We indicate this by writing  $b \equiv c$ , and we say that  $b$  and  $c$  are *equivalent* in the preorder. Notice that *equivalent* does not imply *identical* in a preorder. The picture of a preorder is called its *Hasse diagram*. As the lines above show, a picture is worth several lines of explicit information. We draw the preorder by putting  $a$  at the bottom, since  $a \leq x$  for all  $x \in P$ . We put  $b$  and  $c$  together, since  $b \equiv c$ , and since there is nothing between them and  $a$ . We do the same for  $d$ . There is no significance to the fact that  $b, c$  is left of  $d$ . Notice that the reflexivity condition is implicit in the picture. Similarly, the transitivity condition is also implicit. So we would never draw a separate line from  $a$  to  $e$ , for example.

### Definition 0.8: Opposite and flattening

For any preorder  $\mathbb{P}$ , there is an *opposite preorder*  $\mathbb{P}^{op}$ . Its domain set is  $P$ , the same domain set as for  $\mathbb{P}$ . The order is defined by:

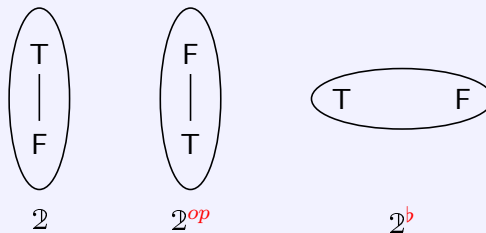
$$p \leq q \text{ in } \mathbb{P}^{op} \quad \text{iff} \quad q \leq p \text{ in } \mathbb{P}$$

There is also a *flattened version*  $\mathbb{P}^b$ , and this time

$$p \leq q \text{ in } \mathbb{P}^b \quad \text{iff} \quad p = q$$

### Example 0.9: Three preorders

An important preorder for logic  $\mathbb{2}$ , also called “Bool”. Its universe is  $\{\mathsf{T}, \mathsf{F}\}$ , and the ordering is as shown on the left below.



We also illustrate its opposite and flattened versions.

**Definition 0.10: product of two preorders**

For any preorders  $\mathbb{P}$  and  $\mathbb{Q}$ , there is a *product preorder*  $\mathbb{P} \times \mathbb{Q}$ . Its domain set is the cartesian product  $P \times Q$ .

$$(p, q) \leq (p', q') \text{ in } \mathbb{P} \times \mathbb{Q} \quad \text{iff} \quad p \leq p' \text{ in } \mathbb{P}, \text{ and } q \leq q' \text{ in } \mathbb{Q}$$

**Definition 0.11: Monotone and antitone functions  $f : \mathbb{P} \rightarrow \mathbb{Q}$** 

Let  $\mathbb{P}$  and  $\mathbb{Q}$  be preorders. A function  $f : P \rightarrow Q$  is *monotone* if the following holds: For all  $p, q \in P$ , if  $p \leq q$  in  $\mathbb{P}$ , then  $f(p) \leq f(q)$  in  $\mathbb{Q}$ .

A function  $f : P \rightarrow Q$  is *antitone* if the following holds: For all  $p, q \in P$ , if  $p \leq q$  in  $\mathbb{P}$ , then  $f(q) \leq f(p)$  in  $\mathbb{Q}$ .

**Example 0.12: Monotone and antitone functions**

Let  $\mathbb{N}$  be the preorder of natural numbers with the usual ordering. Here are some functions from  $\mathbb{N}$  to itself, and discussions of whether they are monotone, antitone, both, or neither. First,  $f(x) = x + 1$ . This is monotone because for all  $n, m \in \mathbb{N}$ , if  $n \leq m$ , then also  $n + 1 \leq m + 1$ .

Second  $g(x) = x^2 - 3x + 2$ . Note that  $g(0) = 2$ ,  $g(1) = 0$ ,  $g(2) = 0$ ,  $g(3) = 2$ . We can see right away that  $g$  is not monotone, since  $0 \leq 2$ , but  $g(0) \not\leq g(2)$ . (As soon as we have one exception, we know that  $g$  is not monotone.) Also,  $g$  is not antitone, since  $1 \leq 3$  but  $g(3) \not\leq g(1)$ . So this function is neither monotone nor antitone.

The function  $h(x) = 577$  is a constant function. It is both monotone and antitone. In fact, every constant function is both monotone and antitone.

Next,  $i(x) = x$  is called the *identity function*. It is monotone but not antitone.

Finally, here is a function that is antitone but not monotone:  $j(0) = 1$ , and  $j(x) = 0$  for  $x > 0$ .

**Example 0.13: The four functions from  $2$  to itself**

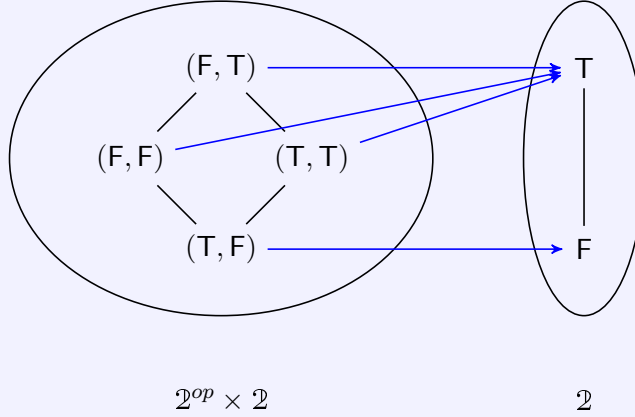
|       | $f(\text{T})$ | $f(\text{F})$ |
|-------|---------------|---------------|
| $f_1$ | T             | T             |
| $f_2$ | T             | F             |
| $f_3$ | F             | T             |
| $f_4$ | F             | F             |

Notice that  $f_1$  is the constant function with value T, and  $f_4$  is the constant function with values F. Also,  $f_2$  is the identity function. Finally,  $f_3$  is called *negation*: it flips the truth values.

Of these,  $f_1$ ,  $f_2$ , and  $f_4$  are monotone, and  $f_1$ ,  $f_3$ , and  $f_4$  are antitone.

**Example 0.14**

The *material conditional* is the function  $\rightarrow: 2 \times 2 \rightarrow 2$  defined by  $T \rightarrow T = T$ ,  $T \rightarrow F = F$ ,  $F \rightarrow T = T$ , and  $F \rightarrow F = T$ . This function is monotone when we take it to be a function from  $2^{op} \times 2$  to  $2$ .



We have some important facts about *composition of functions*. Let  $g: \mathbb{P} \rightarrow \mathbb{Q}$ , and let  $f: \mathbb{Q} \rightarrow \mathbb{R}$  be any functions on preorders. We have a *composition function*  $f \circ g: \mathbb{P} \rightarrow \mathbb{R}$  defined by

$$(f \circ g)(p) = f(g(p))$$

In words, the function  $f \circ g$  takes an input  $p \in P$ , first applies  $f$  to get  $f(p) \in Q$ , and finally applies  $g$  to this to get  $g(f(p))$ .

**Proposition 0.15**

If  $f$  and  $g$  are both monotone, then so is  $f \circ g$ .

*Proof.* Suppose that  $p_1 \leq p_2$  in  $\mathbb{P}$ . Then since  $g$  is monotone,  $g(p_1) \leq g(p_2)$  in  $\mathbb{Q}$ . And since  $f$  also is monotone,  $f(g(p_1)) \leq f(g(p_2))$  in  $\mathbb{R}$ . (This is the important point, so be sure that you see why this is true.) But  $(f \circ g)(p_1) = f(g(p_1))$ , and  $(f \circ g)(p_2) = f(g(p_2))$ . So  $(f \circ g)(p_1) \leq (f \circ g)(p_2)$ . This for all  $p_1, p_2 \in P$  shows what we want.  $\square$

## 0.4 The logic of identity, illustrating soundness and completeness

This section presents a start-to-finish example of the kind of work that will be done in the book. We present this work partly as an illustration of what our work will be like, with comments to that effect, and partly because some of the ideas are going to be needed throughout the book.

**Syntax** We start with a collection of *names*

$$\text{Names} = \{\text{Amina}, \text{Bao}, \text{Clarisse}, \text{Damir}, \dots\}.$$

We are going to use letters like  $a, b, c$  as *variables over names*. Our sentences are the expressions  $a$  is  $b$ , where  $b$  and  $a$  are names.



**Definition 0.17: rules of the proof system E for  $\mathcal{A}$** 

$$\frac{}{a = a} \text{ R} \qquad \frac{a = b}{b = a} \text{ S} \qquad \frac{a = b \quad b = c}{a = c} \text{ T}$$

**Semantics** A *model*  $\mathcal{M}$  is a set  $M$  together with an *interpretation function*  $\llbracket \cdot \rrbracket : \text{Names} \rightarrow M$ . We say that  $\mathcal{M} \models a \text{ is } b$  if  $\llbracket a \rrbracket = \llbracket b \rrbracket$ . We then get derived semantic notions by Definition 0.3.

**Example 0.16**

Let  $M = \{1, 2, 3, 4\}$ , with  $\llbracket a \rrbracket = 1$ ,  $\llbracket b \rrbracket = 1$ ,  $\llbracket c \rrbracket = 2$ . Then  $\mathcal{M} \models a \text{ is } b$ , but  $\mathcal{M} \not\models a \text{ is } c$ .  
Two names can refer to the same person, as in real life:

$$\begin{array}{ll} \llbracket \text{Belcalis Marlenis Almánzar} \rrbracket &= \llbracket \text{Cardi B} \rrbracket \\ \llbracket \text{Eminem} \rrbracket &= \llbracket \text{Marshall Mathers III} \rrbracket \\ \llbracket \text{Superman} \rrbracket &= \llbracket \text{Clark Kent} \rrbracket \\ \llbracket \text{Jorge Mario Bergoglio} \rrbracket &= \llbracket \text{Pope Francis} \rrbracket \end{array}$$

And for us, not every item in a model has to have a name.

So we have facts like the ones below:

1.  $\models a = a$ .
2.  $a = b \models b = a$ .
3.  $a = b, b = c \models a = c$ .
4.  $b = c, d = a, a = c \models d = b$ .

These are basically translations “obvious” facts about the identity relation in mathematics.

The basic goal is to *characterize* the relation between sets  $\Gamma$  of equations (on the left) and single equations (on the right). We do this by formulating a *logical system* using *inference rules*. We list our rules in Logic 0.17.

**Example 0.18:  $b = c, d = a, a = c \vdash d = b$** 

We saw above that  $b = c, d = a, a = c \models d = b$ . Here is a corresponding formal proof of this in our system:

$$\frac{\frac{d = a \quad a = c}{d = c} \text{ T} \quad \frac{b = c}{c = b} \text{ S}}{d = b} \text{ T}$$

In order to prove the completeness of the logic, we take a detour into very general mathematics. We call upon the notion of an *equivalence relation*.

**Definition 0.19: Equivalence relation**

A relation  $E$  on a set  $X$  is an *equivalence relation* provided the following conditions are met:

1. For all  $x \in X$ ,  $x E x$ . (reflexive)
2. For all  $x, y \in X$ , if  $x E y$ , then  $y E x$ . (symmetric)
3. For all  $x, y, z \in X$ , if  $x E y$  and  $y E z$ , then  $x E z$ . (transitive)

If  $E$  is an equivalence relation on  $X$ , and  $x \in X$ , then we define

$$[x] = \{y \in X : x E y\}$$

We call this the *equivalence class of  $x$* . Finally the *quotient set*  $X/E$  is the set of all equivalence classes of members of  $X$ .

**Example 0.20: Equivalence relations and quotient sets**

Let  $X = \{1, 2, 3, 4\}$  and let

$$\begin{aligned} R_1 &= \{(1, 1), (1, 2), (1, 3), (2, 1), (3, 3), (3, 4)\} \\ R_2 &= \{(1, 1), (2, 2), (1, 2), (2, 1), (3, 3), (4, 4)\} \\ R_3 &= \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (1, 3), (3, 1), (2, 3), (3, 2), (4, 4)\} \end{aligned}$$

Then  $R_1$  is not an equivalence relation; it is not even reflexive. Both  $R_2$  and  $R_3$  are equivalence relations. In  $R_2$ ,  $[1] = \{1, 2\} = [2]$ ,  $[3] = \{3\}$ , and  $[4] = \{4\}$ .  $R_3$  has two equivalence classes:  $[1] = [2] = [3]$ , and this set is  $\{1, 2, 3\}$ . In addition,  $[4] = \{4\}$ .

The quotient set  $X/R_2$  is  $\{[1], [3], [4]\}$ . The quotient set of  $X/R_3$  is  $\{[1], [4]\}$ .

**Remark 0.21: Proper definitions of relations and functions on quotient sets**

Frequently we'll want to define functions on quotient sets, or relations on them. There's an important check that one has to do whenever one does this. It's called *making sure that the definition was independent of the choice of representatives*. Please don't be scared by this long locution; the idea is fairly simple.

Here's a very classic example. Suppose that we are considering the set  $X = \{0, 1, \dots, 23\}$ . We define  $x \pmod{12}$  to be the remainder when  $x$  is divided by 12. Let us define

$$x R y \quad \text{iff} \quad x \pmod{12} = y \pmod{12}$$

We have the 24 hours in a day, and the representation of time on a standard 12-hour clock. Then  $R$  is an equivalence relation. Some examples of the equivalence classes:  $[0] = \{0, 12\}$ ,  $[1] = \{1, 13\}$ ,  $[2] = \{2, 14\}$ , .... The quotient  $X/R$  is  $\{[0], [1], \dots, [11]\}$ .

Now we might like to define an addition operation by

$$[x] + [y] = [x + y \pmod{12}]. \quad (0.1)$$

For example, we want to say that  $[3] + [11] = [2]$ , since  $3 + 11 = 14$ , and  $14 \pmod{12} = 2$ . The hitch is that  $[3] = [15]$  and  $[11] = [23]$ , and so when we add  $[3] + [11]$ , what would happen if we chose to pick 15 to “represent”  $[3]$  and 23 to “represent”  $[11]$ ? We'd get  $15 + 23 = 38$ . Fortunately,  $38 \pmod{12} = 2$ . So in this case, it didn't matter which representative we picked when we wanted to add equivalence classes. The general fact is that if  $x = x' \pmod{12}$  and  $y = y' \pmod{12}$ , then also  $x + y = x' + y' \pmod{12}$ . So defining  $+$  on the quotient by (0.1) is a proper definition; it can be made independently of which representatives we call on in the definition itself.

**Definition 0.22**

Let  $\Gamma$  be a set of sentences. We define a relation  $\equiv_\Gamma$  by

$$x \equiv_\Gamma y \quad \text{iff} \quad \Gamma \vdash x = y$$

**Proposition 0.23**

For every set  $\Gamma$ , the relation  $\equiv_\Gamma$  is an equivalence relation.

**Lemma 0.24**

The logical system is sound: if  $\Gamma \vdash x = y$ , then  $\Gamma \models x = y$ .

In other words: if  $\Gamma \vdash x = y$ , and the model  $\mathcal{M}$  happens to satisfy all sentences in  $\Gamma$ , then also  $\mathcal{M} \models x = y$ .

*Proof.* We prove this by induction on the proof system E. □

**Theorem 0.25**

The logical system is complete: if  $\Gamma \models x = y$ , then  $\Gamma \vdash x = y$ .

*Proof.* Suppose that  $\Gamma \models x = y$ . Consider the equivalence relation  $\equiv$  on our set **Names** determined by  $\Gamma$ . We define a model  $\mathcal{M}$  in the following way: Let  $M = \mathbf{Names} / \equiv$ . Let  $\llbracket x \rrbracket = [x]$  for a name  $x$ .

We claim that  $\mathcal{M} \models \Gamma$ . To see this, take an equation in  $\Gamma$ , say  $v = w$ . Since this equation is in  $\Gamma$ ,  $v \equiv w$ . So  $[v] = [w]$ . Thus  $\llbracket v \rrbracket = \llbracket w \rrbracket$ . Thus, our equation is satisfied.

Recalling that  $\Gamma \models x = y$ , we see that  $\mathcal{M} \models x = y$ . Thus,  $[x] = \llbracket x \rrbracket = \llbracket y \rrbracket = [y]$ . But this means that  $x \equiv y$ . And now we see that  $\Gamma \vdash x = y$ , by the definition of  $\equiv_\Gamma$ .  $\square$

## 0.5 Exercises

**Exercise 0.1.** We said that a formal proof is a caricature of an informal proof. Many people believe that an informal proof is a caricature of a formal one. What would be the reasons for both positions? Which statement do you think is correct, and what are your reasons?

**Exercise 0.2.** Tell which is bigger without a calculator:

$$\left(-\frac{7}{4}\right)^{-3} \quad \text{or} \quad \left(-\frac{7}{\pi}\right)^{-3}$$

Use monotonicity reasoning.

**Exercise 0.3.** Example 0.14 shows  $2^{op} \times 2$ . Draw  $2 \times 2$ ,  $2 \times 2^{op}$ , and  $2^{op} \times 2^{op}$ .

**Exercise 0.4.** In Example 0.12, we saw that the function  $g(x) = x^2 - 3x + 2$ . We saw that this function was not monotone on the preorder  $\mathbb{N}$ . Is  $g$  a monotone function when considered as a function of the following form?

$$g: [1.5, \infty) \rightarrow \mathbb{R}$$

**Exercise 0.5.** Let  $\mathbb{P} = (P, \leq)$  be any preorder, and let  $\downarrow$  be the function from  $P$  to its power-set preorder  $(\mathcal{P}(P), \subseteq)$  given by

$$\downarrow p = \{q \in P : q \leq p\}$$

Show that  $\downarrow$  is a *monotone* function. That is, show that if  $p \leq q$ , then  $\downarrow p \subseteq \downarrow q$ .

**Exercise 0.6.** Let  $\mathbb{P}$  be the preorder with three points that looks like a  $\vee$ , and let  $\mathbb{Q}$  be  $\mathbb{P}^{op}$ .

1. How many monotone functions are there from  $\mathbb{P}$  to  $\mathbb{Q}$ ?
2. How many monotone functions are there from  $\mathbb{Q}$  to  $\mathbb{P}$ ?
3. Which functions from  $\mathbb{P}$  to itself are both monotone and antitone?

**Exercise 0.7.** Put  $\uparrow$  and  $\downarrow$  notations on all of the occurrences of *bear(s)* and *dance(s)* in the sentences below. It may be that some occurrences can get both notations, and it also may be that some occurrences get no notation at all.

1. Some bears dance.
2. No bears dance.
3. Most bears dance.
4. Not every bear dances.
5. John sees every bear.
6. Mary sees no bear.
7. Any bear in Hawaii would prefer to move to Alaska.
8. If any bear dances, Harry will be happy.
9. If you play loud enough music, any bear will start to dance.
10. Doreen didn't see any bears in her dorm room.

**Exercise 0.8.** Let  $\mathbb{P}$  be the set of sets of people, considered as a preorder in the usual way, under inclusion. Consider the following functions  $f_1, \dots, f_6 : \mathbb{P} \rightarrow \mathbb{P}$ :

$$\begin{aligned}
 f_1(s) &= \{x : x \text{ is bigger than some person in } s\} \\
 f_2(s) &= \{x : x \text{ is bigger than every person in } s\} \\
 f_3(s) &= \{x : x \text{ is bigger than no person in } s\} \\
 f_4(s) &= \{x : x \text{ is bigger than some person not in } s\} \\
 f_5(s) &= \{x : x \text{ is bigger than every person not in } s\} \\
 f_6(s) &= \{x : x \text{ is bigger than no person not in } s\}
 \end{aligned}$$

Which of these are monotone and which are antitone?

**Exercise 0.9.** We build on Proposition 0.15. Let  $g : \mathbb{P} \rightarrow \mathbb{Q}$  and  $f : \mathbb{Q} \rightarrow \mathbb{R}$ .

1. Prove that if  $f$  is monotone and  $g$  is antitone, then  $g \circ f$  is antitone.
2. Prove that if  $f$  is antitone and  $g$  is monotone, then  $g \circ f$  is antitone.
3. Prove that if  $f$  is antitone and  $g$  is antitone, then  $g \circ f$  is monotone.

**Exercise 0.10.** This problem builds on the card stories which we saw earlier. Suppose we deal the three cards face down, and in reality we have the deal  $\spadesuit \clubsuit \heartsuit$ . After the deal,  $A$  turns her  $\spadesuit$  card over and everyone sees it.

1. Find a model (with a labeled real world) that you think represents the situation described in this problem, after  $A$  turns over  $\spadesuit$ .

2. Give two English sentences about knowledge that are intuitively true about the described situation, and each of which is intuitively false about the face down deal, the face up deal, and the private viewing.
3. Translate your sentences to the logical language of Section ???. Your translations may use the boolean connectives  $\neg$ ,  $\wedge$ ,  $\vee$ ,  $\rightarrow$ , and  $\leftrightarrow$ .
4. Check that the translations of your two sentences are true at the real world of your model. (They should be false at the real worlds of the other models, but you don't have to give all the details.)

**Exercise 0.11.** Suppose that we consider  $M_{face\ up}$ , but with  $B$  blind. (Also assume that  $B$ 's blindness is common knowledge to all.) What would the model look like? Be sure to defend your model by mentioning sentences that hold at the real world.

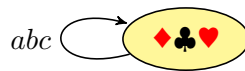
**Exercise 0.12.** Example ?? presents four models. Show that at the real world of all four of these models, each sentence of the following forms is true:

1.  $K_i \varphi \rightarrow \varphi$
2.  $K_i \varphi \rightarrow K_i K_i \varphi$
3.  $K_i \varphi \wedge K_i \psi \rightarrow K_i(\varphi \wedge \psi)$ .

Here  $\varphi$  and  $\psi$  are any sentences in the language that we used for presenting sentences in the first part of this chapter, and  $K_i$  is  $K_a$ ,  $K_b$ , or  $K_c$ . (To do this problem in full would take an inordinate amount of checking. You should do as much explicit checking as it takes to convince yourself that each of the relevant sentences does indeed hold. Hopefully you will also have the intuition and vocabulary to state some general reasons as to *why* these sentences all hold.)

**Exercise 0.13.** This problem is about our representations of the face down and face up deals.

1. What is wrong with the following reasoning about the face down deal of  $\spadesuit\clubsuit\heartsuit$ ? We said that the deal is  $\spadesuit\clubsuit\heartsuit$ , and so it is impossible for it to be anything else. Thus, we should represent it as



2. What do you think about the following reasoning concerning the face up deal? We represented it as six worlds with no connections at all between them. But once we show the deal to be  $\spadesuit\clubsuit\heartsuit$ , all players should discard the other five possible deals. So we should again get the one-world model shown above. And indeed, this model and the previous six-world model agree on all modal sentences. That is, for all sentences  $\varphi$ ,  $\spadesuit\clubsuit\heartsuit \models \varphi$  in the one-world model if and only if  $\spadesuit\clubsuit\heartsuit \models \varphi$  in the six-world model.

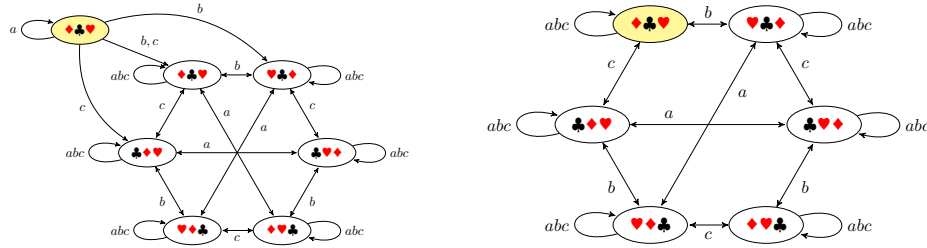
**Exercise 0.14.** The basic language of knowledge used in this chapter is quite spare; the only operators are the three knowledge operators  $K_a$ ,  $K_b$ , and  $K_c$ . Nevertheless, with some circumlocutions, it can express a few additional interesting sentences. For example, to say *Each of the three players has at least one card* we could say

$$(a \heartsuit \vee a \spadesuit \vee a \clubsuit) \wedge (b \heartsuit \vee b \spadesuit \vee b \clubsuit) \wedge (c \heartsuit \vee c \spadesuit \vee c \clubsuit).$$

Find a way to express the following, and write them out in full. (Alternatively, if you use abbreviations, you should explain how yours work. You should not use quantifiers.)

1. Each of the three players has at most one card.
2. Nobody knows that  $B$  has  $\heartsuit$ , except  $B$  himself.
3. If  $B$  knows one of the other two player's cards, then they ( $B$ ) know everyone's card.
4. If any player knows someone else's card, then they know everyone's card.

**Exercise 0.15.** Suppose that after everyone looks privately at their card,  $A$  looks at  $B$ 's card. But she ( $A$ ) does so in such a way that nobody suspects. The correct representation is one of the following two pictures:



The first model adds a world. The second severs the connection for  $a$  from  $\heartsuit, \spadesuit, \clubsuit$  to  $\heartsuit, \spadesuit, \clubsuit$ . Our task is to collect some evidence that will suggest which model is the correct one.

In the real worlds of the two models,  $K_a(b\clubsuit \wedge c\heartsuit)$ , so both models have something going for them. Consider

$B$  knows that  $A$  does not know that he ( $B$ ) has  $\clubsuit$ .

1. Translate the sentence above to logic.
2. Is the English sentence above *intuitively true* about the situation where  $A$  unobtrusively cheats and peeks at  $B$ 's card? (It might be more sensible to replace  $B$  knows with  $B$  believes.)
3. Is the formal sentence true or false in the real world of the first model? How about the second model?
4. Based on your answers above, which model seems like a better representation of the scenario?

**Exercise 0.16.** Let  $A \subseteq \mathbb{R}$  be  $(0, 1) \cup [1, 2) \cup (3, 4] \cup [5, 6]$ .

1. Find  $\Box A$ .
2. Find  $\Box \neg A$ .
3. Find  $\Diamond A$ .
4. Find  $\Diamond \neg A$ .
5. Find  $\Diamond A \wedge \Diamond \neg A$ .

**Exercise 0.17.** Show the following duals to aspects of the proof system pertaining to  $\Box$ :

1.  $\vdash \Diamond \Diamond A \leq \Diamond A$ .
2.  $A \leq B \vdash \Diamond A \leq \Diamond B$ .

**Exercise 0.18.** Fill in the missing steps in Example ?? on page ??.

**Exercise 0.19.** A set  $A$  is called *open* if  $A = \Box A$ .

1. Show (semantically) that the intersection of two open sets is open.
2. Show that  $A = \Box A, B = \Box B \vdash A \wedge B = \Box(A \wedge B)$ .

The second part is a version of the first in our proof theory. The next parts repeat this for unions.

3. Show (semantically) that the union of two open sets is open.
4. Show that  $A = \Box A, B = \Box B \vdash A \vee B = \Box(A \vee B)$ .

**Exercise 0.20.** A set  $A$  is *dense* if  $\Diamond A = \top$ .

1. Check that for a subset  $A \subseteq \mathbb{R}$ ,  $A$  is dense iff for every real number  $r$  and every  $\varepsilon > 0$ ,  $(r - \varepsilon, r + \varepsilon) \cap A \neq \emptyset$ .
2. Use the previous part to check that when  $A$  is the set of rational numbers, then  $\Diamond A = \top$ .
3. Give an example of two dense sets whose intersection is empty.
4. Use the last part to argue that we cannot derive  $\Diamond A = \top, \Diamond B = \top \vdash \Diamond(A \wedge B) = \top$ .
5. Show semantically that if  $A$  is open and dense, and  $B$  is dense, then  $A \wedge B$  is dense.
6. Show syntactically that  $\vdash \Box A = A, \Diamond A = \top, \Diamond B = \top \vdash \Diamond(A \wedge B) = \top$ . [Hints: here are two facts that you might want to use. The hypotheses easily imply  $\vdash \Diamond \Box A = \top$ . Also Example ?? can be used to show that

$$\vdash \Diamond(A \wedge B) \geq \Box A \wedge \Diamond B \geq \Box A.$$

Now apply  $\Diamond$  to both sides of the inequality  $\vdash \Diamond(A \wedge B) \geq \Box A$ . ]



**Exercise 0.21.** This problem presents a difference between the logic of knowledge as we saw it in Section ?? and the logic of the interior operator in Section ??.

1. As in Exercise 0.12, show that each sentence of the form  $\varphi \rightarrow K_i P_i \varphi$  is true at the real worlds of the models in Example ??.
2. Show that there is a set  $A \subseteq \mathbb{R}$  such that  $A$  is not a subset of  $\Box \Diamond A$ .
3. Infer from the last part that  $A \leq \Box \Diamond A$  is not provable in the logical system of Section ??.

**Exercise 0.22.** This exercise puts together the two sections of the chapter. Suppose we take an assertion in the logic of the interior operation, change  $\Box$  to  $K_a$ , replace the sentence letters  $A, B, \dots$ , by any expressions in the logic of knowledge, and then replace  $\leq$  by  $\rightarrow$ , and also  $=$  by  $\leftrightarrow$ . Show that the resulting sentence is true at the real worlds of the models in Example ??.

# Chapter 1

## $\mathcal{A}$ : the Logic of All $p$ are $q$

This book studies a variety of logical systems, starting in this chapter with what must be the simplest logical system “of all,” and then adding more and more features to it. The logical language has only one kind of sentence: All  $p$  are  $q$ . That’s it. No “if”s, “and”s, or “but”s. So the *syntax* here is going to be very small. We will then see a *semantics* consisting of a short list of definitions that explain the meaning of the language in *models* and some related points. And then we turn to a *proof system*.

If you have seen other presentations of logic, you might think that the subject matter of this chapter is boring or uninteresting. We encourage you to take the chapter as something to master and find interesting even though it might be simple. We also hope that you will return to it as you read further in the book, since we want to use the chapter as a first example of the many logical systems that we shall see in later chapters.

### 1.1 Syntax and semantics

All of our logical systems have a *syntax* and a *semantics*. We will discuss what these terms mean later, but for now, let us say that syntax is about the form of sentences, and semantics is about meaning.

For the syntax of our first system, we start with a collection  $\mathbf{P}$  of *nouns*. The elements of  $\mathbf{P}$  might be actual nouns (usually we use plural nouns for animals). But our nouns could be anything. The set  $\mathbf{P}$  might be finite or infinite. We write the nouns as  $p, q, \dots$ ; occasionally we use subscripts or other devices. In the exercises at the end of the chapter (see page 37), we use actual animal names.

#### Definition 1.1: the language $\mathcal{A}$

We take as *sentences* the expressions

$$\text{All } p \text{ are } q$$

where  $p$  and  $q$  are any nouns in  $\mathbf{P}$ . There is nothing else in the language. We call this language  $\mathcal{A}$ .

$\mathcal{A}$  stands for “all.” Note that  $\mathcal{A}$  is really a family of languages, one for each set  $\mathbf{P}$  of nouns at

the outset. This dependence on primitives is true for practically all logical systems. For most of the work in this book, we suppress mention of the primitive syntactic items because it makes the notation lighter and because we rarely need to call attention to the primitives in the first place.

### Example 1.2: a logical language

Suppose that  $\mathbf{P} = \{n, p, q\}$ . In this case, the language  $\mathcal{A}$  would have exactly nine sentences:

|                 |                 |                 |
|-----------------|-----------------|-----------------|
| All $n$ are $n$ | All $p$ are $n$ | All $q$ are $n$ |
| All $n$ are $p$ | All $p$ are $p$ | All $q$ are $p$ |
| All $n$ are $q$ | All $p$ are $q$ | All $q$ are $q$ |

### Definition 1.3: models

A model  $\mathcal{M}$  for this language  $\mathcal{A}$  is a pair

$$\mathcal{M} = (M, \llbracket \cdot \rrbracket)$$

consisting of a set  $M$ , together with an *interpretation function*

$$\llbracket \cdot \rrbracket : \mathbf{P} \rightarrow \mathcal{P}(M)$$

which assigns to each and every noun  $p \in \mathbf{P}$  a subset  $\llbracket p \rrbracket \subseteq M$ . We read  $\llbracket p \rrbracket$  as “the meaning of  $p$  in the model  $\mathcal{M}$ .”

$M$  is called the *universe* of  $\mathcal{M}$ . Sometimes we use a letter like  $N$  for a model, and in that case we would of course write the universe as  $N$ . The difference between  $M$  and  $\mathcal{M}$  is that the former is a set, and the latter is a *structure*, a set together with an interpretation function. But frequently we name a model by its universe. And when we do that, we blur the notational difference between  $M$  and  $\mathcal{M}$ .

### Definition 1.4: truth in a model

The main semantic definition is *truth in a model*:

$$\mathcal{M} \models \text{All } p \text{ are } q \quad \text{iff} \quad \llbracket p \rrbracket \subseteq \llbracket q \rrbracket.$$

We read this notation in various ways, such as  $\mathcal{M}$  *satisfies* All  $p$  are  $q$ , or All  $p$  are  $q$  is *true* in  $\mathcal{M}$ .

If  $\mathcal{M}$  does not satisfy All  $p$  are  $q$ , we would write  $\mathcal{M} \not\models \text{All } p \text{ are } q$ . And of course we would also say that All  $p$  are  $q$  is *false* in  $\mathcal{M}$ .

**Example 1.5: how the semantics works**

We exemplify the semantics using the language from Example 1.2. Let  $M = \{1, 2, 3, 4, 5\}$ . Let  $\llbracket n \rrbracket = \emptyset$ ,  $\llbracket p \rrbracket = \{1, 3, 4\}$ , and  $\llbracket q \rrbracket = \{1, 3\}$ . This is all we need to specify a model. We'll call this model  $\mathcal{M}$ .

The following sentences are true in  $\mathcal{M}$ : All  $n$  are  $n$ , All  $n$  are  $p$ , All  $n$  are  $q$ , All  $p$  are  $p$ , All  $q$  are  $p$ , and All  $q$  are  $q$ . The other three sentences in  $\mathcal{A}$  are false in  $\mathcal{M}$ .

Let us emphasize that the first two of our example sentences are true in  $\mathcal{M}$ . For this, we use the fact that *the empty set  $\emptyset$  is a subset of every set*.

As soon as we have a formal semantics for a logical system, we get two further notions. These are used with every logic in this book, and we have seen them already in Definition 0.3 on page 6.

In this chapter,  $\Gamma$  denotes a set of sentences in  $\mathcal{A}$ . We sometimes call  $\Gamma$  a set of *assumptions*, or a *theory*, following the usage in modern logic.

**Example 1.6: a semantic consequence**

$$\{\text{All } p \text{ are } q, \text{All } n \text{ are } p\} \models \text{All } n \text{ are } q.$$

To justify what we say in Example 1.6 just above, we give a straightforward mathematical proof.<sup>1</sup> Let  $\mathcal{M}$  be any model for  $\mathcal{A}$ , assuming that the underlying set  $\mathbf{P}$  contains  $n$ ,  $p$ , and  $q$ . Assume that  $\mathcal{M}$  satisfies All  $p$  are  $q$  and All  $n$  are  $p$ . We must prove that  $\mathcal{M}$  also satisfies All  $n$  are  $q$ . From our first assumption,  $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket$ . From our second,  $\llbracket n \rrbracket \subseteq \llbracket p \rrbracket$ . It is a general fact about sets that the inclusion relation (written as  $\subseteq$  here) is *transitive*, and so we conclude that  $\llbracket n \rrbracket \subseteq \llbracket q \rrbracket$ . This verifies that indeed  $\mathcal{M}$  satisfies All  $n$  are  $q$ . And since  $\mathcal{M}$  was arbitrary, we are done.

Incidentally, when  $\Gamma$  is a finite set that is written out explicitly on the left side of a  $\models$  or  $\vdash$  symbol, we usually omit the set braces  $\{$  and  $\}$  from the notation. So instead of writing what we saw in Example 1.6, we would usually write

$$\text{All } p \text{ are } q, \text{All } n \text{ are } p \models \text{All } n \text{ are } q.$$

**Example 1.7: a failure of semantic consequence**

$$\text{All } p \text{ are } q \not\models \text{All } q \text{ are } p.$$

To show that a given set  $\Gamma$  does not logically entail another sentence  $\varphi$ , we need to build a model  $\mathcal{M}$  of  $\Gamma$  which is not a model of  $\varphi$ . In this example,  $\Gamma$  is  $\{\text{All } p \text{ are } q\}$ , and  $\varphi$  is All  $q$  are  $p$ . We can get a model  $\mathcal{M}$  that does the trick by setting  $M = \{1, 2\}$ ,  $\llbracket p \rrbracket = \{1\}$ , and  $\llbracket q \rrbracket = \{1, 2\}$ . For that matter, we could also use a different model, say  $\mathcal{N}$ , defined by  $N = \{61\}$ ,  $\llbracket p \rrbracket = \emptyset$ , and  $\llbracket q \rrbracket = \{61\}$ .

<sup>1</sup>Our study of logic will involve *formal proofs* done inside a logical system and also *informal proofs* done as they usually are in mathematics. We'll see soon what formal proofs are all about. Frequently people assume that logic is done using formal proofs, but this isn't true. Proofs are much more readable when done informally and this is what we prefer to do.

**Example 1.8: Challenge**

Here is a more complicated example. Let

$$\Gamma = \left\{ \begin{array}{lll} \text{All } j \text{ are } k, & \text{All } j \text{ are } l, & \text{All } k \text{ are } l, \\ \text{All } l \text{ are } k, & \text{All } l \text{ are } m, & \text{All } k \text{ are } n, \\ \text{All } m \text{ are } q, & \text{All } p \text{ are } q, & \text{All } q \text{ are } p \end{array} \right\}$$

True or false:  $\Gamma \models \text{All } p \text{ are } n$ ? If true, give a reason. If false, give a model of  $\Gamma$  where  $\llbracket p \rrbracket \not\subseteq \llbracket n \rrbracket$ .

The point is not to simply solve this particular problem, but to give an *algorithm* which would solve all problems of this type, and to prove your answer. We'll do this in Section 1.6.

## 1.2 Proof system

Up until now, we have dealt with *semantic concepts*, concepts defined in terms of models. At this point, we move to the proof system. But before we do this, we want to say a few words about syntax and semantics.



Because the discussion is controversial, and potentially tricky, we mark it off with a “dangerous bend” sign. The basic difference between syntax and semantics is that semantic concepts are supposed to relate a formal language to abstract entities that stand in for the real world, and they are supposed to be models of human understanding, while syntactic concepts may be understood in a purely formal way, without any reference to meaning.

Definition 1.9 presents a *proof system*<sup>2</sup> for the language  $\mathcal{A}$ . The name of the proof system is  $\mathbf{A}$ . (This is the same letter as the language, but in a different font.)

This proof system  $\mathbf{A}$  has two *inference rules*. Each rule allows conclusions to be inferred from some set of premises. The conclusion of a rule is the sentence below the horizontal line, and the premise(s) are above the line.

Let us discuss the second rule, (BARBARA) first. The rule says that *All  $p$  are  $q$*  may be inferred from the two premises *All  $p$  are  $n$*  and *All  $n$  are  $q$* . That is, a node in a tree may be labeled *All  $p$  are  $q$*  provided it has two nodes above it in the tree, one labeled *All  $p$  are  $n$*  for some noun  $n$ , and the other labeled *All  $n$  are  $q$* .

The first rule, named (AXIOM), says that *All  $p$  are  $p$*  may label a node in a proof tree provided that node have *no nodes above it*. So no premises are required, and none may appear.

The name (BARBARA) comes from medieval logic. The reason for this is that logicians used  $\mathcal{A}$  for universal assertions. (Of course, they did not use English, so it is purely a coincidence that  $\mathcal{A}$  stands for *All*.) The rule has three  $\mathcal{A}$ s. Classical logicians interpolated consonants “randomly” to get the names of their rules, hence BARBARA. Our rule (AXIOM) is so-named because an axiom in a proof system is like a rule of inference with no premises.

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<sup>2</sup>We also call this a *logical system*.

**Definition 1.9: rules of the proof system A for  $\mathcal{A}$** 

$$\frac{}{\text{All } p \text{ are } p} \text{ AXIOM} \qquad \frac{\text{All } p \text{ are } n \quad \text{All } n \text{ are } q}{\text{All } p \text{ are } q} \text{ BARBARA}$$

**Definition 1.10: proof tree over a set  $\Gamma$** 

1. If the sentence All  $x$  are  $y$  belongs to  $\Gamma$ , then we have a *one-point proof tree*

$$\text{All } x \text{ are } y$$

This one point is both the root of the tree, and also the (only) leaf. We do not include either of our rules in this tree.

2. For all nouns  $x$ , the the one-point tree

$$\frac{}{\text{All } x \text{ are } x} \text{ AXIOM}$$

is a proof tree over  $\Gamma$ . As above, this is a one-point proof tree, and its point All  $x$  are  $x$  is both the root and the leaf of the tree.

3. Suppose that  $\mathcal{T}_1$  and  $\mathcal{T}_2$  are proof trees over  $\Gamma$ , and that there are nouns  $x$ ,  $y$ , and  $z$  such that the roots of  $\mathcal{T}_1$  and  $\mathcal{T}_2$  are All  $x$  are  $y$  and All  $y$  are  $z$ , respectively. Let  $\mathcal{T}$  be the proof tree shown below:

$$\frac{\begin{array}{c} \vdots \\ \text{All } x \text{ are } y \end{array} \quad \begin{array}{c} \vdots \\ \text{All } y \text{ are } z \end{array}}{\text{All } x \text{ are } z} \text{ BARBARA} \quad (1.1)$$

(We mean that  $\mathcal{T}_1$  is the subtree on the left, where the  $\vdots$  are shown, and  $\mathcal{T}_2$  is the analogous proof tree on the right.) Then  $\mathcal{T}$  is a proof tree over  $\Gamma$ .

We write  $\Gamma \vdash \varphi$  to mean that there is a proof tree over  $\Gamma$  whose root is  $\varphi$ . We read this as  $\Gamma$  *proves*  $\varphi$ , or  $\Gamma$  *derives*  $\varphi$ , or that  $\varphi$  *follows from*  $\Gamma$  *in our proof system A*.

**Example 1.11: an example of a proof tree**

Let  $\Gamma$  be

$$\{\text{All } l \text{ are } m, \text{All } q \text{ are } l, \text{All } m \text{ are } p, \text{All } n \text{ are } p, \text{All } l \text{ are } q\}$$

Let  $\varphi$  be All  $q$  are  $p$ .

We show in (1.2) below a proof tree showing that  $\Gamma \vdash \varphi$ .

We take

$$\begin{array}{c}
 \frac{\frac{\text{All } l \text{ are } m \quad \overline{\text{All } m \text{ are } m}}{\text{All } l \text{ are } m} \text{ BARBARA} \quad \text{All } m \text{ are } p}{\text{All } l \text{ are } p} \text{ BARBARA} \\
 \frac{\text{All } q \text{ are } l \quad \text{All } l \text{ are } p}{\text{All } q \text{ are } p} \text{ BARBARA}
 \end{array} \quad (1.2)$$

You might like to check that what we have in (1.2) really is a proof tree according to our definition. That is, all of the non-axiom leaves of our proof tree belong to  $\Gamma$ . Except for the leaves, the nodes of the tree all are applications of (BARBARA). The complete verification would be a fairly long and very boring to read.

Proof trees come with the information of which rule is used at the various nodes of the tree, the way we do it in (1.2). But frequently we drop all the rules, in order to fit wider trees on the page. So our proof tree would look like

$$\begin{array}{c}
 \frac{\frac{\text{All } l \text{ are } m \quad \text{All } m \text{ are } m}{\text{All } l \text{ are } m} \quad \text{All } m \text{ are } p}{\text{All } l \text{ are } p} \\
 \frac{\text{All } q \text{ are } l \quad \text{All } l \text{ are } p}{\text{All } q \text{ are } p}
 \end{array} \quad (1.3)$$

Finally, note also that some sentences from  $\Gamma$  are not used as leaves. This is permitted according to our definition. Also, there is a smaller proof tree that also shows that  $\Gamma \vdash \varphi$ : we could drop *All  $m$  are  $m$* . (The reason why have the rule (AXIOM) is so that that we can have one-element trees labeled with sentences of the form *All  $l$  are  $l$* .)



We now have a proof system  $\mathbf{A}$  for the language  $\mathcal{A}$  of this chapter. As suggested by the two different fonts in the letters  $\mathbf{A}$  and  $\mathcal{A}$ , the language does *not* automatically come with the proof system. The proof system is an “extra.” For that matter, there could well be two different proof systems for the same language. As an artificial example, we could take our system  $\mathbf{A}$  and add a rule that is redundant. For example, we could add

$$\frac{\text{All } w \text{ are } x \quad \text{All } x \text{ are } y \quad \text{All } y \text{ are } z}{\text{All } w \text{ are } z} \text{ NEW} \quad (1.4)$$

Then we would have a *different proof system*. You might be interested to work Exercise 1.12 about the new proof system.

**Soundness and completeness: the definitions** Earlier, we defined the semantics of  $\mathcal{A}$ , using models. From the models and the definition of truth in a model, we get the derived semantic notion  $\Gamma \models \varphi$  as in Definition 0.3. The main work in this chapter connects the semantic notion “ $\Gamma \models \varphi$ ” with the proof-theoretic notion “ $\Gamma \vdash \varphi$ .” This connection happens in two directions. The main definitions pertaining to this connection are found in Definition 1.12 just below. Like much of the work in this chapter, these definitions will be used in all of the logical system we shall see.

Suppose we are given a logical language and a semantics for it, defining a notion  $\Gamma \models \varphi$ . Suppose we also have a proof system for the same language, defining a notion  $\Gamma \vdash \varphi$ .

**Definition 1.12: soundness and completeness**

The proof system is *sound for the semantics* if whenever  $\Gamma \vdash \varphi$ , we also have  $\Gamma \models \varphi$ .

The proof system is *complete for the semantics* if whenever  $\Gamma \models \varphi$ , we also have  $\Gamma \vdash \varphi$ .

We shall prove the soundness in the next section, see Proposition 1.14; the completeness will come later in the chapter. The techniques involved in the two results are rather different.

### 1.3 Induction on proof trees

Our definition of proof trees for  $\mathbf{A}$  was *inductive*: it started with the most basic proof trees (one-point trees) and then told us how to take some given proof trees and build a new proof tree from them. Anytime a class of objects is defined by an inductive definition, then the main tool used to prove properties of those objects is *proof by induction*. This is a general mathematical fact, more of a truism than a theorem, and one worth knowing.

This section is devoted to exploring this issue with proof trees. We defined proof trees over  $\Gamma$  in Definition 1.10. The definition was *inductive*. There is a matching principle of *proof by induction on proof trees over  $\Gamma$*  is stated in Definition 1.15.

We are not going to discuss the general theory of inductive definitions and their matching induction principles in this book; that would take us rather afar from our goals. So this chapter will mainly see examples. The results and exercises that use induction have two overall goals. The first general learning goal is that by seeing many examples of induction, you will be able to see your way to similar applications in new settings. The second, and more specific goal, is that we need induction to prove the soundness of all of the proof systems in the book.

Before we turn to the soundness result, we present a simpler example of proof by induction on proof trees.

#### Proposition 1.13

For all proof trees  $\mathcal{T}$  over  $\Gamma$ , every noun which occurs in  $\mathcal{T}$  also occurs in some leaf of  $\mathcal{T}$ .

*Proof.* This is our first proof by induction, and so we are going to be much more expansive with the details than one usually is with this kind of proof. We want to show

For all proof trees  $\mathcal{T}$  over  $\Gamma$ ,  
every noun which occurs in  $\mathcal{T}$  also occurs in some leaf of  $\mathcal{T}$ .

We go through the steps in Definition 1.15. First, let's take a sentence in  $\Gamma$ , say *All  $u$  are  $v$* . We have a one-point proof tree over  $\Gamma$ , namely

$$\frac{}{\text{All } u \text{ are } v} \Gamma$$

Let us call this tree  $\mathcal{T}$  for a moment. We need to show that every noun which occurs in  $\mathcal{T}$  also occurs in some leaf of  $\mathcal{T}$ . The nouns which occur in this tree are  $u$  and  $v$ . There is one leaf of this tree, namely the node itself. And clearly every noun which occurs in the tree occurs in the leaf.

Our second step concerns a one-point proof tree over  $\Gamma$  justified by the rule (AXIOM). To be concrete, let's say we have

$$\frac{}{\text{All } k \text{ are } k} \text{AXIOM}$$

Now let us call *this* tree  $\mathcal{T}$ . We need to show that every noun which occurs in  $\mathcal{T}$  also occurs in some leaf of  $\mathcal{T}$ . There is just one noun which occur in this tree,  $k$ . There is one leaf of this tree, namely



the node itself. And clearly  $k$  occurs in the leaf. So every noun which occurs in the tree occurs in the leaf.

We turn to the induction step. Let  $\mathcal{T}$  be a tree justified by (BARBARA) at the root, with subtrees of the root being  $\mathcal{T}_1$  and  $\mathcal{T}_2$ . Assume about  $\mathcal{T}_1$  that every noun which occurs in it occurs in one of its leaves, and assume the same thing about  $\mathcal{T}_2$ . Now take a noun  $n$  which occurs in  $\mathcal{T}$ . It must occur in either the root of  $\mathcal{T}$ , or in  $\mathcal{T}_1$ , or in  $\mathcal{T}_2$ . If  $n$  occurs in  $\mathcal{T}_1$ , then it occurs in a leaf of  $\mathcal{T}_1$ . (This exactly is where we use our assumption about  $\mathcal{T}_1$ .) Every leaf of  $\mathcal{T}_1$  is also a leaf of  $\mathcal{T}$ , and we are done in this case. If  $n$  occurs in  $\mathcal{T}_2$ , the same argument shows that it occurs in some leaf of  $\mathcal{T}$ . We are left with the case that  $n$  occurs in the root. So the root is either All  $n$  are  $m$ , or All  $m$  are  $n$ . We only discuss the first case, since the second is similar. The left node above the root of  $\mathcal{T}$  is of the form All  $n$  are  $l$  for some  $p$ . (This is how *barbara* works.) In particular,  $n$  occurs in  $\mathcal{T}_1$ . So  $n$  occurs as a leaf of  $\mathcal{T}_1$  (again we use the induction hypothesis), and hence as a leaf of  $\mathcal{T}$ .

This completes the induction step, and hence the proof overall.  $\square$

**Soundness of the proof system A** Our main goal in this section is to prove the soundness of the A with respect to the semantics. This proof uses *induction*, and so we turn to that topic. To prove something about all proof trees over a set  $\Gamma$ , we use *induction on proof trees over  $\Gamma$* <sup>3</sup>.

We'll see examples of induction on proof trees in Proposition 1.14 and in Exercises 1.5 and 1.13.

#### Proposition 1.14: Soundness

If  $\Gamma \vdash \varphi$ , then  $\Gamma \models \varphi$ .

*Proof.* We prove this by induction on proof trees over  $\Gamma$  as stated in Definition 1.15. We take  $\underline{\mathcal{T}}$  to be the assertion:

$$\text{if the root of } \mathcal{T} \text{ is labeled } \varphi, \text{ then every } \mathcal{M} \models \Gamma \text{ also satisfies } \varphi \quad (1.5)$$

First, we show that  $\underline{\mathcal{T}_1}$  when  $\mathcal{T}$  is a one-point tree labeled with a sentence from  $\Gamma$ . In this case,  $\varphi$  belongs to  $\Gamma$ . And so every model  $\mathcal{M}$  of *all* sentences in  $\Gamma$  is a fortiori a model of  $\varphi$ . Second, we show that  $\underline{\mathcal{T}_1}$  when  $\mathcal{T}$  is proof tree over  $\Gamma$  whose root is justified by (AXIOM). In this case, the root of  $\mathcal{T}$  is a sentence of the form All  $p$  are  $p$ . (Also, in this case, the entire tree is the root. But this is irrelevant.) Every sentence All  $p$  are  $p$  is true in all models  $\mathcal{M}$  whatsoever, regardless of whether  $\mathcal{M}$  satisfies  $\Gamma$  or not. So  $S(\mathcal{T})$  holds in this case.

Now let  $\mathcal{T}^*$  be a tree where (BARBARA) is used at the root. Let's say that the root of  $\mathcal{T}^*$  is All  $n$  are  $q$ , and the two principal subtrees have as roots All  $n$  are  $p$  and All  $p$  are  $q$ , respectively. We assume that every model of  $\Gamma$  satisfies All  $n$  are  $p$ , and also that every model of  $\Gamma$  satisfies All  $p$  are  $q$ . It is our task to show that every model of  $\Gamma$  satisfies All  $n$  are  $q$ . Fix a model  $\mathcal{M}$  of  $\Gamma$ . Then in  $\mathcal{M}$ ,  $\llbracket n \rrbracket \subseteq \llbracket p \rrbracket$ , and also  $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket$ . Continuing to think about this fixed model, we also have  $\llbracket n \rrbracket \subseteq \llbracket q \rrbracket$ . But  $\mathcal{M}$  was an arbitrary model of  $\Gamma$ . So we have shown that every model of  $\Gamma$  satisfies the root of  $\mathcal{T}$ . That is,  $\underline{\mathcal{T}}$  holds.

<sup>3</sup>If you don't know about induction, you might wish to consult some other sources to learn about induction in other contexts, especially *induction on numbers*. Proofs by induction will only play a minor role in what we do, so if you don't mind taking a few results on faith, it would be possible to continue reading, forgetting induction completely. However, the settings that we have are simple enough that you might even be able to pick up induction from the rest of the book.

**Definition 1.15: induction principle for proof trees over  $\Gamma$** 

Suppose that we wish to prove a sentence of the form

$$\text{For all proof trees } \mathcal{T} \text{ for the logic } \mathcal{A} \text{ over } \Gamma, \text{ } \underline{\hspace{2cm}} \mathcal{T} \underline{\hspace{2cm}} \quad (1.6)$$

That is, a sentence about proof trees over  $\Gamma$  that uses  $\mathcal{T}$  as a variable over such trees. Here is a step-by-step procedure that does this.

1. Prove  $\underline{\hspace{2cm}} \text{All } x \text{ are } y \underline{\hspace{2cm}}$ , whenever the sentence All  $x$  are  $y$  belongs to  $\Gamma$ . (That is, we prove our statement for a one-point tree  $\mathcal{T}$  over  $\Gamma$ , a one-point tree labeled with a sentence from  $\Gamma$ .)
2. Prove  $\underline{\hspace{2cm}} \text{All } x \text{ are } x \underline{\hspace{2cm}}$ . (That is, we prove our statement for a one-point tree  $\mathcal{T}$  justified by (AXIOM).)
3. Let  $\mathcal{T}$  be a tree justified by (BARBARA) as shown in (1.1). We assume  $\underline{\hspace{2cm}} \mathcal{T}_1 \underline{\hspace{2cm}}$  and  $\underline{\hspace{2cm}} \mathcal{T}_2 \underline{\hspace{2cm}}$ , and then on the basis of these assumptions, prove  $\underline{\hspace{2cm}} \mathcal{T} \underline{\hspace{2cm}}$ .

Once we have done these steps, we have proved (1.6) by *induction on proof trees over  $\Gamma$* .

The paragraphs above show that for all proof trees  $\mathcal{T}$  over  $\Gamma$ , every model of  $\Gamma$  satisfies the root of  $\mathcal{T}$ . Put another way, if  $\Gamma \vdash \varphi$ , then  $\Gamma \models \varphi$ .  $\square$

**Remark 1.16: on soundness proofs**

The soundness proofs of all the logical systems in this book are all pretty much the same as the one in Proposition 1.14. They are always inductions, and the crux of the matter is usually a simple fact about sets. (In Proposition 1.14, the crux of the matter is that the inclusion relation  $\subseteq$  on subsets of a given set is always a transitive relation.) We almost never present the soundness proof in any detail.

Proposition 1.14 tells us that the formal logical system for  $\mathcal{A}$  is not going to give us any bad results. (This is what *soundness* ought to mean.) Now this is a fairly weak point. If we dropped some of the rules, it would still hold. Even if we decided to be conservative and say that  $\Gamma \vdash \varphi$  *never* holds, Proposition 1.14 would still hold. So the more interesting question to ask is whether the logical system is *strong* enough to prove everything it *should prove*. We want to know if  $\Gamma \models \varphi$  implies that  $\Gamma \vdash \varphi$ . If this implication does hold for all  $\Gamma$  and  $\varphi$ , then we say that our system is *complete*.

## 1.4 Graphs

At this point, we have presented a language called  $\mathcal{A}$ , a semantics for it, and a sound proof system to go along with the semantics. The main theoretical work is to show that the system is complete: if  $\Gamma \models \varphi$ , then  $\Gamma \vdash \varphi$ . We also discuss the algorithmic properties of the system. That is, if someone

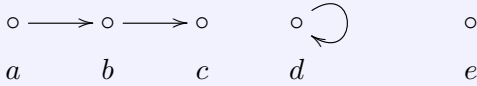
gives us a (finite) set  $\Gamma$  and a sentence  $\varphi$ , how can we determine whether or not  $\Gamma \vdash \varphi$ ? We shall study these soon, but first we need some preliminary notions from general mathematics.

**Definition 1.17: graph**

A *graph* is a pair  $\mathcal{G} = (G, \rightarrow)$ , where  $G$  is a set and  $\rightarrow$  is a relation on  $G$ . The elements of  $G$  are called *nodes*, *vertices*, or *points*.

If  $g$  and  $h$  are nodes of  $\mathcal{G}$ , we usually write  $g \rightarrow h$  to mean that  $g$  and  $h$  are related by the relation  $\rightarrow$ . But sometimes the style of exposition dictates other notation, such as  $(g, h) \in \rightarrow$ . Alternately, we might say that  $g$  is the *parent* of  $h$ . The relation is also called the *edge relation* of the graph. There is no requirement on this relation:  $G$  might have *loops* (i.e., we might have  $g \rightarrow g$  for some, or even all nodes of  $G$ ). But a graph need not have any loops at all. (For that matter, a graph need not have *any* nodes at all! The empty graph is perfectly fine for us. But as with all similar situations, other authors may differ on this point of usage.)

**Example 1.18: a graph**



Technically, we have drawn a picture of a graph  $\mathcal{G}$ , where the node set  $G$  is  $\{a, b, c, d, e\}$ , and the edge relation  $\rightarrow$  is  $\{(a, b), (b, c), (d, d)\}$ .

**Definition 1.19: the all-graph of a set**

For each set  $\Gamma$  of sentences of  $\mathcal{A}$  and each set  $S$  of nouns, we get a graph  $\mathcal{G}_{\Gamma, S}$ . The nodes of  $\mathcal{G}_{\Gamma, S}$  are the elements of the given set  $S$ , and the relation  $\rightarrow$  is defined by

$$x \xrightarrow{\Gamma} y \quad \text{iff} \quad \text{the sentence All } x \text{ are } y \text{ belongs to } \Gamma$$

We call this the *all-graph of  $\Gamma$  on  $S$* . Graphs of this form are the main reason why we introduce graphs in this section.

There are two special cases of this construction which interest us. One is where  $S = \mathbf{P}$ , the set of all nouns. The second is where we have a set  $\Gamma$  and another sentence  $\varphi$ , and  $S$  is the set of all nouns which occur in  $\Gamma$  or in  $\varphi$ .

**Definition 1.20:  $g \xrightarrow{*} h$  in a graph**

Let  $\mathcal{G}$  be a graph, and let  $g$  and  $h$  be points in  $G$ .

A *path* from  $g$  to  $h$  is a finite sequence of points in the graph, each connected to the next by an edge.

We write  $g \xrightarrow{*} h$  to mean that there is a path from  $g$  to  $h$  in  $\mathcal{G}$ .

The relation  $\xrightarrow{*}$  is called the *reflexive-transitive closure* of the graph relation  $\rightarrow$ .

For example, in the graph  $\mathcal{G}$  in Example 1.18, we have  $a \xrightarrow{*} c$ , using the path  $a \rightarrow b \rightarrow c$ . But paths can be of length 1 or even 0. The full listing of  $\xrightarrow{*}$  is

$$a \xrightarrow{*} b \quad b \xrightarrow{*} c \quad a \xrightarrow{*} c \quad a \xrightarrow{*} a \quad b \xrightarrow{*} b \quad c \xrightarrow{*} c \quad d \xrightarrow{*} d \quad e \xrightarrow{*} e$$

When  $g = h$ , we have an *empty path* from  $g$  to  $h$ . It has no edges. So we always have  $g \xrightarrow{*} g$ , for all points  $g$  in all graphs.

**Trees** We defined proof trees for our logic in Section 1.2. But we neglected at that point to say what *trees* actually are.

**Definition 1.21: tree**

A *tree* is a graph with a designated node called the *root* with the property that every node is reachable from the root by a unique path. A *leaf* of a tree is a node with no successors (no outgoing edges).

There is a natural way of turning every graph  $\mathcal{G}$  into a preorder  $\mathbb{P}$ .

**Proposition 1.22**

For any graph  $(\mathcal{G}, \rightarrow)$ , the pair  $(\mathcal{G}, \xrightarrow{*})$  is a preorder.

*Proof.* The fact that we allow paths to have length 0 tells us that for each point  $g$  in  $\mathcal{G}$ , we have  $g \xrightarrow{*} g$ . Thus,  $\xrightarrow{*}$  is reflexive. For the transitivity, suppose that  $g \xrightarrow{*} h$  and  $h \xrightarrow{*} i$ . Then there is a path from  $g$  and  $h$ , and another path from  $h$  to  $i$ . Chaining the paths gives a path from  $g$  to  $i$ , showing that  $g \xrightarrow{*} i$ .  $\square$

We need one more general definition concerning preorders.

**Definition 1.23: down-set**

Let  $\mathbb{P}$  be a preorder, and let  $p \in P$ . Then we define the *down-set* of  $p$ , written  $\downarrow p$ , to be

$$\downarrow p = \{q \in P : q \leq p\}.$$

We read  $\downarrow p$  as “the down-set of  $p$ .” The reason for the name is that in the Hasse diagram of the preorder, the down-set of an element is the set of points below (or equivalent to) that element.

For example, in the preorder shown on page 8,

$$\begin{aligned} \downarrow a &= \{a\} \\ \downarrow b &= \{a, b, c\} \\ \downarrow e &= \{a, b, c, d, e, f, g\} \end{aligned}$$

## 1.5 Completeness

Our work on trees and preorders in the last section was a digression from our main work in this part of the chapter, the completeness theorem for the logical system  $\mathbf{A}$ .

**Definition 1.24:**  $u \leq_{\Gamma} v$ 

Let  $\Gamma$  be a set of sentences in  $\mathcal{A}$ , or in any logical language containing  $\mathcal{A}$ . For nouns  $u$  and  $v$ , we write  $u \leq_{\Gamma} v$  to mean that

$$\Gamma \vdash \text{All } u \text{ are } v.$$

As always, we simplify the notation by dropping the subscript  $\Gamma$  if it is clear from the context.

**Proposition 1.25**

For all sets  $\Gamma$ ,  $(\mathbf{P}, \leq_{\Gamma})$  is a preorder.

*Proof.* To check that  $\leq$  is reflexive, let  $u \in \mathbf{P}$ . Then we have a one-line proof of All  $u$  are  $u$ . The justification is the rule (AXIOM). Indeed, this is the point of including (AXIOM) in the proof system. For the transitivity, we put together a proof of  $u \leq v$  with a proof of  $v \leq w$  to show that  $u \leq w$ .  $\square$

We are going to use the fact that  $\leq$  is a preorder frequently in this book, and often without explicitly mentioning it.

We now have an important model construction. Starting with a set  $\mathbf{P}$  of nouns and a theory  $\Gamma$  in our language  $\mathcal{A}$ , we build a model  $\mathcal{M}$  *from the syntax and the proof theory*.

**Definition 1.26:** the canonical model of a set  $\Gamma$ 

$$\begin{aligned} M &= \mathbf{P} \\ \llbracket u \rrbracket &= \downarrow u \\ &= \{v \in \mathbf{P} : \Gamma \vdash \text{All } v \text{ are } u\} \end{aligned}$$

It is important to see that second equals sign in the definition of  $\llbracket u \rrbracket$  above is a consequence of down-sets in general from Definition 1.23, and also of the definition of the special preorder  $\leq_{\Gamma}$  that we used in the preorder  $\mathbf{P}$ .

Here is a verbal description of the model: we are taking the set of nouns underlying the language  $\mathcal{A}$  to be the universe of our model, and we interpret a given noun  $u$  as the set of all nouns  $v$  which we can prove to be included in  $u$ , using  $\Gamma$  as a set of assumptions.  $\mathcal{M}$  is called the *canonical model* of  $\Gamma$ .

**Lemma 1.27**

The canonical model of  $\Gamma$  satisfies  $\Gamma$ .

*Proof.* Suppose All  $p$  are  $q$  belongs to  $\Gamma$ . Then  $p \leq q$ . We must show that  $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket$ . Let  $v \in \llbracket p \rrbracket$ . Then by Definition 1.26,  $v \leq p$ . Since  $\leq$  is transitive,  $v \leq q$ . That is,  $v \in \llbracket q \rrbracket$ . This for all  $v$  shows our result.  $\square$

**Theorem 1.28: Completeness**

The logic  $\mathcal{A}$  of Definition 1.9 is complete for  $\mathcal{A}$ : If  $\Gamma \models \text{All } p \text{ are } q$ , then  $\Gamma \vdash \text{All } p \text{ are } q$ .

*Proof.* Suppose that  $\Gamma \models \text{All } p \text{ are } q$ . We must prove that  $\Gamma \vdash \text{All } p \text{ are } q$ . Consider  $\mathcal{M}$  from Definition 1.26. By Lemma 1.27,  $\mathcal{M}$  satisfies all sentences in  $\Gamma$ . By hypothesis, we see that  $\text{All } p \text{ are } q$  is true in  $\mathcal{M}$ . Thus  $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket$ . We always have  $p \leq p$ , and so  $p \in \llbracket p \rrbracket$ . Hence  $p \in \llbracket q \rrbracket$ . This means that  $p \leq q$ , so that  $\Gamma \vdash \text{All } p \text{ are } q$ , as desired.  $\square$

#### Remark 1.29

The proof in fact shows that if  $\Gamma \not\models \text{All } p \text{ are } q$ , then  $\text{All } p \text{ are } q$  is false in  $\mathcal{M}$ . So we have a very special and important fact: to see whether a given sentence  $\varphi$  follows from  $\Gamma$  or not, we only have to see whether  $\varphi$  is true or false in one model,  $\mathcal{M}$ . We say that this model  $\mathcal{M}$  is a *characteristic model* of  $\Gamma$ .

## 1.6 Algorithmic analysis

The original definition of the entailment relation  $\Gamma \models \varphi$  involves looking at *all* models of the language. If we are given  $\Gamma$  and  $\varphi$  and we want to know whether or not  $\Gamma \models \varphi$ , we can say “no” by producing a *counter-model*: a model of  $\Gamma$  where  $\varphi$  fails. If we want to say “yes”, the easiest way would be to provide a proof tree in our proof system of  $\varphi$  from  $\Gamma$ . This would show that  $\Gamma \vdash \varphi$ , and then by soundness (Proposition 1.14), we would know that indeed  $\Gamma \models \varphi$ .

But suppose that we are given  $\Gamma$  and  $\varphi$ , and we don’t know whether or not  $\Gamma \models \varphi$ . For example, suppose we are faced with  $\Gamma$  and  $\varphi$  from Example 1.8. What happens now? We could try building a counter-model and searching for a proof at the same time. This would work, but we would like something better. That “something better” would be an algorithm that was more detailed and more organized than a blind search, and that either gave a proof tree or a counter-model. This is the topic of this section. The centerpiece of this section is Theorem 1.30 just below, a *refined version* of the completeness argument which we saw in Section 1.5. Theorem 1.30 leads to Algorithm 1.32.

#### Theorem 1.30

Let  $\Gamma$  be any set of sentences in  $\mathcal{A}$ , let  $S$  be any set of nouns which includes the nouns in  $\Gamma$ , and let  $\mathcal{G}$  be the all-graph of  $\Gamma$  on  $S$ . Let  $\xrightarrow{\Gamma}^*$  be the reflexive-transitive closure of the graph relation in  $\mathcal{G}$ . Let  $p, q \in S$ . Then the following are equivalent:

1.  $p \xrightarrow{\Gamma}^* q$  in  $\mathcal{G}$ .
2.  $\Gamma \vdash \text{All } p \text{ are } q$ .
3.  $\Gamma \models \text{All } p \text{ are } q$ .

*Proof.* (1) $\Rightarrow$ (2): we show by induction on the number  $n$  that if  $q$  is reachable from  $p$  in  $\mathcal{G}$  by a path of length  $n$ , then  $\Gamma \vdash \text{All } p \text{ are } q$ . If  $n = 0$ , then  $p = q$ , and we have a one-point proof tree of  $\text{All } p \text{ are } p$ . Suppose our result is true for  $n$ , and also that  $q$  is reachable from  $p$  in  $\mathcal{G}$  by a path of length  $n + 1$ . Fix such a path. Suppose that  $q'$  is the point on the path right before  $q$ . Then  $q'$  is reachable from  $p$  in  $\mathcal{G}$  by a path of length  $n$ . So by induction hypothesis, we have  $\Gamma \vdash \text{All } p \text{ are } q'$ .

Since there is an edge from  $q'$  to  $q$ , we know that the sentence All  $q'$  are  $q$  belongs to  $\Gamma$ . So we can take  $\mathcal{T}$  and add a step at the end:

$$\frac{\begin{array}{c} \vdots \\ \text{All } p \text{ are } q' \quad \text{All } q' \text{ are } q \end{array}}{\text{All } p \text{ are } q} \text{ BARBARA}$$

This is a proof tree over  $\Gamma$ , and it shows that  $\Gamma \vdash \text{All } p \text{ are } q$ .

(2) $\Rightarrow$ (3): this is the soundness result for the logic which we saw in Proposition 1.14.

(3) $\Rightarrow$ (1): we show the contrapositive. Assume that  $p \not\stackrel{*}{\vdash}_{\Gamma} q$ . Let  $\mathcal{N}$  be the following model: the universe is the set  $S$ , and the interpretation is given by

$$\llbracket u \rrbracket = \{v \in S : v \stackrel{*}{\vdash}_{\Gamma} u\} \quad (1.7)$$

We claim that  $\mathcal{N} \models \Gamma$ . To see this, suppose that  $\Gamma$  contains the sentence All  $x$  are  $y$ . We show that  $\llbracket x \rrbracket \subseteq \llbracket y \rrbracket$ . Let  $z \in \llbracket x \rrbracket$ . That is,  $z \in S$  and there is a path from  $z$  to  $x$  in  $\mathcal{G}_{\Gamma, S}$ . The noun  $y$  occurs in  $\Gamma$  and hence belongs to  $S$ . Then taking the path from above and adding the edge from  $x$  to  $y$  shows that  $z \stackrel{*}{\vdash}_{\Gamma} y$ . This for all  $x$  shows that indeed  $\llbracket x \rrbracket \subseteq \llbracket y \rrbracket$ ; and the preceding observation for all sentences in  $\Gamma$  shows that  $\mathcal{N} \models \Gamma$ . By definition of  $\stackrel{*}{\vdash}_{\Gamma}$ ,  $p \stackrel{*}{\vdash}_{\Gamma} p$ . Also,  $p \in S$  by hypothesis. So  $p \in \llbracket p \rrbracket$ . Our overall assumption that  $p \not\stackrel{*}{\vdash}_{\Gamma} q$  tells us that  $p \notin \llbracket q \rrbracket$ . Hence  $\mathcal{N} \not\models \text{All } p \text{ are } q$ .

This completes the proof.  $\square$

### Corollary 1.31

Let  $\Gamma \cup \{\text{All } p \text{ are } q\}$  be a set of sentences in  $\mathcal{A}$ . Let  $S$  be the set of nouns in  $\Gamma$  together with  $p$  and  $q$ . Let  $\mathcal{G}$  be the all-graph of  $\Gamma$  on  $S$ . Then the following are equivalent:

1.  $\Gamma \models \text{All } p \text{ are } q$ .
2.  $p \stackrel{*}{\vdash}_{\Gamma} q$  in  $\mathcal{G}$ .

Corollary 1.31 leads to the algorithm presented in Algorithm 1.32. More to the point, Corollary 1.31 shows that the algorithm is correct. Given a finite set  $\Gamma$  and a sentence  $\varphi$  (say, All  $p$  are  $q$ ), we first take the set of nouns in  $\Gamma$ . We add  $p$  and  $q$  to this set if they are not there already, and then the resulting set is called  $G$ . We construct the all-graph of  $\Gamma$  on  $G$ . The all-graph is directly read from  $\Gamma$ . There is a well-known algorithm which, given any relation  $R$  finds the reflexive-transitive closure  $R^*$ . So we apply this algorithm to the all-graph of  $\Gamma$  on  $G$ , and we ask whether or not  $p \stackrel{*}{\vdash}_{\Gamma} q$ . If so, then the proof of (1)  $\Rightarrow$  (2) in Theorem 1.30 shows how to turn the path into a proof tree in the logic. If not, then the model  $\mathcal{N}$  constructed in (3)  $\Rightarrow$  (1) is a *counter-model*: a model of  $\Gamma$  where the putative conclusion All  $p$  are  $q$  is false.

**Algorithm 1.32:** given a finite set  $\Gamma \subseteq \mathcal{A}$  and a sentence  $\varphi = \text{All } p \text{ are } q$ , tell whether or not  $\Gamma \vdash \varphi$

```

1: let  $S$  be the set of nouns in  $\Gamma \cup \{\varphi\}$ .
2: let  $\mathcal{G} = (S, \overrightarrow{\Gamma})$  be the all-graph of  $\Gamma$  on  $S$ . ▷ see Definition 1.19.
3: let  $\overrightarrow{\Gamma}^*$  be the reflexive-transitive closure of  $\overrightarrow{\Gamma}$ .
4: if  $p \overrightarrow{\Gamma}^* q$  then
5:   a path in  $\mathcal{G}$  gives a proof tree in the logic, showing that  $\Gamma \vdash \varphi$ .
6: else ▷ we build a counter-model  $\mathcal{N} = (N, \llbracket \cdot \rrbracket)$ 
7:    $N \leftarrow S$ 
8:   for all  $u \in S$  do
9:      $\llbracket u \rrbracket = \{v : v \overrightarrow{\Gamma}^* u\}$ .
10:  end for
11: end if

```

### Example 1.33: a return to Example 1.8

We challenged you to see whether  $\Gamma \vdash \text{All } p \text{ are } n$  or not. At this point, we can meet the challenge. See Figure 1.1.

The relevant set  $S$  here is  $\{j, k, l, m, n, p, q\}$ . We show  $\Gamma$  below, for convenience, and its all-graph, the preorder associated to that graph, and finally the structure of the canonical model.

Notice that the preorder is displayed as a Hasse diagram. For example, since  $j \leq k$ , we draw  $j$  below  $k$ . Note also that  $k \leq l$  and  $l \leq k$ . We indicate this in the picture by situating  $k$  and  $l$  together.

At the end, we interpreted each noun  $u$  according to (1.7): the interpretation of each  $u$  is the set of nouns  $x$  such that there is a path from  $x$  to  $u$  in the all-graph. Equivalently, this is the set of  $x$  which are  $\leq u$  in the preorder.

**Comments** There are a few comments to be made at this point. These are based on questions which I have received in teaching this material, or similar material, to bright students.

(1) *Doesn't this proof confuse syntax and semantics?* This is a good question, since we are building a model out of the “material from the syntax” (in this case, the nouns). But when one thinks about it, there is nothing wrong with building a model out of chairs, numbers, abstract objects, or even the same objects that we used in the syntax. It is more interesting that the interpretation function  $\llbracket \cdot \rrbracket$  in the model was defined in terms of a syntactic notion, the proof system. Again, we are free to define the semantics of a model any way we like. It is *interesting* that we prove completeness in this way. But in a sense it should not be such a surprise. For completeness is about a relation between syntax and semantics, and so it makes sense that it involves a single structure that has aspects of both.

(2) *I thought that the semantic assertion  $\Gamma \models \varphi$  meant that all models of  $\Gamma$  are again models of  $\varphi$ . How is it that we only argue completeness on one particular model rather than many models?* It is true that our semantic assertion  $\Gamma \models \varphi$  is a statement about *all* models. But this does not mean that in proving something we need to *use* all the models. In a sense, the question can be turned around to make an observation: the model  $\mathcal{M}$  we built from a set  $\Gamma$  is as “bad” as any model could



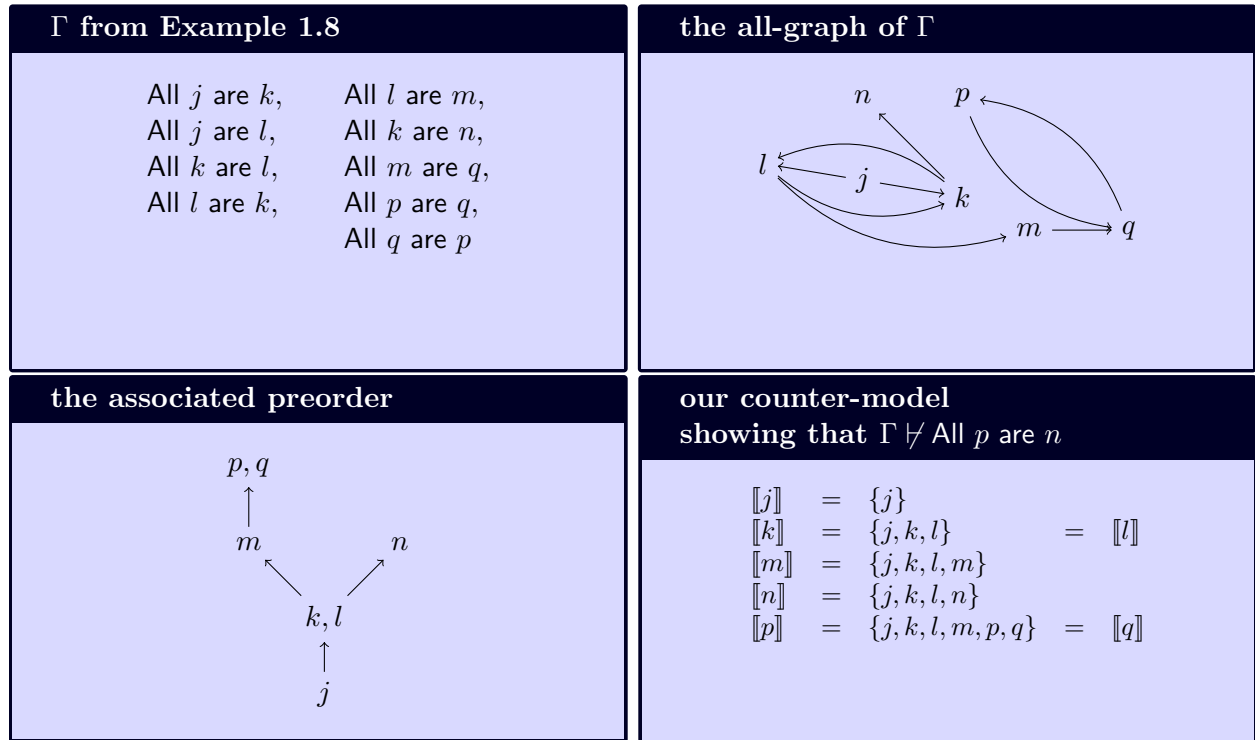


Figure 1.1: Solution to the problem posed in Example 1.8.

possibly be! It makes true any sentence which does not follow from  $\Gamma$ . So once we have built a single model that covers for all the models, we can use that one model to prove completeness.

**What you need to know to go on** At this point, you should understand a few things well: the *syntax* of the language in this chapter, beginning with the set  $\mathbf{P}$  of nouns, the *semantics* of the language, including the definition of a *model* and the definition of when a model *satisfies* a given sentence; the *proof theory*, defined in terms of two *rules of inference*; the statements of *soundness* and *completeness*; the proof that our logical system is complete (it was done twice); the general issue of *algorithms for proof search in a given logic*; how the algorithm works for the particular logic in this chapter; the definition of a *counter-model* to an assertion of the form  $\Gamma \vdash \varphi$ .

The reason that you should understand all of this is because in most of the remaining chapters, we shall see an ever-increasing array of logics. Each time, we'll see all of the *italicized* points; of course, the specific details will vary with the logic. It would be good to really understand all of the points above. One way to be sure would be to write out explanations of everything mentioned in the last paragraph. Be sure to use your own words, but also to use all of the symbols that we introduced in this chapter.

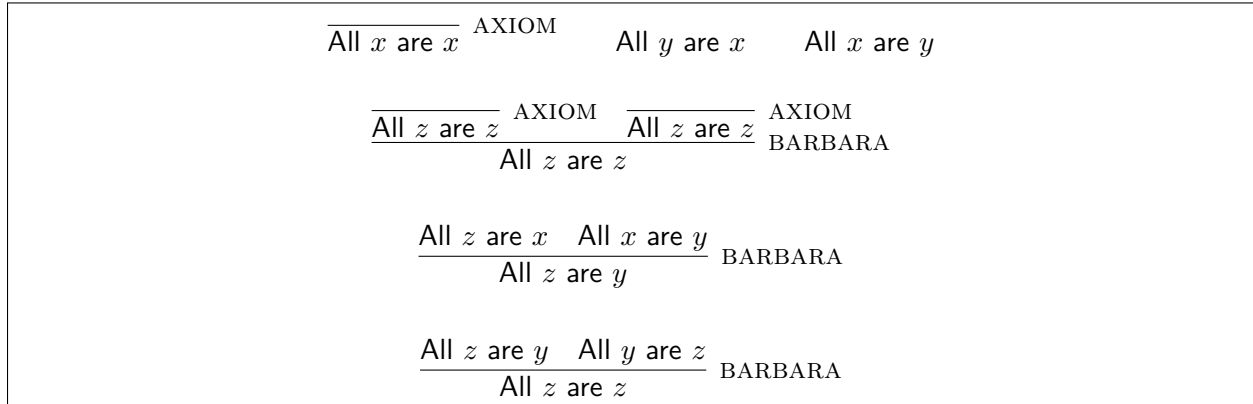


Figure 1.2: Trees from Exercise 1.2.

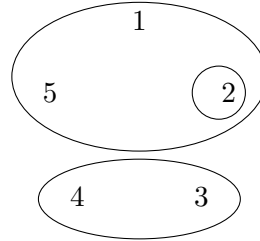
## 1.7 Exercises

**Exercise 1.1.** Suppose that our set  $\mathbf{P}$  of nouns has the three words *hawks*, *birds*, and *turtles*. Here is a model which we'll call  $\mathcal{M}$ . The universe is  $\{1, 2, 3, 4, 5\}$ . The rest of the structure is shown below.

$$\llbracket \text{hawks} \rrbracket = \{2\}$$

$$\llbracket \text{birds} \rrbracket = \{1, 2, 5\}$$

$$\llbracket \text{turtles} \rrbracket = \{3, 4\}$$



Say whether each of the following sentences is true or false in this model  $\mathcal{M}$ .

1. All hawks are birds
2. All hawks are hawks
3. All hawks are turtles

Then find a single model (different from  $\mathcal{M}$ ) where the sentence *All turtles are birds* is true and *All hawks are birds* is false.

**Exercise 1.2.** Here is an exercise on proof trees. Let

$$\Gamma = \{\text{All } x \text{ are } y, \text{All } y \text{ are } z, \text{All } z \text{ are } x\}.$$

Figure 1.2 shows six trees. Which are proof trees over  $\Gamma$  according to the strict letter of Definition 1.2?

**Exercise 1.3.** Here is an exercise for people getting started with proof trees and models. Let  $\mathbf{P} = \{x, y, z, w\}$ . Let

$$\Gamma = \{\text{All } x \text{ are } y, \text{All } x \text{ are } z, \text{All } w \text{ are } z\}.$$

1. Since  $\mathbf{P}$  has 4 elements, there are  $4^2 = 16$  sentences that can be built from  $\mathbf{P}$ . List all 16.
2. Which of your 16 sentences  $\varphi$  in the last part are provable from  $\Gamma$ ? For each such sentence  $\varphi$ , give a proof tree which has  $\varphi$  as the root, and with all leaves in  $\Gamma$ .
3. For each  $\varphi$  from part (1) which is *not* on your list in part (2), give a model of  $\Gamma$  where  $\varphi$  is false.

[This exercise would be long if done completely. One way to make it shorter would be to drop  $w$  from the nouns in  $\mathbf{P}$  and also to drop one of the sentences in  $\Gamma$ . Then the number of sentences in part (1) would be 9 instead of 16.]

**Exercise 1.4.** Here is an exercise just to see if you got the main points of this chapter. Let

$$\Gamma = \{\text{All } x \text{ are } y, \text{All } z \text{ are } y, \text{All } y \text{ are } w\}.$$

Let  $\varphi$  be  $\text{All } x \text{ are } z$ . For each of the following assertions, say whether true or false, with a reason:

1.  $\Gamma \models \varphi$ .
2.  $\Gamma \vdash \varphi$ .
3.  $x \leq_{\Gamma} w$ .
4.  $x \xrightarrow{\Gamma} w$ .
5.  $x \overset{*}{\xrightarrow{\Gamma}} w$ .

It would be best to do this by recalling *the original definitions* of the notation as much as possible. So if you are using the soundness or completeness of the logic  $\mathcal{A}$  in any part, please be sure to note this. That is, please be aware when you are using significant facts!

**Exercise 1.5.** Let  $\Gamma$  be a set of sentences in  $\mathcal{A}$ . Prove that for all  $p$  and  $q$ ,

$$p \xrightarrow{\Gamma} q \quad \text{iff} \quad p \leq_{\Gamma} q$$

[Hint: you will need to use induction, in two forms. One direction of this exercise requires induction on numbers; the other uses induction on proofs in our proof system.]

**Exercise 1.6.** Suppose that  $\Gamma \vdash \varphi$ . For each of the following assertions, tell whether it is True or False, with a reason.

1. There are infinitely many proof trees over  $\Gamma$  which show that  $\Gamma \vdash \varphi$ .
2. Any two proof trees over  $\Gamma$  which show that  $\Gamma \vdash \varphi$  must share a leaf. (The leaves of a proof tree are the sentences at the top which come from  $\Gamma$ .)

3. If a sentence of the form *All  $p$  are  $p$*  occurs in a proof tree, then that occurrence must be at a leaf.

**Exercise 1.7.** This exercise and the next ones develop a different way to prove the completeness of the logic **A**. Let  $\Gamma$  be the set in Exercise 1.4. As you saw in that exercise,  $\Gamma \not\models \text{All } y \text{ are } x$ . Now we make a model  $\mathcal{N}$  where  $N = \{*\}$ , some one-element set, and

$$\begin{array}{ll} \llbracket x \rrbracket &= \emptyset & \llbracket z \rrbracket &= \emptyset \\ \llbracket y \rrbracket &= \{*\} & \llbracket w \rrbracket &= \{*\} \end{array}$$

1. Does  $\mathcal{N} \models \Gamma$ ? That is, are all sentences in  $\Gamma$  true in  $\mathcal{N}$ ?
2. Does  $\mathcal{N} \models \text{All } y \text{ are } x$ ?
3. Check that for all variables  $u \in \{x, y, z, w\}$ ,

$$\llbracket u \rrbracket = \begin{cases} \{*\} & \text{if } \Gamma \vdash \text{All } y \text{ are } u \\ \emptyset & \text{otherwise} \end{cases} \quad (1.8)$$

Note that  $u$  is a variable ranging over set of nouns  $\{x, y, z, w\}$ . But  $y$  on the right of (1.8) is just  $y$ , it's not a variable ranging over anything.

To solve this problem plug the nouns  $x, y, z$  and  $w$  in for  $u$  in (1.8), and then check that each time, the two sides of the equation really are the same set.

[You may use the results which you obtained in Exercise 1.4.]

**Exercise 1.8.** Here is a set  $\Gamma$ :

$$\begin{array}{ll} \text{All } a \text{ are } b & \text{All } c \text{ are } d \\ \text{All } a \text{ are } c & \text{All } a \text{ are } e \\ \text{All } c \text{ are } e & \end{array}$$

Then  $\Gamma \not\models \text{All } d \text{ are } b$ . The point of this problem is to give *two* models of  $\Gamma$  where *All  $d$  are  $b$*  is false.

1. Find the canonical model of  $\Gamma$ , and check that *All  $d$  are  $b$*  is false in that model.
2. Find a model  $\mathcal{M}$  *with just one element* such that  $\mathcal{M} \models \Gamma$  but  $\mathcal{M} \not\models \text{All } d \text{ are } b$ . [Hint: You can do this by modifying the model in Exercise 1.8. That is, you use a model  $\mathcal{M}$  with  $M = \{*\}$ , and with the interpretation function given by something like (1.8). The only difference is that we don't want  $y$  on the right, we want  $\dots$ . You can also get a one-point model this by using Exercise 1.9 just below. On the other hand, some people might find Exercise 1.9 easier to think about after working on *this* exercise.]

**Exercise 1.9.** This exercise is a continuation of Exercises 1.7 and 1.8. Let  $\Gamma$  be any set, not necessarily the one in Exercise 1.4. Fix nouns  $p$  and  $q$ . Suppose that  $\Gamma \models \text{All } p \text{ are } q$ . Consider the model  $\mathcal{N}$  where  $N = \{*\}$ , some one-element set, and

$$\llbracket u \rrbracket = \begin{cases} \{*\} & \text{if } \Gamma \vdash \text{All } p \text{ are } u \\ \emptyset & \text{otherwise} \end{cases} \quad (1.9)$$

Please note that in (1.9), the noun  $p$  is the same one as in the sentence  $\text{All } p \text{ are } q$  that we are considering throughout this problem.

1. Show that  $\mathcal{N} \models \Gamma$ .
2. Show that  $\mathcal{N} \models \text{All } p \text{ are } q$ . [Hint: Use part (1).]
3. Use part (2) to show that  $\Gamma \vdash \text{All } p \text{ are } q$ .

Please note that this problem is more abstract and more general than Exercises 1.7 and 1.8. We don't know anything about  $\Gamma$  except that  $\Gamma \models \text{All } p \text{ are } q$ . So this is good: a more abstract and general result is a better one because it can be used in more settings.

**Exercise 1.10.** This exercise has to do with an aspect of the canonical model defined in Definition 1.26. Recall that we took  $M$  to be  $\mathbf{P}$ , the set of all nouns with underlies our language in this chapter. If we had taken  $M$  to be the set of nouns which occur in  $\Gamma$ , would our construction have worked? That is, what (if anything) would have to change in the rest of this section?

**Exercise 1.11.** True or false: if a logical system is complete, one cannot delete any rules and still have a complete system. Be sure to give a reason for your answer.

**Exercise 1.12.** In (1.4) on page 26, we mentioned a new rule which can be used with the language  $\mathcal{A}$  of this chapter. Let us write  $\mathbf{A}^+$  for the new proof system. And let us write  $\Gamma \vdash_{\text{new}} \varphi$  if there is a proof tree using the three rules of  $\mathbf{A}^+$  whose root is  $\varphi$  and whose leaves belong to  $\Gamma$ .

1. Formulate a principle of proof by induction on trees in our new proof system  $\mathbf{A}^+$ . [This is an easy generalization of Definition 1.15.]
2. Prove that if  $\Gamma \vdash_{\text{new}} \varphi$ , then  $\Gamma \vdash \varphi$ . [Hint: Fix  $\Gamma$ . We want to show by induction on the bigger proof system that if  $\mathcal{T}$  is a proof tree showing that  $\Gamma \vdash_{\text{new}} \varphi$ , then there is a proof tree  $\mathcal{T}'$  in the original proof system  $\mathbf{A}$  showing that  $\Gamma \vdash \varphi$ .]
3. Show that the new proof system is sound and complete.

**Exercise 1.13.** In this problem, we are interested in what happens when  $\Gamma$  is the empty set of sentences. So we are interested in statements of the form  $\emptyset \vdash \varphi$  and  $\emptyset \models \varphi$ . Note that  $\emptyset \vdash \varphi$  means that there is a proof tree whose root is  $\varphi$  and where all leaves are sentences of the form  $\text{All } p \text{ are } p$ . Also,  $\emptyset \models \varphi$  means that  $\varphi$  is true in all models.

Note also that  $\emptyset \vdash \varphi$  means that there is a proof tree all of whose leaves are from the rule (AXIOM), so the leaves are all  $\text{All } x \text{ are } x$  for some noun  $x$ . That is, the leaves in the tree have to be sentences with both nouns the same.

1. Which one of the following conveys in English the same meaning as  $\emptyset \not\models \varphi$ ? *Some model does not satisfy  $\varphi$*  or or else *Every model does not satisfy  $\varphi$* .
2. If  $p = q$ , show that  $\emptyset \vdash \text{All } p \text{ are } q$ . That is, if  $p$  and  $q$  are the same noun, show that we can prove  $\text{All } p \text{ are } q$  from no assumptions.
3. If  $p = q$ , show that  $\emptyset \models \text{All } p \text{ are } q$ . That is, if  $p$  and  $q$  are the same noun, show that in all models  $\mathcal{M}$ , the sentence  $\text{All } p \text{ are } q$  is true. (This just uses a very basic fact about sets.)
4. If  $p \neq q$ , show that  $\emptyset \not\models \text{All } p \text{ are } q$ . That is, if  $p$  and  $q$  are different nouns, show that  $\emptyset \not\models \text{All } p \text{ are } q$ .
5. If  $p \neq q$ , show that  $\emptyset \not\vdash \text{All } p \text{ are } q$ . You can use soundness or completeness if you like. But if you do, you must say which you are using.

**Exercise 1.14.** Let  $\Gamma$  be a theory in any logical system extending **A**. Let  $\mathcal{M}$  be any model with the property that

$$[\ ] : \mathbf{P} \rightarrow \mathcal{P}(M)$$

is a monotone function (see Definition 0.11). That is, we are assuming about  $\mathcal{M}$  that if  $p \leq_{\Gamma} q$ , then  $[p] \subseteq [q]$ . Prove that  $\mathcal{M} \models \Gamma$ . [This is not a hard exercise, but it is one that will be used all the time.]

**Exercise 1.15.** Here is another logical system. Instead of nouns denoting subsets of a given set, our syntax starts with a set **N** of *names*. We use letters like  $a, b$ , etc. for names here. Then we take as sentences the expressions  $a = b$ , where  $a, b \in \mathbf{N}$ . For the semantics, we start with a set  $M$  and interpret a name  $a$  by an *element*  $\llbracket a \rrbracket \in M$ . (Again, we are not interpreting names using subsets the way we did for nouns in this chapter.) This gives us the definition of a *model*  $\mathcal{M}$ .

1. As soon as we have a semantics, we get notions like  $\Gamma \models \varphi$  for this new language. (Note that for purposes of this problem,  $\varphi$  is a sentence like  $a = b$ .) State what  $\Gamma \models \varphi$  means, and then give an example of a set  $\Gamma$  and a sentence  $\varphi$  where  $\Gamma \models \varphi$  holds, and another example where it does not hold.
2. Find a logical system that defines a notion  $\Gamma \vdash \varphi$ , and give an example of a proof in your system. That is, what are the rules of your logic?
3. Prove that your system is complete.

**Exercise 1.16.** If you have worked Exercise 1.15, then the next step is to combine your system from that exercise with the logical system for **A**. We start with sets **P** and **N** of nouns and names, respectively. In the syntax, we use sentences of the following forms:

All  $p$  are  $q$   
 $a = b$   
 $a$  isa  $p$

The semantics is just what you would expect. In particular, in a given model  $\mathcal{M}$ , we say that  $a \text{ isa } p$  is true if  $\llbracket a \rrbracket \in \llbracket p \rrbracket$ . The proof theory for this system should contain the rules in Definition 1.9, the rules in your system from Exercise 1.15, and the two extra rules below:

$$\frac{a \text{ isa } p \quad \text{All } p \text{ are } q}{a \text{ isa } q} \qquad \frac{a = b \quad a \text{ isa } p}{b \text{ isa } p}$$

Prove the completeness of the system.

**Exercise 1.17.** Suppose that  $\mathbf{P}$  is the infinite set  $\{x_1, x_2, \dots\}$ , and let

$$\Gamma = \{\text{All } x_1 \text{ are } x_2, \text{All } x_2 \text{ are } x_3, \dots, \text{All } x_n \text{ are } x_{n+1}, \dots\}$$

We wish to prove that if  $\Gamma \vdash \text{All } x_i \text{ are } x_j$ , then  $i \leq j$ .

1. Use induction on proof trees over  $\Gamma$  to show that for all  $i$  and  $j$ , if  $\Gamma \vdash \text{All } x_i \text{ are } x_j$ , then  $i \leq j$ .
2. Exhibit a model of  $\Gamma$  where  $\llbracket x_1 \rrbracket \subset \llbracket x_2 \rrbracket \subset \llbracket x_3 \rrbracket \subset \dots$ . (The  $\subset$  notation means *proper subset*.) Then use either soundness or completeness to get the same result as in part (1).

## Chapter 2

# $\mathcal{A}(\mathcal{RC})$ : Verbs and Relative Clauses

The logical language  $\mathcal{A}$  of Chapter 1 is perhaps the simplest possible logic, both in terms of the syntax and semantics. The proof system  $\mathbf{A}$  is also probably the simplest possible logical system, having just two rules. Now is the time to enrich the language and the proof system. This chapter thus builds on Chapter 1. In fact, nearly *all* of the chapters in this book extend the work we did in Chapter 1 to more and more complex languages.

### 2.1 Verbs and relative clauses

The main addition in this section concerns *verbs* and *relative clauses*. The idea is that we would like to consider sentences such as *All cats chase all rats*. To put this in the mold of  $\mathcal{A}$ , we think of this as *All cats (chase all rats)*. So our syntax in the chapter will arrange that *chase all rats* is a *constituent*: part of the syntax in a significant way. We'll call it a *term*. And our semantics will arrange that a term like *chase all rats* is interpreted by a set, just as *cats* is; and then we shall say that *All cats (chase all rats)* is true in a given model iff  $\llbracket \text{cats} \rrbracket \subseteq \llbracket \text{chase all rats} \rrbracket$ . And to interpret *chase all rats* by a set invites us to go further. We can just as well have *chase all  $x$*  where  $x$  is “the same type of object as” *rats*. For example,  $x$  could be *see all birds*. And then we would want to have another *term*, *chase all (see all birds)*. The idea is that this would denote the entities which chase everything which sees all birds. This is where relative clauses enter. In English, relative clauses are the snippets of language which frequently (but not always) start with a *relative pronoun* such as *who* or *which* and which modify and follow some *head noun*. Our formal language in this section is not an adequate treatment of relative clauses: it is only the beginning. It only covers *subject relative clauses* rather than *object relative clauses*. For example, we'll have sentences corresponding to *Everyone who likes all plumbers sits* but not *Everyone who all plumbers like sits*. Worse, we'll have no way to modify head nouns at all. To extend what we do in this section to cover sentences like *Every carpenter who likes all plumbers sits* would involve mixing the work of this section with that of Section 2.5. So overall, what we have is a formal language which is not going to read like English at all (despite what we would want). It is going to serve as a language in which to interpret a small number of sentences involving verbs and relative clauses. But despite its meager linguistic interest, it gives us an opportunity to build a new (and, we hope, interesting) logical system.



**Definition 2.1: Syntax of  $\mathcal{A}(\mathcal{RC})$** 

We start with one collection  $\mathbf{P}$  of *nouns*, just as we did with  $\mathcal{A}$  in Section 1.1. But we also start with another collection,  $\mathbf{R}$  of *verbs*,

The verbs here are what we would call in grammar *transitive verbs*: the verbs which take a direct object.

We define the *terms* of  $\mathcal{A}(\mathcal{RC})$  to be the smallest collection containing the nouns and with the property that if  $x$  is a term and  $r$  is a verb, then

$$r \text{ all } x$$

is a term.

The *sentences* of  $\mathcal{A}(\mathcal{RC})$  are the expressions  $\text{All } x \ y$ , where  $x$  and  $y$  are terms.

The “ $\mathcal{RC}$ ” in  $\mathcal{A}(\mathcal{RC})$  stands for “relative clause.”

Note that terms allow recursion. So we get terms like

see all (like all (hate all dogs)).

(Of course, this is on the assumption that  $\mathbf{P}$  contains *dogs* and  $\mathbf{R}$  contains *see*, *like*, and *hate*.) Although we are not yet done with the syntax of the language, the interpretation of the term above in a given model would be the set of individuals who see all who like all who hate all dogs.

Please note that our official sentences in this chapter do not have the word *are* which sentences in English really would have.

We frequently use parentheses in the syntax to increase the readability.

**Definition 2.2: induction on terms**

Because the set of terms of  $\mathcal{A}(\mathcal{RC})$  is defined by recursion, to prove a statement about all terms, it is frequently necessary to use *proof by induction on terms*. To prove some assertion about all terms, do the following:

1. Prove the assertion about all nouns.
2. Fix a term  $t$ , assume the assertion about  $t$ , and prove it for a term  $r \text{ all } t$ , where  $r$  is any verb.

If you do those two things, you may conclude the assertion for all terms.

Step 1 is the *base step*, and Step 2 is the *induction step*.

**Definition 2.3: semantics**

A *model*  $\mathcal{M}$  for  $\mathcal{A}(\mathcal{RC})$  is a set  $M$ , called (as before) the *universe*, together with interpretations of the nouns and verbs. For each noun  $p$ , we have an interpretation  $\llbracket p \rrbracket \subseteq M$ . And for each verb  $r$ , we have an interpretation  $\llbracket r \rrbracket \subseteq M \times M$ . (Here and below,  $M \times M$  means the set of ordered pairs  $(x, y)$  such that both  $x$  and  $y$  belong to  $M$ .)

We use recursion to interpret the terms. The model comes with interpretations of the nouns, the “base case” of terms. And the general case is

$$\llbracket r \text{ all } x \rrbracket = \{m \in M : \text{for all } n \in \llbracket x \rrbracket, m \llbracket r \rrbracket n\}. \quad (2.1)$$

In other words, we are extending the interpretation function from the nouns to the set of all terms.

Finally, we follow Definition 1.4 to define truth of sentences in a model.

$$\mathcal{M} \models \text{All } x \ y \quad \text{iff} \quad \llbracket x \rrbracket \subseteq \llbracket y \rrbracket. \quad (2.2)$$

Please see Example 2.4 on page 46 for an example of the semantics. Here are a few words of explanation. The picture shows a model which we call  $\mathcal{M}$ . The universe  $M$  of the model is  $\{1, 2, 3, 4, 5, 6, 7\}$ . The nouns are **hawks**, **birds**, and **turtles**. Their interpretations are shown on the left of Example 2.4. There is only one verb, **see**. It is shown on the right. An arrow  $m \rightarrow n$  means that  $m$  sees  $n$  in the model.<sup>1</sup> For example, 3 sees 1, but 1 does not see 3. A loop means that the object sees itself.

We should also mention that there are lots of mathematical notations which say “ $m$  sees  $n$  in the model.” We could think of  $\llbracket \text{see} \rrbracket$  as a set of ordered pairs, and then we would write  $(m, n) \in \llbracket \text{see} \rrbracket$ . But we could also write  $\llbracket \text{see} \rrbracket$  as a relation between  $m$  and  $n$ :  $m \llbracket \text{see} \rrbracket n$ . This would be a parallel to the way we write  $3 > -4$  for an assertion between numbers, putting the relation symbol  $>$  in the middle. We use either notation  $(m, n) \in \llbracket \text{see} \rrbracket$  or  $m \llbracket \text{see} \rrbracket n$ .

Example 2.4 also gives the semantics of some of the sets. For example,  $\llbracket \text{see all birds} \rrbracket$  is given as  $\{3, 4, 5, 7\}$ . We get this by looking at (2.1) in Definition 2.3. In this case,  $r$  is **see** and  $x$  is **birds**. So we get

$$\begin{aligned} \llbracket \text{see all birds} \rrbracket &= \{m \in M : \text{for all } n \in \llbracket \text{birds} \rrbracket, m \llbracket \text{see} \rrbracket n\} \\ &= \{m \in M : \text{for all } n \in \{1, 6\}, m \llbracket \text{see} \rrbracket n\} \end{aligned} \quad (2.3)$$

Again, we claim that  $\llbracket \text{see all birds} \rrbracket = \{3, 4, 5, 7\}$ . And here is why: 3 sees all elements of  $\{1, 6\}$ , 4 sees all elements of  $\{1, 6\}$ , 5 sees all elements of  $\{1, 6\}$ , and 7 sees all elements of  $\{1, 6\}$ . Just as important, it also means that no other objects see both 1 and 6. Let’s check this: 1 does not see either 1 or 6, 2 does not see 6, and 6 does not see either 1 or 6. So we have checked exhaustively that the set  $\llbracket \text{see all birds} \rrbracket$  is exactly  $\{3, 4, 5, 7\}$ . In general, if we are given a picture of a model, then an exhaustive (and sometimes exhausting) examination is the only way to get the semantics of the terms.

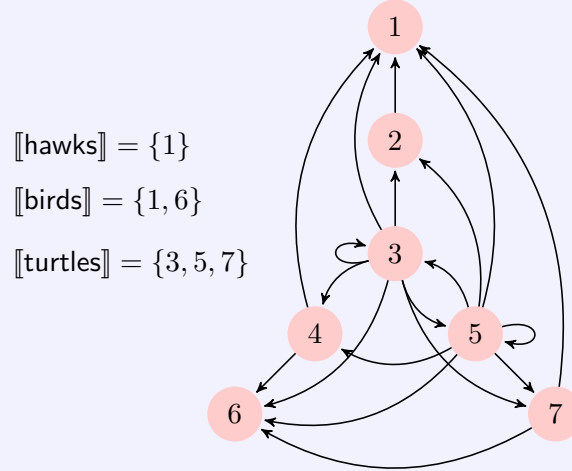
As an illustration of the recursive nature of the semantics, let us look at

$$\llbracket \text{see all (see all hawks)} \rrbracket.$$

<sup>1</sup>If you know about propositional logic, you may confuse this use of  $\rightarrow$  with logical implication. Please don’t do this!

**Example 2.4**

Consider the model  $\mathcal{M}$  with universe  $\{1, \dots, 7\}$ , and with interpretations of three nouns, hawks, birds, and turtles, and one verb, see, shown below.



In this model,

$$\begin{array}{ll}
 \llbracket \text{see all hawks} \rrbracket &= \{2, 3, 4, 5, 7\} & \llbracket \text{see all turtles} \rrbracket &= \{3, 5\} \\
 \llbracket \text{see all birds} \rrbracket &= \{3, 4, 5, 7\} & \llbracket \text{see all (see all hawks)} \rrbracket &= \{3, 5\}
 \end{array}$$

Here are some examples of sentences false and true in the model:

$$\begin{array}{l}
 \mathcal{M} \not\models \text{All (see all hawks) (see all birds)} \\
 \mathcal{M} \models \text{All (see all turtles) (see all (see all hawks))} \\
 \mathcal{M} \models \text{All (see all (see all hawks)) (see all turtles)}
 \end{array}$$

Here, we need to *first* check that  $\llbracket \text{see all hawks} \rrbracket$  is  $\{2, 3, 4, 5, 7\}$ . We leave this to you. And then we ask who are the ones who see all of 2, 3, 4, 5, and 7? It's 3 and 5, and only these two. And so this is why  $\llbracket \text{see all (see all hawks)} \rrbracket = \{3, 5\}$ .

As for the semantics of sentences, we use (2.2) in Definition 2.3. For example, let's look at the first sentence in Example 2.4,

$$\text{All (see all hawks) (see all birds)}.$$

In terms of (2.2),  $x$  is see all hawks, and  $y$  is see all birds. So before we can see whether our sentence is true or false in  $\mathcal{M}$ , we need to evaluate  $\llbracket x \rrbracket$  and  $\llbracket y \rrbracket$ . We have seen that  $\llbracket y \rrbracket = \{3, 4, 5, 7\}$ . A similar verification shows that  $\llbracket x \rrbracket = \{2, 3, 4, 5, 7\}$ . Since  $\llbracket x \rrbracket \not\subseteq \llbracket y \rrbracket$ , we see that the sentence under discussion is false in  $\mathcal{M}$ .

**Example 2.5: For more practice**

You should also work Exercises 2.1 and 2.2 at this point to be sure that you understand what the models look like and how the semantics works.

**Derived semantic notions** Review Definition 0.3 for some general definitions, especially the definition of the semantic consequence relation  $\Gamma \models \varphi$ .

### Example 2.6: An example of a semantic consequence

One very natural example of  $\Gamma \models \varphi$  is

$$\text{All dogs (chase all animals), All cats animals} \models \text{All dogs (chase all cats)}.$$

So here  $\Gamma$  is the set of two sentences on the left, and  $\varphi$  is the sentence on the right. To check that we really have  $\Gamma \models \varphi$ , let  $\mathcal{M} \models \Gamma$ . So in  $\mathcal{M}$ ,  $\llbracket \text{dogs} \rrbracket \subseteq \llbracket \text{chase all animals} \rrbracket$ , and also  $\llbracket \text{cats} \rrbracket \subseteq \llbracket \text{animals} \rrbracket$ . We shall show that  $\llbracket \text{dogs} \rrbracket \subseteq \llbracket \text{chase all cats} \rrbracket$ . For this, let  $x \in \llbracket \text{dogs} \rrbracket$ . We show that  $x \in \llbracket \text{chase all cats} \rrbracket$ . So let  $y \in \llbracket \text{cats} \rrbracket$ . So  $y \in \llbracket \text{animals} \rrbracket$  as well. (Why?) And thus  $x \llbracket \text{chase} \rrbracket y$ . Since  $y$  is arbitrary, we see that  $x \in \llbracket \text{chase all cats} \rrbracket$ . And since  $x$  is arbitrary, we are done.

### Example 2.7: A further example

Here is another example of semantic consequences:

$$\text{All skunks mammals} \models \text{All (love all mammals) (love all skunks)}$$

We'll leave it to you to think about why this holds.

### Example 2.8: A non-example

In contrast,

$$\text{All skunks mammals} \not\models \text{All (love all skunks) (love all mammals)} \quad (2.4)$$

To see this, we need only construct example of a model of the assumption where the conclusion is false. One example would be  $M = \{1, 2, 3\}$  with

$$\begin{aligned} \llbracket \text{skunks} \rrbracket &= \{1\} \\ \llbracket \text{mammals} \rrbracket &= \{1, 2\} \\ \llbracket \text{love} \rrbracket &= \{(3, 1)\} \end{aligned}$$

This is just one way to show (2.4). There are many others, of course.

## 2.2 Two logical systems

At this point, we have stated the syntax and semantics of our language  $\mathcal{A}(\mathcal{RC})$ . The next step is to find some simple instances of semantic entailment; in other words, we want simple examples of  $\Gamma \models \varphi$ . The idea is that we should find a few simple instances in the hopes that we found “all” of them. More accurately, we hope to find a few which, when we use them in a logical system, give a system which is complete. In fact, we already have seen one simple instance in Example 2.5 and

another in Example 2.6. We stated these with concrete nouns and verbs, and the more abstract formulation would use letters as variables ranging over terms. We use letters like  $x, y, z, \dots$  for terms in this chapter, and all of our work to come uses variables in this way.

Since our syntax includes the syntax of Chapter 1, it is natural to assume that our logic would include (AXIOM) and (BARBARA). Our work in Chapter 1 stated these with variables ranging over nouns and verbs, but now we want the same rules but with variables ranging over terms. Generalizing Examples 2.5 and 2.6 would seem to point to *two* extra rules. However, it turns out that adding *either* of those rules to (BARBARA) gives a complete system. We shall see this below. The upshot is that we have two different logical systems for the same semantics, each of which is sound and complete.

**The logics  $A(\mathcal{RC})$  and  $A(\mathcal{RC})'$**  There are *two different* logical systems in this chapter. They are called  $A(\mathcal{RC})$  and  $A(\mathcal{RC})'$ , shown in Definition 2.10. Let us look at  $A(\mathcal{RC})$ , since it is the system which we shall use most when we study completeness.

The new rule is (DOWN). (We shall discuss the reason for this much later, when we discuss *monotonicity*.<sup>2</sup>)

**Example 2.9: The first difficult derivation in this book.**

Let us assume the following:

All who hate all ducks are cats  
All who love all pigs see all cats  
All who see all who hate all ducks are cats

and then prove that All who love all pigs are cats.

Here is a derivation in  $A(\mathcal{RC})$ :

$$\frac{\frac{\text{All (hate all d) cats} \quad \text{All (love all pigs) (see all cats)}}{\text{All (love all pigs) (see all (hate all d))}} \text{DOWN} \quad \text{All (see all (hate all d)) cats}}{\text{All (love all pigs) cats}} \text{BARB}$$

As you can see from trying it yourself, finding a derivation is difficult. You will construct more derivations in  $A(\mathcal{RC})$  in Examples 2.12 and 2.11.

At this point, we wish to turn to the system  $A(\mathcal{RC})'$ . Instead of (DOWN), we have a rule called (ANTI). Please be sure that you see that we have two different rules, hence two different logical systems. Example 2.11 is a whimsical example, but it shows how the logical system works. The next example of a proof in  $A(\mathcal{RC})'$  shows that the rule (DOWN) is provable in  $A(\mathcal{RC})'$ . Using Example 2.12, it follows by induction on proof trees in  $A(\mathcal{RC})$  that if  $\Gamma \vdash \varphi$  in  $A(\mathcal{RC})$ , then also  $\Gamma \vdash \varphi$  in  $A(\mathcal{RC})'$ . And the converse is also true; see Exercise 2.7.

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<sup>2</sup>Be sure to come back to this

**Definition 2.10: rules of A(RC) and A(RC)'**

Here are the rules of A(RC):

$$\frac{}{\text{All } x \ x} \text{AX} \quad \frac{\text{All } x \ y \quad \text{All } y \ z}{\text{All } x \ z} \text{BARB} \quad \frac{\text{All } x \ (r \ \text{all } y) \quad \text{All } z \ y}{\text{All } x \ (r \ \text{all } z)} \text{DOWN}$$

And here are the rules of A(RC)':

$$\frac{}{\text{All } x \ x} \text{AX} \quad \frac{\text{All } x \ y \quad \text{All } y \ z}{\text{All } x \ z} \text{BARB} \quad \frac{\text{All } y \ x}{\text{All } (r \ \text{all } x) \ (r \ \text{all } y)} \text{ANTI}$$

Note:  $x$ ,  $y$ , and  $z$  may be any terms; they need not be nouns.

Note also that we have shortened the names of the rules; we do this frequently in wide proof trees as well.

**Example 2.11: skunks and mammals in A(RC)'**

$$\frac{\frac{\text{All skunks mammals}}{\text{All (love all mammals) (love all skunks)}} \text{ANTI}}{\text{All (hate all (love all skunks)) (hate all (love all mammals))}} \text{ANTI}$$

**Example 2.12: A derivation of (DOWN) in A(RC)'**

$$\frac{\text{All } x \ (\text{see all } y) \quad \frac{\text{All } z \ y}{\text{All (see all } y)(\text{see all } z)} \text{ANTI}}{\text{All } x \ (\text{see all } z)} \text{BARBARA}$$

For the remainder of this chapter, the logical system A(RC) is the main object of study. The soundness of A(RC) is an elaboration of what we saw in Example 2.5 above. For more details, see Exercise 2.4.

**Formal definition**  $\Gamma \vdash \varphi$  means that there is a tree  $\mathcal{T}$  labeled by sentences in the language such that: (1) the root of  $\mathcal{T}$  is labeled with  $\varphi$ ; (2) all of the leaves of the tree belong to  $\Gamma$  or are (AXIOM) nodes; and (3) all non-leaves follows from their parents by applying one of the rules in the system.

This definition is a parallel to Definition 1.10 in Chapter 1. Similarly, we have a principle of *proof by induction on proof trees*. It is stated just like the principle in Definition 1.15, except that we need an extra clause for applications of (DOWN).

**Soundness** The second order of business is to check that the logical systems are sound. We are only going to discuss A(RC). There are several steps here. First, we check the *soundness of the*

*individual rules*. This means that we check the following facts:

$$\begin{array}{ll} & \models \text{All } x \ x \\ \text{All } x \ y, \text{All } y \ z & \models \text{All } x \ z \\ \text{All } x \ (r \ \text{all } y), \text{All } z \ y & \models \text{All } x \ (r \ \text{all } z) \end{array}$$

The first two of these should be familiar from Chapter 1, and the third is essentially done in Example 2.6.

This point done, we can check the *soundness of the entire logical system*. This means that we prove that if  $\Gamma$  is any set of sentences and  $\mathcal{T}$  is any proof tree over  $\Gamma$ , and  $\mathcal{M} \models \Gamma$ , then  $\mathcal{M}$  satisfies the root of  $\mathcal{T}$ . The proof of this is a straightforward induction on proof trees in the logic. The induction steps follow from the soundness of the individual rules, using the facts which we noted in the previous paragraph.

**Starting in on completeness** With the proof system at hand, the main remaining work concerns the completeness of the proof system and the development of an algorithm to tell whether or not  $\Gamma \vdash \varphi$ .

Suppose that we are given a set  $\Gamma$  in this language  $\mathcal{A}(\mathcal{RC})$ . We aim to build a model  $\mathcal{M} = \mathcal{M}_\Gamma$  of  $\Gamma$  with the property that if  $\mathcal{M} \models \varphi$ , then  $\Gamma \vdash \varphi$ . This would prove the completeness theorem for our logic  $\mathcal{A}(\mathcal{RC})$ , because if  $\Gamma \not\vdash \varphi$ , then  $\mathcal{M} \not\models \varphi$ . (In other words, every sentence which is true in all models of  $\Gamma$  (hence in  $\mathcal{M}$ ) would be provable from  $\Gamma$ .)

**Definition 2.13: the canonical model  $\mathcal{M}_\Gamma$  of a set of assertions  $\Gamma$  in  $\mathcal{A}(\mathcal{RC})$**

We take  $M$  to be the set of all terms. And then we define for  $p$  be a noun and  $r$  a verb,

$$\begin{aligned} \llbracket p \rrbracket &= \{x : \Gamma \vdash \text{All } x \ p\} \\ \llbracket r \rrbracket &= \{(x, y) : \Gamma \vdash \text{All } x \ (r \ \text{all } y)\} \end{aligned} \tag{2.5}$$

The clause for nouns is what we saw in Section 1.5. But the clause for the verbs should not be obvious. As we discuss things, please note that the interpretation of verbs  $r$  in (2.5) is **not**  $\{(x, y) : \Gamma \vdash \text{All } x \ y\}$ .



You will find an example of the canonical model construction in Example 2.14. But that example has a bit of cheating. It is fairly easy to see that the arrows above are *included* in the canonical model. To do this, we only would need to exhibit the derivations corresponding to the arrows present in the figure. But to show that no others arrows are present, one would have to do additional work. One way is to show by induction on proof trees over  $\Gamma$  that if  $\Gamma \vdash \text{All } x \ (r \ \text{all } y)$ , then there is an arrow  $x \rightarrow y$  in the graph above. This is clear for the axioms  $\text{All } p_{n+1} \ (r \ \text{all } p_n)$  and  $\text{All } q_{n+1} \ (r \ \text{all } q_n)$ . Further, assume our result for a given proof tree  $\mathcal{T}$ , and consider what happens when we add an application of (ANTI) at the root. The root of  $\mathcal{T}$  corresponds to one of the four types of arrows:  $p_{n+1} \rightarrow p_n$ ,  $p_{2n+2} \rightarrow q_{2n+1}$ ,  $q_{n+1} \rightarrow q_n$ , and  $q_{2n+1} \rightarrow p_{2n}$ . Applying (ANTI) to each of the sentences corresponding to these gives sentences which correspond to the arrows  $p_{n+2} \rightarrow p_{n+1}$ ,  $q_{2n+3} \rightarrow p_{2n+2}$ ,  $q_{n+2} \rightarrow q_{n+1}$ , and  $p_{2n+2} \rightarrow q_{2n+1}$ . We also have to do an induction step for attaching two trees using (BARBARA), but we omit those details.

We now return to our development, showing the completeness of the logic using the canonical model. Lemma 2.17 is the key result in this direction.

**Remark 2.15**

Before we turn to the proof, it might be useful to answer a question that comes up a lot. Why didn't we just *define* the canonical model so that (2.6) holds? After all, this would save us the trouble of proving (2.6).

The problem is that we can only define a model by giving an interpretation of the nouns and verbs. Once we do that, we get the interpretations of the terms automatically, by recursion. So we can't simply *define* models to have desirable features like (2.6). We have to define the model some other way and *prove* what we want.

**Theorem 2.16: If  $\Gamma \not\vdash \varphi$ , then  $\varphi$  is false in the canonical model  $\mathcal{M}$  of  $\Gamma$** 

Until further notice,  $\mathcal{M}$  is the canonical model of  $\Gamma$ .

**Lemma 2.17: For all terms  $x$** 

$$\llbracket x \rrbracket = \{y : \Gamma \vdash \text{All } y \ x\}. \quad (2.6)$$

*Proof.* The proof is by induction on  $x$ . When  $x$  is a noun, our result is by the definition of our model. Assume that  $x$  is a term and that (2.6) holds for  $x$ . Now fix a verb  $r$ . We prove that

$$\llbracket r \text{ all } x \rrbracket = \{y : \Gamma \vdash \text{All } y \ (r \text{ all } x)\}.$$

There are two directions. First, let  $y \in \llbracket r \text{ all } x \rrbracket$ . By induction hypothesis,  $x \in \llbracket x \rrbracket$ . So  $(y, x) \in \llbracket r \rrbracket$ . That is,  $\Gamma \vdash \text{All } y \ (r \text{ all } x)$ . In the other direction, assume that  $\Gamma \vdash \text{All } y \ (r \text{ all } x)$ . Suppose that  $z \in \llbracket x \rrbracket$ . Then by induction hypothesis,  $\Gamma \vdash \text{All } z \ x$ . Using (DOWN),

$$\Gamma \vdash \text{All } y \ (r \text{ all } z).$$

So  $(y, z) \in \llbracket r \rrbracket$ . This for all  $z \in \llbracket x \rrbracket$  shows that  $y \in \llbracket r \text{ all } x \rrbracket$ . □

**Lemma 2.18:  $\mathcal{M} \models \Gamma$** 

*Proof.* Suppose that  $\Gamma$  contains  $\text{All } u \ v$ . We show that  $\llbracket u \rrbracket \subseteq \llbracket v \rrbracket$  in  $\mathcal{M}$ . For this, let  $y \in \llbracket u \rrbracket$ . By Lemma 2.17,  $\Gamma \vdash \text{All } y \ u$ . And then using (BARBARA), we have  $\Gamma \vdash \text{All } y \ v$ . Therefore,  $y \in \llbracket v \rrbracket$ , just as desired. □

**Lemma 2.19: For all sentences  $\varphi$ , if  $\mathcal{M} \models \varphi$ , then  $\Gamma \vdash \varphi$ .**

*Proof.* Take a sentence  $\varphi$ , and write it as  $\text{All } a \ b$ . Suppose that  $\llbracket a \rrbracket \subseteq \llbracket b \rrbracket$  in  $\mathcal{M}$ . Now  $\mathcal{M} \models \Gamma$ , by Lemma 2.18. But  $a \in \llbracket a \rrbracket$ , using (AXIOM) in our system and also Lemma 2.17. And so  $a \in \llbracket b \rrbracket$ . By Lemma 2.17 again, we have  $\Gamma \vdash \text{All } a \ b$ . □



## 2.3 Algorithmic analysis

Theorem 2.16 shows the completeness of  $A(\mathcal{RC})$ . Our next order of business is the algorithmic analysis.

If we have a *finite* set  $\Gamma$ , and another sentence  $\varphi$ , how can we determine whether or not  $\Gamma \vdash \varphi$ ? We cannot directly answer this question by constructing the canonical model, since it is infinite. However, given a finite set  $S$ , we *can* answer the question

Is there a derivation showing  $\Gamma \vdash \varphi$  all of whose terms belong to  $S$ ?

The reason is that for a finite set  $S$ , there are only finitely many sentences that can be made from  $S$ , hence only finitely many of those  $\varphi$  can follow from  $\Gamma$ . We still need a way to find all of those sentences. We do this by generating all derivations that only use terms from  $S$ . (Actually, since there are infinitely many of those, we'll only generate the sentences at the roots.) It might come as a surprise that this is a sensible way of doing things, but our analysis will support this. And even more importantly, assuming that  $S$  is “big enough”, we'll show that every sentence which follows from  $\Gamma$  according to our original definitions follows under all the restrictions that we have mentioned.

On top of all of this, we still have to say what  $S$  is, and why it is “big enough”. It turns out that we can take  $S$  to be all terms in  $\Gamma$ , together with all terms in  $\varphi$ , and also the subterms of the terms in  $\varphi$ .

### Definition 2.20: subterms

We define the *subterms* of a term  $t$  by recursion: the only subterm of a noun is itself, and the subterms of  $r$  all  $x$  are this term  $r$  all  $x$  together with the subterms of  $x$ . For example, the set of subterms of

see all (like all buffaloes)

is {buffaloes, like all buffaloes, see all (like all buffaloes)}.

Now let's take a sentence like All  $x$  are  $y$ . The *principal terms* of this sentence are the terms  $x$  and  $y$ . (That is, we do not include the proper subterms of  $x$  and  $y$ .)

As pertinent example, consider the case when

$$\Gamma = \{\text{All } x \text{ } y, \text{All } y \text{ } z\},$$

and when we want to see if

$$\Gamma \vdash \text{All (see all } z) \text{ (see all } x)$$

or not. We might go hunting for a proof that only used the terms  $x$ ,  $y$ ,  $z$ , see all  $z$  and see all  $x$ . There are some proof trees that do *not* meet this criterion, as the proof on the left shows:

$$\frac{\frac{\text{All } y \text{ } z}{\text{All (see all } z) \text{ (see all } y)}}{\text{All (see all } z) \text{ (see all } x)}} \quad \frac{\text{All } x \text{ } y}{\text{All (see all } y) \text{ (see all } x)}} \quad \frac{\frac{\text{All } x \text{ } y \quad \text{All } y \text{ } z}{\text{All } x \text{ } z}}{\text{All (see all } z) \text{ (see all } x)}}$$

The term see all  $y$  is neither a principal term in the leaves nor a subterm of the root. On the other hand, the proof on the right does what we want. It turns out that in general, if  $\Gamma \vdash \varphi$ , then there

is a derivation which is “confined” in the sense that all terms in it belong to  $S$ , where  $S$  is the set of all principal terms of the leaves, together with the subterms of the root.

This is an important fact about the proof system  $A(RC)$ . However, we are not going to prove it (We’ll sketch a proof in the exercises.) In effect, we use a set  $S$  which is bigger than the one mentioned above. Our set will be the principal terms of  $\Gamma \cup \{\varphi\}$ , together with all subterms. (So this set includes the subterms of the principal terms in  $\Gamma$ .) So our result is weaker, but it still does what we want. And the work is easier.

Up until now, our discussion in this section has been purely motivational. Now we turn to the details.

**Algorithm 2.21:** given a finite set  $\Gamma \subseteq \mathcal{A}(\mathcal{RC})$  and a sentence  $\varphi = \text{All } p \ q$ , tell whether or not  $\Gamma \vdash \varphi$ .

```

1:  $M \leftarrow$  the set of principal terms in  $\Gamma \cup \{\varphi\}$ , plus the subterms of these terms
2:  $AX_M \leftarrow \{\text{All } x \ x : x \in M\}$ 
3:  $\Gamma^* \leftarrow$  the smallest set such that
    $\Gamma \cup AX_M \subseteq \Gamma^*$ 
   If  $\Gamma^*$  contains  $\text{All } x \ y$  and  $\text{All } y \ z$ , then  $\Gamma^*$  contains  $\text{All } x \ z$ .
   If  $\Gamma^*$  contains  $\text{All } x \ y$  and  $\text{All } z \ (r \text{ all } y)$ , then  $\Gamma^*$  contains  $\text{All } z \ (r \text{ all } x)$ .
4: if  $\varphi \in \Gamma^*$  then
5:   output “Yes”  $\triangleright \Gamma \vdash \varphi$ 
6: else  $\triangleright$  we construct a counter-model with universe  $M$ 
7:   for all  $u \in \mathbf{P}, r \in \mathbf{R}$  do
8:      $\llbracket u \rrbracket = \{v \in M : \Gamma^* \text{ contains } \text{All } v \ u\}$ .
9:      $\llbracket r \rrbracket = \{(u, v) \in M \times M : \Gamma^* \text{ contains } \text{All } u \ (r \text{ all } v)\}$ .
10:   end for
11:   output “No” and the model  $\mathcal{M} = (M, \llbracket \cdot \rrbracket)$  in lines 7–10.
12: end if
```

The idea in the algorithm is to take  $\Gamma$  and build a larger set called  $\Gamma^*$ . This set  $\Gamma^*$  is built by using the proof rules in our logic, but in a restricted way. In particular,  $\Gamma^*$  will be finite (unlike the set of all sentences provable from  $\Gamma$ ). The “restricted way” here has a specific meaning: see lines 2 and 3. The set  $M$  plays the role of  $S$  in our previous discussion. We have changed letters because eventually  $M$  will be the universe of our counter-models. In any case, when reading the proof of Theorem 2.22, you should think of “ $\varphi \in \Gamma^*$ ” as an algorithmic surrogate for “ $\Gamma \vdash \varphi$ ”. Once we have  $\Gamma^*$ , we see whether the input sentence  $\varphi$  is in or out. If it is in, then the original  $\Gamma$  will imply  $\varphi$ . If not, then the work we did in building  $\Gamma^*$  gives us enough “stuff” to make a finite version of the canonical model using  $M$  as the universe.

**Theorem 2.22**

Algorithm 2.21 works: Given a finite set  $\Gamma$  in  $\mathcal{A}(\mathcal{RC})$  and a sentence  $\varphi$  in  $\mathcal{A}(\mathcal{RC})$ , the algorithm halts in finite time. Moreover,

1. If the output is “Yes”, then  $\Gamma \vdash \varphi$ .
2. If the output is “No” and a model  $\mathcal{M}$ , then  $\mathcal{M} \models \Gamma$  but  $\mathcal{M} \not\models \varphi$ .

*Proof.* Here is the reason that the algorithm eventually terminates. Let  $M$  be as in line 1 of Algorithm 2.21. Let  $M^+$  be all terms  $r$  all  $x$  with  $x$  in  $M$  and  $r$  occurring in  $\Gamma \cup \{\varphi\}$ . Then  $\Gamma^*$  is a subset of  $\{\text{All } a \ b : a \in M, b \in M^+\}$ . This is a finite set. In order to build this set, we take the finite set  $\Gamma$  and add the finite set  $\text{AX}_M$ , and then we keep adding more and more consequences using (BARBARA) and (DOWN). Eventually we reach a point when nothing new can be deduced from what we have. At that point, we go to line 4.

We divide the rest of the proof of Theorem 2.22 into a series of lemmas. After the next two lemmas, the lemmas here are refined versions of the results we saw in the previous section concerning the canonical model.

**Lemma 2.23**

Every sentence in  $\Gamma^*$  is provable from the original set  $\Gamma$ .

*Proof.* This is simply because the sentences in  $\text{AX}_M$  are provable, and because the rules which we use to build  $\Gamma^*$  are special cases of the logical rules in our system  $\mathcal{A}(\mathcal{RC})$ .  $\square$

**Lemma 2.24**

For all terms  $x \in M$ ,  $\llbracket x \rrbracket = \{y \in M : \Gamma^* \text{ contains the sentence } \text{All } y \ x\}$ .

*Proof.* The proof is by induction on the term  $x$ . For  $x$  a noun in  $M$ , the assertion in our lemma is Step 8 in the algorithm. Assume that  $r$  all  $x$  belongs to  $M$  and that our lemma holds for  $x$ ; we show that our result holds for the larger term  $r$  all  $x$ . Before we begin, note that since  $M$  is closed under subterms,  $x$  also belongs to  $M$ .

Let  $y \in \llbracket r \text{ all } x \rrbracket$ . Then  $y \in M$ , and  $y$  is related by  $\llbracket r \rrbracket$  to all elements of  $\llbracket x \rrbracket$ . One of those elements is  $x$  itself. This is by our induction hypothesis, and by the fact that  $\text{All } x \ x$  belongs to  $\text{AX}_M \subseteq \Gamma^*$ . So by definition of  $\llbracket r \rrbracket$ , the sentence  $\text{All } y \ (r \text{ all } x)$  belongs to  $\Gamma^*$ . In the other direction, assume that  $y \in M$  and that  $\text{All } y \ (r \text{ all } x)$  belongs to  $\Gamma^*$ . We show that  $y$  is related by  $\llbracket r \rrbracket$  to all  $z \in M$  such that  $z \in \llbracket x \rrbracket$ . So let  $z \in \llbracket x \rrbracket$ . Then by induction hypothesis,  $\Gamma^*$  contains  $\text{All } z \ x$ . So by (DOWN),  $\Gamma^*$  contains  $\text{All } y \ (r \text{ all } z)$ . This last point tells us what we want:  $y \llbracket r \rrbracket z$ .  $\square$

**Lemma 2.25**

$\mathcal{M} \models \Gamma$ .

*Proof.* Take one of the sentences in  $\Gamma$ , say  $\text{All } u \ v$ . First of all, notice that  $u$  and  $v$  belong to  $M$ . (Why is this?) To see that  $\llbracket u \rrbracket \subseteq \llbracket v \rrbracket$  in  $\mathcal{M}$ , let  $y \in \llbracket u \rrbracket$ . Then  $y$  again belongs to  $M$ . By Lemma 2.24,  $\Gamma^*$  contains  $\text{All } y \ u$ . Also,  $\Gamma^*$  contains  $\text{All } u \ v$ . Using (BARBARA),  $\Gamma^*$  contains  $\text{All } y \ v$ . Thus,  $y \in \llbracket v \rrbracket$ , just as desired.  $\square$

### Lemma 2.26

Let  $\psi$  be a sentence  $\text{All } u \ v$  with  $u, v \in M$ . If  $\mathcal{M} \models \psi$ , then  $\Gamma^*$  contains  $\text{All } u \ v$ .

*Proof.* Again, we emphasize that  $u$  and  $v$  belong to  $M$  in this lemma. We assume that  $\llbracket u \rrbracket \subseteq \llbracket v \rrbracket$  in  $\mathcal{M}$ . Now  $u \in \llbracket u \rrbracket$ , by Lemma 2.24. And so  $u \in \llbracket v \rrbracket$ . By Lemma 2.24 again,  $\Gamma^*$  contains  $\text{All } u \ v$ .  $\square$

The previous lemmas enable us to complete the proof of Theorem 2.22. Let us take things from the top. We begin with  $\Gamma$  and  $\varphi$ , where  $\varphi$  is the sentence  $\text{All } p \ q$ . The nouns  $p$  and  $q$  belong to  $M$ . Now we run the algorithm. If  $\Gamma^*$  contains  $\text{All } p \ q$ , then Algorithm 2.21 says “Yes,” and Lemma 2.26 shows that  $\Gamma \vdash \varphi$ . But if  $\Gamma^*$  does not contain  $\text{All } p \ q$ , the algorithm outputs “No” together with a model  $\mathcal{M}$ . Lemma 2.25 shows that  $\mathcal{M}$  indeed satisfies  $\Gamma$ . Furthermore,  $\mathcal{M} \not\models \varphi$ , since if we did have  $\mathcal{M} \models \varphi$ , then Lemma 2.26 would show that  $\Gamma^*$  contains  $\text{All } p \ q$ , contrary to this case.  $\mathcal{M}$  shows us that  $\Gamma \not\models \text{All } p \ q$ . By soundness of the logic,  $\Gamma \not\vdash \text{All } p \ q$ .  $\square$

## 2.4 Extra stuff

### Remark 2.27: NOTE

The material in this section is not needed for what follows, and probably it should all be deleted. Maybe some of it could survive as a set of exercises.

### Definition 2.28: cut-term and normal proof tree

In an application of (BARBARA), say

$$\frac{\text{All } x \text{ are } y \quad \text{All } y \text{ are } z}{\text{All } x \text{ are } z}$$

the *cut-term* is  $y$ .

A proof tree  $\mathcal{T}$  over  $\Gamma$  is *normal* if every term which occurs in  $\mathcal{T}$  is either a principal term of a leaf, or else is a subterm of the root.

### Theorem 2.29

If  $\Gamma \vdash \varphi$ , then there is a normal proof tree  $\mathcal{T}$  which shows this.

*Proof.* We prove this by induction on proof trees over  $\Gamma$ . The trees which are simply elements of  $\Gamma$  or instances of (AXIOM) are normal. If we take a normal proof tree and add a root using (ANTI), the result is also normal. So the main argument concerns (BARBARA). Suppose that we have proof

trees  $\mathcal{T}_1$  and  $\mathcal{T}_2$ , and the roots of these trees are suitable for an application of (BARBARA). By induction hypothesis, we assume that  $\mathcal{T}_1$  and  $\mathcal{T}_2$  are normal. (That is, if need be, replace  $\mathcal{T}_1$  and  $\mathcal{T}_2$  by such normal trees.) So we assume that all terms in each  $\mathcal{T}_i$  are either principal terms of the leaves of  $\mathcal{T}_i$  or else subterms of the root of  $\mathcal{T}_i$ . Let us say that the root of  $\mathcal{T}_1$  is  $\text{All } x \ t$ , and the root of  $\mathcal{T}_2$  is  $\text{All } t \ z$ . We combine the two trees to get a proof tree  $\mathcal{T}^*$  whose root is  $\text{All } x \ z$ . The only term which needs to be considered is  $t$ , the cut term at the root. If  $t$  were a principal term in a leaf of either  $\mathcal{T}_1$  or  $\mathcal{T}_2$ , we are done because then  $\mathcal{T}$  would be normal. term,  $\mathcal{T}^*$  is not normal, and we need to modify it. Consider what happens when we follow the term  $t$  up the tree in both  $\mathcal{T}_1$  and  $\mathcal{T}_2$ . The occurrence in  $\mathcal{T}_1$  goes up the right-branch of  $\mathcal{T}_1$ , but it cannot go all the way to the leaves. Similarly, the occurrence in  $\mathcal{T}_2$  goes up the left-branch of  $\mathcal{T}_2$ . In both trees, we eventually reach an instance of ANTI). This means that  $t$  is of the form  $\text{see all } y$ , where  $\text{see}$  is just a possible verb here. Going back on the two sides, we eventually reach places where  $\text{see all } y$  is introduced. To make life easier, we are going to show what happens when  $\text{see all } y$  is introduced two steps up from the roots of  $\mathcal{T}_1$  and  $\mathcal{T}_2$ . So  $\mathcal{T}^*$  looks like

$$\begin{array}{c}
 \begin{array}{c} (1) \\ \vdots \\ \text{All } x \text{ (see all } p) \end{array} \quad \frac{\begin{array}{c} (2) \\ \vdots \\ \text{All } y \ p \end{array}}{\text{All (see all } p) \text{ (see all } y)} \quad \frac{\begin{array}{c} (3) \\ \vdots \\ \text{All } q \ y \end{array}}{\text{All (see all } y) \text{ (see all } q)} \quad \begin{array}{c} (4) \\ \vdots \\ \text{All (see all } q) \ z \end{array} \\
 \hline
 \text{All } x \text{ (see all } y) \quad \text{All (see all } y) \ z \\
 \hline
 \text{All } x \ z
 \end{array}$$

Since every subtree of a normal tree is normal, the trees marked (1)–(4) are normal. Further,  $\text{see all } p$  and  $\text{see all } q$  occur in the leaves of  $\mathcal{T}^*$ . Consider the following tree:

$$\begin{array}{c}
 \begin{array}{c} (1) \\ \vdots \\ \text{All } x \text{ (see all } p) \end{array} \quad \frac{\begin{array}{c} (3) \quad (2) \\ \vdots \quad \vdots \\ \text{All } q \ y \quad \text{All } y \ p \end{array}}{\text{All } q \ p} \quad \begin{array}{c} (4) \\ \vdots \\ \text{All (see all } q) \ z \end{array} \\
 \hline
 \text{All (see all } p) \text{ (see all } q) \quad \text{All (see all } q) \ z \\
 \hline
 \text{All (see all } p) \ z \\
 \hline
 \text{All } x \ z
 \end{array}$$

This is a proof tree over  $\Gamma$ , and by our reasoning above, it is normal.  $\square$

Theorem 2.29 leads to an algorithm which will tell whether or not  $\Gamma \vdash \varphi$ . This is presented in Algorithm 2.21. The main points are that the algorithm is correct and that it halts in finite time. The correctness is mainly due to the definition of *normality* and to the result in Theorem 2.29. And here is the reason that the algorithm halts. Each of the  $\Gamma_i$  sets is a set of sentences of the form  $\text{All } x \ y$  with  $x$  and  $y$  coming from  $M$ . There are only finitely many such sentences. So eventually, the sequence of sets

$$\Gamma_0 \subseteq \Gamma_1 \subseteq \dots \subseteq \Gamma_{2n} \subseteq \dots$$

must stabilize.

**Example 2.30**

[enhanced, watermark color=red!15!white, watermark text=Challenge]Challengechallenge  
 Here is a question: does the conclusion follow?

$$\begin{array}{l} \text{All hawks birds} \\ \text{All (see all turtles) (see all (see all hawks))} \\ \text{All (see all (see all hawks)) turtles} \\ \text{All turtles (see all birds)} \\ \hline \text{All (see all hawks) (see all birds)} \end{array}$$

One way to see would be to follow Algorithm 2.21. But in this case, one can see that set  $\Gamma$  of premises has a special feature:

$$\Gamma \vdash \text{All } x \text{ hawks} \quad \text{iff} \quad x \text{ is hawks}$$

This is an easy induction on proofs. The main point is that none of the assumptions have hawks as their second noun. Using this fact, the same kind of argument shows that

$$\Gamma \vdash \text{All (see all hawks) } x \quad \text{iff} \quad x \text{ is see all hawks}$$

This shows that the conclusion *does not* follow. But it does not show us how to build a model of the assumptions where the conclusion fails. For this, we need some additional work.

**Normal proofs** At the beginning of this section, we raised the following issue: to check whether  $\Gamma \vdash \varphi$  or not, is there a way to state ahead of time a set  $S$  of terms (definable from  $\Gamma$  and  $\varphi$ ) with the property that  $\Gamma \vdash \varphi$  if and only if there is a proof tree all of whose terms are in  $S$ ? Theorem 2.22 contains an answer to this question:  $S$  can be taken to be the terms in  $\Gamma$  and  $\varphi$ , together with their subterms. But this is not the best possible answer. We can make  $S$  smaller.

**Theorem 2.31**

If  $\Gamma \vdash \varphi$ , then there is a derivation all of whose terms are either principal terms of the leaves or else are subterms of the root.

This does not seem to follow from the work we did. In fact, we aren't aware of any model-theoretic argument for it. Instead, there is a proof-theoretic proof which we outline in Exercise ??.

**Remark 2.33**

Algorithm 2.21 had the feature that if  $\Gamma \vdash \varphi$ , the algorithm will output “Yes”. But it will not output a derivation in our proof system  $\mathbf{A(RC)}$ . However, it is easy to modify the algorithm to arrange this. At each stage when we build up  $\Gamma^*$ , we should include not only sentences but also proof trees in the system. We did not do this in our statement of the algorithm because it makes an already-complicated proof even harder to understand.

**Background assumptions on verbs** There are a few additional interesting phenomena that happen with more than one verb. The first is that we might want to work with *background assumptions that are not expressible in our logic so far*. For example, consider the following purported inference:

$$\begin{array}{l} \text{All bicyclists are football players} \\ \text{All football fans love all football players} \\ \hline \text{All football fans like all bicyclists} \end{array} \quad (2.7)$$

The way in which we are doing things so far, there is no relation between like and love in our models. And so the inference above fails. Yet there is a sense in which a competent speaker *would* agree to the conclusion given the premises. And we cannot add a premise such as “loving involves liking” because this is not expressible in the language. For this reason, we expand our overall framework.

In addition to our assumptions  $\Gamma$ , we can adopt a set  $\Delta$  of *background assumptions* of the form  $r \sqsubseteq s$  for  $r, s \in \mathbf{R}$ . ( $\Delta$  is the Greek letter Delta.) In this case above, we might have

$$\Delta = \{\text{loves} \sqsubseteq \text{likes}\}.$$

#### Definition 2.34: a model respecting a set of assumptions

A model  $\mathcal{M}$  *respects*  $\Delta$  if whenever  $r \sqsubseteq s$  belongs to  $\Delta$ , then  $\llbracket r \rrbracket \subseteq \llbracket s \rrbracket$  in  $\mathcal{M}$ .

In the proof theory, we define  $\Gamma; \Delta \vdash \varphi$  by adding to the proof system in Definition 2.10 the following rules:

$$\frac{\text{All } x (r \text{ all } y)}{\text{All } x (s \text{ all } y)} r \sqsubseteq s$$

We have one rule for each inequality assertion  $r \sqsubseteq s$  in  $\Delta$ .

Turning back to (2.7), we take  $\Delta = \{\text{like} \sqsubseteq \text{love}\}$ . And then the inference in (2.7) works out. It corresponds to the following derivation, where we have dropped the word **football** to save space:

$$\frac{\frac{\text{All fans (love all players)}}{\text{All fans (like all players)}} \text{LOVES} \sqsubseteq \text{LIKES} \quad \frac{\text{All bicyclists players}}{\text{All (like all players) (like all bicyclists)}} \text{ANTI}}{\text{All fans (like all bicyclists)}} \text{B}$$

The main result on this system is left to you in Exercise 2.9.

## 2.5 Binary intersection terms

This section presents another addition to  $\mathcal{A}$ , a language  $\mathcal{A}_2$  where we can talk about *intersections of two sets*.

**Syntax, semantics, and proof theory** The language in this section is called  $\mathcal{A}_2$ . Recall that  $\mathcal{A}$  is constructed on top of a set  $\mathbf{P}$  of nouns. In contrast,  $\mathcal{A}_2$  uses a different (but related) set  $\mathbf{P}_2$  whose elements are the formal pairs  $x \sqcap y$  of elements of  $\mathbf{P}$ . We call these items *terms*. We allow  $x$  and  $y$  to be the same, and so we have terms  $x \sqcap x$  for all nouns  $x$ . We take sentences of the form

$$\text{All } x \sqcap y \text{ are } z \sqcap w.$$

**Definition 2.35:  $\mathcal{A}_2$ , the logic of binary intersections**

$$\begin{array}{c}
\frac{}{\text{All } x \sqcap y \text{ are } y \sqcap x} \qquad \frac{\text{All } x \sqcap y \text{ are } z \sqcap w \quad \text{All } z \sqcap w \text{ are } a \sqcap b}{\text{All } x \sqcap y \text{ are } a \sqcap b} \\
\\
\frac{}{\text{All } x \sqcap y \text{ are } x \sqcap x} \qquad \frac{\text{All } x \sqcap y \text{ are } z \sqcap w \quad \text{All } x \sqcap y \text{ are } a \sqcap b}{\text{All } x \sqcap y \text{ are } z \sqcap a}
\end{array}$$

We could read this as

$$\text{All } x \text{ which are } y \text{ are also } z \text{ which are } w.$$

The  $\sqcap$  symbol is not a term-forming operator in the usual sense; we cannot form terms like  $x \sqcap (y \sqcap z)$ . We read  $\sqcap$  as “meet” (following lattice theory) or as “intersection” (since that is how we interpret it). All of the terms are “binary meets”. We also don’t have variables on their own as terms in this language. So we cannot say  $\text{All } x \sqcap y \text{ are } z$ . Instead, we would say  $\text{All } x \sqcap y \text{ are } z \sqcap z$ . This detail is a choice on our part, and we made it to simplify some of the work below. We could have made other choices for the syntax, and then these other choices would be reflected in different formulations of the proof theory.

Technically, the terms  $x \sqcap y$  and  $y \sqcap x$  are different. However, the idea is that they should denote the same set, and so they will be semantically identical. That is, for all models  $\mathcal{M}$ ,

$$\llbracket x \sqcap y \rrbracket = \llbracket y \sqcap x \rrbracket \tag{2.8}$$

(We have not given the semantics yet, but you might try to guess it and then to check that (2.8) holds.) Moreover, the proof theory will also identify them. That is, we’ll have  $\vdash \text{All } x \sqcap y \text{ are } y \sqcap x$ , and  $\vdash \text{All } y \sqcap x \text{ are } x \sqcap y$ .

The semantics of  $\mathcal{A}_2$  works as follows. Recall that the set  $\mathbf{P}_2$  is built from some set  $\mathbf{P}$  of nouns. We use models of the form

$$\mathcal{M} = (M, \llbracket \cdot \rrbracket : \mathbf{P} \rightarrow \mathcal{P}(M)),$$

where  $\mathcal{P}(M)$  is the *power-set* of  $M$ , the set of all subsets of  $M$ . Then we define truth in such a model by:

$$\mathcal{M} \models \text{All } x \sqcap y \text{ are } z \sqcap w \quad \text{iff} \quad \llbracket x \rrbracket \cap \llbracket y \rrbracket \subseteq \llbracket z \rrbracket \cap \llbracket w \rrbracket.$$

**Derived semantic notions** Just as with all our logical languages, once we have a definition of  $\mathcal{M} \models \varphi$ , we get the derived notions and  $\mathcal{M} \models \Gamma$  and  $\Gamma \models \varphi$  automatically. (See Definition 0.3.) The main technical work is in crafting a proof system so as to define  $\Gamma \vdash \varphi$  in a syntactic way, and then to prove the soundness and completeness of this logical system.

**Logic** We have a logical system  $\mathcal{A}_2$  for this language, using the rules in Definition 2.35. The first two rules are the preorder property of the order relation. The third rule tells us that  $\text{All } x \sqcap y \text{ are } x \sqcap x$ . As another instance of this rule, we get  $\text{All } x \sqcap y \text{ are } y \sqcap y$ . The final rule in the figure is a variant on saying that  $z \sqcap a$  is the greatest lower bound of  $z$  and  $a$ . It says that if an arbitrary element  $(x \sqcap y)$  is below both  $z$  and  $a$ , then it is below  $z \sqcap a$ .



**Soundness/Completeness** Fix a set  $\Gamma$  of sentences in the logic. We define an order  $\leq_\Gamma$  on  $\mathbf{P}_2$  by

$$x \sqcap y \leq_\Gamma a \sqcap b \quad \text{iff} \quad \Gamma \vdash \text{All } x \sqcap y \text{ are } a \sqcap b$$

As before, we sometimes drop the  $\Gamma$  from the notation. It is important to note that  $\leq_\Gamma$  is always a preorder.

### Theorem 2.36

$\Gamma \vdash \text{All } x \sqcap y \text{ are } z \sqcap w$  iff  $\Gamma \models \text{All } x \sqcap y \text{ are } z \sqcap w$ .

*Proof.* The soundness being trivial, we only detail the completeness half of our theorem. Suppose that  $\Gamma \models \text{All } x \sqcap y \text{ are } z \sqcap w$ . Let  $\mathcal{M}$  be the structure defined by

$$\begin{aligned} M &= \mathbf{P}_2 \\ \llbracket u \rrbracket &= \{a \sqcap b : \Gamma \vdash \text{All } a \sqcap b \text{ are } u \sqcap u\} \end{aligned}$$

We claim that  $\mathcal{M} \models \Gamma$ . For this, let  $\text{All } u \sqcap v \text{ are } a \sqcap b$  belong to  $\Gamma$ . Let  $c \sqcap d \in \llbracket u \rrbracket \cap \llbracket v \rrbracket$ . From  $\Gamma$  we have  $\text{All } c \sqcap d \text{ are } u \sqcap u$  and also  $\text{All } c \sqcap d \text{ are } v \sqcap v$ . We have a derivation from  $\Gamma$ :

$$\frac{\frac{\frac{\vdots}{\text{All } c \sqcap d \text{ are } u \sqcap u} \quad \frac{\vdots}{\text{All } c \sqcap d \text{ are } v \sqcap v}}{\text{All } c \sqcap d \text{ are } u \sqcap v} \quad \text{All } u \sqcap v \text{ are } u \sqcap u}{\text{All } c \sqcap d \text{ are } u \sqcap u}$$

So  $c \sqcap d \in \llbracket u \rrbracket$ . Similarly,  $c \sqcap d \in \llbracket v \rrbracket$ , and so  $c \sqcap d \in \llbracket u \rrbracket \cap \llbracket v \rrbracket$ . In this way, the sentence  $u \sqcap v \leq a \sqcap b$  holds in  $\mathcal{M}$ .

We now know that  $\mathcal{M} \models \Gamma$ . By our overall assumption that  $\Gamma \models \text{All } x \sqcap y \text{ are } z \sqcap w$ , we know that  $\text{All } x \sqcap y \text{ are } z \sqcap w$  holds in  $\mathcal{M}$ . Now  $x \sqcap y \in \llbracket x \rrbracket \cap \llbracket y \rrbracket$ , easily. So  $x \sqcap y \in \llbracket z \rrbracket \cap \llbracket w \rrbracket$ . Thus we have a derivation from  $\Gamma$ :

$$\frac{\frac{\vdots}{\text{All } x \sqcap y \text{ are } z \sqcap z} \quad \frac{\vdots}{\text{All } x \sqcap y \text{ are } w \sqcap w}}{\text{All } x \sqcap y \text{ are } z \sqcap w}$$

This completes the proof. □

## 2.6 Exercises

**Exercise 2.1.** To get practice with the semantics, Example 2.4 shows a model. The universe is  $\{1, 2, 3, 4, 5\}$ , and the interpretations of the nouns are shown as sets. For the verbs we show the relation by drawing arrows. For example, in  $\mathcal{M}$ , 5 respects 2, but 2 does not respect 5. And 3 and 4 see each other, and also they see themselves.

Find the following sets:

1.  $\llbracket \text{respect all hawks} \rrbracket$
2.  $\llbracket \text{respect all birds} \rrbracket$

3.  $\llbracket \text{fear all (respect all birds)} \rrbracket$
4.  $\llbracket \text{see all (respect all birds)} \rrbracket$
5.  $\llbracket \text{fear all (see all (respect all birds))} \rrbracket$
6.  $\llbracket \text{respect all (fear all (respect all birds))} \rrbracket$

**Exercise 2.2.** Are the following true or false in the model  $\mathcal{M}$  from Exercise 2.1?

1. all hawks birds
2. all (respect all birds) (respect all hawks)
3. all hawks (see all hawks)
4. all turtles (see all turtles)
5. all (see all (respect all birds)) turtles
6. all hawks (see all (respect all birds))

**Exercise 2.3.** Let  $\Gamma$  be the set of sentences below:

All hawks birds  
 All (see all turtles) (see all (see all hawks))  
 All (see all (see all hawks)) turtles  
 All turtles (see all birds)

Show that each of the following is derivable from  $\Gamma$  by constructing a formal proof in either  $\mathcal{A}(\text{RC})$  or  $\mathcal{A}(\text{RC})'$ .

1. All (see all (see all hawks)) (see all hawks)
2. All (see all turtles) (see all (see all birds))
3. All (see all turtles) (see all (see all turtles))

Doing formal proofs in systems like this is mostly a matter of trial and error. It takes patience and luck. Reading through Section 2.3 will make this easier.

**Exercise 2.4.** This exercise has to do with the soundness of  $\mathcal{A}(\text{RC})$ .

1. First, prove an easy fact about the semantics: If  $\llbracket z \rrbracket \subseteq \llbracket y \rrbracket$  and  $\llbracket x \rrbracket \subseteq \llbracket r \text{ all } y \rrbracket$ , then  $\llbracket x \rrbracket \subseteq \llbracket r \text{ all } z \rrbracket$ .
2. Let  $\Gamma$  be any set of sentences in  $\mathcal{A}(\text{RC})$ . Show by induction on proof trees  $\mathcal{T}$  over  $\Gamma$  that if  $\varphi$  is the root of  $\mathcal{T}$ , then  $\Gamma \models \varphi$ .

3. Conclude from the last part that the logic is sound.

**Exercise 2.5.** Re-read Example 2.14, and check that the picture shown there really is the canonical model of  $\{\text{All } p \text{ are } q\}$ . That is, fill in the missing step in the inductive argument that was started in the text.

**Exercise 2.6.** Find the canonical model of

$$\Gamma = \{\text{All } p \text{ are } q, \text{All } q \text{ (see all } p)\} .$$

It's fine to draw a picture as we did in Example 2.14, and you don't have to give a detailed verification that no edges are missing from your picture.

**Exercise 2.7.** Recall that we have two logical systems,  $\mathcal{A}(\mathcal{RC})$  and  $\mathcal{A}(\mathcal{RC})'$ . We have seen that every instance of (DOWN) is provable in  $\mathcal{A}(\mathcal{RC})'$ . It follows that if  $\Gamma \vdash \varphi$  in  $\mathcal{A}(\mathcal{RC})$  then  $\Gamma \vdash \varphi$  in  $\mathcal{A}(\mathcal{RC})'$ .

Your problem: show that every instance of (ANTI) is provable in  $\mathcal{A}(\mathcal{RC})$ . Use this to show that if  $\Gamma \vdash \varphi$  in  $\mathcal{A}(\mathcal{RC})'$  then  $\Gamma \vdash \varphi$  in  $\mathcal{A}(\mathcal{RC})$ .

**Exercise 2.8.** This exercise sketches a way to make the models that our algorithm gives a little smaller (sometimes).

1. First, review the details of Example 2.32. We had a set  $\Gamma$  and a sentence  $\varphi$ , and we built a model which satisfies  $\Gamma$  and falsifies  $\varphi$ . The model at the end is shown in Example 2.4 on page 46. Let us declare the points *see all turtles* and *see all see all hawks* to be the same; in pictures, this means declaring points 3 and 5 to be the same. Let's call this point 8. So the points in the model are  $\{1, 2, 4, 6, 7, 8\}$ . The main point about the smaller model is that it is "well-defined". This means that it makes sense to talk about which nouns are true of 8, and also which points "see" 8, and which are seen by 8. Check all of this.
2. Draw the resulting model, and check that it satisfies  $\Gamma$  and falsifies  $\varphi$ .
3. Now it is your job to generalize this. Under what conditions can we declare two points  $x$  and  $y$  in a model to be the same? And how do we define the smaller model that arises from identifying  $x$  and  $y$ ?
4. Suppose that we identify two points according to what you said just above. Do the smaller model and the bigger model agree on all sentences  $\varphi$ ?

**Exercise 2.9.** This exercise concerns the addition of background assumptions on verbs which we saw on page 57. Prove the following completeness theorem: Let  $\Gamma \cup \varphi$  be a set of sentences in  $\mathcal{A}(\mathcal{RC})$ , and let  $\Delta$  be a set of background assumptions. Then the following are equivalent:

1. Every model of  $\Gamma$  which respects the assumptions in  $\Delta$  satisfies  $\varphi$ .
2.  $\Gamma; \Delta \vdash \varphi$ .

**Exercise 2.10.** Fill in the details in the last paragraph of the proof of Theorem 2.36 on page 60.

**Exercise 2.11.** Carry out the algorithmic analysis of the logic of binary intersections. That is, find an algorithm to tell whether or not  $\Gamma \vdash \varphi$ : if so, it gives a proof; if not, a counter-model. Prove that your algorithm is correct.

**Exercise 2.12.** Suppose we dropped the last rule in Definition 2.35 and instead used the following rule:

$$\frac{\text{All } x \sqcap y \text{ are } z \sqcap z \quad \text{All } x \sqcap y \text{ are } a \sqcap a}{\text{All } x \sqcap y \text{ are } z \sqcap a}$$

This rule is a special case of the rule which was dropped. Let us write  $\Gamma \vdash^* \varphi$  to mean that there is a derivation of  $\varphi$  from  $\Gamma$  in the second proof system. This problem addresses the issue of whether  $\Gamma \vdash \varphi$  is the same as  $\Gamma \vdash^* \varphi$ . The matter boils down to whether the two rules are in fact equivalent in the presence of the other rules.

Here are two ways which you can show this:

1. Check that the arguments in Section 2 go through for the relation  $\Gamma \vdash^* \varphi$ . Therefore, that system is complete. Then say why we must have

$$\text{All } x \sqcap y \text{ are } z \sqcap w, \text{All } x \sqcap y \text{ are } a \sqcap b \vdash^* \text{All } x \sqcap y \text{ are } z \sqcap a. \quad (2.9)$$

2. Show directly that (2.9) holds by giving a derivation.

**Another variation** We conclude this section with a few exercises on a logic for sentences of the form

All  $x$  which are  $y$  are  $z$ .

To save space, we abbreviate this by  $(x, y, z)$ . We take this sentence to be true in a given model  $\mathcal{M}$  if  $\llbracket x \rrbracket \cap \llbracket y \rrbracket \subseteq \llbracket z \rrbracket$ . Note that All  $x$  are  $y$  is semantically equivalent to  $(x, x, y)$ .

**Exercise 2.13.** Prove that the logic of All  $x$  which are  $y$  are  $z$  in Definition 2.38 is complete. [Hint: you can do this with a one-point model, just as in Exercise 1.9.]

**Definition 2.38**

$$\frac{}{(x, y, x)} \quad \frac{}{(x, y, y)} \quad \frac{(x, y, u) \quad (x, y, v) \quad (u, v, z)}{(x, y, z)}$$

**Exercise 2.14.** Let  $\Gamma$  be a set of  $(x, y, z)$  sentences.

1. Show that if  $\Gamma \models (x, y, z)$ , then  $\Gamma \models (y, x, z)$ .
2. Show that if  $\Gamma \vdash (x, y, z)$ , then  $\Gamma \vdash (y, x, z)$ .

3. Suppose that we remove the axiom  $(x, y, y)$ , and in its place take the symmetry rule

$$\frac{(y, x, z)}{(x, y, z)}$$

Show that the new system is complete.

**Exercise 2.15.** Let  $\varphi$  be a sentence of  $\mathcal{A}$ , say  $\varphi = \text{All } x \text{ are } y$ . Let  $\varphi^* = (x, x, y)$ . If  $\Gamma$  is a set of sentences of the first fragment, let  $\Gamma^* = \{\varphi^* : \varphi \in \Gamma\}$ . It is easy to check that if  $\Gamma \vdash \varphi$ , then  $\Gamma^* \vdash \varphi^*$ . Prove the converse. We say that the proof system for assertions  $(x, y, z)$  is a *conservative extension* of the system for All.

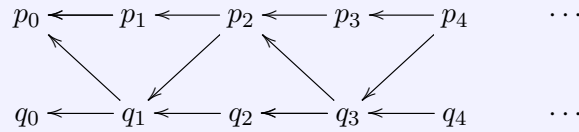
**Exercise 2.16.** Find a way to show that the logic of binary intersections  $x \sqcap y \leq z \sqcap w$  is a conservative extension of the logic of assertions  $(x, y, z)$ .

**Example 2.14**

Suppose we have two nouns  $p$  and  $q$ , and one verb,  $r$ . Let

$$\Gamma = \{\text{All } p \text{ } q\}$$

To make the notation simpler, let us write  $p_0$  for  $p$ , and  $p_{n+1}$  for  $r$  all  $p_n$ ; we adopt similar notation for  $q$ . Then in the canonical model of  $\Gamma$ , we have  $\llbracket p \rrbracket = \{p_0\}$ ,  $\llbracket q \rrbracket = \{p_0, q_0\}$ , and the interpretation of  $r$  is shown below:



The way we got this structure was to start generating all of the proof trees over our set  $\Gamma$ . Whenever  $\Gamma \vdash \text{All } x (r \text{ all } y)$ , we get an edge  $x \rightarrow y$ . For example,

$$\Gamma \vdash \text{All } p_2 \text{ } p_2$$

i.e.,  $\Gamma \vdash \text{All } p_2 (r \text{ all } p_1)$ . So we have  $p_2 \rightarrow p_1$  above. And

$$\Gamma \vdash \text{All } (r \text{ all } q) (r \text{ all } p).$$

This gives us the arrow  $q_1 \rightarrow p_0$ . Then based on the repetitive patterns in small examples, we drew the graph above.

**Example 2.32**

Let  $\Gamma$  be the set of four sentences below:

All hawks birds,  
 All (see all turtles) (see all (see all hawks)),  
 All (see all (see all hawks)) turtles,  
 All turtles (see all birds)

And let  $\varphi$  be

All (see all hawks) (see all birds).

We wish to see whether or not  $\Gamma \vdash \varphi$  using the work in this section.

$M$  is then the terms listed below:

- |                            |                    |
|----------------------------|--------------------|
| 1. hawks                   | 5. see all turtles |
| 2. see all hawks           | 6. birds           |
| 3. see all (see all hawks) | 7. turtles         |
| 4. see all birds           |                    |

These are just the terms that occur in  $\Gamma$  and  $\varphi$ , together with their subterms.

The set  $\Gamma^*$  turns out to be

|                                                           |                                                                   |
|-----------------------------------------------------------|-------------------------------------------------------------------|
| All hawks birds                                           | All turtles (see all hawks)                                       |
| All (see all turtles) (see all (see all hawks))           | All (see all (see all hawks)) (see all hawks)                     |
| All (see all (see all hawk)) turtles                      | All (see all turtles) (see all (see all turtles))                 |
| All turtles (see all birds)                               | All (see all (see all hawks)) (see all (see all birds))           |
| All (see all turtles) turtles                             | All (see all turtles) (see all hawks)                             |
| All (see all (see all hawks)) (see all birds)             | All (see all turtles) (see all (see all birds))                   |
| All (see all birds) (see all hawks)                       | All (see all (see all hawks)) (see all turtles)                   |
| All (see all turtles) (see all (see all (see all hawks))) | All (see all (see all hawks)) (see all (see all (see all hawks))) |
| All (see all turtles) (see all birds)                     | All (see all (see all hawks)) (see all (see all turtles))         |

and the seven sentences All  $x$   $x$  for  $x \in M$ . We can see that  $\varphi$  is not in the list above, and so we know that  $\Gamma \not\vdash \varphi$ . We want to make a counter-model. The points are the seven elements of  $M$ . We interpret the nouns using  $\Gamma^*$ , following what we saw in Step 8 of Algorithm 2.21.

$\llbracket \text{hawks} \rrbracket = \{\text{hawks}\}$   
 $\llbracket \text{birds} \rrbracket = \{\text{hawks, birds}\}$   
 $\llbracket \text{turtles} \rrbracket = \{\text{turtles, see all (see all hawks), see all turtles}\}$

$\llbracket \text{see} \rrbracket$  is defined from  $\Gamma^*$  in Step 9 of Algorithm 2.21. It is

(see all turtles, see all hawks), (turtles, birds), (see all (see all hawks), birds), (see all birds, hawks),  
 (see all turtles, see all (see all hawks)), (see all turtles, birds), (turtles, hawks), (see all (see all hawks), hawks),  
 (see all turtles, see all turtles), (see all (see all hawks), see all birds), (see all turtles, hawks), (see all turtles, see all birds),  
 (see all (see all hawks), turtles), (see all (see all hawks), see all hawks), (see all (see all hawks), see all turtles),  
 (see all hawks, hawks), (see all birds, birds), (see all turtles, turtles), (see all (see all hawks), see all (see all hawks))

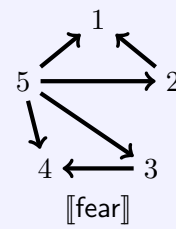
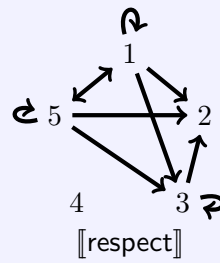
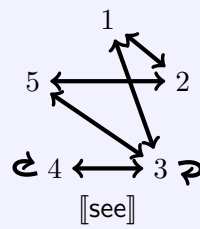
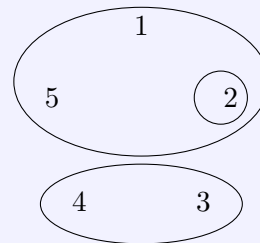
In drawing pictures, it makes sense to replace the elements of  $M$  with numbers in some fashion or other. We gave one numbering when we listed  $M$  above. And then the picture of the model is shown in Example 2.4 on page 46.

Example 2.37: A model for Exercises 2.1 and 2.2

$$\llbracket \text{hawks} \rrbracket = \{2\}$$

$$\llbracket \text{birds} \rrbracket = \{1, 2, 5\}$$

$$\llbracket \text{turtles} \rrbracket = \{3, 4\}$$







## Chapter 3

# $\mathcal{S}$ and $\mathcal{S}^\dagger$ : Syllogistic Logics

The best-known language in the history of logic is the language of syllogistic sentences in *All*, *Some*, *No*. It is not too much of an exaggeration to say that reasoning in this language was the central activity of logic for 1,500 years. The language of this chapter is a little bit larger than this language.

Despite its classical feel, the logical system of this chapter is largely missing from contemporary logic. This is because one cannot say very much with sentences of the restricted forms that we meet in this chapter. And since one has to go much further anyways, it seems that there is no point in stopping to study syllogistic logic. But in the rush to move on, we feel that something has been lost. As we'll see, although one cannot say much *in* syllogistic logic, there is a lot to be said *about* it.

This chapter follows on Chapter 1 rather than Chapter 2 in the sense that we have no verbs. (So it could be read right after Chapter 1.) We first add the sentence formers *Some* and *No* and look at the logic of those. The main work of the chapter comes when we add *noun-level negation*. So this would be like taking natural language nouns like *farmer*, *artist*, and *senator* and allowing one to say *non-farmer*, *non-artist*, and *non-senator*.

### 3.1 $\mathcal{S}$ : the classical syllogistic

At this point, you might like to review the syntax and semantics of the language  $\mathcal{A}$  from Section 1.1.<sup>1</sup> We want to extend  $\mathcal{A}$  by adding two new kinds of sentences

Some  $p$  are  $q$   
No  $p$  are  $q$

The definition of a model in Definition 1.3 stays the same. And for the semantics of our new sentences, we take

$$\begin{array}{lll} \mathcal{M} \models \text{Some } p \text{ are } q & \text{iff} & \llbracket p \rrbracket \cap \llbracket q \rrbracket \neq \emptyset \\ \mathcal{M} \models \text{No } p \text{ are } q & \text{iff} & \llbracket p \rrbracket \cap \llbracket q \rrbracket = \emptyset \end{array}$$

---

<sup>1</sup>Chapter 2 adds verbs and relative clauses to  $\mathcal{A}$ . That chapter is not relevant here, except for very few remarks.

**Example 3.1: Based on the model  $\mathcal{M}$  from Exercise 1.1.**

1.  $\mathcal{M} \not\models$  Some hawks are turtles because  $\llbracket \text{hawks} \rrbracket = \{2\}$ ;  $\llbracket \text{turtles} \rrbracket = \{3, 4\}$ ; and  $\{2\} \cap \{3, 4\} = \emptyset$ .
2.  $\mathcal{M} \not\models$  No hawks are birds because  $\llbracket \text{hawks} \rrbracket \cap \llbracket \text{birds} \rrbracket = \{2\} \neq \emptyset$ .
3.  $\mathcal{M} \models$  No hawks are turtles because  $\llbracket \text{hawks} \rrbracket \cap \llbracket \text{turtles} \rrbracket = \emptyset$ .
4.  $\mathcal{M} \not\models$  No birds are birds because  $\llbracket \text{birds} \rrbracket \cap \llbracket \text{birds} \rrbracket = \{1, 2, 5\}$ ; and  $\{1, 2, 5\} \neq \emptyset$ .

These examples only mention sentences of the form *Some  $p$  are  $q$*  and *No  $p$  are  $q$* . The semantics of sentences of the form *All  $p$  are  $q$*  is exactly as we have seen it in Chapter 1.

Now that we have seen the semantics, we get derived notions such as  $\Gamma \models \varphi$  from Definition 0.3.

**Example 3.2: All  $p$  are  $q$ , Some  $p$  are  $r \models$  Some  $q$  are  $r$ .**

This is perhaps the first general semantic fact to check in this logic. To see this, let  $\mathcal{M}$  be a model and assume that the two hypotheses are true in  $\mathcal{M}$ . Then  $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket$ , and also  $\llbracket p \rrbracket \cap \llbracket r \rrbracket \neq \emptyset$ . Let  $x \in \llbracket p \rrbracket \cap \llbracket r \rrbracket$ . Then  $x \in \llbracket p \rrbracket$  and also  $x \in \llbracket r \rrbracket$ . Since  $x \in \llbracket p \rrbracket$  and  $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket$ , we also have  $x \in \llbracket q \rrbracket$ . So  $x \in \llbracket q \rrbracket \cap \llbracket r \rrbracket$ . As a result,  $\llbracket q \rrbracket \cap \llbracket r \rrbracket \neq \emptyset$ . This means that  $\mathcal{M} \models$  Some  $q$  are  $r$ . This argument holds for all  $\mathcal{M}$ , and so we have established our semantic fact.

**Example 3.3: All  $p$  are  $v$ , All  $q$  are  $w$ , No  $v$  are  $w \models$  No  $p$  are  $q$ .**

To see this, fix a model  $\mathcal{M}$  where the hypotheses all hold. That is,  $\llbracket p \rrbracket \subseteq \llbracket v \rrbracket$ ,  $\llbracket q \rrbracket \subseteq \llbracket w \rrbracket$ , and  $\llbracket v \rrbracket \cap \llbracket w \rrbracket = \emptyset$ . By elementary reasoning involving sets, we see that  $\llbracket p \rrbracket \cap \llbracket q \rrbracket = \emptyset$ . This means that  $\mathcal{M} \models$  No  $p$  are  $q$ . Since  $\mathcal{M}$  was an arbitrary model of the hypotheses, we are done.

**Example 3.4: Some  $p$  are  $q$ , Some  $q$  are  $n$ , All  $q$  are  $m \not\models$  Some  $p$  are  $n$** 

We need only build a model in which the hypotheses hold and the conclusion fails. Let  $\varphi$  be *Some  $p$  are  $q$* , and let  $\psi$  be *Some  $q$  are  $n$* . (That is, we take the points of the model to be sentences.) We take for a model  $\mathcal{M} = \{\varphi, \psi\}$  with  $\llbracket p \rrbracket = \{\varphi\}$ ,  $\llbracket q \rrbracket = \{\varphi, \psi\}$ ,  $\llbracket n \rrbracket = \{\psi\}$ ,  $\llbracket m \rrbracket = \{\varphi, \psi\}$ . It is clear that  $\mathcal{M}$  has the desired properties.

Here is a comment on the uses of *sentences as points in a model*, something we just saw in Example 3.4. This is unexpected and strange at first glance. (In fact, it might be downright confusing!) The reason that we do that is that we want *some object or other* to serve as “witnesses” in our model that a *some* sentence holds. That is, to insure that *Some  $x$  are  $y$*  holds in a model, we need an object to put into both  $\llbracket x \rrbracket$  and  $\llbracket y \rrbracket$ . What could be more natural than the *sentence* *Some  $x$  are  $y$*  itself? And it is not a perfect novelty for us to use syntactic objects in our semantic objects. We have done the same thing all along, starting with our work on  $\mathcal{A}$  in Chapter 1: the points in the canonical model in Chapter 1 were nouns of the language.

## 3.2 Proof system

We define a proof system  $\mathcal{S}$  for  $\mathcal{S}$  using the rules in Logic 3.8. We have seen (AXIOM) and (BARBARA). Here is a discussion of the soundness of the rest of the rules other than (X). Fix a model  $\mathcal{M}$ . Assume that  $\llbracket p \rrbracket \cap \llbracket q \rrbracket \neq \emptyset$ . Let  $m \in \llbracket p \rrbracket \cap \llbracket q \rrbracket$ . This same  $m$  shows that  $\llbracket q \rrbracket \cap \llbracket p \rrbracket \neq \emptyset$ . What is more,  $\llbracket p \rrbracket$  itself must be non-empty. If in addition,  $\llbracket p \rrbracket \subseteq \llbracket n \rrbracket$ , then also  $\llbracket p \rrbracket \cap \llbracket n \rrbracket \neq \emptyset$ , due to the same point  $m$ . This leads to the soundness of (SOME<sub>1</sub>), (SOME<sub>2</sub>), and (DARII).

The rules (ZERO) is concerned with what is often called vacuous universal quantification. Please keep in mind that a sentence **All**  $x$  are  $y$  is true in a given model whenever  $\llbracket x \rrbracket = \emptyset$ . And for the same reason, if  $\llbracket q \rrbracket = \emptyset$  in a model, then every sentence of the form **No**  $q$  are  $p$  is true in the model. This is what is behind (ZERO). (NO) is similar: if  $\llbracket p \rrbracket = \emptyset$ , then  $\llbracket p \rrbracket \cap \llbracket q \rrbracket = \emptyset$  as well.



We turn to (CAMESTRES)<sup>2</sup>: let us assume that  $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket$  and  $\llbracket q \rrbracket \cap \llbracket r \rrbracket = \emptyset$ . To show that  $\llbracket r \rrbracket \cap \llbracket p \rrbracket = \emptyset$ , we assume the opposite and aim towards a contradiction. Let  $m \in \llbracket r \rrbracket \cap \llbracket p \rrbracket$ . Then  $m \in \llbracket r \rrbracket$  and  $m \in \llbracket p \rrbracket$ . Since  $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket$ , we also have  $m \in \llbracket q \rrbracket$ . But now  $m \in \llbracket q \rrbracket \cap \llbracket r \rrbracket$ . This contradicts our assumption that  $\llbracket q \rrbracket \cap \llbracket r \rrbracket = \emptyset$ . So we see that  $\llbracket r \rrbracket \cap \llbracket p \rrbracket = \emptyset$ , after all.

Before we discuss (X), we look at examples of the rules we have discussed so far. We frequently do not display the names of rules in our proof trees except when to emphasize some point or other.

### Example 3.5

Here is the first important derivation in the logic:

$$\frac{\frac{\text{All } n \text{ are } p \quad \text{Some } n \text{ are } n}{\text{Some } n \text{ are } p}}{\frac{\text{All } n \text{ are } q \quad \text{Some } p \text{ are } n}{\text{Some } p \text{ are } q}}$$

That is, if there is an  $n$ , and if all  $ns$  are  $ps$  and also  $qs$ , then some  $p$  is a  $q$ .

### Example 3.6

Corresponding to Example 3.1, let us check that

$\text{All } p \text{ are } v, \text{All } q \text{ are } w, \text{No } v \text{ are } w \vdash \text{No } p \text{ are } q.$

$$\frac{\frac{\frac{\text{All } p \text{ are } v \quad \text{No } v \text{ are } w}{\text{No } p \text{ are } w}}{\text{All } q \text{ are } w \quad \text{No } w \text{ are } p}}{\frac{\text{No } p \text{ are } q}{\text{No } q \text{ are } p}} \quad (3.1)$$

<sup>2</sup>The reason for the exclamation point is to mark this paragraph; we shall formalize this reasoning a little later (see page 76).

**Definition 3.8: rules of  $\mathcal{S}$** 

|                                                                                                                        |                                                                                                                        |                                                                                      |
|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| $\frac{}{\text{All } p \text{ are } p} \text{ AXIOM}$                                                                  | $\frac{\text{Some } p \text{ are } q}{\text{Some } p \text{ are } p} \text{ SOME}_1$                                   | $\frac{\text{Some } p \text{ are } q}{\text{Some } q \text{ are } p} \text{ SOME}_2$ |
| $\frac{\text{All } p \text{ are } n \quad \text{All } n \text{ are } q}{\text{All } p \text{ are } q} \text{ BARBARA}$ | $\frac{\text{All } q \text{ are } n \quad \text{Some } p \text{ are } q}{\text{Some } p \text{ are } n} \text{ DARII}$ |                                                                                      |
| $\frac{\text{No } q \text{ are } q}{\text{All } q \text{ are } p} \text{ ZERO}$                                        | $\frac{\text{No } p \text{ are } p}{\text{No } p \text{ are } q} \text{ NO}$                                           |                                                                                      |
| $\frac{\text{All } p \text{ are } q \quad \text{No } q \text{ are } r}{\text{No } r \text{ are } p} \text{ CAMESTRES}$ | $\frac{\text{No } p \text{ are } q \quad \text{Some } p \text{ are } q}{\varphi} \text{ X}$                            |                                                                                      |

**Example 3.7: (ANTI):  $\text{No } p \text{ are } q \vdash \text{No } q \text{ are } p$** 

|                                                       |                                                                                     |
|-------------------------------------------------------|-------------------------------------------------------------------------------------|
| $\frac{}{\text{All } p \text{ are } p} \text{ AXIOM}$ | $\frac{\text{No } p \text{ are } q}{\text{No } q \text{ are } p} \text{ CAMESTRES}$ |
|-------------------------------------------------------|-------------------------------------------------------------------------------------|

We come to (x), a rule relating **Some** and **No**. There are no models which satisfy both **No**  $p$  are  $q$  and **Some**  $p$  are  $q$ . So all models of both satisfy  $\varphi$ , *no matter what  $\varphi$  is*. We incorporate this observation into the proof system by adopting the rule of *ex falso quodlibet*. (x) is also called *ex contradictione quodlibet*: anything follows from a contradiction. It means that if you have proof tree with a node  $\varphi$ , and the nodes right above it are contradictories – one is **No**  $p$  are  $q$ , and the other is **Some**  $p$  are  $q$ , then for all  $\varphi$ , we can conclude  $\varphi$ . From a contradiction, we may have anything we like.

**The partial completeness result** The next theorem is “half” of the completeness of the proof system for  $\mathcal{S}$ . The second half is postponed to Exercises 3.18 and 3.19.

We begin with a bit of notation.  $\mathcal{S}$  has sentences using three difference sentence formers, **All**, **Some**, and **No**. We sometimes want to focus attention on subsets that only use one or two of these words. We’ll continue to write  $\mathcal{A}$  for the sentences using **All**. We write  $\mathcal{S}(\text{all, some})$  for the sentences of  $\mathcal{S}$  using **All** or **Some**, and similarly for  $\mathcal{S}(\text{all, no})$ . If  $\Gamma \subseteq \mathcal{S}$ , then we write

$$\begin{aligned}
 \Gamma_{\text{all}} &= \Gamma \cap \mathcal{A} \\
 \Gamma_{\text{all, some}} &= \Gamma \cap \mathcal{S}(\text{all, some}) \\
 \Gamma_{\text{some}} &= \Gamma \cap \mathcal{S}(\text{some})
 \end{aligned}$$

**Definition 3.9: the second canonical model**

Suppose that  $\Gamma \subseteq \mathcal{S}(\text{all}, \text{some})$ . The *second canonical model*  $\mathcal{M}_2$  is defined as follows: We take  $M_2$  to be  $\Gamma_{\text{some}}$ . (In words, the elements of  $M_2$  are the sentences in  $\Gamma$  that use the word *Some*.) For each noun  $u$ , define  $\llbracket u \rrbracket$  by

$$\llbracket u \rrbracket = \{ \varphi \in M_2 : \varphi \text{ contains a noun } v \leq u \}. \quad (3.2)$$

(The preorder  $\leq$  is defined from  $\Gamma$  in Definition 1.24.) Another way to put it:

$$\text{Some } x \text{ are } y \text{ belongs to } \llbracket u \rrbracket \quad \text{iff} \quad \text{either } x \leq u \text{ or } y \leq u.$$

**Lemma 3.10**

$$\mathcal{M}_2 \models \Gamma_{\text{all}, \text{some}}.$$

*Proof.* First, take a sentence in  $\Gamma$  using *All*, say *All  $x$  are  $y$* . We show that  $\llbracket x \rrbracket \subseteq \llbracket y \rrbracket$  in  $\mathcal{M}_2$ . Let  $\varphi \in \llbracket x \rrbracket$ ; let's say  $\varphi$  is *Some  $p$  are  $q$* . Then either  $p \leq x$  or  $q \leq x$ . But  $x \leq y$ . And so either  $p \leq y$  or  $q \leq y$ . Thus, our sentence  $\varphi$  belongs to  $\llbracket y \rrbracket$ . This just says that  $\varphi$  is true in  $\mathcal{M}_2$ .

Second, take a sentence in  $\Gamma$  using *Some*, say  $\varphi = \text{All } x \text{ are } y$ . Then  $\varphi \in \llbracket x \rrbracket \cap \llbracket y \rrbracket$ . So  $\llbracket x \rrbracket \cap \llbracket y \rrbracket \neq \emptyset$ . So in this case we again see that  $\varphi$  is true in  $\mathcal{M}_2$ .  $\square$

**Lemma 3.11**

For all sentences  $\varphi = \text{Some } p \text{ are } q$ , if  $\mathcal{M}_2 \models \varphi$ , then  $\Gamma \vdash \varphi$ .

*Proof.* Suppose that  $\mathcal{M}_2 \models \varphi$ , and thus  $\llbracket p \rrbracket \cap \llbracket q \rrbracket \neq \emptyset$ . We sketch a proof that  $\Gamma \vdash \varphi$ . To get started, let *Some  $x$  are  $y$*  be any sentence which belongs to  $\llbracket p \rrbracket \cap \llbracket q \rrbracket$ . We have four cases depending on whether  $x \leq p$  or  $y \leq p$ , and also whether  $x \leq q$  or  $y \leq q$ . Now breaking in to those cases shows that  $\Gamma \vdash \varphi$ .  $\square$

**Theorem 3.12: Partial completeness theorem for  $\mathcal{S}$** 

Let  $\Gamma$  be any set of sentences in  $\mathcal{S}$ , and suppose that  $\Gamma \models \text{Some } p \text{ are } q$ . Then  $\Gamma \vdash \text{Some } p \text{ are } q$ .

*Proof.* Let  $\mathcal{M}_2$  be defined from  $\Gamma$  as above. We know that this model satisfies  $\Gamma_{\text{all}, \text{some}}$ , by Lemma 3.10. We break into two cases, depending on whether  $\mathcal{M}_2 \models \Gamma_{\text{no}}$ .

If  $\mathcal{M}_2 \models \Gamma_{\text{no}}$ , then  $\mathcal{M}_2 \models \Gamma$ . Then by assumption,  $\mathcal{M}_2 \models \text{Some } p \text{ are } q$ . By Lemma 3.11,  $\Gamma \vdash \text{Some } p \text{ are } q$ .

If  $\mathcal{M}_2 \not\models \Gamma_{\text{no}}$ , there is some sentence, say *No  $m$  are  $n$* , which is in  $\Gamma$  but is false in  $\mathcal{M}_2$ . Then  $\mathcal{M}_2 \models \text{Some } m \text{ are } n$ . And by Lemma 3.11 again,  $\Gamma \vdash \text{Some } m \text{ are } n$ . But  $\Gamma \vdash \text{No } m \text{ are } n$ , too. (Why is this?) So now we may use (x) to see that  $\Gamma$  proves anything; in particular,  $\Gamma \vdash \text{Some } p \text{ are } q$ .  $\square$

At this point we have a partial completeness result for our proof system  $\mathcal{S}$ . It is possible to prove the full completeness theorem, but that result cannot simply use the model  $\mathcal{M}_2$  which we have seen (see Exercise 3.4). That is, Lemma 3.11 is false when  $\varphi$  is a sentence *All  $p$  are  $q$* . In order to rectify matters, one has to do some other work. And for that work, we find it convenient to move to the larger logic  $\mathcal{S}^\dagger$  which we introduce just below. After seeing the completeness theorem for that logic, we'll be in a better position to return to our work on  $\mathcal{S}$ .

### 3.3 $\mathcal{S}^\dagger$ : syllogistic logic with full negation on nouns

At this point we reformulate our logical language  $\mathcal{S}$ . We do this for a few reasons. First, we want to be able to say a little bit more than we can with  $\mathcal{S}$ : we want to say *Some  $a$  are not  $b$* . We could do this just by enlarging our syntax to add sentences of that very form. But we also want to put a “negative sign” on the first noun, to get *Some non- $a$  are not  $b$* . In effect, this is what we will do. To make things more mathematically suggestive, we add a *complementation operation* to our nouns.

We again begin with a set  $\mathbf{P}$  of nouns. We use  $p, q, \dots$ , for the elements of  $\mathbf{P}$ . Let

$$\mathbf{Lit} = \mathbf{P} \cup \{\bar{p} : p \in \mathbf{P}\}.$$

In other words, we have two copies of  $\mathbf{P}$ , using the “bar” symbol  $\bar{\phantom{x}}$  to distinguish the copies. We call the elements of this set *literals* following uses in areas of logic. Once again, the elements of  $\mathbf{Lit}$  as either nouns  $p, q$ , etc., or as *complemented nouns*  $\bar{p}, \bar{q}, \dots$



Moreover, we extend this idea of complementation to a *complementation operation*  $\bar{\phantom{x}} : \mathbf{Lit} \rightarrow \mathbf{Lit}$  on the literals such that  $\bar{\bar{p}} = p$  for all literals  $p$ . (Yes, we use the same letters  $p, q$ , etc. to range over literals in this section.) This is called the *involution* property. It implies that if  $\bar{p}$  and  $\bar{q}$  are the same literal, then  $p$  and  $q$  also are the same. (To see this, take the complement of both  $\bar{p}$  and  $\bar{q}$ . Then  $\bar{\bar{p}}$  is the same as  $\bar{\bar{q}}$ . That is,  $p$  is the same as  $q$ .)

**Syntax** We now come to the syntax of the language  $\mathcal{S}^\dagger$ . We take as sentences *All  $p$  are  $q$*  and *Some  $p$  are  $q$* , where  $p$  and  $q$  are any literals, allowing them to be complemented nouns. As before, we use letters like  $\varphi$  to denote sentences.

**Semantics** A *model*  $\mathcal{M}$  is a set  $M$  and a subset  $\llbracket p \rrbracket \subseteq M$  for each basic noun  $p$ , extended to the complemented nouns by

$$\llbracket \bar{p} \rrbracket = M \setminus \llbracket p \rrbracket \tag{3.3}$$

for all  $p$ . That is, a model  $\mathcal{M}$  is a pair  $(M, \llbracket \phantom{x} \rrbracket)$ , just as before. Then we extend the interpretation function to the complemented nouns by taking complement sets. Notice that (3.3) will hold not only when  $p$  is a noun but also when it is a complemented noun; so (3.3) holds for all literals.

We then define the satisfaction relation  $\mathcal{M} \models \varphi$  just as in the last section.

We again get derived notions such as  $\Gamma \models \varphi$  from Definition 0.3.

#### Example 3.13: illustration of complement sets and the semantics of $\mathcal{S}^\dagger$

Let  $M$  be the following model:  $M = \{1, 2, 3, 4, 5\}$ ,  $\llbracket p \rrbracket = \{1, 2\}$ ,  $\llbracket q \rrbracket = \{1, 2, 3\}$ , and  $\llbracket r \rrbracket = \{1, 3, 4, 5\}$ . Then we automatically have  $\llbracket \bar{p} \rrbracket = \{3, 4, 5\}$ ,  $\llbracket \bar{q} \rrbracket = \{4, 5\}$ , and  $\llbracket \bar{r} \rrbracket = \{2\}$ . Let  $\Gamma = \{\text{All } p \text{ are } q, \text{All } \bar{p} \text{ are } r\}$ . Then  $\mathcal{M} \models \Gamma$ .

**Example 3.14**

We claim that  $\Gamma \models \text{All } x \text{ are } z$ , where

$$\Gamma = \{ \text{All } \bar{y} \text{ are } p, \text{All } p \text{ are } q, \text{All } q \text{ are } y, \text{All } y \text{ are } p, \text{All } q \text{ are } z \}.$$

Here is an explanation. Since all  $y$  and all  $\bar{y}$  are  $p$ , everything whatsoever is a  $p$ . And since all  $p$  are  $q$ , and all  $q$  are  $y$ , we see that everything is a  $y$ . In particular, all  $x$  are  $y$ . But the last two premises and the fact that all  $p$  are  $q$  also imply that all  $y$  are  $z$ . So all  $x$  are  $z$ .

**Example 3.15**

Here is an example which we mention mostly as a challenge. Let  $\Gamma$  be the set of the sentences below:

$$\text{All } y \text{ are } x, \text{All } \bar{y} \text{ are } x, \text{All } \bar{z} \text{ are } y, \text{All } z \text{ are } \bar{y}, \text{All } z \text{ are } w.$$

It is *not* true that

$$\Gamma \vdash \text{All } y \text{ are } w.$$

Find a model  $\mathcal{M} \models \Gamma$  where  $\llbracket y \rrbracket \not\subseteq \llbracket w \rrbracket$ . The point of this exercise is that the details of the completeness proof for our logic will give us a way of *automatically* solving problems like this!

We shall see a solution later. But you might try to do this now in order to discover the ideas that we'll see in due course.

**No** We may regard **No**  $p$  are  $q$  as a variant notation for **All**  $p$  are  $\bar{q}$ . In other words, since we have sentences like **All**  $p$  are  $\bar{q}$ , there is no need for **No**  $p$  are  $q$ .

**Example 3.16**

For every literal  $p$ , there are no models of **Some**  $p$  are  $\bar{p}$ . We write  $\perp$  for every sentence of this form, and we think of it as a *contradiction*. Notice that

$$\text{Some } p \text{ are } \bar{p} \models \varphi \tag{3.4}$$

for *every* sentence  $\varphi$ . That is, if we write out what it means for (3.4) to hold, we see that it does hold *vacuously*.

**Semantic negations** The main difference between  $\mathcal{S}$  and  $\mathcal{S}^\dagger$  is every sentence  $\varphi$  in  $\mathcal{S}^\dagger$  has a *semantic negation*  $\bar{\varphi}$ . It is defined as follows:<sup>3</sup>

| $\varphi$        | $\bar{\varphi}$        |       |
|------------------|------------------------|-------|
| All $x$ are $y$  | Some $x$ are $\bar{y}$ | (3.5) |
| Some $x$ are $y$ | All $x$ are $\bar{y}$  |       |

<sup>3</sup>For people who have seen propositional logic, we should mention the difference between a negation  $\neg\varphi$  and our notation  $\bar{\varphi}$ . In this chapter, there is no negation symbol on sentences. The notation  $\bar{\varphi}$  is “meta-linguistic” in the sense that the overline symbol is not part of the language  $\mathcal{S}^\dagger$  of this section.



**Definition 3.17: rules of  $\mathcal{S}^\dagger$ :  $n$ ,  $p$ , and  $q$  are literals**

$$\begin{array}{c}
\frac{}{\text{All } p \text{ are } p} \text{ AXIOM} \qquad \frac{\text{All } p \text{ are } n \quad \text{All } n \text{ are } q}{\text{All } p \text{ are } q} \text{ BARBARA} \\
\\
\frac{\text{Some } p \text{ are } q}{\text{Some } p \text{ are } p} \text{ SOME}_1 \qquad \frac{\text{Some } p \text{ are } q}{\text{Some } q \text{ are } p} \text{ SOME}_2 \qquad \frac{\text{All } q \text{ are } n \quad \text{Some } p \text{ are } q}{\text{Some } p \text{ are } n} \text{ DARII} \\
\\
\begin{array}{c}
[\varphi] \\
\vdots \\
\perp \\
\overline{\overline{\varphi}} \text{ RAA}
\end{array}
\end{array}$$

We call  $\overline{\varphi}$  the *semantic negation* of  $\varphi$ . The main feature of this is that for all models  $\mathcal{M}$ ,

$$\mathcal{M} \not\models \varphi \quad \text{iff} \quad \mathcal{M} \models \overline{\varphi}$$

Another feature of this is that  $\overline{\overline{\varphi}} = \varphi$  for all  $\varphi$ .

**Proof by contradiction** The logic in Definition 3.17 has something we have not seen so far, the rule (RAA). This rule gives us the ability to do *proof by contradiction*. We discuss this by connecting to our earlier informal proof of the soundness of (CAMESTRES) on page 71. We turn to the translation of that informal proof into our current language and proof system. We

assume All  $p$  are  $q$  and All  $q$  are  $\bar{r}$ , also assume toward a contradiction Some  $r$  are  $\bar{p}$

The overall proof first obtains a contradiction as shows below:

$$\frac{\frac{\text{All } p \text{ are } q \quad \text{All } q \text{ are } \bar{r}}{\text{All } p \text{ are } \bar{r}} \text{ BARBARA} \quad \text{Some } r \text{ are } p}{\text{Some } r \text{ are } \bar{r}} \text{ DARII}$$

The sentence Some  $r$  are  $\bar{r}$  is our central example of a contradiction. At this point, we “take back” the assumption that was made “toward a contradiction.” In fact, our taking this back entitles us to *withdraw* that assumption and indeed to conclude the opposite. So we put brackets around the withdrawn assumption, and we add a new root to the tree:

$$\frac{\frac{\frac{\text{All } p \text{ are } q \quad \text{All } q \text{ are } \bar{r}}{\text{All } p \text{ are } \bar{r}} \text{ BARBARA} \quad [\text{Some } r \text{ are } p]}{\text{Some } r \text{ are } \bar{r}} \text{ DARII}}{\text{All } r \text{ are } \bar{p}} \text{ RAA} \tag{3.6}$$

The name (RAA) is short for *reductio ad absurdum*.

**Definitions concerning (RAA)** The sentences of the form Some  $p$  are  $\bar{p}$  are called *contradictions* in  $\mathcal{S}^\dagger$ . We use  $\perp$  (“bottom”) as a symbol for any of these contradictions. The rule (RAA) tells us that if we can prove a contradiction with some tree  $\mathcal{T}$ , then we may take any sentence  $\varphi$ , withdraw some or all of the occurrences of  $\varphi$  in the leaves of our derivation by putting brackets around them, and then using the rule (RAA) to infer  $\bar{\varphi}$  at the root. We obtain a new tree  $\mathcal{T}^+$ . Here  $\bar{\varphi}$  is defined from  $\varphi$  in (3.5). We allow the case when  $\varphi$  does not actually occur in the leaves of the tree  $\mathcal{T}$ . So in this case,  $\mathcal{T}$  and  $\mathcal{T}^+$  would have the same set of non-withdrawn leaves.

We write  $\Gamma \vdash \varphi$  if there is a proof tree  $\mathcal{T}$  whose root is  $\varphi$  and all of whose *non-withdrawn* leaves belong to  $\Gamma$ .

**What is the difference between (x) and *reductio ad absurdum* (RAA)?** Here again is (RAA), stated slightly differently than before:

$$\begin{array}{l} \text{If } \Gamma \cup \{\varphi\} \vdash \text{Some } p \text{ are } \bar{p}, \\ \text{then } \Gamma \vdash \bar{\varphi} \end{array} \quad (3.7)$$

The important point is that we change the set on the left of the turnstile symbol  $\vdash$ . This deviates from everything which we have done up until now.

For comparison, here is (x):

$$\begin{array}{l} \text{If } \Gamma \cup \{\varphi\} \vdash \text{Some } p \text{ are } \bar{p}, \\ \text{then for all } \psi \text{ whatsoever } \Gamma \cup \{\varphi\} \vdash \psi \end{array} \quad (3.8)$$

**Example 3.18: a derivation of (No):** No  $p$  are  $p \vdash$  No  $p$  are  $q$

$$\frac{\frac{\frac{\text{All } p \text{ are } \bar{p} \quad \frac{\text{[Some } p \text{ are } \bar{q}] \text{ SOME}}{\text{Some } p \text{ are } p} \text{ DARII}}{\text{Some } p \text{ are } \bar{p}} \text{ RAA}}{\text{All } p \text{ are } q}$$

**Remark** We claim that (RAA) is at least as strong as (x): anything that (3.8) can prove can also be proved with (3.7). Let us prove this. Fix  $\Gamma$ ,  $p$ ,  $q$ ,  $\psi$  and assume that  $\Gamma \cup \{\varphi\} \vdash \text{Some } p \text{ are } q$  and also that  $\Gamma \cup \{\varphi\} \vdash \text{No } p \text{ are } q$ . Our proof systems are *monotonic* in the sense that if we have a set  $\Delta$  and if  $\Delta \vdash \chi$ , then any  $\Delta' \supseteq \Delta$  also has  $\Delta' \vdash \chi$ . To see this, go back and look at the definition of what it means to be a *proof tree over*  $\Delta$ : it means that the leaves of the tree *belong to*  $\Delta$ . But then they belong to any larger set  $\Delta'$ . The upshot is that we can add  $\bar{\psi}$  to  $\Gamma \cup \{\varphi\}$  to get a bigger set, namely  $\Gamma \cup \{\varphi, \bar{\psi}\}$ . And then we would have  $\Gamma \cup \{\varphi, \bar{\psi}\} \vdash \text{Some } p \text{ are } q$  and also that  $\Gamma \cup \{\varphi, \bar{\psi}\} \vdash \text{No } p \text{ are } q$ . Now  $\Gamma \cup \{\varphi, \bar{\psi}\} = (\Gamma \cup \{\varphi\}) \cup \{\bar{\psi}\}$ . Then by (3.7) applied to the right-hand side, we see that  $\Gamma \cup \{\varphi\} \vdash \psi$ . So in this way, we have *simulated* (x) using (RAA).

**Lemma 3.19**

Let  $\Gamma \subseteq \mathcal{S}^\dagger$ . If  $\Gamma \cup \{\varphi\} \vdash \perp$ , then  $\Gamma \vdash \bar{\varphi}$ .

*Proof.* Assuming that  $\Gamma \cup \{\varphi\} \vdash \perp$ , we have a proof tree ending in  $\perp$  (a contradiction) whose non-withdrawn leaves belong to  $\Gamma \cup \{\varphi\}$ . We take all non-withdrawn leaves which are  $\varphi$ , and withdraw them. (There might not be any such leaves, and this is not a problem.) We apply (RAA) at the root, inferring  $\bar{\varphi}$ . This new tree has all of its non-withdrawn leaves in  $\Gamma$ . And it shows that  $\Gamma \vdash \bar{\varphi}$ .  $\square$

This leads to the following definition.

**Definition 3.20: inconsistent sets**

A set  $\Gamma$  is *inconsistent in  $\mathcal{S}^\dagger$*  if  $\Gamma \vdash \perp$ . This is equivalent, given the rest of the rules of the logic to some other statements: first,  $\Gamma \vdash \varphi$  for all  $\varphi$ . And second: there is some proof tree from  $\Gamma$  that uses (RAA). If  $\Gamma$  is not inconsistent, we say that it is *consistent*.

We should emphasize that consistency is a property of the proof system, not the semantics. That is, if we change the proof system, the definition changes. For example, if we were to replace (RAA) with (x), what counts as consistent would change.

### 3.4 Orthoposets and states

An important step in our work is to develop an *algebraic semantics* for  $\mathcal{S}^\dagger$ . There are several definitions, and we encourage you to read this section slowly.

**Definition 3.21: poset**

A *poset* (*partially ordered set*) is a preorder  $(P, \leq)$  which has the additional property of *antisymmetry*: if  $x \leq y$  and  $y \leq x$ , then  $x = y$ .

**Definition 3.22: orthoposet**

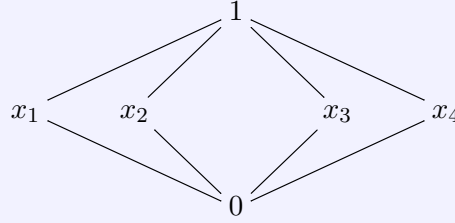
An *orthoposet* is a tuple  $\mathbb{P} = (P, \leq, 0, \bar{\phantom{x}})$  such that

1.  $(P, \leq)$  is a poset.
2. 0 is a minimum element:  $0 \leq p$  for all  $p \in P$ .
3. the function  $x \mapsto \bar{x}$  is antitone: if  $x \leq y$ , then  $\bar{y} \leq \bar{x}$ .
4.  $x \mapsto \bar{x}$  is *involution*:  $\bar{\bar{x}} = x$ .
5. *complement inconsistency*: If  $x \leq y$  and  $x \leq \bar{y}$ , then  $x = 0$ .

Please do not get worried by the scary word *orthoposet*. It might be better to call it a *poset with negation*, but the terminology is already in place. It comes from papers on quantum logic.

**Example 3.23: The Chinese lantern orthoposet**

The example below is sometimes called the *Chinese lantern*, and we'll call it  $\mathbb{P}_2$ .<sup>a</sup>



What we mean here is that the poset is the set of six points above, with the order as shown. The 0 is at the bottom. We define the operation  $-$  by:  $\bar{0} = 1$ ,  $\bar{1} = 0$ ,  $\bar{x}_1 = x_2$ ,  $\bar{x}_2 = x_1$ ,  $\bar{x}_3 = x_4$ , and  $\bar{x}_4 = x_3$ . You might like to verify the conditions in Definition 3.22.

<sup>a</sup>The more standard name for this seems to be *MO2*, with *MO* standing for *modular ortholattice*.

**Example 3.24: Power set orthoposets**

For all sets  $X$  we have an orthoposet  $(\mathcal{P}(X), \subseteq, \emptyset, -)$ , where  $\mathcal{P}(X)$ , the *power set of  $X$*  is the set of all subsets of  $X$ ,  $\subseteq$  is the inclusion relation,  $\emptyset$  is the empty set, and  $\bar{a} = X \setminus a$  for all subsets  $a$  of  $X$ .

The main examples of orthoposets for us are the ones coming from Definition 3.25 just below.<sup>4</sup>

**Definition 3.25: orthoposets defined from sets in the logic**

Let  $\Gamma$  be any set of sentences in  $\mathcal{S}^\dagger$ . Definition 1.24 defines the fundamental relation  $\leq$  on all nouns (complemented or not). The relation  $\leq$  is defined from  $\Gamma$  and the logic, and Proposition 1.25 shows this relation to be a preorder. We have an *equivalence relation*  $\equiv$  defined by

$$p \equiv q \quad \text{iff} \quad p \leq q \text{ and } q \leq p$$

We take  $\mathbb{P}_\Gamma$  to be the *quotient set*  $\mathbf{Lit}/\equiv$ ; this is the set of equivalence classes of elements of  $\mathbf{P}$ .

This set  $\mathbb{P}_\Gamma$  is a poset under the induced relation: if  $[u] \leq [v]$  and  $[v] \leq [u]$ , then  $u \equiv v$  so that  $[u] = [v]$ . If there is some  $p$  such that  $p \leq \bar{p}$ , then for all  $q$  we have  $[p] \leq [q]$  in  $\mathbf{Lit}/\equiv$ . In this case, set 0 to be  $[p]$  for any such  $p$ . (If such  $p$  exists, its equivalence class is unique.) We finally define  $[\bar{p}] = [\bar{p}]$ . If there is no  $p$  such that  $p \leq \bar{p}$ , we add fresh elements 0 and 1 to  $\mathbf{Lit}/\equiv$ . We then stipulate that  $\bar{0} = 1$ , and that for all  $x \in \mathbb{P}_\Gamma$ ,  $0 \leq x \leq 1$ .

We consider  $\mathbf{Lit}/\equiv$  as an orthoposet. (So we mean the structure  $(\mathbf{Lit}/\equiv, \leq, 0, -)$ .) The antitone property comes from the axiom with the same name, and the complement inconsistency is verified

<sup>4</sup>If you do not know or remember *equivalence relations*, now is a good time to look this up. What you need to know at this point could be gleaned from Example 3.26 just below. The idea is that in that example, we can prove that *All  $y$  are  $\bar{z}$*  and *All  $\bar{z}$  are  $y$* . While  $y$  and  $\bar{z}$  are different (unequal), they are *equivalent*. The notation  $[y]$  stands for the set of literals that is equivalent to  $y$ ; in this example, it is  $\{y, \bar{z}\}$ . Equivalence classes are always sets, and the idea is that  $a \equiv b$  is the same as  $[a] = [b]$ .

using the similarly-named part of Lemma 3.6.

**Example 3.26:  $\Gamma$  from Example 3.15**

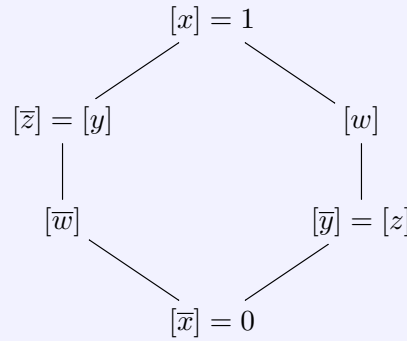
We again have

$$\Gamma = \{\text{All } y \text{ are } x, \text{All } \bar{y} \text{ are } x, \text{All } \bar{z} \text{ are } y, \text{All } z \text{ are } \bar{y}, \text{All } z \text{ are } w\}.$$

Then

$$\begin{array}{llll} [x] = \{x\} & [\bar{x}] = \{\bar{x}\} & [z] = \{\bar{y}, z\} & [\bar{z}] = \{y, \bar{z}\} \\ [y] = \{y, \bar{z}\} & [\bar{y}] = \{\bar{y}, z\} & [w] = \{w\} & [\bar{w}] = \{\bar{w}\} \end{array}$$

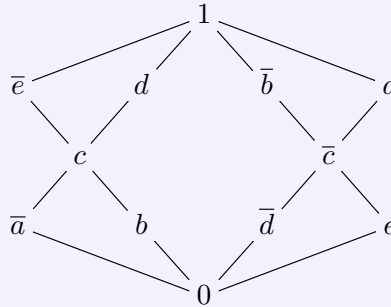
Here is a picture of the orthoposet  $\mathbb{P}_\Gamma$ :



As with all Hasse diagrams of posets, the order  $\leq$  is indicated by going up on the lines. The complement operation is defined in the obvious way, and it turns out to be the same as passing through the center of the hexagon. The 0 is  $[\bar{x}]$ .

**Example 3.27: Another example:  $\Gamma = \{\text{All } \bar{a} \text{ are } c, \text{All } b \text{ are } c, \text{All } c \text{ are } d, \text{All } c \text{ are } \bar{e}\}$**

All of the equivalence classes are singletons, and so we simplify notation by writing  $x$  for  $[x]$  throughout. The orthoposet  $\mathbb{P}_\Gamma$  is shown below:



We add a fresh 0 and 1 because there is no literal  $x$  provably  $\leq$  all literals.

**Definition 3.28:** Let  $\mathbb{P}$  be an orthoposet and  $S \subseteq P$ .

$S$  is *up-closed* if whenever  $p \in S$  and  $p \leq q$ , then  $q \in S$ .

$S$  is *complete* if for all  $p$ , either  $p \in S$  or  $\bar{p} \in S$ .

$S$  is *consistent* if for all  $p$ ,  $S$  does not contain both  $p$  and  $\bar{p}$ .

$S$  is *extendible* if for all  $p, q \in S$ ,  $p \not\leq \bar{q}$ .

$S$  is a *state* if  $S$  is non-empty, up-closed, complete, and consistent.

We next present a number of examples, ranging from the concrete to the abstract. We encourage you to check that each of them really satisfies the conditions in Definition 3.28,

**Example 3.29:** points in models of  $\Gamma$  give states of  $\mathbb{P}_\Gamma$

Let  $\Gamma = \{\text{All } p \text{ are } q, \text{All } \bar{p} \text{ are } r\}$ . Let  $M$  be the following model of  $\Gamma$ :  $M = \{1, 2, 3, 4, 5\}$ ,  $\llbracket p \rrbracket = \{1, 2\}$ ,  $\llbracket q \rrbracket = \{1, 2, 3\}$ , and  $\llbracket r \rrbracket = \{1, 3, 4, 5\}$ . Each point  $m \in M$  gives a state of  $\mathbb{P}_\Gamma$ , namely the state of all literals  $x$  such that  $m \in S_x$ . Explicitly,

$$\begin{array}{lll} S_1 & = & \{p, q, r\} \\ S_2 & = & \{p, q, \bar{r}\} \end{array} \quad \begin{array}{lll} S_3 & = & \{\bar{p}, q, r\} \\ S_4 & = & \{\bar{p}, \bar{q}, r\} \end{array} \quad \begin{array}{lll} S_5 & = & \{\bar{p}, \bar{q}, r\} \end{array}$$

The last example will be generalized in Exercise 3.15. In fact, the exercise points to the source of the definition. Suppose that we have a model  $\mathcal{M} \models \Gamma$ , and also some  $m \in M$ . Then what can we say about  $S_m = \{x : m \in \llbracket x \rrbracket\}$ , where  $x$  here ranges over the literals? A moment's thought shows that this set  $S_m$  must be closed upwards in  $\leq_\Gamma$ , and it must be complete and consistent as well. So it is a state. The way that we'll use states turns this observation around. We are going to need to build models as part of proving completeness. In order to build a model, we need to arrange that each  $S_m$  be a state. And indeed our models are going to be built out of the states: see Definition 3.37.

**Example 3.30**

Let  $X = \{1, 2, 3\}$ , and let  $\mathcal{P}(X)$  be the power set orthoposet from Example 3.24. Then  $S$  is a state, where

$$S = \{\{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

(More generally, if  $p$  is any finite set, then the collection of subsets of  $p$  containing more than half of the elements of  $p$  is a state of  $\mathcal{P}(X)$ .) Also, one can check that the states on this orthoposet are exactly  $S$  as above and the three sets  $T_1$ ,  $T_2$ , and  $T_3$ , where

$$T_i = \{A \subseteq X : i \in A\}.$$

**Example 3.31**

The orthoposet in Example 3.26 has three states. They are

$$\begin{aligned} S_1 &= \{[\bar{w}], [y], [x]\} \\ S_2 &= \{[y], [x], [w]\} \\ S_3 &= \{[z], [w], [x]\} \end{aligned}$$

You might like to find the states in Example 3.27 above.

**Proposition 3.32**

Every state  $S$  in an orthoposet is extendible.

*Proof.* Suppose that  $S$  is a state, and (towards a contradiction) that  $S$  is not extendible. We have  $p, q \in S$  such that  $p \leq \bar{q}$ . Since  $p \in S$  and  $p \leq \bar{q}$ , we have  $\bar{q} \in S$  (since  $S$  is up-closed). But then  $S$  contains  $q$  and  $\bar{q}$ , and we contradict the consistency of  $S$ .  $\square$

**Proposition 3.33**

A set  $\{p, \bar{q}\}$  is not extendible iff  $p \leq q$ .

*Proof.* Assume that  $\{p, \bar{q}\}$  is not extendible. Then either  $p \leq \bar{p}$ , or  $p \leq \bar{\bar{q}} = q$ , or  $\bar{q} \leq \bar{p}$ , or  $\bar{q} \leq \bar{\bar{q}}$ . In the first case,  $p \leq p$  and  $p \leq \bar{p}$ , so  $p = 0$  by complement inconsistency. Hence  $p \leq q$ . In the third case,  $p \leq q$  by the antitone property. And in the last case,  $\bar{q} = 0$ , so  $q = 1$  and  $p \leq \bar{q}$ .  $\square$

**Lemma 3.34**

Let  $S$  be an extendible subset in an orthoposet  $\mathbb{P}$ , and let  $p \in P$ . Then either  $S \cup \{p\}$  is extendible or  $S \cup \{\bar{p}\}$  is.

*Proof.* Suppose towards a contradiction that neither set were extendible. Since  $S \cup \{p\}$  is not extendible, we have  $x, y \in S \cup \{p\}$  such that  $x \leq \bar{y}$ . And since  $S$  is extendible, at least one of these has to be  $p$ . From the second, we have  $a, b \in S \cup \{\bar{p}\}$  such that  $a \leq \bar{b}$ , and at least one has to be  $\bar{p}$ . There are a number of cases here, and we'll just give one: when  $x = p$ ,  $y \in S$ ,  $b = \bar{p}$ , and  $a \in S$ . In this case,  $y \leq \bar{x} = \bar{p} = b \leq \bar{a}$ . And this means that  $S$  is not extendible, a contradiction.  $\square$

**Lemma 3.35**

For a subset  $S_0$  of an orthoposet  $\mathbb{P} = (P, \leq, 0, \bar{\phantom{x}})$ , the following are equivalent:

1.  $S_0$  is a subset of some state  $S$  of  $\mathbb{P}$ .
2.  $S_0$  is extendible.

*Proof.* We showed in Proposition 3.32 that states are extendible, and it is easy to check that every subset of an extendible set is extendible. This proves (1)  $\implies$  (2). In the other direction, let's list the elements of  $P$  as  $p_1, \dots, p_n$ .<sup>5</sup> Then form a sequence of extendible sets

$$S_0 \subseteq S_1 \subseteq \dots \subseteq S_n$$

in the following way:  $S_0$  is the given set. Given  $S_i$ , let  $S_{i+1}$  be either  $S_i \cup \{p_i\}$  or  $S_i \cup \{p_{i+1}\}$ , whichever is extendible. (If both are extendible, take it to be  $S_i \cup \{p_i\}$ .) Let  $S = S_n$ . So  $S$  is extendible, and it contains each element of  $P$  or its complement.

We need only check that it is up-closed. Suppose that  $p \in S$  and  $p \leq q$ . Suppose that  $q \notin S$ , so that  $\bar{q} \in S$ . Let  $i, j \leq n$  be such that  $p = p_i$  and  $\bar{q} = p_j$ . Then  $p \notin S_j$ , since otherwise  $S_j \cup \{\bar{q}\}$  would not be extendible; in that case, we would have  $S_{j+1} = S_j \cup \{q\}$ , putting  $q \in S$  after all. Now  $p \in S_{i+1}$ , and so  $j \leq i + 1$ . But  $S_{i+1}$  then contains  $p$  and  $\bar{q}$ , contradicting the fact that it is extendible.  $\square$

Lemma 3.35 leads to an algorithm for building states in orthoposets, starting with elements  $p \not\leq \bar{q}$ . This implies that  $\{p, q\}$  is extendible, by Proposition 3.33.

**Algorithm 3.36:** to build a state  $S$  which includes  $\{p, q\}$ , provided  $p \not\leq \bar{q}$

```

1:  $S \leftarrow \{p, q\}$ 
2: for all  $x \in \mathbf{P}$  do
3:   if  $S \cup \{x\}$  is extendible then
4:      $S \leftarrow S \cup \{x\}$ 
5:   else
6:      $S \leftarrow S \cup \{\bar{x}\}$  ▷ see Lemma 3.34
7:   end if
8: end for

```

## 3.5 Completeness for $\mathcal{S}^\dagger$

Our work on states of orthoposets will now be put to use.

**Definition 3.37:** the third canonical model

Suppose that  $\Gamma \subseteq \mathcal{S}^\dagger$ . The *third canonical model*  $\mathcal{M}_3$  is defined as follows: Recall  $\mathbb{P}_\Gamma$  from Definition 3.25. Let  $M_3$  be the set of states of  $\mathbb{P}_\Gamma$ . We interpret nouns by

$$\llbracket u \rrbracket = \{S \in M_3 : [u] \in S\}. \quad (3.9)$$

Here is an important observation about this model  $\mathcal{M}_3$ . For all nouns  $u$ ,  $\llbracket u \rrbracket$  is defined by (3.9). Thus  $\llbracket \bar{u} \rrbracket = \{S \in M_3 : [u] \notin S\}$ . But since states are complete and consistent (see Definition 3.28), we see that  $\{S \in M_3 : [u] \notin S\} = \{S \in M_3 : [\bar{u}] \in S\}$ . The point is that (3.9) holds not only for nouns but also for complemented nouns as well.

<sup>5</sup>This proof assumes that  $P$  is a finite set. We invite you to think about the case when  $P$  is infinite in an exercise.



**Example 3.38:**  $\mathcal{M}_3$  when  $\Gamma$  is from Example 3.26.

$\mathcal{M}_3$  is defined as follows:  $M_3 = \{S_1, S_2, S_3\}$ , and

$$\begin{array}{ll} \llbracket x \rrbracket &= \{S_1, S_2, S_3\} & \llbracket z \rrbracket &= \{S_3\} \\ \llbracket y \rrbracket &= \{S_1, S_2\} & \llbracket w \rrbracket &= \{S_2, S_3\} \end{array}$$

Note that this answers the challenge raised in Example 3.15.

**Lemma 3.39**

Let  $\Gamma \subseteq \mathcal{S}^\dagger$  be a consistent set in the logic  $\mathcal{S}$ . Then  $\mathcal{M}_3 \models \Gamma$ .

*Proof.* First, let's take a sentence in  $\Gamma_{all}$ , say **All  $x$  are  $y$** . Let  $S \in \llbracket x \rrbracket$ . Then  $x \in S$ . But since  $x \leq y$  and  $S$  is closed upwards, we see that  $y \in S$ . This proves that  $\mathcal{M}_3 \models \text{All } x \text{ are } y$ .

Now consider a sentence in  $\Gamma_{some}$ , say **Some  $x$  are  $y$** . Consider  $\{x, y\}$ . Suppose for a moment that this set is not extendible. We have that either  $x \leq \bar{y}$  or  $y \leq \bar{x}$ . Either way,  $\Gamma$  is inconsistent in  $\mathcal{S}^\dagger$ . And this contradiction means that  $\{x, y\}$  really is extendible. By Lemma 3.35, let  $S$  be a state with  $\{x, y\} \subseteq S$ . Then  $S \in \llbracket x \rrbracket \cap \llbracket y \rrbracket$ . And so  $\mathcal{M}_3 \models \text{Some } x \text{ are } y$ .  $\square$

**Theorem 3.40: Soundness/completeness of  $\mathcal{S}^\dagger$** 

For a set of sentences  $\Gamma$  of  $\mathcal{S}^\dagger$ ,  $\Gamma \vdash \varphi$  iff  $\Gamma \models \varphi$ .

*Proof.* As always, the soundness half is an easy induction on proof trees, and we omit this verification. The main step is Exercise 3.3. Concerning the completeness half, we show the contrapositive. Suppose that  $\Gamma \not\models \varphi$ . By Lemma 3.19,  $\Gamma \cup \{\bar{\varphi}\} \not\models \perp$ . That is,  $\Gamma \cup \{\bar{\varphi}\}$  is consistent. So by Lemma 3.39,  $\Gamma \cup \{\bar{\varphi}\}$  has a model. Thus,  $\Gamma \not\models \varphi$ .  $\square$

**Definition 3.41: satisfiable set**

A set  $\Gamma$  is *satisfiable* if it has a model. Otherwise,  $\Gamma$  is *unsatisfiable*.

You should compare Definitions 3.20 and 3.41. The concepts of *consistency* and *satisfiability* turn out to be the same, but it took work to see this.

**Corollary 3.42**

A set  $\Gamma \subseteq \mathcal{S}^\dagger$  is consistent iff it is satisfiable.

*Proof.* We apply Theorem 3.40 with  $\varphi$  being some contradiction. That result would tell us that  $\Gamma$  is inconsistent iff  $\Gamma$  is unsatisfiable. Taking the negation of both sides:  $\Gamma \subseteq \mathcal{S}^\dagger$  is consistent iff it is satisfiable.  $\square$

### 3.6 Algorithmic analysis

We have the completeness of the logical system  $S^\dagger$ . We'd like to go a bit further and give an efficient algorithm to tell whether, given a finite  $\Gamma \subseteq S^\dagger$  and some  $\varphi \in S^\dagger$ ,  $\Gamma \vdash \varphi$  or not. This is not immediate, due to the rule (RAA). Unlike (x), (RAA) complicates the proof search. In a different direction, if  $\Gamma \not\vdash \varphi$ , we'd like an algorithm to construct a counterexample.

We tackle all of these problems here. We need the ideas from earlier in the chapter, but we need a few additional results also.

Here is an outline of the work in this section. Fix a finite set  $\Gamma \subseteq S^\dagger$ . We construct a graph  $H_\Gamma$  (different from the graph  $G_\Gamma$ ) in several steps. See Algorithm 3.43. This graph may be computed efficiently from  $\Gamma$ ; indeed, this is the point of it. We then make an orthoposet from  $H_\Gamma$ , and then the states of this orthoposet give us a new canonical model which establish the correctness of a few algorithms at the end of the section. The algorithms themselves do not need anything beyond the graph  $H_\Gamma$ , but their correctness relies on the theory that we develop.

**Definition 3.43: to build a graph  $H_\Gamma$  from a finite set  $\Gamma \subseteq S^\dagger$**

Let  $H = \{p \in \mathbf{Lit} : p \text{ occurs in } \Gamma\}$ .

Define a relation  $\xrightarrow{\Gamma}$  by  $p \xrightarrow{\Gamma} q$  if  $\Gamma$  contains either *All  $p$  are  $q$*  or *All  $\bar{q}$  are  $\bar{p}$* .

Take its reflexive-transitive closure  $\xrightarrow{\Gamma^*}$ , defined by  $p \xrightarrow{\Gamma^*} q$  if there is a path of length at least zero from  $p$  to  $q$  following  $\xrightarrow{\Gamma}$ .

Define the *zero-points* to be  $\{p : p \xrightarrow{\Gamma^*} \bar{p}\}$ , and define the *one-points* to be  $\{p : \bar{p} \xrightarrow{\Gamma^*} p\}$ .

Define a second relation  $\preceq$  by  $p \preceq q$  if either  $p \xrightarrow{\Gamma^*} q$ , or if  $p$  is a zero-point, or if  $q$  is a one-point.

Finally, let  $H_\Gamma$  be the graph with node set  $H$  and edge relation  $\preceq$

**Lemma 3.44**

Concerning  $\xrightarrow{\Gamma^*}$  and  $\preceq$ :

1. The relation  $\preceq$  is a preorder.
2. If  $p \xrightarrow{\Gamma^*} q$ , then  $\bar{q} \xrightarrow{\Gamma^*} \bar{p}$ .
3. If  $p \preceq q$ , then  $\bar{q} \preceq \bar{p}$ .
4. If  $p \preceq \bar{p}$ , then  $p \xrightarrow{\Gamma^*} \bar{p}$ .

*Proof.* For (1), it is easy to check that  $\preceq$  is reflexive. Assume that  $p \preceq q$  and  $q \preceq r$ . Then there are nine cases, and in each of these, we see that  $p \preceq r$ . Part (2) is an easy induction on the lengths of paths showing  $p \xrightarrow{\Gamma^*} q$ . Part (3) is a direct verification, using the fact that  $p$  is a zero-point iff  $\bar{p}$  is a one-point. Part (4) also follows easily from the definitions.  $\square$

Lemma 3.44 shows that  $\preceq$  has almost all of the properties of an orthoposet. But it is not anti-symmetric. To remedy this, let's write  $x \equiv y$  for  $x \preceq y \preceq x$ . Then  $\equiv$  is an equivalence relation. We want to take the quotient, leading to the following definitions.

**Definition 3.45:  $\mathbb{Q}_\Gamma$  and  $\mathcal{M}_4$** 

Suppose that  $\Gamma \subseteq \mathcal{S}^\dagger$ . Let  $\mathbb{Q}_\Gamma$  be the set of equivalence classes of literals, ordered by  $[p] \leq [q]$  iff  $p \preceq q$ . We equip this with an orthoposet structure by defining  $\overline{[x]} = [\bar{x}]$ , and 0 to be the equivalence class of any zero-point. (If there are no zero-points, chose a fresh 0.)

The *fourth canonical model*  $\mathcal{M}_4$  is the set of states of  $\mathbb{Q}$ , with the usual interpretation function

$$\llbracket u \rrbracket = \{S \in \mathcal{M}_4 : [u] \in S\}. \quad (3.10)$$

As before, we have that for all complemented nouns  $u$ ,  $\llbracket u \rrbracket = \{S \in \mathcal{M}_4 : [u] \in S\}$ .

Exercise 3.8 invites you to verify that  $\mathbb{Q}_\Gamma$  actually is an orthoposet.

**Lemma 3.46**

Let  $\Gamma \subseteq \mathcal{S}^\dagger$ , and let  $p$  and  $q$  be any literals. The following are equivalent:

1.  $\Gamma_{all} \models \text{All } p \text{ are } q$ .
2.  $p \preceq q$  in the graph  $H_\Gamma$ .

*Proof.* (1) $\implies$ (2). Let  $\mathcal{M}_4$  be the model in Definition 3.45. The fact that states are up-closed implies that  $\mathcal{M}_4 \models \Gamma_{all}$ ; see the first half of the proof of Lemma 3.39. Hence  $\mathcal{M}_4 \models \text{All } p \text{ are } q$ . Consider  $\{[p], [\bar{q}]\}$  as a subset of  $\mathbb{Q}_\Gamma$ . If this set is extendible, there is some state  $S$  which includes it. This state would belong to  $\llbracket p \rrbracket$  but not to  $\llbracket q \rrbracket$ . This contradicts  $\mathcal{M}_4 \models \text{All } p \text{ are } q$ . So  $\{[p], [\bar{q}]\}$  is not extendible. Thus  $[p] \leq [q]$  by Proposition 3.33. And this tells us what we want:  $p \preceq q$ .

In the other direction, we show that if  $p \preceq q$ , then  $\Gamma \vdash \text{All } p \text{ are } q$ . If  $p \xrightarrow{\Gamma} q$ , then clearly  $\Gamma \vdash \text{All } p \text{ are } q$  with a one-point tree, or perhaps a two-point tree. An easy induction on the number  $n$  shows that if there is a path in  $\xrightarrow{\Gamma}$  of length  $n$  from  $p$  to  $q$ , then again  $\Gamma \vdash \text{All } p \text{ are } q$ . This shows that if  $p \xrightarrow{\Gamma}^* q$ , then  $\Gamma \vdash \text{All } p \text{ are } q$ . So if  $p$  is a zero-point, then  $\Gamma \vdash \text{All } p \text{ are } \bar{p}$ . This implies that for all  $q$ ,  $\Gamma \vdash \text{All } p \text{ are } q$ . Similarly, if  $q$  is a one-point, then for all  $p$ ,  $\Gamma \vdash \text{All } p \text{ are } q$ . From these last two observations, if  $p \preceq q$ ,  $\Gamma \vdash \text{All } p \text{ are } q$ .  $\square$

**Lemma 3.47: Pratt-Hartmann**

Let  $\Gamma \subseteq \mathcal{S}^\dagger$ . Suppose that  $\Gamma \models \text{Some } p \text{ are } q$ . Then there is a sentence in  $\varphi \in \Gamma_{some}$  such that

$$\Gamma_{all} \cup \{\varphi\} \models \text{Some } p \text{ are } q.$$

*Proof.* If not, then for every  $\varphi \in \Gamma_{some}$ , there is a model  $\mathcal{M}_\varphi \models \Gamma_{all} \cup \{\varphi\}$  such that  $\llbracket p \rrbracket \cap \llbracket q \rrbracket = \emptyset$  in  $\mathcal{M}_\varphi$ . Let  $\mathcal{M}$  be the disjoint union of the models  $\mathcal{M}_\varphi$ . Since each  $\mathcal{M}_\varphi$  satisfies  $\Gamma_{all}$ , so does  $\mathcal{M}$ . And for each  $\varphi \in \Gamma_{some}$ , say  $\text{Some } a \text{ are } b$ ,  $\mathcal{M} \models \varphi$  because  $\mathcal{M}_\varphi$  has an element in  $\llbracket a \rrbracket \cap \llbracket b \rrbracket$ . Thus  $\mathcal{M} \models \Gamma_{all} \cup \Gamma_{some} = \Gamma$ . And  $\text{Some } p \text{ are } q$  fails in  $\mathcal{M}$ , since it fails in each  $\mathcal{M}_\varphi$ .  $\square$

**Lemma 3.48**

For every set  $\Gamma \subseteq S^\dagger$ , the following are equivalent:

1.  $\Gamma$  is inconsistent in  $S^\dagger$ .
2.  $\Gamma \models \text{Some } x \text{ are } \bar{x}$ .
3. There is a sentence  $\text{Some } p \text{ are } q$  in  $\Gamma_{\text{some}}$  such that  $\Gamma_{\text{all}} \cup \{\text{Some } p \text{ are } q\} \models \perp$ .
4. There is a sentence  $\text{Some } p \text{ are } q$  in  $\Gamma_{\text{some}}$  such that  $\Gamma_{\text{all}} \models \text{All } p \text{ are } \bar{q}$ .
5. There is a sentence  $\text{Some } p \text{ are } q$  in  $\Gamma_{\text{some}}$  such that  $p \preceq \bar{q}$ .

*Proof.* (1) $\iff$ (2) is by the soundness and completeness of the logic. (2) $\iff$ (3) is by Lemma 3.47 with  $q = \bar{p}$ . (3) $\iff$ (4) is by Lemma 3.19. (4) $\iff$ (5) is by Lemma 3.46.  $\square$

**An algorithm to see if a set  $\Gamma$  is inconsistent in  $S^\dagger$  or not** Construct the graph  $H_\Gamma$  via Algorithm 3.43. Then go through the sentences  $\text{Some } p \text{ are } q$  in  $\Gamma_{\text{some}}$ , and for each of them, ask whether  $p \preceq \bar{q}$  or not. If any of these answers are Yes, we know that  $\Gamma$  is inconsistent. Otherwise, it is consistent.

**An algorithm to tell if  $\Gamma \vdash \text{All } p \text{ are } q$  or not** We start by seeing whether  $\Gamma$  is consistent. If not, then  $\Gamma$  proves anything. Otherwise, we continue. We again construct the graph  $H_\Gamma$  and ask if  $p \preceq q$  or not. This gives the answer to our original question.

Work is needed to see that this is correct. This work may be found in Exercise 3.9. And another exercise, Exercise 3.10, implies the correctness of the next algorithm.

**An algorithm to tell if  $\Gamma \vdash \text{Some } p \text{ are } q$  or not** As above we may assume that  $\Gamma$  is consistent. By Lemma 3.47 there is a sentence  $\text{Some } a \text{ are } b$  in  $\Gamma_{\text{some}}$  such that

$$\Gamma_{\text{all}} \cup \{\text{Some } a \text{ are } b\} \vdash \text{Some } p \text{ are } q$$

Define  $\preceq$  from  $\Gamma_{\text{all}}$ . By Exercise 3.10, we have  $a \preceq p$  or  $b \preceq p$ , and also that  $a \preceq q$  or  $b \preceq q$ .

Thus to see whether a consistent  $\Gamma$  derives  $\text{Some } p \text{ are } q$ , look for a sentence  $\text{Some } a \text{ are } b$  in  $\Gamma$  such that either  $a \preceq p$  or  $b \preceq p$ , and also either  $a \preceq q$  or  $b \preceq q$ . If such a sentence  $\text{Some } a \text{ are } b$  in  $\Gamma$  exists, then  $\Gamma \vdash \text{All } p \text{ are } q$ , and vice-versa.

**Remark** The exercises below have a proof of the completeness of  $S$ , something which we only did half of earlier. The work that we did on  $\mathcal{M}_4$  in Section 3.6 points to yet another completeness result, this time for  $S^\dagger$ : we can trade in (RAA) for (x), provided we also include a few of the logical laws that we used (RAA) to prove: (ZERO), (ONE), and (CAMESTRES).

### 3.7 Exercises

**Exercise 3.1.** What does the rule (ONE) mean? Why is this rule (ONE) sound?

**Exercise 3.2.** Concerning  $\mathcal{S}^\dagger$  and the notations in this chapter: True or false?

1. For every model  $\mathcal{M}$ ,  $\mathcal{M} \models \varphi$  or  $\mathcal{M} \models \bar{\varphi}$ .
2. For every set  $\Gamma$ ,  $\Gamma \models \varphi$  or  $\Gamma \models \bar{\varphi}$ .

**Exercise 3.3.** The main point in the soundness of  $\mathcal{S}^\dagger$  concerns (RAA). Suppose that every model of  $\Gamma \cup \{\varphi\}$  is a model of **Some  $p$  are  $\bar{p}$** . Show that every model of  $\Gamma$  is a model of  $\bar{\varphi}$  as defined in (3.5).

**Exercise 3.4.** In  $\mathcal{S}$ , let  $\Gamma$  contain just the sentence **Some  $p$  are  $q$** .

1. Show that in this case,  $\mathcal{M}_2$  from Definition 3.9 satisfies **All  $p$  are  $q$** , but this sentence is not provable from  $\Gamma$ .
2. What can we say about the **No** sentences in the models  $\mathcal{M}_2$ ? If we construct  $\mathcal{M}_2$  from  $\Gamma$ , would every **No** sentence which happens to be true in  $\mathcal{M}_2$  be provable from  $\Gamma$ ?

**Exercise 3.5.** Prove the rules (ZERO) and (ONE) in  $\mathcal{S}^\dagger$ . (That is, translate them from  $\mathcal{S}$  to  $\mathcal{S}^\dagger$  and prove them in the logical system  $\mathcal{S}^\dagger$  from Logic 3.17.)

**Exercise 3.6.** Show the following in  $\mathcal{S}^\dagger$ .<sup>6</sup>

1. **No  $p$  are  $q$   $\vdash$  No  $q$  are  $p$**  (E-CONVERSION)
2. **Some  $p$  are  $q$ , No  $q$  are  $n \vdash$  Some  $p$  are  $\bar{n}$**  (FERIO)
3. **All  $q$  are  $n$ , All  $q$  are  $\bar{n} \vdash$  No  $q$  are  $q$**  (COMPLEMENT INCONSISTENCY)
4. **All  $y$  are  $p$ , All  $\bar{y}$  are  $p$ , All  $q$  are  $z$ , Some  $x$  are  $\bar{z} \vdash$  Some  $p$  are  $\bar{q}$ .**

**Exercise 3.7.** Let  $\mathcal{M}$  and  $\mathcal{N}$  be models. Define  $\mathcal{M} + \mathcal{N}$  to be the disjoint union. We are going to pretend  $M$  and  $N$  are disjoint sets to simplify matters. Then the universe of  $\mathcal{M} + \mathcal{N}$  is  $M \cup N$ , and for each noun  $u$ ,  $\llbracket u \rrbracket = \llbracket u \rrbracket_{\mathcal{M}} \cup \llbracket u \rrbracket_{\mathcal{N}}$ . Prove the following:

1. For a complemented noun  $\bar{u}$ ,  $\llbracket \bar{u} \rrbracket_{\mathcal{M} + \mathcal{N}} = \llbracket \bar{u} \rrbracket_{\mathcal{M}} \cup \llbracket \bar{u} \rrbracket_{\mathcal{N}}$ . [Hint: draw a picture. In any model,  $\llbracket \bar{x} \rrbracket$  is always everything in the universe which is not in  $\llbracket x \rrbracket$ .]
2.  $\mathcal{M}$  and  $\mathcal{N}$  both satisfy **All  $x$  are  $y$**  if and only if  $\mathcal{M} + \mathcal{N} \models$  **All  $x$  are  $y$** .

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<sup>6</sup>In some parts of this exercise, you will need to remember that **No  $p$  are  $q$**  is a re-statement of **All  $p$  are  $\bar{q}$** .

3. Either  $\mathcal{M}$  or  $\mathcal{N}$  satisfies **Some  $x$  are  $y$**  if and only if  $\mathcal{M} + \mathcal{N} \models \text{Some } x \text{ are } y$ .

**Exercise 3.8.** Verify that  $\mathbb{Q}_\Gamma$  as defined in Definition 3.45 actually has all the properties of an orthoposet.

**Exercise 3.9.** Let  $\Gamma \subseteq \mathcal{S}^\dagger$  be a consistent set, and suppose that  $\Gamma \models \text{All } x \text{ are } y$ . Prove that  $\Gamma_{all} \models \text{All } x \text{ are } y$ . [The first thing to see is that this is not trivial. And for a hint: Let  $\mathcal{M} \models \Gamma_{all}$ . We need to see that  $\Gamma \models \text{All } x \text{ are } y$ . Let  $\mathcal{N} \models \Gamma$ , and use  $\mathcal{M} + \mathcal{N}$  from just above.]

**Exercise 3.10.** Let  $\Gamma$  be consistent, and assume that

$$\Gamma_{all} \cup \{\text{Some } a \text{ are } b\} \vdash \text{Some } p \text{ are } q.$$

Show that  $a \preceq p$  or  $b \preceq p$ , and similarly that  $a \preceq q$  or  $b \preceq q$ . [Hint: the easiest way is probably to give a semantic argument: if  $a \not\preceq_\Gamma p$  and  $b \not\preceq_\Gamma p$ , get a one-point model of  $\Gamma_{all} \cup \{\text{Some } a \text{ are } b\}$  where  $\llbracket p \rrbracket = \emptyset$ .]

**Exercise 3.11.** In an orthoposet, let  $q$  be such that for all  $p$ ,  $p \leq q$ . Show that  $q = \bar{0}$ .

**Exercise 3.12.** This exercise asks us to think about empty models in the logic of this chapter. We permit the universe of our models to be empty. Thus, the rule below is not sound in  $\mathcal{S}^\dagger$ :

$$\frac{\text{All } \bar{x} \text{ are } x}{\text{Some } x \text{ are } x} \text{ NON-EMPTY}$$

1. Suppose we change the definition of  $\Gamma \models \varphi$  to say that all *non-empty* models of  $\Gamma$  are also models of  $\varphi$ . Is the rule above sound?
2. When we discussed turning a set  $\Gamma$  into an orthoposet  $\mathbb{P}_\Gamma$ , we neglected the possibility that we'd want  $0 = 1$  in  $\mathbb{P}_\Gamma$ . When would we want this?
3. Let  $\mathbb{P}$  be an orthoposet. Show that  $\mathbb{P}$  has no states iff  $0 = 1$ .
4. Suppose we add (NON-EMPTY) above to  $\mathcal{S}^\dagger$ . Show that every consistent set has a non-empty model.

**Exercise 3.13.** Let  $\mathbb{P} = (P, \leq, 0, \neg)$  be an orthoposet. Let  $\mathbb{Q} = (P, \geq, 1, \neg)$ . Show that  $\mathbb{Q}$  is also an orthoposet. [In other words, suppose that we have an orthoposet  $\mathbb{P}$ . Let  $\mathbb{Q}$  be the opposite preorder  $\mathbb{P}^{op}$  (see Definition 0.8), along with the same complement operation as in  $\mathbb{P}$ , and taking the 0 of  $\mathbb{Q}$  to be the 1 of  $\mathbb{P}$ . Show that  $\mathbb{Q}$  really is an orthoposet.]

**Exercise 3.14.** Let  $\mathbb{P}$  be an orthoposet. Show that a set  $S \subseteq P$  is a state if and only if (a)  $S$  is up-closed, and (b) for each  $p \in P$ , exactly one of  $\{p, \bar{p}\}$  belongs to  $S$ .

**Exercise 3.15.** Let  $\Gamma \subseteq \mathcal{S}^\dagger$ , let  $\mathbb{P}_\Gamma$  be the orthoposet defined from  $\Gamma$  in Definition 3.25, and let  $\mathcal{M} \models \Gamma$ .

1. Show that each  $m \in M$  gives us a state of  $\mathbb{P}_\Gamma$

$$S_{\mathcal{M},m} = \{[p] : m \in \llbracket p \rrbracket \text{ in } \mathcal{M}\}$$

2. Show that every state of  $\mathbb{P}_\Gamma$  is of the form  $S_{\mathcal{M},m}$  for some model  $\mathcal{M}$  and some  $m \in M$ .

**Exercise 3.16.** Find the orthoposet  $\mathbb{P}_\Gamma$  for the set  $\Gamma$  below:

$$\begin{array}{ll} \text{All } a \text{ are } b & \text{All } w \text{ are } x \\ \text{All } b \text{ are } c & \text{All } x \text{ are } y \\ \text{All } c \text{ are } d & \text{All } y \text{ are } z \\ \text{All } d \text{ are } e & \end{array}$$

How many states does  $\mathbb{P}_\Gamma$  have?

**Exercise 3.17.** Suppose that  $\mathbf{P}$  is  $\{p_1, \dots, p_n\}$ . Which  $\Gamma$  maximizes the number of states in  $\mathbb{P}_\Gamma$ ?

**Exercise 3.18.** Let  $\Gamma$  be any set of sentences in  $\mathcal{S}(\text{all}, \text{no})$ . Following the steps below to show that there is a model  $\mathcal{M}$  with the following properties:

1.  $\mathcal{M} \models \Gamma$ .
2. If  $\varphi$  is any sentence in  $\mathcal{S}(\text{all}, \text{no})$ , and  $\mathcal{M} \models \varphi$ , then  $\Gamma \vdash \varphi$ .

Take for  $M$  the set of sets  $A \subseteq \mathbf{P}$  satisfying the following conditions:

- a.  $A$  is *up-closed*: If  $v \in A$  and  $v \leq w$ , then  $w \in A$ .
- b. If  $v, w \in A$ , then  $\Gamma \not\vdash \text{No } v \text{ are } w$ .

(Note as a special case of (3.18) that if  $v \in A$ , then  $\Gamma \not\vdash \text{No } v \text{ are } v$ .) We set

$$\llbracket v \rrbracket = \{A \in M : v \in A\}. \quad (3.11)$$

Here are your tasks:

1. Show that  $\mathcal{M} \models \Gamma$ .
2. Next, let  $\mathcal{M} \models \varphi$ . If  $\varphi$  is of the form *All*  $p$  are  $q$ , show that  $\Gamma \vdash \varphi$ . The hint here is to break into cases depending on whether or not the set  $A$  belongs to the model  $M$ , where

$$A = \{z : p \leq z\}.$$

3. Continuing with our proof of (2), suppose that  $\varphi$  is *No*  $p$  are  $q$ . Let

$$A = \{z : p \leq z \text{ or } q \leq z\}.$$

Argue that  $A$  cannot belong to the universe  $M$  of our model. Then use condition (b) in the definition of  $M$  to show that  $\Gamma \vdash \varphi$ . As usual, you'll need a case-by-case analysis.

**Exercise 3.19.** Finish the completeness proof of the logic for  $\mathcal{S}$  in Definition 3.8. Here is an outline.

1. Suppose that  $\Gamma \models \varphi$ , with  $\varphi$  of the form *Some  $p$  are  $q$* . In this case, we use Theorem 3.12.
2. Suppose that  $\Gamma \models \varphi$ , with  $\varphi$  of the form *All  $p$  are  $q$*  or *No  $p$  are  $q$* . Let  $\Gamma_{all,no}$  be the sentences in  $\Gamma$  using *All* or *No*. Let  $\mathcal{M}$  be the model from Exercise 3.18 for this set  $\Gamma_{all,no}$ . We have two cases, depending on whether  $\mathcal{M} \models \Gamma_{some}$  or not. Argue that either way,  $\Gamma \vdash \varphi$ .

**Exercise 3.20.** Example 3.7 shows that (ANTI) is derivable in  $\mathcal{S}$ . If one replaces (CAMESTRES) by (ANTI) in  $\mathcal{S}$ , is the logic still complete?

**Exercise 3.21.** As we know from the chapter, the logic  $\mathcal{S}^\dagger$  is complete. In this exercise, we want to see what happens if we replace (RAA) by (X) in  $\mathcal{S}^\dagger$ . Let  $\mathcal{S}_0^\dagger$  be the proof system with rules (AXIOM), (BARBARA), (SOME<sub>1</sub>), (SOME<sub>2</sub>), (DARII), and (X).

1. Suppose we add (ZERO) and (ONE) as rules, and call the resulting system  $\mathcal{S}_0^\dagger + (\text{ZERO}) + (\text{ONE})$ . Show that in this system,  $\text{All } x \bar{y} \not\vdash \text{All } y \bar{x}$ . So  $\mathcal{S}_0^\dagger + (\text{ZERO}) + (\text{ONE})$  is not complete.
2. Show that  $\mathcal{S}_0^\dagger + (\text{ANTI})$  is also not complete.
3. Finally, show that  $\mathcal{S}_0^\dagger + (\text{ANTI}) + (\text{ZERO})$  is complete.





## Chapter 4

# $\mathcal{R}$ : the relational syllogistic

At this point, we return to logics with verbs, a topic which we first encountered in Section 2.1.<sup>1</sup> The basic goal is to have a logical system in which we may represent a valid argument such as

$$\begin{array}{l} \text{Every porter recognizes every porter} \\ \text{No quarterback recognizes any quarterback} \\ \hline \text{No porter is a quarterback} \end{array} \quad (4.1)$$

### 4.1 Syntax and semantics of $\mathcal{R}$

As in our previous work, we adopt as minimal a syntax as needed. In fact, the English sentences of interest are listed in Definition 4.1. below, using  $p$  and  $q$  for nouns and  $r$  for verbs.

The name of this fragment comes from the word “relation”, since a transitive verb – that is, a verb which takes a direct object – will be interpreted as a relation on the universe  $M$ .

The syntax of  $\mathcal{R}$  starts the same way as the syntax of  $\mathcal{A}(\mathcal{RC})$  (see page 44). All of our work builds on collections  $\mathbf{P}$  of unary atoms and  $\mathbf{R}$  of binary atoms. This time we include a complement operation on *both* unary and binary atoms. (So this is a difference with what we saw earlier with  $\mathcal{A}(\mathcal{RC})$ .) As in Section 3.3, we understand this operation to be involutive, so  $\bar{\bar{p}}$  and  $p$  are taken to be identical, as are  $\bar{\bar{r}}$  and  $r$ . We call the items  $p$  and  $\bar{p}$  *unary literals*. Similarly, if  $r$  is a binary atom, then  $r$  and  $\bar{r}$  are called *binary literals*.

**Terms** At times, it is convenient to adopt a bit of extra notation to the syntax to simplify presentational matters. As in Chapter 2, we speak of *terms*; these are syntactic items which denote sets in our models. So for example, unary atoms are terms. The terms of  $\mathcal{R}$  are

$$p \quad \bar{p} \quad r \text{ all } p \quad r \text{ some } p \quad \bar{r} \text{ all } p \quad \bar{r} \text{ some } p$$

We use  $c$  as a variable ranging over terms. The sentences of  $\mathcal{R}$  are given by

$$\text{all}(p, c) \quad \text{some}(p, c)$$

---

<sup>1</sup>However, our treatment here is different in significant ways, and you will not need to use any of the results on the language  $\mathcal{A}(\mathcal{RC})$  from Section 2.1. The differences are that here we have **Some** and **No**, and we also have complements of the binary atoms, not just the unary ones. And the language of this chapter does not make use one key feature of  $\mathcal{A}(\mathcal{RC})$ , the fact that terms might appear inside other terms. So in contrast to  $\mathcal{A}(\mathcal{RC})$ , if we formulate  $\mathcal{R}$  over a finite set of atoms, then the resulting language is itself finite.

**Definition 4.1: reading the syntax of  $\mathcal{R}$** 

|                                     |                       |                                           |                              |
|-------------------------------------|-----------------------|-------------------------------------------|------------------------------|
| $\text{all}(p, q)$                  | all $p$ are $q$       | $\text{some}(p, \bar{q})$                 | some $p$ aren't $q$          |
| $\text{some}(p, q)$                 | some $p$ are $q$      | $\text{all}(p, \bar{q})$                  | no $p$ are $q$               |
| $\text{all}(p, r \text{ all } q)$   | all $p$ $r$ all $q$   | $\text{some}(p, \bar{r} \text{ some } q)$ | some $p$ don't- $r$ some $q$ |
| $\text{all}(p, r \text{ some } q)$  | all $p$ $r$ some $q$  | $\text{some}(p, \bar{r} \text{ all } q)$  | some $p$ don't- $r$ any $q$  |
| $\text{some}(p, r \text{ all } q)$  | some $p$ $r$ all $q$  | $\text{all}(p, \bar{r} \text{ some } q)$  | all $p$ don't- $r$ some $q$  |
| $\text{some}(p, r \text{ some } q)$ | some $p$ $r$ some $q$ | $\text{all}(p, \bar{r} \text{ all } q)$   | all $p$ don't- $r$ any $q$   |

**Expansions of  $\mathcal{R}$**  There are several ways in which we could expand this language  $\mathcal{R}$ . First, notice that the syntax is not symmetric. In  $\mathcal{R}$ , we have a sentence  $\forall(\text{dogs, see all animals})$ , but we do not have  $\forall(\text{see all animals, dogs})$ . So one extension of  $\mathcal{R}$  would be to allow for symmetric syntax, allowing for sentences  $\forall(c, p)$  and  $\exists(c, p)$ .

Next, we could allow complemented atoms to be the subjects of sentences. That is, we could double the possible sentence types in Definition 4.1 by allowing sentences like  $\text{all}(\bar{p}, \bar{q})$  and  $\text{some}(\bar{p}, r \text{ all } q)$ . The resulting language would be called  $\mathcal{R}^\dagger$ . (The name should suggest the way we expanded  $\mathcal{S}$  to  $\mathcal{S}^\dagger$  in Chapter 3.)

Another way in which we could expand  $\mathcal{R}$  would be to make the collection of terms a recursive construct. This is what we did in Chapter 2. When we do this we would get a language  $\mathcal{RCA}$ ; it is studied in Chapter ??.

This book would be repetitive and boring if we had a detailed treatment of every possible logical system. So we have tried to organize things pedagogically, using each chapter to highlight a particular issue. In this chapter, the issue of interest is proof by contradictions, leading to the rule of *reductio ad absurdum* (RAA) in the proof system.

**Positive and negative sentences; negations** The sentences on the left of Definition 4.1 are called *positive*, and the ones on the right are called *negative*. The way that the figure is arranged, each sentence  $\varphi$  has a *semantic negation*  $\bar{\varphi}$  on the other side. For example, if  $\varphi$  is  $\text{all}(p, \bar{q})$ , then  $\bar{\varphi}$  is  $\text{some}(p, q)$ . Conversely,

$$\overline{\text{some}(p, q)} = \text{all}(p, \bar{q})$$

Note that  $\bar{\bar{\varphi}} = \varphi$  for all sentences  $\varphi$ .

However, please note that the “negation bar” is not part of the language  $\mathcal{R}$ . It is part of our *meta-language*, the language that we use to talk about  $\mathcal{R}$ .

**Definition 4.2: model for  $\mathcal{R}$** 

A *model*  $\mathcal{M}$  for  $\mathcal{R}$  is a set  $M$ , together with *interpretation functions*

$$\begin{aligned} \llbracket \cdot \rrbracket &: \mathbf{P} \rightarrow \mathcal{P}(M) \\ \llbracket \cdot \rrbracket &: \mathbf{R} \rightarrow \mathcal{P}(M \times M) \end{aligned}$$

Both functions are written with the same symbol, and it usually will not cause confusion to do this. This means that for each unary atom  $p \in \mathbf{P}$ ,  $\llbracket p \rrbracket \subseteq M$ , and for each binary atom  $r$ ,  $\llbracket r \rrbracket \subseteq M \times M$ . We interpret literals  $\bar{p}$  and  $\bar{r}$  using complements:

$$\llbracket \bar{p} \rrbracket = M \setminus \llbracket p \rrbracket \quad \llbracket \bar{r} \rrbracket = M^2 \setminus \llbracket r \rrbracket.$$

We then interpret terms by subsets of  $M$  in the following way

$$\begin{aligned} \llbracket s \text{ all } p \rrbracket &= \{m \in M : \text{for all } n \in \llbracket p \rrbracket, (m, n) \in \llbracket s \rrbracket\} \\ \llbracket s \text{ some } p \rrbracket &= \{m \in M : \text{for some } n \in \llbracket p \rrbracket, (m, n) \in \llbracket s \rrbracket\} \end{aligned}$$

This should be familiar from Section 2.1.

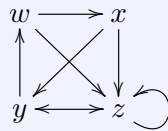
**Definition 4.3: truth and consequence**

$$\begin{aligned} \mathcal{M} \models \text{all}(p, c) &\quad \text{iff} \quad \llbracket p \rrbracket \subseteq \llbracket c \rrbracket \\ \mathcal{M} \models \text{some}(p, c) &\quad \text{iff} \quad \llbracket p \rrbracket \cap \llbracket c \rrbracket \neq \emptyset \end{aligned}$$

Finally, we have definitions of semantic consequence relation  $\Gamma \models \varphi$ , just as we have seen it for other logical languages: see Definition 0.3 on page 6.

**Example 4.4**

We consider a simple case, with one unary atom  $p$ , and one binary atom  $s$ . A model  $\mathcal{M}$  might have  $M = \{w, x, y, z\}$ ,  $\llbracket p \rrbracket = \{w, x, y\}$  and  $s$  as shown below:



In this model,  $\llbracket \bar{p} \rrbracket = \{z\}$ ,  $\llbracket s \text{ all } p \rrbracket = \emptyset$ ,  $\llbracket s \text{ some } \bar{p} \rrbracket = M$ , and  $\llbracket \bar{s} \text{ some } p \rrbracket = M$ . Here are two  $\mathcal{R}$ -sentences true in  $\mathcal{M}$ :  $\text{some}(p, p)$ , and also  $\text{all}(p, s \text{ some } \bar{p})$ .

**Remark 4.5: scope ambiguity**

Sentences like “All porters recognize some quarterback” are ambiguous in English. This sentence could mean that there is one particular quarterback that all porters recognize. Or, it could mean that every porter recognizes some quarterback or other, with different porters possibly recognizing different quarterback. The first reading is rarer, and one sometimes needs to emphasize the words in a special way to get it. (This phenomenon in linguistics is called *focus*.) The second, more common, reading is called the *subject wide scope reading*. In  $\mathcal{R}$ , we can represent the subject wide scope reading: every porter has the property of recognizing some quarterback or other. It is not possible to represent the object wide scope reading, where there is one particular quarterback who every porter recognizes.

**Remark 4.6**

The language  $\mathcal{R}$  is built from a set  $\mathbf{P}$  of unary atoms and a set  $\mathbf{R}$  of binary atoms. In all of our examples, and in most of the discussion, we only had one binary atom. Why is this? Let us suppose that  $\varphi$  has a binary atom, say *see*. If  $\Gamma \models \varphi$  and  $\Gamma$  is consistent, then  $\Gamma_0 \models \varphi$ , where the sentences in  $\Gamma_0$  have the same verb, or else have no verb at all. Similarly, if  $\Gamma \vdash \varphi$  and  $\Gamma$  is consistent, then  $\Gamma_0 \vdash \varphi$ . (We ask you to prove these in Exercise 4.3.) The point is that even though our language may be formulated with many binary atoms, in any given (semantic or proof-theoretic) inference, or tree of inferences, only one binary atom may appear.

## 4.2 Proof system



Our rules appear in Definition 4.7. We would like to take all of the variables to range over *literals* of the appropriate arity, but the syntax of  $\mathcal{R}$  has a restriction: the head nouns (unary atoms) of sentences and terms must be positive. This means that we want the rules of our system to be formulated in a way which insures that the rules only involve sentences in the fragment. This boils down to the following: in Logic 4.7, we should take  $p$ ,  $q$ , and  $n$  to be *positive literals*; i.e., unary atoms without negation. We permit  $c$  to be an *arbitrary* term, and we allow  $t$  to be a positive or negative binary literal.

Rules (D1), (D2), (D3), (B), (ZERO), (T) and (I) are natural generalizations of their namesakes in  $\mathcal{S}$ . In contrast,  $(\forall\forall)$ ,  $(\exists\exists)$ ,  $(\forall\exists)$  and (II) express genuinely relational logical principles. In some settings, these last rules are called *monotonicity principles*.

In addition to syllogistic rules,  $\mathcal{R}$  contains the rule of (RAA), whereby one derives  $\varphi$  from  $\Gamma$  by temporarily adding the semantic negation  $\bar{\varphi}$  of  $\varphi$  to  $\Gamma$ , thereby obtaining  $\Gamma \cup \{\bar{\varphi}\}$ , and then deriving a contradiction from this larger set. This rule (RAA) is *not the same* as *ex falso quodlibet*, the rule we have called (X) in Section 3.3. To see why, we mention the semantic justifications for both rules.

**The bottom symbol  $\perp$**  We also define  $\perp$  to be any contradiction. In  $\mathbf{R}$ , this means a sentence of the form *some*( $p, \bar{p}$ ).

Here are the ideas behind (RAA) and (X):

1. If  $\Gamma \cup \{\bar{\varphi}\} \models \perp$ , then  $\Gamma \models \varphi$ .

**Definition 4.7: rules of R**

$$\begin{array}{ll}
\frac{\text{some}(p, q) \quad \text{all}(q, c)}{\text{some}(p, c)} \text{ (D1)} & \frac{\text{all}(p, q) \quad \text{all}(q, c)}{\text{all}(p, c)} \text{ (B)} \\
\\
\frac{\text{all}(p, q) \quad \text{some}(p, c)}{\text{some}(q, c)} \text{ (D2)} & \frac{}{\text{all}(p, p)} \text{ (T)} \quad \frac{\text{some}(p, c)}{\text{some}(p, p)} \text{ (I)} \\
\\
\frac{\text{all}(q, \bar{c}) \quad \text{some}(p, c)}{\text{some}(p, \bar{q})} \text{ (D3)} & \frac{\text{all}(p, \bar{p})}{\text{all}(p, c)} \text{ ZERO} \quad \frac{\text{some}(p, t \text{ some } q)}{\text{some}(q, q)} \text{ (II)} \\
\\
\frac{\text{all}(p, t \text{ all } n) \quad \text{some}(q, n)}{\text{all}(p, t \text{ some } q)} \text{ (all}\exists\text{)} & \frac{\text{some}(p, t \text{ some } q) \quad \text{all}(q, n)}{\text{some}(p, t \text{ some } n)} \text{ (}\exists\exists\text{)} \\
\\
\frac{\text{all}(p, t \text{ some } q) \quad \text{all}(q, n)}{\text{all}(p, t \text{ some } n)} \text{ (allall)} & \frac{\begin{array}{c} [\bar{\varphi}] \\ \vdots \\ \perp \end{array}}{\varphi} \text{ RAA}
\end{array}$$

2. If  $\Gamma \models \perp$ , then  $\Gamma \models \varphi$

The (RAA) justification changes the assumptions in the derivation from  $\Gamma \cup \{\bar{\varphi}\}$  to  $\Gamma$ . This is the exact difference between *reductio ad absurdum* and *ex falso quodlibet*.

We must incorporate this observation into our proof system, and we do so in (RAA). We now can display this rule in natural-deduction-style:

$$\frac{\begin{array}{c} [\bar{\varphi}] \\ \vdots \\ \perp \end{array}}{\varphi} \text{ RAA}$$

**Definition 4.8: proof tree**

A *proof tree over*  $\Gamma$  is a pair  $(\mathcal{T}, \text{Canc})$ , where finite tree  $\mathcal{T}$  whose nodes are labeled with sentences, and Canc is a set of labeled leaves of  $\mathcal{T}$  called the *canceled leaves*. Each node  $n$  in the tree must satisfy one of the following conditions:

1.  $n$  is a leaf labeled by an element of  $\Gamma$ .
2.  $n$  comes from its parent(s) by an application of a rule other than (RAA).
3.  $n \in \text{Canc}$ , and there is some node  $m$  on the path from  $n$  to the root such that the parent of  $m$  is labeled  $\perp$ , and the label of  $m$  is the semantic negation of the label of  $n$ .

We write  $\Gamma \vdash \varphi$  if there is a proof tree  $\mathcal{T}$  with  $\varphi$  at the root whose uncanceled leaves all belong to  $\Gamma$ .

**Example 4.9:**  $\text{all}(x, \bar{x}) \vdash \text{all}(y, r \text{ all } x)$ 

In words, if there are no  $x$ s, then all  $y$ 's have any relation whatsoever to all of them. Note that the conclusion does not follow from (ZERO). Instead, we use (RAA):

|   |                                           |             |
|---|-------------------------------------------|-------------|
| 1 | $\text{all}(x, \bar{x})$                  | Assumption  |
| 2 | $\text{some}(y, \bar{r} \text{ some } x)$ | Assumption  |
| 3 | $\text{some}(x, x)$                       | II, 2       |
| 4 | $\text{some}(x, \bar{x})$                 | DARII, 1, 3 |
| 5 | $\text{all}(y, r \text{ all } x)$         | RAA, 2–4    |

**Example 4.10:**  $\text{all}(x, r \text{ all } y), \text{all}(p, y), \text{some}(p, q) \vdash \text{all}(x, r \text{ some } y)$ 

|                                                                                                                                                                                                                                                                         |  |   |                                    |                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|------------------------------------|-------------------------|
| $\frac{\text{all}(p, y) \quad \frac{\text{all}(x, r \text{ all } y) \quad \frac{\text{some}(p, q) \quad \text{all}(p, y)}{\text{some}(p, y)} \text{ (D1)}}{\text{all}(x, r \text{ some } p)} (\forall\forall) \quad \text{all}(x, r \text{ some } y)} (\forall\exists)$ |  | 1 | $\text{some}(p, q)$                | Ass'n                   |
|                                                                                                                                                                                                                                                                         |  | 2 | $\text{all}(x, r \text{ all } y)$  | Ass'n                   |
|                                                                                                                                                                                                                                                                         |  | 3 | $\text{all}(p, y)$                 | Ass'n                   |
|                                                                                                                                                                                                                                                                         |  | 4 | $\text{some}(p, y)$                | D1, 1, 3                |
|                                                                                                                                                                                                                                                                         |  | 5 | $\text{all}(x, r \text{ some } p)$ | DOWN, 2, 3              |
|                                                                                                                                                                                                                                                                         |  | 6 | $\text{all}(x, r \text{ some } y)$ | $\forall\exists$ , 3, 5 |

**Example 4.11:**  $\text{all}(x, \bar{x}) \vdash \text{all}(y, r \text{ all } x)$ 

In words, if there are no  $x$ s, then all  $y$ 's have any relation whatsoever to all of them. As in Example 4.9, this does not follow from the rule (ZERO). But here is a derivation:

|   |                                           |             |
|---|-------------------------------------------|-------------|
| 1 | $\text{all}(x, \bar{x})$                  | assumption  |
| 2 | $\text{some}(y, \bar{r} \text{ some } x)$ | assumption  |
| 3 | $\text{some}(x, x)$                       | II, 2       |
| 4 | $\text{some}(x, \bar{x})$                 | DARII, 1, 3 |
| 5 | $\text{all}(y, r \text{ all } x)$         | RAA, 2–4    |

As we shall see much later in Theorem ??, the rule (RAA) is essential.

Although (RAA) may be used at any point in a derivation, our proof system in this section has the extra property that if  $\Gamma \vdash \varphi$  using (RAA), then there is a proof using (RAA) at most once. In this book we are not going to be concerned with this stronger property, and for more on it, see [?].

Recall that  $\mathcal{R}$  has a *semantic negation*: for each  $\varphi$  there is some  $\bar{\varphi}$  such that every model satisfies  $\varphi$  or  $\bar{\varphi}$  but not both.

At this point, we mention a few features of the logical system which play a role in our completeness proof. The first is *proof by cases*.

**Lemma 4.12: Cases**

Suppose that  $\Gamma \cup \{\varphi\} \vdash \psi$  and also  $\Gamma \cup \{\bar{\varphi}\} \vdash \psi$ . Then  $\Gamma \vdash \psi$ .

*Proof.* The hypotheses imply that  $\Gamma \cup \{\varphi, \bar{\psi}\} \vdash \perp$ . Also  $\Gamma \cup \{\bar{\varphi}, \bar{\psi}\} \vdash \perp$ , and by (RAA) we also have  $\Gamma \cup \{\bar{\psi}\} \vdash \varphi$ . Let  $\mathcal{T}_1$  be a proof tree showing that  $\Gamma \cup \{\varphi, \bar{\psi}\} \vdash \perp$ , let  $\mathcal{T}_2$  be a proof tree showing that  $\Gamma \cup \{\bar{\psi}\} \vdash \varphi$ . Take  $\mathcal{T}_1$  and replace every leaf labeled  $\varphi$  with a copy of  $\mathcal{T}_2$ . This gives a proof tree with root  $\perp$  and whose leaves are labeled in  $\Gamma \cup \{\bar{\psi}\} \vdash \perp$ . Using (RAA) again, we see that  $\Gamma \vdash \psi$ .  $\square$

**Definition 4.13: Consistency**

A set  $\Gamma$  is *consistent* if  $\Gamma \not\vdash \perp$ . That is, no sentence of the form **some**( $p, \bar{p}$ ) is derivable from  $\Gamma$ .

**Lemma 4.14: a useful sufficient condition for completeness**

Let  $\mathcal{L}$  be a logic with a semantic negation, and let  $\vdash$  be a proof system with RAA. The proof system is complete iff every consistent set is satisfiable.

*Proof.* Assume that the logic is complete. Let  $\Gamma$  be unsatisfiable, so  $\Gamma \models \perp$ . By completeness,  $\Gamma \vdash \perp$ . Thus,  $\Gamma$  is inconsistent.

Assume that every consistent set in the logic is satisfiable. Suppose that  $\Gamma \models \varphi$ . Let  $\bar{\varphi}$  be the semantic negation of  $\varphi$ . Then  $\Gamma \cup \{\bar{\varphi}\}$  has no models. By our assumption,  $\Gamma \cup \{\bar{\varphi}\}$  is inconsistent. Using RAA, we see that  $\Gamma \vdash \varphi$ . Thus the logic is complete.  $\square$

**Remark 4.15**

Lemmas 4.12–4.14 have nothing to do with the particular proof system, and indeed they hold for all logical systems which have (RAA). So versions of these results hold for all of the logical systems in the rest of the book.

**Lemma 4.16: soundness of the proof system for  $\mathcal{R}$** 

If  $\Gamma \vdash \varphi$ , then  $\Gamma \models \varphi$ .

**Theorem 4.17: Completeness of the proof system for  $\mathcal{R}$** 

If  $\Gamma \models \varphi$ , then  $\Gamma \vdash \varphi$ .

The main work of this section is the proof of this Theorem.

Definition 4.18 defines a model  $\mathcal{M}$  from a set  $\Gamma$ . It will turn out that this model  $\mathcal{M}$  is a model of  $\Gamma$ , provided that  $\Gamma$  be consistent.



**Definition 4.18: model construction for  $\mathcal{R}$** 

Let  $\Gamma$  be a set of sentences in  $\mathcal{R}$ . We define a model  $\mathcal{M} = \mathcal{M}_\Gamma$  by giving the universe  $M$  and the interpretation of all unary and binary atoms.

Let  $M$  consist of  $\Gamma_{\text{some}}$ , the “some” sentences in  $\Gamma$ , together with two copies of all the terms  $p$  such that  $\Gamma \vdash \text{some}(p, p)$ . We write these two copies as  $p_\forall$  and  $p_\exists$ .

Here is how we define the structure:

$$\begin{array}{ll}
 p_\forall \in \llbracket u \rrbracket & \text{iff } p \leq u \\
 p_\exists \in \llbracket u \rrbracket & \text{iff } p \leq u \\
 \text{some}(p, c) \in \llbracket u \rrbracket & \text{iff } p \leq u \\
 & \text{or } c \text{ is a unary atom, and } c \leq u
 \end{array}$$


---


$$\begin{array}{l}
 x \llbracket r \rrbracket y \text{ iff one of the following conditions holds:} \\
 (1) \Gamma \vdash \text{all}(u, v \text{ all } r) \text{ for some } u \text{ and } v, \text{ and } x \in \llbracket u \rrbracket, \text{ and } y \in \llbracket v \rrbracket \\
 (2) \Gamma \vdash \text{all}(u, r \text{ some } v) \text{ for some } u \text{ and } v, \text{ and } x \in \llbracket u \rrbracket, \text{ and } y = v_\exists \\
 (3) x = \text{some}(u, r \text{ some } v) \text{ for some } u \text{ and } v, \text{ and } y = v_\exists \\
 (4) x = \text{some}(u, r \text{ all } v) \text{ for some } u \text{ and } v, \text{ and } y \in \llbracket v \rrbracket
 \end{array}$$

**Lemma 4.19**

If  $\llbracket p \rrbracket \cap \llbracket q \rrbracket \neq \emptyset$ , then  $\Gamma \vdash \text{some}(p, q)$ .

*Proof.* There are several cases, depending on the nature of the common element of  $\llbracket p \rrbracket$  and  $\llbracket q \rrbracket$ , and also the reasons for having this in both  $\llbracket p \rrbracket$  and  $\llbracket q \rrbracket$ . Here are some of the cases.

Suppose that  $u_\forall \in \llbracket p \rrbracket \cap \llbracket q \rrbracket$  or that  $u_\exists \in \llbracket p \rrbracket \cap \llbracket q \rrbracket$ . Then by construction  $\Gamma \vdash \text{some}(u, u)$ . We also have  $u \leq p, q$ ; that is,  $\Gamma \vdash \text{all}(u, p)$  and  $\Gamma \vdash \text{all}(u, q)$ . Then easily  $\Gamma \vdash \text{some}(p, q)$ .

Suppose that  $\text{some}(u, v) \in \llbracket p \rrbracket \cap \llbracket q \rrbracket$ . This sentence  $\text{some}(u, v)$  belongs to  $\Gamma$ , and we also have two additional facts: first, either  $u \leq p$ , or  $v \leq p$ ; second, either  $u \leq q$ , or  $v \leq q$ . In all cases, we can put things together to see that indeed  $\Gamma \vdash \text{some}(p, q)$ .

There are similar arguments when the common element of  $\llbracket p \rrbracket$  and  $\llbracket q \rrbracket$  is of the form  $\text{some}(u, \bar{v})$ , or if it is  $\text{some}(u, c)$  for a term  $c$ .  $\square$

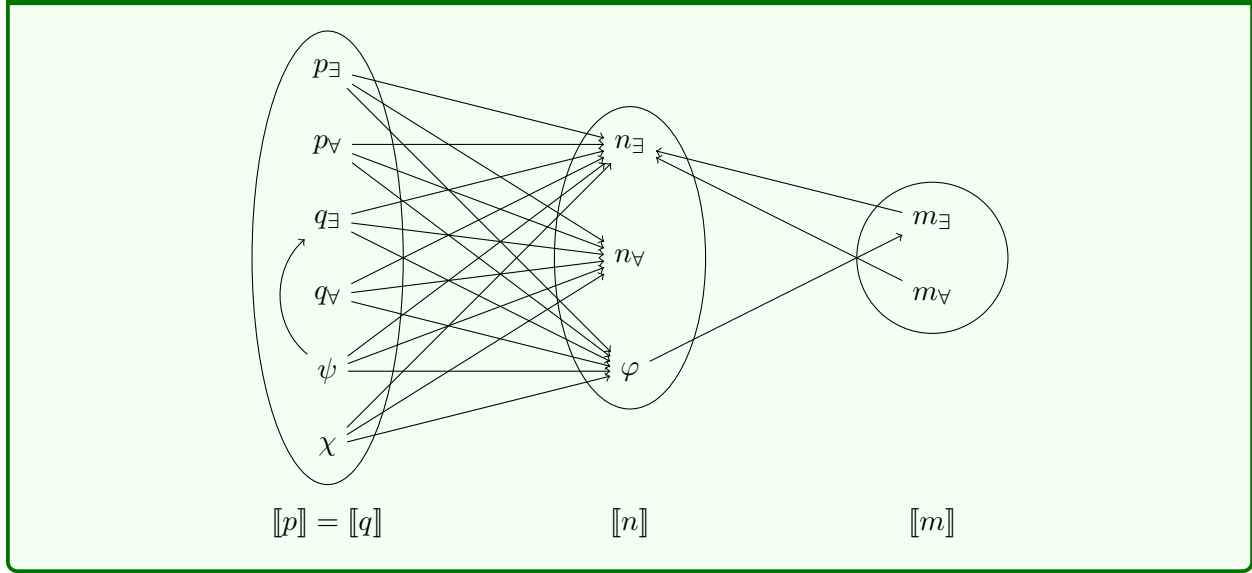
We note a special case of Lemma 4.19.

**Lemma 4.20**

If  $\llbracket p \rrbracket \neq \emptyset$ , then  $\Gamma \vdash \text{some}(p, p)$ .

We return to the proof of Theorem 4.17.

*Proof.* We are going to use Lemma 4.14. Fix a consistent set  $\Gamma$ . Now it is not hard to check that every positive sentence in  $\Gamma$  holds in  $\mathcal{M}$ . For the most part, the details are straightforward.

**Definition 4.21: example of the model construction for  $\mathcal{R}$** 

In the other direction, we shall show that for all negative sentences  $\varphi \in \Gamma$ ,  $\mathcal{M} \models \varphi$ . However, we do this in an indirect fashion. We take a negative sentence  $\varphi \in \Gamma$  with the property that  $\mathcal{M} \not\models \varphi$ , and we obtain a contradiction. And for this, we argue case-by-case as to  $\varphi$ .

First, let  $\varphi$  be  $\text{some}(p, \bar{q})$ . Note that  $\varphi \in M$ , and indeed  $\varphi \in \llbracket p \rrbracket$ . Since we assume that  $\mathcal{M} \not\models \varphi$ ,  $\varphi \in \llbracket q \rrbracket$ . By construction,  $p \leq q$ . Thus  $\Gamma$  is inconsistent.

Second, let  $\varphi$  be  $\text{all}(p, \bar{q})$ . Since  $\mathcal{M} \not\models \varphi$ , we have some  $x \in \llbracket p \rrbracket \cap \llbracket q \rrbracket$ . We have several cases depending on whether  $x$  is a subscripted atom  $u\forall$  or  $u\exists$  or a sentence  $\varphi$ . If  $x$  is  $u\forall$  or  $u\exists$ , then we see that  $u \leq p$  and  $u \leq q$ , and also  $\Gamma \vdash \text{some}(u, u)$ . So  $\Gamma$  is inconsistent. If  $x$  is of the form  $\text{some}(u, c)$ , then the same argument works for this  $u$ .

Third, let  $\varphi$  be  $\text{all}(p, \bar{r} \text{all } q)$ . Assume that  $\mathcal{M} \not\models \varphi$ . Let  $x \in \llbracket p \rrbracket$  be related by  $\llbracket r \rrbracket$  to  $y \in \llbracket q \rrbracket$ . There are four subcases, corresponding to the four clauses of the definition of  $\llbracket r \rrbracket$  in  $\mathcal{M}$ .

Subcase 1: There is a sentence  $\text{all}(a, r \text{all } b)$  provable from  $\Gamma$  such that  $x \in \llbracket a \rrbracket$  and  $y \in \llbracket b \rrbracket$ . Then  $\Gamma \vdash \text{some}(a, a)$ , and similarly for  $b$ , and  $\Gamma$  is inconsistent.

Subcase 2:  $x \in \llbracket p \rrbracket$ ,  $y$  is  $q\exists$ , and the sentence  $\text{all}(p, r \text{some } q)$  belongs to  $\Gamma$ . This time, we again have  $\Gamma \vdash \text{some}(p, p)$ . And so we easily get a contradiction.

Subcase 3: for some  $u$  and  $v$ ,  $x = \text{some}(u, r \text{some } v)$ ,  $y = v\exists$ , and  $\Gamma$  contains  $\text{some}(u, r \text{some } v)$ . Since  $x \in \llbracket p \rrbracket$ , we have  $u \leq p$ . And as  $y = v\exists \in \llbracket q \rrbracket$ , we have  $v \leq q$ . Thus  $\Gamma \vdash \text{some}(p, r \text{some } q)$ , and this shows it to be inconsistent.

Fourth, let  $\varphi$  be  $\text{all}(p, \bar{r} \text{some } q)$ . If  $\mathcal{M} \not\models \varphi$ , we fix some  $x \in \llbracket p \rrbracket$  is related by  $\llbracket r \rrbracket$  to all elements of  $\llbracket q \rrbracket$ . Since  $\varphi \in \Gamma$ , and also  $\Gamma \vdash \text{some}(p, p)$ , we indeed have  $\Gamma \vdash \text{some}(q, q)$ . The point of this is that  $M$  contains  $q\forall$ . Hence  $\Gamma \vdash \text{all}(u, r \text{all } v)$  for some  $u, v$  such that  $x \in \llbracket u \rrbracket$  and  $y \in \llbracket v \rrbracket$ . By Lemma 4.19,  $\Gamma \vdash \text{some}(u, p)$ . Thus  $\Gamma \vdash \text{some}(p, r \text{all } v)$ . We also have  $y \leq v$ . So  $\Gamma \vdash \text{some}(p, r \text{all } y)$ . Thus  $\Gamma$  is inconsistent.

Fifth, let  $\varphi$  be  $\text{some}(p, \bar{r} \text{all } q)$ . This sentence belongs to  $\Gamma$ , hence to  $M$  and indeed to  $\llbracket p \rrbracket$ . Assuming that  $\mathcal{M} \not\models \varphi$ , we must have some  $y \in \llbracket q \rrbracket$  such that  $\varphi[r]y$ . Also,  $\Gamma \vdash \text{some}(p, p)$ . By

examining the definition of  $\llbracket r \rrbracket$ , we have two cases. The first case is when  $\Gamma \vdash \text{all}(u, r \text{ all } v)$  for some  $u$  and  $v$  such that  $p \leq u$  and  $y \in \llbracket v \rrbracket$ . In this case,  $\Gamma \vdash \text{some}(v, v)$ , and so  $\Gamma \vdash \text{all}(p, r \text{ some } v)$ . The second case is when  $\Gamma \vdash \text{all}(p, r \text{ some } v)$  and  $y = v_{\exists}$ . Since  $y = v_{\exists} \in \llbracket q \rrbracket$ , we have  $v \leq q$ , and so  $\Gamma \vdash \text{all}(p, r \text{ some } q)$ .

Finally, let  $\varphi$  be  $\text{some}(p, \bar{r} \text{ some } q)$ . Assume that  $\mathcal{M} \not\models \varphi$ . Then  $\varphi \in M$  and also  $q_{\forall} \in M$ . We have  $\varphi \in \llbracket p \rrbracket$  and  $q_{\forall} \in \llbracket q \rrbracket$ . Since  $\mathcal{M} \not\models \varphi$ ,  $\varphi \llbracket r \rrbracket q_{\forall}$ . And so there is some sentence  $\text{all}(u, r \text{ all } v)$  derived from  $\Gamma$  such that  $\varphi \in \llbracket u \rrbracket$  and  $q_{\forall} \in \llbracket v \rrbracket$ . This means that  $p \leq u$  and  $q \leq v$ . So  $\Gamma \vdash \text{all}(p, r \text{ all } q)$ .  $\Gamma$  is therefore inconsistent.

This completes the proof. □

### Example 4.22

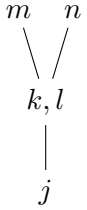
Here is an illustration of the model construction for  $\mathcal{R}$ . Let  $\Gamma$  be the set below

|                    |                    |                                                   |                                                          |
|--------------------|--------------------|---------------------------------------------------|----------------------------------------------------------|
| $\text{all}(j, k)$ | $\text{all}(l, m)$ | $\frac{\text{all}(q, p)}{\text{all}(n, \bar{m})}$ | $\text{all}(m, \bar{\text{see}} \text{ some } n)$        |
| $\text{all}(l, k)$ | $\text{all}(p, q)$ | $\text{all}(p, \text{see all } n)$                | $\varphi : \text{some}(n, \text{see some } m)$           |
| $\text{all}(m, q)$ | $\text{all}(k, l)$ | $\text{all}(m, \text{see some } n)$               | $\psi : \text{some}(p, \text{see some } q)$              |
| $\text{all}(j, l)$ | $\text{all}(k, n)$ |                                                   | $\chi : \text{some}(p, \bar{\text{see}} \text{ all } j)$ |

We have seen the first nine sentences in Example 1.8 on page 24, but changing the notation to match this chapter.  $\Gamma$  also has three  $\exists$  sentences at the end, and so these sentences will be elements of our model.

Recall that  $\mathbb{P}_{\Gamma}$  is given by the picture on the left. The unary atoms whose existence can be proved by  $\Gamma$  are  $m, n, p$ , and  $q$ . So we have two copies of each of these, one subscripted  $\forall$  and the other subscripted  $\exists$ . In addition, we have the three existential sentences in  $\Gamma$ ,  $\varphi$ ,  $\psi$ , and  $\chi$ . Definition 4.18 arranges that  $\varphi \in \llbracket n \rrbracket$ , and that  $\psi$  and  $\chi$  belong to  $\llbracket p \rrbracket = \llbracket q \rrbracket$ . To make  $\text{all}(p, \text{see all } n)$  true, we simply insist that all elements of  $\llbracket p \rrbracket$  be related to all elements of  $\llbracket n \rrbracket$ . (These two sets happen to be the same in this model, but this is not what usually happens.) To make  $\text{all}(m, \bar{n})$  true, nothing needs to be done. To make  $\text{all}(m, \text{see some } n)$  true, we relate all elements of  $\llbracket m \rrbracket$  to the single element  $m_{\exists}$ . To make  $\text{all}(m, \bar{\text{see}} \text{ some } n)$  true, nothing has to be done: all elements of  $\llbracket m \rrbracket$  are un-related to the single element  $m_{\forall}$ . To get  $\varphi$  to be true, the sentence  $\varphi$  itself will be taken to belong to  $\llbracket n \rrbracket$ , and it will be related to  $m_{\exists}$ . In order that  $\psi$  be true in the model, we put  $\psi \in \llbracket p \rrbracket$  and then take  $\psi \llbracket \text{see} \rrbracket q_{\exists}$ . Finally, to make  $\chi$  true, we again take  $\chi \in \llbracket p \rrbracket$  and there is nothing else to do. In fact, in the model  $\llbracket j \rrbracket = \emptyset$ , and  $\chi$  thus holds. A picture of the model may be found in Figure 4.21.

$p, q$



## 4.3 Algorithmic analysis

As with our previous logical systems, the next goal is to extend the work we did on completeness to provide an algorithm which, when given a finite set  $\Gamma$  of sentences in  $\mathcal{R}$  and another sentence  $\varphi$ , tells whether or not  $\Gamma \models \varphi$ . Since we have shown the completeness of the logical system  $\mathcal{R}$ , this is the same as asking whether or not  $\Gamma \vdash \varphi$ .

Let us highlight two issues. First of all, logical systems with rules like (RAA) are algorithmically more challenging than ones without it. The reason is that with (RAA), to know whether or not  $\Gamma \vdash \varphi$ , one might need to know whether some larger set of sentences  $\Delta$  has the property that  $\Delta \vdash \psi$  for some particular  $\psi$ . And since (RAA) may be used more than once, if our underlying language is infinite, this means that there are infinitely many assertions  $\Delta \vdash \psi$  to consider. Now the underlying language of this chapter may be taken to be finite, and so this is not a serious problem here. (But in later chapters, we shall consider languages with recursive concepts, and these always are infinite.)

The second issue pertains to Definition 4.18 and some of the work we did in the proof of Theorem 4.17. The model  $\mathcal{M}$  built in Definition 4.18 seemed to use the provability relation at all points. So it would seem that we cannot use Definition 4.18 in order to see whether  $\Gamma \vdash \varphi$  or not. Similarly, in the proof of Theorem 4.17 we used the notion of consistency. And to tell whether  $\Gamma$  is consistent or not, we need to understand the very provability relation  $\vdash$  that is at issue here.

Fortunately, there is a way around both of these problems. First, we modify Definition 4.18 so as to only use provability *without* (RAA). This means that we change the meaning of  $\leq$  and  $\vdash$  for the purposes constructing  $\mathcal{M}_\Gamma$ . This makes a difference: given any syllogistic proof system which does not change the premise sets in the course of a derivation has the property that the derivability relation  $\Gamma \vdash \varphi$  may be computed in time polynomial in the number of atoms in  $\Gamma \cup \{\varphi\}$ . This very general result will be discussed in Chapter ??.

The same proof that we saw in Theorem 4.17 shows that  $\mathcal{M}_\Gamma$  satisfies all of the positive sentences in  $\Gamma$ , and that if  $\Gamma$  is consistent, then  $\mathcal{M}_\Gamma$  satisfies all of the negative sentences in  $\Gamma$ , too. So to tell whether or not  $\Gamma$  is consistent or not, we need only build  $\mathcal{M}_\Gamma$  and check to see if it satisfies the full set  $\Gamma$ . If so,  $\Gamma$  is satisfiable and hence consistent. If not, then we have some  $\psi$  in  $\Gamma$  such that  $\mathcal{M} \not\models \psi$ . From the proof of Theorem 4.17, we can even turn the fact that  $\mathcal{M} \not\models \psi$  into a proof in our system that  $\Gamma$  is inconsistent.

Going a little further, we can use these observations to get an algorithm to decide whether or not  $\Gamma \vdash \varphi$ . See Algorithm 4.23.

**Algorithm 4.23:** given a finite set  $\Gamma$  and a sentence  $\varphi$  in  $\mathcal{R}$ , tell whether or not  $\Gamma \vdash \varphi$ . If  $\Gamma \vdash \varphi$ , output a proof; otherwise, output a counter-model.

```

1:  $\Delta \leftarrow \Gamma \cup \{\bar{\varphi}\}$ .
2:  $\mathcal{M} = \mathcal{M}_\Delta \leftarrow$  the model from Definition 4.18. ▷ using derivations sans (RAA).
3: if  $\mathcal{M} \models \Delta$  then
4:    $\Gamma \not\models \varphi$ , and output  $\mathcal{M}$  as a counter-model.
5: else
6:   Get a derivation of a contradiction from  $\Delta$ 
7:   Add a step using (RAA), withdrawing  $\bar{\varphi}$  and using  $\varphi$  as a conclusion.
8: end if
```

## 4.4 Incorporating background facts in R

Suppose we have a stock of background facts about verbs, such as

$$\text{hitting entails touching} \tag{4.2}$$

**Definition 4.24: Rules of inference corresponding to the background assertion**  
 $r \sqsubseteq s$

$$\begin{array}{cc} \frac{\text{all}(d, r \text{ all } c)}{\text{all}(d, s \text{ all } c)} r \sqsubseteq s & \frac{\text{all}(d, r \text{ some } c)}{\text{all}(d, s \text{ some } c)} r \sqsubseteq s \\ \frac{\text{some}(d, r \text{ all } c)}{\text{some}(d, s \text{ all } c)} r \sqsubseteq s & \frac{\text{some}(d, r \text{ some } c)}{\text{some}(d, s \text{ some } c)} r \sqsubseteq s \end{array}$$

We mean that every act of hitting is also an act of touching. This background fact cannot be stated in any of the languages which we have so far studied. Nevertheless, it can be made into a semantic requirement: we would require of a model that the interpretation of *hitting* be a sub-relation of the interpretation of *touching*. And then we might like to study the entailment relation on this smaller class of models.

Even though we cannot state (4.2) as an *axiom*, it does yield a rule of inference. To state it more abstractly, suppose we have a rule like

$$r \sqsubseteq s \tag{4.3}$$

and we restrict attention to the models where  $\llbracket r \rrbracket \subseteq \llbracket s \rrbracket$ . Then Figure ?? lists sound rules of inference.

We add these to the system  $\mathcal{R}$  as listed in Definition 4.7.

**Proposition 4.25**

The system  $\mathcal{R}$  together with the rules in Figure ?? give a sound and complete logic:  $\Gamma \vdash \varphi$  iff  $\Gamma \models \varphi$ .

**Sources** The first syllogistic logic to employ verbs was Nishihara, Morita, and Iwata [?]. The logical system for  $\mathcal{R}$  comes from Moss [?] and from Pratt-Hartmann and Moss [?]. The work on incorporating background facts is new here.

## 4.5 Exercises

**Exercise 4.1.** Let  $\Gamma$  be the set of sentences in Example 4.22 and shown again below:

$$\begin{array}{llll} \text{all}(j, k) & \text{all}(l, m) & \frac{\text{all}(q, p)}{\text{all}(n, \overline{m})} & \text{all}(m, \overline{\text{see some } n}) \\ \text{all}(l, k) & \text{all}(p, q) & & \varphi : \text{some}(n, \text{see some } m) \\ \text{all}(m, q) & \text{all}(k, l) & \text{all}(p, \text{see all } n) & \psi : \text{some}(p, \text{see some } q) \\ \text{all}(j, l) & \text{all}(k, n) & \text{all}(m, \text{see some } n) & \chi : \text{some}(p, \overline{\text{see all } j}) \end{array}$$

For each of the following sentences  $\varphi$ , do the following: if  $\Gamma \vdash \varphi$ , then give a derivation in our system. If  $\Gamma \not\vdash \varphi$ , then give a model of  $\Gamma$  where  $\varphi$  is false.

1.  $\text{some}(n, n)$
2.  $\text{all}(p, n)$
3.  $\text{some}(q, \text{see some } p)$
4.  $\text{all}(j, \bar{k})$

**Exercise 4.2.** In this section we gave a precise definition of the derivability relation  $\Gamma \vdash \varphi$ . Here is another characterization of this relation. Let  $\vdash_{alt}$  be the the smallest relation on  $\mathcal{P}(\mathcal{R}) \times \mathcal{R}$  such that

1. If  $\varphi \in \Gamma$ , then  $\Gamma \vdash_{alt} \varphi$ .
2. If  $\Gamma \vdash_{alt} \psi_1, \dots, \Gamma \vdash_{alt} \psi_n$  and one of the rules of  $\mathbf{R}$  other than (RAA) has as a substitution instance  $\psi_1, \dots, \psi_n \setminus \varphi$ , then  $\Gamma \vdash_{alt} \varphi$ .
3. If  $\Gamma \cup \{\bar{\varphi}\} \vdash_{alt} \perp$ , then  $\Gamma \vdash_{alt} \varphi$ .

Prove that  $\Gamma \vdash \varphi$  iff  $\Gamma \vdash_{alt} \varphi$ .

**Exercise 4.3.** Let  $\text{see}$  be a binary atom in a sentence  $\varphi$ . Given a set  $\Gamma$ , let  $\Gamma_0$  be the set of sentences in  $\Gamma$  which have the same verb,  $\text{see}$  or else have no verb at all. Prove the following:

1. If  $\Gamma \models \varphi$  and  $\Gamma$  is consistent, then  $\Gamma_0 \models \varphi$ .
2. If  $\Gamma \vdash \varphi$  and  $\Gamma$  is consistent, then  $\Gamma_0 \vdash \varphi$ .

The first part should be a semantic argument, and the second a proof-theoretic one.

**Exercise 4.4.** For each of the following, say whether it is true or false, along with a reason.

1. Every set of positive sentences of  $\mathcal{R}$  is satisfiable.
2.  $\text{all}(p, \text{see some } q) \vdash \text{some}(p, \text{see all } q)$ .
3.  $\text{some}(p, \text{see all } q) \vdash \text{all}(p, \text{see some } q)$ .

**Exercise 4.5.** In the proof of Theorem 4.17, we asserted that the model  $\mathcal{M}$  satisfied the positive sentences of  $\Gamma$ . But we didn't check this. Your task is to check it. The argument goes case-by-case as to the syntax, and so there are a lot of cases. But they are all fairly short arguments.

**Exercise 4.6.** This exercise is about a modification of Definition 4.18. Given  $\Gamma$ , let  $N$  be two copies of all  $\text{some}$  sentences derivable from  $\Gamma$ . (This is larger than two copies of  $\Gamma_{\text{some}}$ .) Figure out a reasonable way to make  $N$  into a model  $\mathcal{N}$ , and then use this  $\mathcal{N}$  to prove the completeness theorem for  $\mathbf{R}$ . This means showing two things:

1.  $\mathcal{N}$  satisfies the positive sentences in  $\Gamma$ .

2.  $\mathcal{N}$  satisfies the negative sentences in  $\Gamma$  also, unless  $\Gamma$  is inconsistent.

**Exercise 4.7.** Prove the following derivable rules in the system of this chapter, allowing background assumptions.

$$\frac{\text{all}(d, \bar{s} \text{ all } c)}{\text{all}(d, \bar{r} \text{ all } c)} r \sqsubseteq s \qquad \frac{\text{all}(d, \bar{s} \text{ some } c)}{\text{all}(d, \bar{r} \text{ some } c)} r \sqsubseteq s$$

$$\frac{\text{some}(d, \bar{s} \text{ all } c)}{\text{some}(d, \bar{r} \text{ all } c)} r \sqsubseteq s \qquad \frac{\text{some}(d, \bar{s} \text{ some } c)}{\text{some}(d, \bar{r} \text{ some } c)} r \sqsubseteq s$$

**Exercise 4.8.** Prove Proposition 4.25. [Hint: use Exercise 4.7.]

**Exercise 4.9.** This exercise is about the logical language  $\mathcal{S}^\dagger$  which we saw in Chapter 3. Show that the following gives a different sound and complete proof system for  $\mathcal{S}^\dagger$ : (DARII), (ZERO), (ONE), and (RAA). The point is that by using (RAA) instead of (X) the statement of the rules has become simpler.

## Chapter 5

# Propositional Logic

Up until now, the logical systems in this book have been unusual. That is, syllogistic systems of logic are not in the mainstream of the subject. The most popular systems are *propositional logic* and *first-order logic*. Your study of logic would not be complete without a good grounding on those systems, and now is a good time to begin. We move toward the mainstream by presenting propositional logic in this chapter.

Propositional logic is the study of the word *not*, the sentential connectives *and*, *or*, and *if* ..., *then*, and the phrase *if and only if*. The great majority of logical systems build on *classical propositional logic*.

Propositional logic is sometimes thought of as a trivial logical system, one without much use or interest. Both points are mistaken! Propositional logic could be the subject of several long courses, with many applications. However, we tried hard to make this chapter relatively short by concentrating on the basics of the subject. So this book will not even hint at the richness of propositional logic.

### 5.1 Introduction

We formulate propositional logic as a branch of mathematics about the sentential connectives mentioned above. We'll use symbols for them:  $\neg$  for “not,”  $\wedge$  for “and,”  $\vee$  for “or,”  $\rightarrow$  for “if,” and  $\leftrightarrow$  for “if and only if”. These are going to be *function symbols*, on the same level as  $+$  and  $\times$  in arithmetic. We also will have *constants*  $\top$  and  $\bot$ ; these are like 1 and 0. But unlike arithmetic, the operations are not on the set of numbers but rather on the set

$$\text{Bool} = \{\top, \bot\}.$$

There are five operations, and they work as shown below:

#### Definition 5.1: Truth tables

| $p$    | $\neg p$ | $p$    | $q$    | $p \wedge q$ | $p$    | $q$    | $p \vee q$ | $p$    | $q$    | $p \rightarrow q$ | $p$    | $q$    | $p \leftrightarrow q$ |
|--------|----------|--------|--------|--------------|--------|--------|------------|--------|--------|-------------------|--------|--------|-----------------------|
| $\top$ | $\bot$   | $\top$ | $\top$ | $\top$       | $\top$ | $\top$ | $\top$     | $\top$ | $\top$ | $\top$            | $\top$ | $\top$ | $\top$                |
| $\bot$ | $\top$   | $\top$ | $\bot$ | $\bot$       | $\top$ | $\bot$ | $\top$     | $\top$ | $\bot$ | $\bot$            | $\top$ | $\bot$ | $\bot$                |
|        |          | $\bot$ | $\top$ | $\bot$       | $\bot$ | $\top$ | $\top$     | $\bot$ | $\top$ | $\top$            | $\bot$ | $\top$ | $\bot$                |
|        |          | $\bot$ | $\bot$ | $\bot$       | $\bot$ | $\bot$ | $\bot$     | $\bot$ | $\bot$ | $\top$            | $\bot$ | $\bot$ | $\top$                |



The charts above allow us to compute with our logical symbols on the tiny universe  $\text{Bool} = \{\text{T}, \text{F}\}$ . For example,  $\neg\neg\neg\text{F} = \neg\neg\text{T} = \neg\text{F} = \text{T}$ . And  $\text{F} \wedge (\text{T} \vee \text{T}) = \text{F} \wedge \text{T} = \text{F}$ .

### 5.1.1 On our notation

There is no significance to our use of the symbols  $\text{T}$  for “true” and  $\text{F}$  for “false.” We could just as well have used 1 and 0, or something else. The same is true of our choices for the sentential connectives. If you read other books and papers, you will find several other symbols used.

### 5.1.2 Atomic Sentences

The structure  $\text{Bool}$  is interesting, but it is “too small” to really do much. So we’ll consider a set  $\text{AtSen}$  of *atomic sentences*. Think of atomic sentences as things that denote *propositions*, where propositions are taken to be entities which can be either true or false.

Here is what we mean. Later in the book, we’ll want to do *propositional logic on top of our previous logical systems*. For example, in “propositional logic on top of  $\mathcal{A}(\mathcal{RC})$ ”, we’ll want to have sentences like

$$(\text{some dog (eats some apple)}) \rightarrow (\text{all cats (see some rabbit)})$$

We’ll want to give the semantics and proof theory for this larger logic. In this chapter, we pave the way by focusing only on the propositional logic part. So instead of the sentence above, we’d just say  $p \rightarrow q$ , regarding  $p$  and  $q$  as “atomic”.

To be as general as possible about what we want to do propositional logic “on top of”, we take some set of *atomic sentences*

$$\text{AtSen} = \{p_1, p_2, \dots, p_n, \dots\}. \quad (5.1)$$

Later, or in special contexts, we might decide to “open these up” reveal more structure. But most of the time in this chapter, we simply regard these as “atomic” (indivisible).

We let  $p, q, r, \dots$ , possibly with subscripts, range over  $\text{AtSen}$ .<sup>1</sup> We do this even when  $\text{AtSen}$  only has one or two elements.

### 5.1.3 Sentences

Here is the definition of a *sentence* of propositional logic.

---

<sup>1</sup>The fact that we are using the letters  $p, q$ , and  $r$  again might be confusing, since we used them for other things earlier in the book. In this chapter, those letters are for atomic sentences in propositional logic.

**Definition 5.2: sentences**

We define the *sentences* of propositional logic to be the smallest set  $\text{Sen}$  with the following properties:

1. Each  $p_i$  which belongs to  $\text{AtSen}$  belongs to  $\text{Sen}$ .
2.  $\top$  and  $\text{F}$  belong to  $\text{Sen}$ .
3. If  $\varphi$  belongs to  $\text{Sen}$ , then  $\neg\varphi$  belongs to  $\text{Sen}$  as well.
4. If  $\varphi$  and  $\psi$  both belong to  $\text{Sen}$ , then also the sentences  $\varphi \wedge \psi$ ,  $\varphi \vee \psi$ ,  $\varphi \rightarrow \psi$  and  $\varphi \leftrightarrow \psi$  belong to  $\text{Sen}$ .

We call  $\text{Sen}$  the *language of propositional logic (built over AtSen)*.

**Example 5.3**

Here are three sentences:  $\neg p_3$ ,  $\text{F}$ , and

$$p_{239} \wedge (\neg(p_4 \vee \neg p_{155}) \rightarrow \top).$$

You should notice that the sentence above has parentheses. We are going to be informal about parentheses. If we wanted to be persnickety, we would rewrite Definition 5.2 to put parentheses in the sentences. Then our examples would be  $(\neg p_3)$ ,  $\text{F}$ , and

$$(p_{239} \wedge ((\neg(p_4 \vee (\neg p_{155}))) \rightarrow \top)).$$

Then we would need some conventions about dropping parentheses to make things more readable. We prefer to avoid all such matters.

**Remark 5.4**

We read  $\wedge$  as “and,” and also  $\vee$  as “or.” Incidentally, the word “or” in English has both *inclusive* and *exclusive* readings. Frequently, one intends an exclusive reading, as in *You can have cake or pie for dessert*. However, occasionally we have inclusive or: *If you have taken a beginning-level math logic class or an intermediate-level computer science class, then you can take Algorithms for Nerds*. As is standard practice in formal logic, unless otherwise noted,  $\vee$  is taken to mean *inclusive* or.

We read  $\rightarrow$  as *if*. However, there is a complication here. As we shall see, the rendering of the English “if ... then ...” by the standard conditional arrow  $\rightarrow$  has some bad features. In a sense it is just the *first approximation* to an adequate treatment of conditionals. In this course, we are interested in going further and studying better proposals for the mathematical treatment of conditionals.

**Remark 5.5: Levels of notation**

There are several levels of notation in this chapter. The official set of **AtSen** is given in (5.1) using the notation  $p_i$ , where  $i$  is a positive natural number. We use letters like  $p$  and  $q$  as variables to range over **AtSen**. We do this mostly to avoid subscripts when we don't need them. And Greek letters like  $\varphi, \psi, \dots$ , range over the set **Sen** of sentences. Officially, sentences look like the ones in Example 5.3, not like the letter  $\varphi$ .

**5.1.4 Semantics of propositional logic****Definition 5.6: valuation and their extensions**

A *valuation* is a function  $v : \text{AtSen} \rightarrow \text{Bool}$ .

Along with  $v$  comes a function called  $\llbracket \cdot \rrbracket_v$ . This will be a function from **Sen** to **Bool**:

$$\llbracket \cdot \rrbracket_v : \text{Sen} \rightarrow \text{Bool}.$$

It works in a natural way, using the following *recursion equations* for  $\llbracket \cdot \rrbracket_v$ :

$$\begin{aligned} \llbracket p_i \rrbracket_v &= v(p_i) \\ \llbracket \neg \varphi \rrbracket_v &= \neg \llbracket \varphi \rrbracket_v \\ \llbracket \varphi \wedge \psi \rrbracket_v &= \llbracket \varphi \rrbracket_v \wedge \llbracket \psi \rrbracket_v \end{aligned} \tag{5.2}$$

and similarly for  $\vee, \rightarrow$ , and  $\leftrightarrow$

**Example 5.7**

Suppose that  $v(p_1) = \text{F}$ ,  $v(p_2) = \text{T}$ ,  $v(p_3) = \text{T}$ , and  $v(p_i) = \text{F}$  for  $i \geq 4$ . Then we have

$$\llbracket (p_1 \vee p_2) \wedge p_3 \rrbracket_v = \text{T}.$$

The reasoning is as follows, dropping the subscript  $v$ :

$$\begin{aligned} \llbracket (p_1 \vee p_2) \wedge p_3 \rrbracket &= \llbracket p_1 \vee p_2 \rrbracket \wedge \llbracket p_3 \rrbracket \\ &= (\llbracket p_1 \rrbracket \vee \llbracket p_2 \rrbracket) \wedge \text{T} \\ &= (\text{F} \vee \text{T}) \wedge \text{T} \\ &= \text{T} \wedge \text{T} \\ &= \text{T} \end{aligned} \tag{5.3}$$

This illustrates the general principle that  $v$  gives us the truth values of the atomic sentences, and  $\llbracket \cdot \rrbracket_v$  extends this to the complex sentences in **Sen** by following the tables.

It is important to notice that in the equations (5.2) and in calculations such as (5.3), some of the connectives  $\wedge$  and  $\vee$  are from the syntax of propositional logic, and some are operation symbols that are part of the definition of **Bool**. Can you tell which are which?

**Remark** Here is a query for you. What is the difference between  $v$  and  $\llbracket \cdot \rrbracket_v$ ? If we have a valuation

**Definition 5.8: semantic notions for propositional logic**

$\varphi$  is a (*propositional*) *tautology* if  $\emptyset \models \varphi$ . This means: for all valuations  $v$ ,  $\llbracket \varphi \rrbracket_v = \top$ . We always write this as  $\models \varphi$ . We also say that  $\varphi$  is *valid*, or *logically valid*.

$\varphi$  is *satisfiable* if there is some  $v$  such that  $\llbracket \varphi \rrbracket_v = \top$ .

$\varphi$  *logically implies*  $\psi$  if  $\varphi \models \psi$ . This means: for all  $v$  such that  $\llbracket \varphi \rrbracket_v = \top$ ,  $\llbracket \psi \rrbracket_v = \top$  as well.

Finally,  $\varphi$  and  $\psi$  are *logically equivalent* if each logically implies the other. This amounts to the condition that for all  $v$ ,  $\llbracket \varphi \rrbracket_v = \llbracket \psi \rrbracket_v$ .

$v$ , why do we need to talk about  $\llbracket \cdot \rrbracket_v$  at all?

## 5.2 Semantic notions of propositional logic

At this point, we have defined the meaning of a sentence under a valuation. As in previous chapters, get derived notions such as  $\Gamma \models \varphi$  from Definition 0.3. This would mean that every valuation making all sentences in  $\Gamma$  true also makes  $\varphi$  true. We read this as  $\Gamma$  *semantically implies*  $\varphi$ . But unlike what we had before, the case when  $\Gamma = \emptyset$  is not trivial.<sup>2</sup> The other main semantic notions of propositional logic are listed in Definition 5.8.

### 5.2.1 Truth tables

The definition of  $\Gamma \models \varphi$  is in terms of valuations, and valuations are functions on our set  $\text{AtSen}$  listed in (5.1). In a concrete setting, especially one with three or fewer atomic sentences, it will be easier to study things using *truth tables*. Here is an example, showing that  $\Gamma \models p \vee (q \wedge r)$ , where  $\Gamma = \{p \vee q, p \wedge r\}$ .

|   | $p$ | $q$ | $r$ | $p \vee q$ | $p \wedge r$ | $p \vee (q \wedge r)$ |
|---|-----|-----|-----|------------|--------------|-----------------------|
| ☞ | T   | T   | T   | T          | T            | T                     |
| ☞ | T   | T   | F   | T          | F            | T                     |
| ☞ | T   | F   | T   | T          | T            | T                     |
| ☞ | T   | F   | F   | T          | F            | T                     |

|   | $p$ | $q$ | $r$ | $p \vee q$ | $p \wedge r$ | $p \vee (q \wedge r)$ |
|---|-----|-----|-----|------------|--------------|-----------------------|
| ☞ | F   | T   | T   | T          | F            | T                     |
|   | F   | T   | F   | T          | F            | F                     |
|   | F   | F   | T   | F          | F            | F                     |
|   | F   | F   | F   | F          | F            | F                     |

We make a table, listing the eight ways that to assign truth values to  $p$ ,  $q$ , and  $r$ . These are the three left-most columns. Next, we have columns for the two assumptions:  $p \vee q$  and  $p \wedge r$ . We separated the assumptions from the conclusion  $p \vee (q \wedge r)$  by an extra vertical line. We also put a hand ☞ next to the rows that made all (=both) of the premises true. The main question at this point is: in all the rows with the ☞, is the conclusion also true? In this case, the answer is Yes. So this justifies our saying that  $\Gamma \models p \vee (q \wedge r)$ .

<sup>2</sup>For the logics in the previous chapters, which sentences  $\varphi$  have the property that  $\models \varphi$ ?

**Remark 5.9**

We have defined  $\Gamma \models \psi$  above. It will also be important to study the opposite concept, written

$$\Gamma \not\models \psi.$$

This means that there is some  $v$  which makes all sentences in  $\Gamma$  true and  $\psi$  false. Equivalently, *some* row of the truth table makes all sentences in  $\Gamma$  true and  $\psi$  false.

**Example 5.10**

As a non-example of logical implication, let us show that  $p \vee q \not\models p \wedge q$ . Here is the truth table:

| $p$ | $q$ | $p \vee q$ | $p \wedge q$ |
|-----|-----|------------|--------------|
| T   | T   | T          | T            |
| T   | F   | T          | F            |
| F   | T   | T          | F            |
| F   | F   | F          | F            |

In the second and third rows, the conclusion is false. Since we found at least one row which made the assumption true and the conclusion false, we know that  $p \vee q \not\models p \wedge q$ . In terms of valuations, we could read off two valuations from the second and third rows. These are  $v$  and  $w$  given by

$$\begin{array}{ll} v(p) = \text{T} & w(p) = \text{F} \\ v(q) = \text{F} & w(q) = \text{T} \end{array}$$

Then  $\llbracket p \vee q \rrbracket_v = \text{T}$  while  $\llbracket p \wedge q \rrbracket_v = \text{F}$ , and similarly for  $\llbracket \rrbracket_w$ . Again, mentioning either of these valuations shows that  $p \vee q \not\models p \wedge q$ .

**5.2.2 An example of a non-obvious tautology**

Here is a sentence that we shall call  $\pi$ :

$$(p_1 \wedge p_2) \vee (p_1 \wedge p_3) \vee (p_2 \wedge p_3) \vee (\neg p_1 \wedge \neg p_2) \vee (\neg p_1 \wedge \neg p_3) \vee (\neg p_2 \wedge \neg p_3) \quad (5.4)$$

We claim that  $\models \pi$ . We could check this out using truth tables. This would show *that*  $\models \pi$ , but it would not show *why*  $\models \pi$ . A deeper explanation comes from the connection of  $\pi$  to the “Pigeonhole Principle,” as we shall explain.

**Proposition 5.11:**  $\models \pi$ 

That is, for all valuations  $v$ ,  $\llbracket \pi \rrbracket_v = \text{T}$ .

*Proof.* Let  $v$  be any valuation. We show that  $\llbracket \pi \rrbracket_v = \text{T}$ . The function  $v$  assigns one of two truth values (T or F) to each of the atomic sentences  $p_1$ ,  $p_2$ , and  $p_3$ . Thus it must assign the same truth value to at least two of those three sentences. (This is the “Pigeonhole Principle” at work.) If  $v$  assigns T to at least two of  $p_1$ ,  $p_2$ ,  $p_3$ , then one of the first three disjuncts of  $\pi$  must be made T

by  $\bar{v}$ . If  $v$  assigns F to at least two of  $p_1, p_2, p_3$ , then one of the last three disjuncts of  $\pi$  must be made T by  $\bar{v}$ . Either way,  $\llbracket \pi \rrbracket_v = \text{T}$ .  $\square$

### 5.2.3 Logical equivalence

We say that two sentences  $\varphi$  and  $\psi$  are *logically equivalent* if  $\varphi \models \psi$  and  $\psi \models \varphi$ . We write  $\varphi \equiv \psi$  for this.

Example 5.12 lists a number of standard logical equivalences, along with their names.

#### Example 5.12: Laws of Boolean algebra

|                                     |          |                                                    |                 |                                      |          |            |            |
|-------------------------------------|----------|----------------------------------------------------|-----------------|--------------------------------------|----------|------------|------------|
| $\varphi \wedge \psi$               | $\equiv$ | $\psi \wedge \varphi$                              | commutative     | $\varphi \wedge \varphi$             | $\equiv$ | $\varphi$  | idempotent |
| $\varphi \vee \psi$                 | $\equiv$ | $\psi \vee \varphi$                                | commutative     | $\varphi \vee \varphi$               | $\equiv$ | $\varphi$  | idempotent |
| $\varphi \wedge (\psi \wedge \chi)$ | $\equiv$ | $(\varphi \wedge \psi) \wedge \chi$                | associative     | $\varphi \wedge (\varphi \vee \psi)$ | $\equiv$ | $\varphi$  | absorptive |
| $\varphi \vee (\psi \vee \chi)$     | $\equiv$ | $(\varphi \vee \psi) \vee \chi$                    | associative     | $\varphi \vee (\varphi \wedge \psi)$ | $\equiv$ | $\varphi$  | absorptive |
| $\varphi \wedge (\psi \vee \chi)$   | $\equiv$ | $(\varphi \wedge \psi) \vee (\varphi \wedge \chi)$ | distributive    | $\text{T} \wedge \varphi$            | $\equiv$ | $\varphi$  | top        |
| $\varphi \vee (\psi \wedge \chi)$   | $\equiv$ | $(\varphi \vee \psi) \wedge (\varphi \vee \chi)$   | distributive    | $\text{T} \vee \varphi$              | $\equiv$ | $\text{T}$ | top        |
| $\neg(\varphi \wedge \psi)$         | $\equiv$ | $(\neg\varphi) \vee (\neg\psi)$                    | de Morgan       | $\text{F} \wedge \varphi$            | $\equiv$ | $\text{F}$ | bottom     |
| $\neg(\varphi \vee \psi)$           | $\equiv$ | $(\neg\varphi) \wedge (\neg\psi)$                  | de Morgan       | $\text{F} \vee \varphi$              | $\equiv$ | $\varphi$  | bottom     |
| $\neg\neg\varphi$                   | $\equiv$ | $\varphi$                                          | double negation | $\varphi \vee \neg\varphi$           | $\equiv$ | $\text{T}$ | complement |
|                                     |          |                                                    |                 | $\varphi \wedge \neg\varphi$         | $\equiv$ | $\text{F}$ | complement |

There is no real reason to learn all of these equivalences. You should read them and convince yourself that they are correct; for example, you could take  $\varphi$  to be  $p_3$  and  $\psi$  to be  $p_6$  in the first law and make the truth table in order to see that  $p_3 \wedge p_6 \equiv p_6 \wedge p_3$ .

Please note that there is a difference between  $\leftrightarrow$  and  $\equiv$ . The difference is that  $\leftrightarrow$  is a symbol *in* our language  $\mathcal{L}$  of propositional logic. In contrast,  $\equiv$  is in our *metalanguage*, the language that we are using to talk about propositional logic.

#### Proposition 5.13

$\varphi \equiv \psi$  if and only if  $\models \varphi \leftrightarrow \psi$ .

You might like to think about why this proposition is true, and also about the status of the English phrase “if and only if” in it. Is it part of our object language, our metalanguage, or our metametalanguage?

In addition, we have the following fact which we leave to you to verify.

#### Lemma 5.14: Facts about logical equivalence

For all  $\varphi$  and  $\psi$ , the following hold:

1. If  $\varphi \equiv \psi$ , then  $\neg\varphi \equiv \neg\psi$ .
2. If  $\varphi_1 \equiv \psi_1$  and  $\varphi_2 \equiv \psi_2$ , then  $\varphi_1 \wedge \psi_1 \equiv \varphi_2 \wedge \psi_2$ .
3. If  $\varphi_1 \equiv \psi_1$  and  $\varphi_2 \equiv \psi_2$ , then  $\varphi_1 \vee \psi_1 \equiv \varphi_2 \vee \psi_2$ .
4. If  $\varphi_1 \equiv \psi_1$  and  $\varphi_2 \equiv \psi_2$ , then  $\varphi_1 \rightarrow \psi_1 \equiv \varphi_2 \rightarrow \psi_2$ .
5. If  $\varphi_1 \equiv \psi_1$  and  $\varphi_2 \equiv \psi_2$ , then  $\varphi_1 \leftrightarrow \psi_1 \equiv \varphi_2 \leftrightarrow \psi_2$ .

### 5.3 Using a SAT solver

The point of this section is to emphasize the importance of satisfiability, and to indicate some connections to reasoning. Our discussion is centered on the logic puzzle found<sup>3</sup> in Example 5.15.

#### Example 5.15: A logic puzzle

During a recent police investigation, Chief Inspector Holmes was interviewing five suspects in order to identify who stole his magnifying glass. Below is a summary of their statements:

Arnold: it wasn't Edward; it was Brian.

Brian: it wasn't Charlie; it wasn't Edward.

Charlie: it was Edward; it wasn't Arnold.

Derek: it was Charlie; it was Brian.

Edward: it was Derek; it wasn't Arnold.

Each suspect told exactly one truth and one lie. Who stole the magnifying glass?

**Solution, informally** There are several ways to solve this; here is one. Derek's statement tell us that either Brian or Charlie did it. And Brian's tells us that either Charlie or Edward did it. Now if Charlie didn't do it, then both Brian and Edward would have done it. And as Holmes said, "Once you eliminate the impossible, whatever remains, no matter how improbable, must be the truth." So we see that Charlie did it.

**Solution, with assistance from a SAT solver** Let's symbolize the situation in Example 5.15 using the following logic sentence:

$$(\neg e + b) \wedge (\neg c + \neg e) \wedge (e + \neg a) \wedge (c + b) \wedge (d + \neg a) \quad (5.5)$$

The symbol  $+$  here is for *exclusive or*. That is,  $\varphi + \psi$  means that *exactly one of*  $\varphi, \psi$  is true. So  $\varphi + \psi$  abbreviates  $(\varphi \vee \psi) \wedge \neg(\varphi \wedge \psi)$ . You could also un-abbreviate the sentence above (but this would make it twice as long). I entered this in a SAT solver program found on the internet.<sup>4</sup> The answer came back

Clause set is true if we assign values to variables as: e b -c a d

According to this, the valuation which we want is is

$$v(a) = \text{T}, v(b) = \text{T}, v(c) = \text{F}, v(d) = \text{T}, v(e) = \text{T}$$

We see that Arnold stole the magnifying glass, as did Brian, Derek, and Edward; Charlie didn't do it.

The problem is that this doesn't match our informal solution. In fact, it's completely the opposite! One way to proceed (and also show what is going on) is to ask for a *second* valuation. Let's add  $\neg a$  to the query in (5.5). This time we get back

Clause set is true if we assign values to variables as: -e -b c -a -d

<sup>3</sup>Since this puzzle is about theft, I don't mind saying that I stole this puzzle from another source.

<sup>4</sup>I would mention which one, but these programs come and go. There were a few which could be run without downloading anything.

**Definition 5.16: the induction principle for sentences of propositional logic**

Let  $S(\varphi)$  be an assertion about sentences in propositional logic. Suppose that

1.  $S(p_i)$  for all  $i$ .
2.  $S(\top)$  and  $S(\text{F})$ .
3. If  $S(\varphi)$ , then  $S(\neg\varphi)$ .
4. If  $S(\varphi)$  and  $S(\psi)$ , then  $S(\varphi \wedge \psi)$ , and also  $S(\varphi \vee \psi)$ ,  $S(\varphi \rightarrow \psi)$ , and  $S(\varphi \leftrightarrow \psi)$ .

Assuming these conditions, we have  $S(\varphi)$  for all sentences  $\varphi$  of propositional logic.

This corresponds to the valuation

$$w(a) = \text{F}, w(b) = \text{F}, w(c) = \text{T}, w(d) = \text{F}, w(e) = \text{F}$$

And this *was* the solution.

We can now see that there was a *missing premise* in the way we set up the problem. The missing premise was that there is only one thief. Life has a lot of missing premises.

Incidentally, there is a good reason why the solver gave back an answer which was exactly the opposite of what we wanted: see Exercise 5.5.

## 5.4 Induction on sentences

Many of the important properties of sentences in propositional logic are proved using a *principle of induction*. We formulate this principle in Definition 5.16.

Here is a first result which uses induction.

**Proposition 5.17**

Let  $\varphi$  be any sentence built from atomic sentences,  $\top$ ,  $\wedge$ ,  $\vee$ , and  $\rightarrow$ . (In particular,  $\neg$  does not appear in  $\varphi$ , and neither does  $\text{F}$ .) Let  $v$  be the valuation which is constantly  $\top$ . (That is  $v(p_i) = \top$  for all  $i$ .) Then  $\llbracket \varphi \rrbracket_v = \top$ .

In particular, there is no  $\varphi$  logically equivalent to  $\neg p$  which does not use  $\neg$  or  $\text{F}$ .

*Proof.* In this proof as in our statement,  $v$  denotes the valuation which is constantly  $\top$ . Let  $S(\varphi)$  be the following statement about sentences  $\varphi$ : if  $\varphi$  is built from atomic sentences,  $\top$ ,  $\wedge$ ,  $\vee$ , and  $\rightarrow$ , then  $\llbracket \varphi \rrbracket_v = \top$ . We prove that  $S(\varphi)$  holds for all  $\varphi$ . Note that  $S(\varphi)$  is a conditional, and so if the antecedent is false for a given sentence, then  $S(\varphi)$  is automatically true.

For the base case of an atomic sentence  $p_i$ , we have  $\llbracket p_i \rrbracket_v = \top$  by the definition of  $v$ . For  $\top$ , we have  $\llbracket \top \rrbracket_v = \top$  because every valuation  $w$  has  $\llbracket \top \rrbracket_w = \top$ . For  $\text{F}$ , the antecedent of  $S(\text{F})$  is automatically false.

Assume about  $\varphi$  that  $S(\varphi)$ . Then the antecedent of  $S(\neg)$  is automatically false, and so  $S(\neg\varphi)$  holds.



Assume that  $S(\varphi)$  and  $S(\psi)$  both hold. To show that  $S(\varphi \wedge \psi)$ , assume that  $\varphi \wedge \psi$  is built from atomic sentences,  $\top$ ,  $\wedge$ ,  $\vee$ , and  $\rightarrow$ . Then so are  $\varphi$  and  $\psi$ . So by  $S(\varphi)$ , we have  $\llbracket \varphi \rrbracket_v = \top$ . And by  $S(\psi)$ ,  $\llbracket \psi \rrbracket_v = \top$ . Thus

$$\llbracket \varphi \wedge \psi \rrbracket_v = \llbracket \varphi \rrbracket_v \wedge \llbracket \psi \rrbracket_v = \top \wedge \top = \top.$$

And this is what we want.

The induction steps for  $\vee$ ,  $\rightarrow$ , and  $\leftrightarrow$  are similar. □

We have one final illustration of induction. It is defined in terms of a function which plays a key role later in the chapter.

**Definition 5.18:  $occ(\varphi)$**

We define the set of atomic sentences which occur in a sentence  $\varphi$  in the following way:

$$\begin{aligned} occ(p_i) &= \{p_i\} \\ occ(\top) &= \emptyset \\ occ(\bot) &= \emptyset \\ occ(\neg\varphi) &= occ(\varphi) \\ occ(\varphi \wedge \psi) &= occ(\varphi) \cup occ(\psi) \end{aligned}$$

and similarly for the other boolean connectives

**Example 5.19**

$occ(p_3 \rightarrow p_{12}) = occ(p_3) \cup occ(p_{12}) = \{p_3\} \cup \{p_{12}\} = \{p_3, p_{12}\}$ . We would read

$$occ(p_3 \rightarrow p_{12}) = \{p_3, p_{12}\}$$

as “the sentence letters that occur in  $p_3 \rightarrow p_{12}$  are  $p_3$  and  $p_{12}$ .” Definition 5.18 is an example of a function defined by *recursion on sentences*, and while we shall need this once in a while, it is not terribly important in this book. So for this reason, we shall not formulate the general idea of recursion. But we do want to mention that all of the properties of a function defined by recursion must be verified using a proof involving induction.

**Theorem 5.20: The Occurrence Theorem**

Let  $\varphi$  be any sentence, and let  $v$  and  $w$  be any two valuations with the property that for all  $p_i \in occ(\varphi)$ ,  $v(p_i) = w(p_i)$ . Then  $\llbracket \varphi \rrbracket_v = \llbracket \varphi \rrbracket_w$ .

**Example 5.21**

Here are two valuations, called here  $v$  and  $w$ :

$v(p_1) = \text{T}$ ,  $v(p_2) = \text{F}$ , and for  $i \geq 3$ ,  $v(p_i) = \text{T}$ .

$w(p_1) = \text{T}$ ,  $w(p_2) = \text{F}$ ,  $w(p_3) = \text{T}$ , and for  $i \geq 4$ ,  $w(p_i) = \text{F}$ .

Consider the sentence  $\varphi = (p_1 \rightarrow p_2) \vee p_3$ . Then  $\text{occ}(\varphi) = \{p_1, p_2, p_3\}$ . For all  $p_i \in \text{occ}(\varphi)$ ,  $v(p_i) = w(p_i)$ . In words,  $v$  and  $w$  agree on the sentence letters that occur in our sentence  $\varphi$ .

The theorem tells us that  $\llbracket \varphi \rrbracket_v = \llbracket \varphi \rrbracket_w$ . (In fact, this is easy to check: both truth values are  $\text{T}$ .) The point of the result not to help us calculate anything, but rather to tell us that the truth value of a sentence  $\varphi$  under a valuation will not change if we alter the valuation on atomic sentences that don't occur in  $\varphi$ .

## 5.5 Natural deduction proof system

Up until now, all of our work in this chapter has been entirely *semantic* because it involves valuations. Our proofs of semantic facts have been carried out in the usual style of mathematical proofs, just as one would expect.

The time has come to shift gears a bit, to present a *formal proof system* for propositional logic. Formal proofs differ from the informal kind that we have been presenting because they come with a precise definition. In fact, we regard them as *syntactic objects*, just like sentence. They are a formalized version of very simple reasoning.

This presents a proof system for propositional logic, but we are doing so very quickly. Such systems are presented in most beginning courses in symbolic logic, and so there are many textbook presentations that cover this topic at a very leisurely pace. In fact, there are many, many different styles of such a system. In order that all readers be “on the same page”, we are going to present one formal system in complete (and therefore, painful) detail, a *natural deduction system*. We'll write the main proof relation as  $\Gamma \vdash \varphi$ . We'll present the rules and examples, and then discuss the soundness of the system.

Most formal proof systems are combinatorially intricate in the sense that more than one rule can apply at any one place, and one has to be lucky and/or insightful to find a long formal proof. It is almost always easier for a computer to find such a formal proof than a person. It is also a fairly tedious process for a person to read a formal proof and get any idea of what is going on, since there is no connecting text to expose any ideas. One could almost say that a formal proof is a demonstration that something is correct, given in a way that would convince someone who was unable to understand *anything*. A person absolutely never thinks in a way that corresponds to a formal proof; this is not what formal proofs are about in the first place. They are intended as *mathematical models* of how people reason mathematically. But mathematical models usually oversimplify, and in the case of logic there are many oversimplifications. For this reason, a formal proof could even be regarded as a caricature of a real, humanly convincing proof. However, we should hasten to add that because formal proofs come with a precise syntax, checking one is an automatic (if tedious) procedure. There is much less chance of an error.

To get started, you should read Definition 5.22 on the overall structure of proofs. The formal rules of our natural deduction begin in Logic 5.23, so you should read that now as well, before

**Definition 5.22: Proofs: a first point**

A *proof* in this system is a numbered list of sentences and justifications. A proof of  $\varphi_1, \dots, \varphi_n \vdash \psi$  begins with *premise lines*:  $\varphi_1$  as line 1,  $\varphi_2$  as line 2, etc., all justified as P (premise). The proof ends with  $\psi$  as its *conclusion*. We might have  $n = 0$ ; there might be no premise lines. The justification of each line consists of a rule together with a line number, or an interval of line numbers, or (in one case) a sequence of such intervals.

In making a proof of  $\varphi_1, \dots, \varphi_n \vdash \psi$ , one typically starts by writing the premises and then the conclusion  $\psi$  below them, without knowing what line  $\psi$  will be on when the proof is finished, and also without yet having a justification for  $\psi$ .

**Logic 5.23: Easy rules**

A line  $\varphi \wedge \psi$  may be justified by  $\wedge i$  from two available lines  $\varphi$  and  $\psi$  above it.

A line  $\varphi$  may be justified by  $\wedge e$  from an available line  $\varphi \wedge \psi$  above it. The same  $\varphi \wedge \psi$  also justifies  $\psi$ .

A line  $\psi$  may be justified by  $\rightarrow i$  from two available lines  $\varphi \rightarrow \psi$  and  $\varphi$  above it.

A line  $\varphi$  may be justified by  $\neg e$  from an available line  $\neg \neg \varphi$  above it.

A line  $\varphi \vee \psi$  may be justified by  $\vee i$  from an available line  $\varphi$  above it.  $\varphi \vee \psi$  may also be justified by  $\vee i$  from  $\psi$ .

A line  $\varphi$  may be justified by R from an available line  $\varphi$  above it.

A line F may be justified by Fi from two available lines  $\varphi$  and  $\neg \varphi$  above it.

A line  $\varphi \leftrightarrow \psi$  may be justified by  $\leftrightarrow i$  from two available lines  $\varphi \rightarrow \psi$  and  $\psi \rightarrow \varphi$  above it.

A line  $\varphi \rightarrow \psi$  may be justified by  $\leftrightarrow e$  from an available line  $\varphi \leftrightarrow \psi$  above it. The same  $\varphi \leftrightarrow \psi$  also justifies  $\psi \rightarrow \varphi$ .

looking at an example of most of the easy rules of the system.

**Example 5.24:  $p \wedge u, \neg \neg q, (p \wedge q) \rightarrow s \vdash s \vee t$** 

|   |                              |                       |
|---|------------------------------|-----------------------|
| 1 | $p \wedge u$                 | premise               |
| 2 | $\neg \neg q$                | premise               |
| 3 | $(p \wedge q) \rightarrow s$ | premise               |
| 4 | $p$                          | $\wedge e, 1$         |
| 5 | $q$                          | $\neg \neg e, 2$      |
| 6 | $p \wedge q$                 | $\wedge i, 4, 5$      |
| 7 | $s$                          | $\rightarrow e, 3, 6$ |
| 8 | $s \vee t$                   | $\vee i, 7$           |

This proof illustrates the first six rules in Logic 5.23. Later examples will show the rules R and Fi. (The rules pertaining to the connective  $\leftrightarrow$  will not appear in any of our work.)

Notice that every line is justified. The proof overall shows that  $p \wedge u, \neg \neg q, (p \wedge q) \rightarrow s \vdash s \vee t$

Now we turn to the harder rules of the system, the ones that deal with subproofs. You'll want to read Definition 5.27 first, and see how the points about subproofs are exemplified in our examples

to follow. The hard rules of the system are listed after that. We present examples in turn of  $\rightarrow$ i,  $\neg$ i, and  $\vee$ e.

**Example 5.25:**  $p \vdash q \rightarrow p$

|   |  |                   |                     |
|---|--|-------------------|---------------------|
| 1 |  | $p$               | premise             |
| 2 |  |                   |                     |
| 3 |  | $q$               | A                   |
| 4 |  | $p$               | R 1                 |
| 5 |  | $q \rightarrow p$ | $\rightarrow$ i 2–3 |

This proof has a top-level premise,  $p$ . In or-

der to derive  $q \rightarrow p$ , we start a subproof with the assumption of  $q$ . Then we reiterate  $p$  on line 3. We close the subproof, and on line 4 we have the assumption of the subproof ( $q$ ) implying the concluding line of the subproof  $p$ .

**Example 5.26:**  $\vdash p \rightarrow (q \rightarrow p)$

|   |  |                                   |                   |                     |
|---|--|-----------------------------------|-------------------|---------------------|
| 1 |  |                                   | $p$               | A                   |
| 2 |  |                                   |                   |                     |
| 3 |  |                                   | $q$               | A                   |
| 4 |  |                                   | $p$               | R 1                 |
| 5 |  |                                   | $q \rightarrow p$ | $\rightarrow$ i 2–3 |
| 6 |  | $p \rightarrow (q \rightarrow p)$ |                   | $\rightarrow$ i 1–4 |

This is a different derivation from what we saw in Example 5.25. Note that there are no premise lines, and what is proved has nothing to the left of the turnstile  $\vdash$ . The way

one comes up with this proof is to immediately start a subproof, since what we are proving is an implication. Then we put down the assumption  $p$  and the “inside conclusion”  $p \rightarrow q$ . After that, we fill this in with yet another subproof, starting with the assumption  $q$  and deriving  $p$ . This is lines 2 – 3.

Having noticed the difference with Example 5.25, we can see that there is still a close relationship. Can you formulate a general result along these lines, including cases involving derivations with more than one premise?

**Definition 5.27: Subproof structure**

A proof may have *subproofs*. Those subproofs are proofs, too; thus a subproof may have subproofs of its own. And every subproof must meet all the requirements of a proof, including what is in this passage. The only difference is that there should be just one assumption line in a subproof.

We indicate the subproof structure by vertical lines and indentation. So every line will have a *depth*, starting with depth 1 for the premise lines (if there are any). The overall conclusion of the proof must have depth 1. One can *open* a new subproof at any time. To do this, we start a new level of indenting. So if line  $n$  had depth  $k$  and we decide to start a new subproof right after it, line  $n + 1$  would have depth  $k + 1$ . That line  $n + 1$  is justified as A (assumption). For some  $m \geq n + 1$  we *close* the subproof after line  $m$ . Line  $m + 1$  will then have depth  $k$  again.

A subproof may only be closed if all of the lines inside are justified. After any subproof is closed, the lines inside it are not available for use later on.

**Logic 5.28:  $\rightarrow$ i**

A line  $\varphi \rightarrow \psi$  may be justified as  $\rightarrow$ i  $n - m$  if the following conditions are met. We must have  $m \geq n$ . The line itself must be of some depth, say  $k$ . And prior to the  $\varphi \rightarrow \psi$  line, there must be a closed subproof of depth  $k + 1$  whose assumption is  $\varphi$  on line  $n$  and whose conclusion is  $\psi$  on line  $m$ .

In practice, to have  $\varphi \rightarrow \psi$  on a line, one opens a subproof before that line, with  $\varphi$  justified as A and  $\psi$  as the conclusion, and then one fills additional lines, with justifications, in order to justify  $\psi$ . After that subproof is closed, one may justify the  $\varphi \rightarrow \psi$  using  $\rightarrow$ i, mentioning  $n$  and  $m$  as the beginning and ending lines of the subproof.

**Logic 5.29:  $\neg$ i**

$\neg\varphi$  works the same way as  $\varphi \rightarrow \mathbf{F}$ . So a line  $\neg\varphi$  may be justified as  $\neg$ i  $n - m$  if prior to it there is a closed subproof one level deeper with assumption  $\varphi$  and conclusion  $\mathbf{F}$ .

In practice, to have  $\neg\varphi$  on a line, one opens a subproof before that line, with  $\varphi$  justified as A and  $\mathbf{F}$  as the conclusion, and then one fills in the subproof.

**Logic 5.30:  $\vee$ e**

A line  $\psi$  may be justified from  $\varphi_1 \vee \varphi_2 \vee \cdots \vee \varphi_n$  and  $n$  closed subproofs, each one level deeper, with subproof  $i$  having  $\varphi_i$  as the assumption and  $\psi$  as the conclusion.

In practice, to prove  $\psi$  from  $\varphi_1 \vee \varphi_2 \vee \cdots \vee \varphi_n$ , one opens  $n$  separate subproofs. The  $i$ th subproof would have  $\varphi_i$  as the assumption and  $\psi$  as the conclusion. Once all of those subproofs are closed, one can justify  $\psi$  by  $\vee$ e, mentioning the line that had  $\varphi_1 \vee \varphi_2 \vee \cdots \vee \varphi_n$  and the starting and ending lines of the subproofs.

**Definition 5.31: Proofs in PL: the definition**

The rules of PL are the introduction and elimination rules of  $\wedge$ ,  $\vee$ ,  $\rightarrow$ , and  $\leftrightarrow$ , and also the introduction rules for  $\mathbf{T}$ ,  $\mathbf{F}$ , and  $\neg$ ; the elimination rule for  $\neg\neg$ ; and reiteration.

We write  $\Gamma \vdash \varphi$  if there is a proof  $\Pi$  with the following properties: (1) all of the premise lines of  $\Pi$  are in  $\Gamma$ ; (2) all lines in  $\Pi$  are justified; (3) the last line of  $\Pi$  is  $\varphi$ ; and (4) the last line of  $\Pi$  is on level 1. The last condition is equivalent to saying that  $\Pi$  has no open subproofs in the last line.

**Example 5.32:**  $\vdash (p \wedge \neg q) \rightarrow \neg(p \rightarrow q)$ 

|   |  |                                                       |                       |
|---|--|-------------------------------------------------------|-----------------------|
| 1 |  | $p \wedge \neg q$                                     | A                     |
| 2 |  | $p \rightarrow q$                                     | A                     |
| 3 |  | $p$                                                   | $\wedge e, 1$         |
| 4 |  | $q$                                                   | $\rightarrow e, 2, 3$ |
| 5 |  | $\neg q$                                              | $\wedge e, 1$         |
| 6 |  | F                                                     | Fi, 4, 5              |
| 7 |  | $\neg(p \rightarrow q)$                               | $\neg i, 2-6$         |
| 8 |  | $(p \wedge \neg q) \rightarrow \neg(p \rightarrow q)$ | $\rightarrow i, 1-7$  |

This is our first example of a proof by contradiction. In this logic we have a form of *reductio ad absurdum*. Here is how it works. At any point, one can open a subproof and put an assumption line  $\varphi$  at the top of it. In this example,  $\varphi$  is  $p \rightarrow q$ . One tries to derive F, and this is any sentence  $\psi$  and its negation  $\neg\psi$ . Once we have F, we can close the subproof, and the next line is  $\neg\varphi$ .

**Example 5.33:**  $\vdash p \vee \neg p$ 

|   |  |                           |                  |
|---|--|---------------------------|------------------|
| 1 |  | $\neg(p \vee \neg p)$     | A                |
| 2 |  | $p$                       | A                |
| 3 |  | $p \vee \neg p$           | $\vee i, 2$      |
| 4 |  | F                         | $\wedge i, 1, 3$ |
| 5 |  | $\neg\neg(p \vee \neg p)$ | $\neg i, 2-4$    |
| 6 |  | $p \vee \neg p$           | $\vee i, 5$      |
| 7 |  | F                         | $\wedge i, 1, 6$ |
| 8 |  | $\neg\neg(p \vee \neg p)$ | $\neg i, 1-7$    |
| 9 |  | $p \vee \neg p$           | $\neg\neg e, 8$  |

This is a very tricky derivation because it starts two levels in. In order to get  $p \vee \neg p$ , we sneak around and get  $\neg\neg(p \vee \neg p)$  first. For that, we assume  $\neg(p \vee \neg p)$  and get a contradiction. This leads to  $\neg\neg(p \vee \neg p)$ , and then we use the rule  $\neg\neg e$  to drop the two  $\neg$ 's.

Incidentally, it can be proved that there is no derivation in our system of  $p \vee \neg p$  that does not use  $\neg\neg e$ .

**Example 5.34**

|   |  |                         |                      |    |  |                                                       |                       |
|---|--|-------------------------|----------------------|----|--|-------------------------------------------------------|-----------------------|
| 1 |  | $\neg(p \rightarrow q)$ | A                    | 10 |  | $\neg\neg p$                                          | $\neg i, 1-9$         |
| 2 |  | $\neg p$                | A                    | 11 |  | $p$                                                   | $\neg\neg e, 10$      |
| 3 |  | $p$                     | A                    | 12 |  | $\neg q$                                              | *                     |
| 4 |  | $\neg q$                | A                    | 13 |  | $p \wedge \neg q$                                     | $\wedge i, 11, 12$    |
| 5 |  | F                       | Fi, 2, 3             | 14 |  | $\neg(p \rightarrow q) \rightarrow (p \wedge \neg q)$ | $\rightarrow i, 1-13$ |
| 6 |  | $\neg\neg q$            | $\neg i, 4-5$        |    |  |                                                       |                       |
| 7 |  | $q$                     | $\neg\neg e, 6$      |    |  |                                                       |                       |
| 8 |  | $p \rightarrow q$       | $\rightarrow i, 3-7$ |    |  |                                                       |                       |
| 9 |  | F                       | Fi, 1, 8             |    |  |                                                       |                       |

You might try to fill in a few lines that are missing between lines 11 and 12.

**Example 5.35: an associative law of  $\vee$** 

|   |                     |             |
|---|---------------------|-------------|
| 1 | $p \vee (q \vee r)$ | A           |
| 2 | $p$                 | A           |
| 3 | $p \vee q$          | $\vee i, 2$ |
| 4 | $(p \vee q) \vee r$ | $\vee i, 3$ |

The overall proof uses  $\vee e$ , deriving the conclusion  $(p \vee q) \vee r$  from both  $p$  and  $q \vee r$ . And in the second of these derivations, we again use  $\vee e$  with the same kind of proof. There is no way to say “similarly” in a proof system of this kind.

|    |                     |                        |
|----|---------------------|------------------------|
| 5  | $q \vee r$          | A                      |
| 6  | $q$                 | A                      |
| 7  | $p \vee q$          | $\vee i, 6$            |
| 8  | $(p \vee q) \vee r$ | $\vee i, 7$            |
| 9  | $r$                 | A                      |
| 10 | $(p \vee q) \vee r$ | $\vee i, 9$            |
| 11 | $(p \vee q) \vee r$ | $\vee e, 5, 6-8, 9-10$ |
| 12 | $(p \vee q) \vee r$ | $\vee e, 1, 2-4, 5-11$ |

## 5.6 Completeness of the proof system

The main result in this section is Theorem 5.45. It states that our proof system from Section 5.5 is *complete* in the following sense: if  $\models \varphi$ , then  $\vdash \varphi$ . In other words, the proof system is “strong enough” to derive all of the valid sentences.

This completeness theorem is the second thing that one would want to know about a proof system. This first thing would be soundness with respect to an interesting semantics; without this, there would be little point in studying the proof system. For most of the systems in the book, the soundness is routine, but also a bit tedious. For this reason, we shall usually omit the details of the soundness, or to merely sketch them. Completeness is another matter, and much of the technical content of the course is devoted to completeness results.

Once one has those results, the next step would be to see if there is an algorithm to tell whether  $\vdash \varphi$  or not. In other words, could we write a program which, when given a sentence as an input, would tell for sure whether or not  $\vdash \varphi$ ? (By soundness and completeness, this the same as asking whether or not  $\models \varphi$ .) If there is such an algorithm, we say that the logical system is *decidable*. As we have seen from the discussion of the truth-table method, propositional logic is decidable.

### 5.6.1 State descriptions in propositional logic

**Definition 5.36: state description of order  $n$** 

A *state description of order  $n$*  is a sentence of the form

$$\pm p_1 \wedge \pm p_2 \wedge \cdots \wedge \pm p_n,$$

where each “ $\pm p_i$ ” means “either  $p_i$  or  $\neg p_i$ .” In words, a state description of order  $n$  is a conjunction which consists of the first  $n$  atomic sentences or their negations (but not both). We write  $SD_n$  for the set of state descriptions of order  $n$ .

**Example 5.37**

For  $n = 1$ , there are two state descriptions of order 1, namely  $p_1$  and  $\neg p_1$ . So

$$SD_1 = \{p_1, \neg p_1\}.$$

When dealing with small  $n$ , it is better to drop the subscript. So we would write  $p$  for  $p_1$ , and then  $SD_1 = \{p, \neg p\}$ . With this notation,

$$SD_2 = \{p \wedge q, p \wedge \neg q, \neg p \wedge q, \neg p \wedge \neg q\}.$$

**Example 5.38**

Here are two elements of  $SD_9$ :

$$\begin{array}{cccccccccccc} p_1 & \wedge & p_2 & \wedge & \neg p_3 & \wedge & p_4 & \wedge & \neg p_5 & \wedge & \neg p_6 & \wedge & p_7 & \wedge & \neg p_8 & \wedge & \neg p_9 \\ \neg p_1 & \wedge & p_2 & \wedge & \neg p_3 & \wedge & \neg p_4 & \wedge & p_5 & \wedge & p_6 & \wedge & p_7 & \wedge & \neg p_8 & \wedge & p_9 \end{array}$$

$SD_9$  is a set with  $2^9 = 512$  sentences, so there are too many to list.

**5.6.2 Three lemmas about state descriptions****Lemma 5.39:**  $\vdash \bigvee SD_n$ .

That is, the disjunction of all state descriptions of order  $n$  is provable in propositional logic.

*Proof.* The case  $n = 1$  was shown in Example 5.33.

Next, we see the case  $n = 2$  in Example 5.40. As always with these large derivations, what is shown in the figure is not a derivation in full detail (that would be too large), but rather an approximation to such a derivation. To get a real derivation, we would replace the lines marked “Ex. 5.33” with lines that really those of Example 5.33, or the same ones with  $q$  replacing  $p$ .

For the general case, we use induction on  $n$ . Assume that  $\vdash \bigvee SD_n$ ; we’ll show that  $\vdash \bigvee SD_{n+1}$ . Take the derivation in Example 5.40, and replace all of the lines  $p \vee \neg p$  with derivations of  $\bigvee SD_n$ . The lines  $q \vee \neg q$  should be replaced by  $p_{n+1} \vee \neg p_{n+1}$ . The main change is that before, the lines  $p \vee \neg p$  resulted in two-way case splits: one for  $p$  and one for  $\neg p$ . In the general case, we need a  $2^n$ -way split, one for each disjunct in  $\bigvee SD_n$ . That is, each disjunct in  $\bigvee SD_n$  is a state description of order  $n$ , and so we may conjoin either  $p_{n+1}$  or  $\neg p_{n+1}$  to make a state description of order  $n + 1$ . Each such state description is a disjunct in  $\bigvee SD_{n+1}$ .  $\square$



**Example 5.40:** A derivation showing  $\vdash \bigvee \text{SD}_2$ .

|    |                       |                      |    |                        |                           |
|----|-----------------------|----------------------|----|------------------------|---------------------------|
| 1  | $p \vee \neg p$       | Ex. 5.33             | 11 | $\neg p$               | A                         |
| 2  | $p$                   | A                    | 12 | $q \vee \neg q$        | Ex. 5.33                  |
| 3  | $q \vee \neg q$       | Ex. 5.33             | 13 | $q$                    | A                         |
| 4  | $q$                   | A                    | 14 | $\neg p \wedge q$      | $\wedge$ i 11,13          |
| 5  | $p \wedge q$          | $\wedge$ i 2,4       | 15 | $\bigvee \text{SD}_2$  | $\vee$ i 14               |
| 6  | $\bigvee \text{SD}_2$ | $\vee$ i 6           | 16 | $\neg q$               | A                         |
| 7  | $\neg q$              | A                    | 17 | $\neg p \wedge \neg q$ | $\wedge$ i 11, 16         |
| 8  | $p \wedge \neg q$     | $\wedge$ i 2, 7      | 18 | $\bigvee \text{SD}_2$  | $\vee$ i 18               |
| 9  | $\bigvee \text{SD}_2$ | $\vee$ i 8           | 19 | $\bigvee \text{SD}_2$  | $\vee$ e 12, 13–15, 16–18 |
| 10 | $\bigvee \text{SD}_2$ | $\vee$ e 3, 4–6, 7–9 | 20 | $\bigvee \text{SD}_2$  | $\vee$ e 1, 2–10, 11–19   |

**Lemma 5.41**

Let  $\varphi$  be any sentence, and let  $n$  be large enough so that  $\text{occ}(\varphi) \subseteq \{p_1, \dots, p_n\}$ . For all  $\alpha \in \text{SD}_n$ , either  $\vdash \alpha \rightarrow \varphi$  or  $\vdash \alpha \rightarrow \neg\varphi$ .

In words, if  $\varphi$  is any sentence, and if  $n$  is big enough so that all sentence letters that occur in  $\varphi$  have a subscript which is  $\leq n$ , and if  $\alpha$  is a state description of order  $n$ , then we either can prove that  $\alpha \rightarrow \varphi$ , or we can prove that  $\alpha \rightarrow \neg\varphi$ .

**Remark 5.42**

Before we turn to the proof, let's see why the statement has the extra complication indicated by the phrase *let  $n$  be large enough so that  $\text{occ}(\varphi) \subseteq \{p_1, \dots, p_n\}$* . To see why this is needed, suppose that we drop this point from our statement. The result would say that *For all  $\varphi$  and all  $\alpha \in \text{SD}_n$ , either  $\vdash \alpha \rightarrow \varphi$  or  $\vdash \alpha \rightarrow \neg\varphi$* . This assertion is wrong. For example, take  $\varphi = p_3$ ,  $n = 2$ , and  $\alpha = p_1 \wedge p_2$ . Notice that

$$\not\vdash p_1 \wedge p_2 \rightarrow p_3 \text{ and } \not\vdash p_1 \wedge p_2 \rightarrow \neg p_3$$

Thus by soundness,

$$\not\vdash p_1 \wedge p_2 \rightarrow p_3 \text{ and } \not\vdash p_1 \wedge p_2 \rightarrow \neg p_3$$

The problem here is that neither  $p_3$  nor  $\neg p_3$  is a conjunct in  $\alpha$ . Fixing this problem leads to the correct statement of our lemma.

*Proof.* We argue by induction on  $\varphi$ .<sup>5</sup>

**Step 1** Note that  $\text{occ}(p_i) = \{p_i\}$ . let  $n$  be large enough so that  $\{p_i\} \subseteq \{p_1, \dots, p_n\}$ . This just means that  $i \leq n$ . We must show that for all  $\alpha \in \text{SD}_n$ , either  $\vdash \alpha \rightarrow p_i$  or  $\vdash \alpha \rightarrow \neg p_i$ . So fix

<sup>5</sup>See Definition 5.16 for a review of the steps in an induction on sentences.

$\alpha \in \text{SD}_n$ . If  $p_i$  is a conjunct of  $\alpha$ , then  $\vdash \alpha \rightarrow p_i$  using  $\wedge$ -elimination. If  $\neg p_i$  is a conjunct of  $\alpha$ , then  $\vdash \alpha \rightarrow \neg p_i$ , again using  $\wedge$ -elimination. Either way, we see have the desired conclusion that either  $\vdash \alpha \rightarrow p_i$  or  $\vdash \alpha \rightarrow \neg p_i$ .

Step 2 Note that  $\text{occ}(\top) = \emptyset = \text{occ}(\varphi)$ . Any  $n$  is large enough so that  $\emptyset \subseteq \{p_1, \dots, p_n\}$ . For any sentence  $\varphi$ ,  $\vdash \varphi \rightarrow \top$ . Let  $\alpha \in \text{SD}_n$ . Then  $\vdash \alpha \rightarrow \top$ . Similarly,  $\vdash \alpha \rightarrow \neg \text{F}$ .

Step 3 We assume about  $\varphi$  that if  $n$  is large enough so that  $\text{occ}(\varphi) \subseteq \{p_1, \dots, p_n\}$ , then for all  $\alpha \in \text{SD}_n$ , either  $\vdash \alpha \rightarrow \varphi$  or  $\vdash \alpha \rightarrow \neg \varphi$ .

What we need to prove is that if  $n$  is large enough so that  $\text{occ}(\neg \varphi) \subseteq \{p_1, \dots, p_n\}$ , then for all  $\alpha \in \text{SD}_n$ , either  $\vdash \alpha \rightarrow \neg \varphi$  or  $\vdash \alpha \rightarrow \neg \neg \varphi$ .

So, let  $n$  be large enough so that  $\text{occ}(\neg \varphi) \subseteq \{p_1, \dots, p_n\}$ . Let  $\alpha \in \text{SD}_n$ . But the sentence letters that occur in  $\varphi$  are the same as those in  $\neg \varphi$ . That is,  $\text{occ}(\varphi) = \text{occ}(\neg \varphi)$ . Thus

$$\text{occ}(\varphi) = \text{occ}(\neg \varphi) \subseteq \{p_1, \dots, p_n\}$$

We need to do this calculation in order to know that our assumption on  $\varphi$  “kicks in” to tell us that either  $\vdash \alpha \rightarrow \varphi$  or  $\vdash \alpha \rightarrow \neg \varphi$ . In case  $\vdash \alpha \rightarrow \varphi$ , we see easily that  $\vdash \varphi \rightarrow \neg \neg \varphi$ . The reason is that the logic is strong enough to have a derivation of this sentence; this is an important fact. In the second case,  $\vdash \alpha \rightarrow \neg \varphi$ . In this case, we are done immediately since we only have to show that either  $\vdash \alpha \rightarrow \neg \varphi$  or  $\vdash \alpha \rightarrow \neg \neg \varphi$ .

Step 4 We assume about  $\varphi$  that if  $n$  is large enough so that  $\text{occ}(\varphi) \subseteq \{p_1, \dots, p_n\}$ , and if  $\alpha \in \text{SD}_n$ , then either  $\vdash \alpha \rightarrow \varphi$  or  $\vdash \alpha \rightarrow \neg \varphi$ . And we also assume about  $\psi$  that if  $m$  is large enough so that  $\text{occ}(\psi) \subseteq \{p_1, \dots, p_m\}$ , and if  $\alpha \in \text{SD}_m$ , then either  $\vdash \alpha \rightarrow \psi$  or  $\vdash \alpha \rightarrow \neg \psi$ . (Note that I changed the  $n$  to  $m$  here to avoid confusion with our first assumption.)

Let  $k = \max(m, n)$ . We prove that about  $\varphi \wedge \psi$  that if  $k \geq \max(m, n)$ ,  $\alpha \in \text{SD}_k$  then either  $\vdash \alpha \rightarrow (\varphi \wedge \psi)$  or  $\vdash \alpha \rightarrow \neg(\varphi \wedge \psi)$ .

The choice of  $k$  ensures that  $\text{occ}(\varphi \wedge \psi) \subseteq \{p_1, \dots, p_k\}$ . Notice that

$$\text{occ}(\varphi \wedge \psi) = \text{occ}(\varphi) \cup \text{occ}(\psi).$$

In words, the set of sentence letters that occur in the conjunction  $\varphi \wedge \psi$  is the union (not the intersection) of the set of set of sentence letters that occur in  $\varphi$  and the set of sentence letters that occur in  $\psi$ . Therefore

$$\begin{aligned} \text{occ}(\varphi) &\subseteq \text{occ}(\varphi \wedge \psi) \subseteq \{p_1, \dots, p_k\} \\ \text{occ}(\psi) &\subseteq \text{occ}(\varphi \wedge \psi) \subseteq \{p_1, \dots, p_k\} \end{aligned}$$

Let  $\alpha \in \text{SD}_k$ . By the points above and the induction hypotheses, we know that either  $\vdash \alpha \rightarrow \varphi$  or  $\vdash \alpha \rightarrow \neg \varphi$ ; and also, either  $\vdash \alpha \rightarrow \psi$  or  $\vdash \alpha \rightarrow \neg \psi$ .

We have four cases at this point:

- (a)  $\vdash \alpha \rightarrow \varphi$  and  $\vdash \alpha \rightarrow \psi$ .
- (b)  $\vdash \alpha \rightarrow \neg \varphi$  and  $\vdash \alpha \rightarrow \psi$ .
- (c)  $\vdash \alpha \rightarrow \varphi$  and  $\vdash \alpha \rightarrow \neg \psi$ .
- (d)  $\vdash \alpha \rightarrow \neg \varphi$  and  $\vdash \alpha \rightarrow \neg \psi$ .

Here is the argument in Case (a). In this case, we use  $\wedge$ -introduction to see that  $\vdash \alpha \rightarrow (\varphi \wedge \psi)$ . The other three cases are similar, and so I'll only go into details on (b). In propositional logic, we can prove

$$\neg\varphi \rightarrow \neg(\varphi \wedge \psi).$$

So we take a derivation showing that  $\vdash \alpha \rightarrow \neg\varphi$  and add to it a derivation showing that  $\vdash \neg\varphi \rightarrow \neg(\varphi \wedge \psi)$ , and finally add at the end a few lines giving us  $\vdash \alpha \rightarrow \neg(\varphi \wedge \psi)$ .

The other two cases, (c) and (d), are similar.

Steps 5–7 These induction steps are similar to Step 4, the step for  $\wedge$ .

This completes the proof <sup>6</sup>. □

**Lemma 5.43:** Each  $\alpha \in \text{SD}_n$  is satisfiable.

There is some valuation  $v$  so that  $\llbracket \alpha \rrbracket_v = \text{T}$ .

Before we turn to the proof, let's look at an example.

**Example 5.44**

Here again is the first state description of order 9 mentioned in Example 5.38,

$$p_1 \wedge p_2 \wedge \neg p_3 \wedge p_4 \wedge \neg p_5 \wedge \neg p_6 \wedge p_7 \wedge \neg p_8 \wedge \neg p_9$$

Let's call this state description  $\alpha$ . To get a valuation  $v$  such that  $\llbracket \alpha \rrbracket = \text{T}$ , we would set

$$\begin{array}{lllll} v(p_1) = \text{T} & v(p_2) = \text{T} & v(p_3) = \text{F} & v(p_4) = \text{T} & v(p_5) = \text{F} \\ v(p_6) = \text{F} & v(p_7) = \text{T} & v(p_8) = \text{F} & v(p_9) = \text{F} & \end{array}$$

You should check that indeed  $\llbracket \alpha \rrbracket_v = \text{T}$ .

The idea in this example is now generalized to all state descriptions.

*Proof.* Let  $\alpha$  be any state description of order  $n$ . Define  $v$  from  $\alpha$  by

$$v(p_i) = \begin{cases} \text{T} & \text{if } p_i \text{ occurs as a conjunct in } \alpha, \text{ without negation} \\ \text{F} & \text{if } \neg p_i \text{ occurs as a conjunct in } \alpha \end{cases}$$

We need to check that this works. Recall that  $\alpha$  is a conjunction of choices made from the first  $n$  atomic sentences or their negations; that is, we can write  $\alpha$  as

$$q_1 \wedge q_2 \wedge \cdots \wedge q_n,$$

where each  $q_i$  is either the  $i$ th atomic sentence  $p_i$ , or its negation  $\neg p_i$ . In the first case,  $\llbracket q_i \rrbracket_v = \llbracket p_i \rrbracket_v = \text{T}$ . In the second case,  $\llbracket q_i \rrbracket_v = \llbracket \neg p_i \rrbracket_v = \text{T}$ . Either way,  $\llbracket q_i \rrbracket_v = \text{T}$ . And now  $\llbracket \alpha \rrbracket_v = \text{T} \wedge \cdots \wedge \text{T} = \text{T}$ . This goes to show that the valuation  $v$  that we defined from  $\alpha$  indeed has the property that we want, that  $\llbracket \alpha \rrbracket_v = \text{T}$ . □

<sup>6</sup>In fact, Steps 5–7 are not really needed in this result. Every sentence  $\varphi$  in propositional logic is equivalent to a sentence  $\psi$  whose only connectives are  $\neg$  and  $\wedge$ . (The equivalence here is that  $\vdash \varphi \leftrightarrow \psi$ .) And so if one doesn't want to go through the details of the induction in this proof, it would be possible to avoid them by translating them away.

### 5.6.3 Finishing the proof

We can prove the completeness of propositional logic using Lemmas 5.39–5.43.

**Theorem 5.45:** If  $\models \varphi$ , then  $\vdash \varphi$

*Proof.* Fix a sentence  $\varphi$ , and assume that  $\models \varphi$ . For this  $\varphi$ , let  $n$  be large enough so that  $\text{occ}(\varphi) \subseteq \{p_1, \dots, p_n\}$ . (Again, this just means that  $n$  is at least as big as all of the subscripts on atomic sentences in  $\varphi$ .)

**Claim** For all  $\alpha \in \text{SD}_n$ ,  $\vdash \alpha \rightarrow \varphi$ .

*Proof.* Suppose not. Let  $\alpha \in \text{SD}_n$  be such that  $\not\vdash \alpha \rightarrow \varphi$ . Then by Lemma 5.41,  $\vdash \alpha \rightarrow \neg\varphi$ . By Lemma 5.43, let  $v$  be a valuation so that  $\llbracket \alpha \rrbracket_v = \text{T}$ . By our assumption that  $\models \varphi$ ,  $\llbracket \varphi \rrbracket_v = \text{T}$ . By soundness of the logic,  $\models \alpha \rightarrow \neg\varphi$ . So  $\llbracket \alpha \rightarrow \neg\varphi \rrbracket_v$  must be  $\text{T}$  also. And then we must have  $\llbracket \neg\varphi \rrbracket_v = \text{T}$ , so  $\llbracket \varphi \rrbracket_v = \text{F}$ . This is our contradiction! So our claim is proved.  $\square$

We now return to the proof of Theorem 5.45. We started with a sentence  $\varphi$  which happened to be a tautology; that is,  $\models \varphi$ . We want to show that  $\vdash \varphi$ . At this point we know that for all  $\alpha \in \text{SD}_n$ ,  $\vdash \alpha \rightarrow \varphi$ .

By using  $\vee$ -introduction  $2^n$  times, we see that

$$\vdash \bigvee \text{SD}_n \rightarrow \varphi$$

By Lemma 5.43,  $\vdash \bigvee \text{SD}_n$ . Using  $\rightarrow\text{e}$ , we get  $\vdash \varphi$ .  $\square$

#### Remark 5.46: What does completeness mean?

It is important to see what the Completeness Theorem says, and also what it does not say. To clarify matters, please look back at the sentence  $\pi$  from (5.4) on page 112. We showed in Proposition 5.11 that  $\models \pi$ , and so it follows from Completeness that  $\vdash \pi$ . However, the formal proof that we would get if we followed the method of this section would be much longer than the proof in Proposition 5.11. That proof used “smarts” like the Pigeonhole Principle, and also some high-level reasoning. A proof in our proof system would be more like an examination of all  $2^3 = 8$  lines of the truth-table written in full detail. It would be very painful to read. But it would be extremely easy to check, and a computer could do this in no time. This contrasts with the shorter proof in English; to get a computer to check this would be much more problematic. The Completeness Theorem just says that if a sentence is valid, then there will be a formal proof for it. It moreover gives us a way to find the formal proof. It does not say that this formal proof will be particularly short or illuminating.

#### Remark 5.47

It is possible to prove a strengthening of Theorem 5.45 that allows arbitrary sets of assumptions  $\Gamma$  in both the semantic and syntactic sides.

**Return to this!**

## 5.7 Exercises

**Exercise 5.1.** Suppose that  $v(p) = \top$  and  $v(q) = \text{F}$ . For this  $v$ , compute

$$\llbracket (p \rightarrow \neg q) \leftrightarrow (\neg p \rightarrow \neg q) \rrbracket_v$$

Show all the calculations using the definition of  $\llbracket \cdot \rrbracket_v$ .

**Exercise 5.2.** Here are five valuations, presented in the form of tables:

| valuation | $p_1$      | $p_2$      | $p_3$      | $p_4$      | $p_n$ for $n \geq 5$ |
|-----------|------------|------------|------------|------------|----------------------|
| $v_1$     | $\top$     | $\top$     | $\text{F}$ | $\top$     | $\top$               |
| $v_2$     | $\top$     | $\top$     | $\top$     | $\top$     | $\top$               |
| $v_3$     | $\text{F}$ | $\text{F}$ | $\text{F}$ | $\text{F}$ | $\text{F}$           |
| $v_4$     | $\top$     | $\top$     | $\text{F}$ | $\text{F}$ | $\top$               |
| $v_5$     | $\text{F}$ | $\top$     | $\top$     | $\top$     | $\text{F}$           |

1. Fill in the chart below:

| valuation | $\neg p_1$ | $p_1 \rightarrow p_2$ | $p_3 \leftrightarrow p_3$ | $p_2 \vee p_5$ | $p_4 \rightarrow p_1$ |
|-----------|------------|-----------------------|---------------------------|----------------|-----------------------|
| $v_1$     |            |                       |                           |                |                       |
| $v_2$     |            |                       |                           |                |                       |
| $v_3$     |            |                       |                           |                |                       |
| $v_4$     |            |                       |                           |                |                       |
| $v_5$     |            |                       |                           |                |                       |

For example, the first entry should be  $\top$  or  $\text{F}$ ; it should be the value of  $\llbracket \neg p_1 \rrbracket_{v_1}$ .

2. For any two valuations, say  $w_1$  and  $w_2$ , we define  $w_1 \leq w_2$  if the following condition holds: for all  $i$ , if  $w_1(p_i) = \top$ , then also  $w_2(p_i) = \top$ . Show that this is a preorder of the set of all valuations. (For *preorder*, see Definition 0.4 on page 8.)
3. Make a preorder from  $v_1, \dots, v_5$  in this problem, using  $\leq$ . Draw the Hasse diagram.

**Exercise 5.3.** Look again at the calculation in (5.3) on page 110. For each occurrence of  $\wedge$  and  $\vee$ , tell whether it is part of the syntax of propositional logic, or part of the structure **Bool**.

**Exercise 5.4.** Would it have been possible to prove Proposition 5.11 using truth tables? If so, what would be the difference between the proof we gave, and one that used truth-tables? And which do you think would be easier? If you think it would not be possible to prove Proposition 5.11 using truth tables, why not?

**Exercise 5.5.** Let  $\varphi$  be a sentence which is a conjunction of sentences  $\pm p \pm q$ ; that is, the conjuncts of  $\varphi$  are exclusive-or sentences, using two atomic sentences or their negation. (For example, (5.5) is one such sentence. Let  $v$  and  $w$  be valuations which are *opposites*: for all  $p \in \text{AtSen}$ ,  $v(p) = \neg w(p)$ . Then  $\llbracket \varphi \rrbracket_v = \llbracket \varphi \rrbracket_w$ .

**Exercise 5.6.** Recall that a sentence  $\varphi$  is *satisfiable* if there is some valuation  $v$  so that  $\llbracket \varphi \rrbracket_v = \top$ . Explain in words why this definition is equivalent to each of the following two conditions:

1.  $\not\models \neg\varphi$ .
2.  $\varphi$  is true on some row of the truth table.

This is not a terribly hard question in the sense that the answers should be short and there are no tricks involved. The point is to see why the definition is equivalent to each of the other two statements, and to write clear and succinct sentences that explain matters.

**Exercise 5.7.** Is the following sentence satisfiable or not?

$$(a \vee b \vee c) \wedge (\neg b \vee c \vee e) \wedge (a \vee \neg b \vee d) \wedge (a \vee b \vee \neg d) \wedge (b \vee \neg c \vee d) \wedge (b \vee c \vee f) \wedge (\neg a \vee \neg d \vee \neg f)$$

There are a few ways to go about this: you may guess blindly, you may try out all  $2^6 = 64$  rows of the truth-table systematically, and you may use a computer tool.

**Exercise 5.8.** Let  $\varphi$  and  $\psi$  be sentences in propositional logic. Show that the following are equivalent:

1. There is a valuation  $v$  so that exactly one of  $\varphi, \psi$  is true under  $\llbracket \cdot \rrbracket_v$ .
2.  $(\varphi \vee \psi) \wedge \neg(\varphi \wedge \psi)$  is satisfiable.

**Exercise 5.9.** Show that all of the following are (non-obvious) tautologies.

1.  $(p \rightarrow q) \vee (p \rightarrow \neg q)$ .
2.  $(p \rightarrow q) \vee (q \rightarrow p)$ .
3.  $((p \wedge q) \rightarrow r) \leftrightarrow (p \rightarrow (q \rightarrow r))$ .
4.  $((p \rightarrow q) \rightarrow (q \rightarrow p)) \rightarrow (q \rightarrow p)$ .
5.  $((p \otimes q) \otimes r) \leftrightarrow (p \otimes (q \otimes r))$ . (The symbol  $\otimes$  here is for exclusive “or”.)

**Exercise 5.10.** Find a sentence logically equivalent to  $p \vee q$  that only uses  $p, q$ , and  $\rightarrow$ . For example, you might try something like  $p \rightarrow ((q \rightarrow p) \rightarrow q)$ . This is wrong, but actual answer is shorter than this.

**Exercise 5.11.** For each of the following sentences, tell whether true or false. For the true ones, give a short proof. For the false ones, give a concrete counterexample.

1. Every sentence  $\varphi$  or its negation  $\neg\varphi$  is satisfiable.
2. Both  $\varphi$  and  $\neg\varphi$  are satisfiable.
3. If  $\varphi \wedge \psi$  is satisfiable, then both  $\varphi$  and  $\psi$  are satisfiable.

4. If both  $\varphi$  and  $\psi$  are satisfiable, then  $\varphi \wedge \psi$  is satisfiable.
5. If  $\varphi \vee \psi$  is satisfiable, then either  $\varphi$  or  $\psi$  (or both) are satisfiable.
6. If either  $\varphi$  or  $\psi$  is satisfiable, then  $\varphi \vee \psi$  is satisfiable.
7. If  $\varphi$  and  $\varphi \rightarrow \psi$  are both satisfiable sentences, then  $\psi$  is satisfiable.

[As a hint, I'll solve the first two parts. Part (1) is true for the following reason: Fix the sentence  $\varphi$ . Let  $v$  be any valuation. Either  $\llbracket \varphi \rrbracket = \text{T}$  or  $\llbracket \varphi \rrbracket = \text{F}$ . In the first case,  $v$  shows that  $\varphi$  is satisfiable. In the second case,

$$\llbracket \neg \varphi \rrbracket = \neg \llbracket \varphi \rrbracket = \neg \text{F} = \text{T},$$

and therefore  $v$  shows that  $\neg \varphi$  is satisfiable. On the other hand, part (2) is false. We may take  $\varphi$  to be  $p \wedge \neg p$ . This sentence is not satisfiable.]

**Exercise 5.12.** For each of the following sentences, tell whether true or false. For the true ones, give a short proof. For the false ones, give a concrete counterexample.

1. Every sentence  $\varphi$  or its negation  $\neg \varphi$  is a tautology.
2. If  $\varphi \wedge \psi$  is a tautology, then both  $\varphi$  and  $\psi$  are tautologies.
3. If  $\varphi \vee \psi$  is a tautology, then either  $\varphi$  or  $\psi$  (or both) are tautologies.
4. If  $\varphi$  and  $\varphi \rightarrow \psi$  are tautologies, then  $\psi$  is a tautology.

**Exercise 5.13.** What is the relationship between  $\neg$ i in propositional logic and (RAA) in  $S^\dagger$ ?

**Exercise 5.14.** This exercise is about the concept of *logical equivalence* in propositional logic. We say that two sentences  $\varphi$  and  $\psi$  are *logically equivalent* if  $\models \varphi \leftrightarrow \psi$ .

1. Show that  $\varphi$  and  $\psi$  are logically equivalent if and only if for all valuations  $v$ ,  $\llbracket \varphi \rrbracket_v = \llbracket \psi \rrbracket_v$ .
2. Suppose that  $\varphi$  and  $\psi$  are logically equivalent. Show that  $\neg \varphi$  and  $\neg \psi$  are again logically equivalent.
3. Suppose that  $\varphi_1$  and  $\psi_1$  are logically equivalent, and also that  $\varphi_2$  and  $\psi_2$  are logically equivalent. Show that  $\varphi_1 \wedge \varphi_2$  and  $\psi_1 \wedge \psi_2$  are again logically equivalent.
4. Suppose that  $\varphi_1$  and  $\psi_1$  are logically equivalent, and also that  $\varphi_2$  and  $\psi_2$  are logically equivalent. Show that  $\varphi_1 \rightarrow \varphi_2$  and  $\psi_1 \rightarrow \psi_2$  are again logically equivalent.

**Exercise 5.15.** We continue to study the equivalence relation  $\equiv$  from Exercise 5.14. Let  $P$  be the set of equivalence classes. Define  $\leq$  on  $P$  by

$$[\varphi] \leq [\psi] \quad \text{iff} \quad \models \varphi \rightarrow \psi$$

Part (4) of Exercise 5.14 shows that this definition is sensible.

1. We would like to make  $\mathbb{P} = (P, \leq)$  into an orthoposet (see Definition 3.22 on page 78). What should we take for 0 and the complement operation?
2. Prove that we really have an orthoposet.

**Exercise 5.16.** On page ??, we have the diagram of logic sentences built from  $p$  and  $q$ , up to logical equivalence. For each of the 16 points in the diagram, find a logic sentence that uses just  $p$ ,  $q$  and one or two of the symbols  $\top$ ,  $\bot$ ,  $\neg$ ,  $\wedge$ ,  $\vee$ ,  $\rightarrow$ , or  $\leftrightarrow$ . For example,  $\bigvee\{\alpha_1, \alpha_2\} \equiv p$ .

**Exercise 5.17.** Prove that for every sentence  $\varphi$  there is some sentence  $\psi$  such that (a)  $\varphi$  and  $\psi$  are logically equivalent; (b) the only connectives which are used in  $\psi$  are  $\neg$  and  $\rightarrow$ . [Hint: You will need to use induction on sentences, and also Exercise 5.14, and also some assertions like those in Exercise 5.14 which you must formulate and prove for yourself.]

**Exercise 5.18.** Suppose that we formulate a language  $\mathcal{L}'$  that is different from the language  $\mathcal{L}$  in this chapter because it has the exclusive-or operator  $\oplus$  as its only connective. So one example of a sentence in  $\mathcal{L}'$  is  $p \oplus (q \oplus r)$ .

Prove that the truth table of every sentence  $\varphi$  in  $\mathcal{L}'$  has an even number of rows where  $\varphi$  comes out true, and an even number of rows where it comes out false.

**Exercise 5.19.** Suppose that  $\varphi_1, \varphi_2, \dots, \varphi_n \models \psi$ . Show that  $\models \varphi_1 \rightarrow (\varphi_2 \rightarrow \dots (\varphi_n \rightarrow \psi) \dots)$ .

**Exercise 5.20.** Suppose that we took our proof system and dropped the requirement that no sentence can be copied out of a closed subproof. Show that our system would be unsound. That is, find a derivation showing  $\vdash \varphi$  in this system for some sentence  $\varphi$  such that  $\not\models \varphi$ . [Hint: you can do this by modifying several derivations that occur in this chapter.]

**Exercise 5.21.** Fill in the details on the following proof of the *Compactness Theorem for Propositional Logic*: Let  $\Gamma$  be a set of sentences, and suppose that every finite subset of  $\Gamma$  is satisfiable. Then  $\Gamma$  itself is satisfiable.

We are going to sketch this result in the case that the set  $\text{AtSen}$  of atomic sentences is *countably infinite*.<sup>7</sup> Let  $\Gamma$  be enumerated as  $\varphi_1, \varphi_2, \dots, \varphi_k, \dots$ . For each  $k$ , the subset of  $\Gamma$  consisting of  $\varphi_1, \dots, \varphi_k$  is finite. So by hypothesis, there is some  $v^k$  such that for  $1 \leq i \leq k$ ,  $\llbracket \varphi_i \rrbracket_{v^k} = \top$ .

Define a valuation  $v$  by recursion, defining  $v(p_1)$  to start with, and given  $v(p_1), \dots, v(p_n)$ , defining  $v(p_{n+1})$  in terms of these values. Each  $v(p_n)$  will have the property that

there are infinitely many  $j$  such that  $v^j$  and  $v$  agree on  $p_1, \dots, p_n$

To define  $v(p_1)$ , let it be  $\top$  if there are infinitely many  $j$  such that  $v^j(p_1) = \top$ . If there are only finitely many such  $j$ , then there must be infinitely many  $j$  such that  $v^j(p_1) = \bot$ . (Why is this true?) We let  $v(p_1) = \bot$  in this case.

Now given  $v(p_1), \dots, v(p_n)$ , again we let  $v(p_{n+1}) = \top$  if there are infinitely many  $j$  such that  $v^j$  and  $v$  agree on  $p_1, \dots, p_n$ , and  $v^j(p_{n+1}) = \top$ . If not, then again there must be there must

<sup>7</sup>It holds for all sets  $\text{AtSen}$ , even uncountable sets.



be infinitely many  $j$  such that that  $v^j$  and  $v$  agree on  $p_1, \dots, p_n$ , and  $v^j(p_{n+1}) = \text{F}$ . We define  $v(p_{n+1})$  accordingly.

This defines  $v$ . This is the main point for you to check that  $\llbracket \varphi_i \rrbracket_v = \text{T}$  for all  $i$ ; this proves that  $\Gamma$  is indeed satisfiable. [Hint:  $\text{occ}(\varphi_i)$  is finite for all  $i$ .]

**Exercise 5.22.** It is easy to confuse the following two assertions:

- (i)  $\not\vdash \varphi \rightarrow \psi$
- (ii)  $\vdash \varphi \rightarrow \neg\psi$ .

1. Give an example of two sentences  $\varphi$  and  $\psi$  in propositional logic with the property that  $\not\vdash \varphi \rightarrow \psi$  but  $\not\vdash \varphi \rightarrow \neg\psi$ . This shows that (i) does not in general imply (ii).
2. Give an example of two sentences  $\varphi$  and  $\psi$  in propositional logic with the property that  $\vdash \varphi \rightarrow \psi$  and also  $\vdash \varphi \rightarrow \neg\psi$ . This shows that (ii) does not in general imply (i).

**Exercise 5.23.** Here are some exercises about our proof system.

1. Show that  $\vdash \neg\text{F}$ . [Hint: this is short.]
2. Show that  $\vdash (\text{T} \rightarrow \text{F}) \leftrightarrow \text{F}$ .
3. Show that for all  $\varphi$ ,  $\vdash \varphi \rightarrow \neg\neg\varphi$ .

**Exercise 5.24.** An incorrect formal proof.

1. What is wrong with the following formal proof?

|   |  |  |                              |                            |
|---|--|--|------------------------------|----------------------------|
| 1 |  |  | $p$                          | A                          |
| 2 |  |  | $q$                          | A                          |
| 3 |  |  | $p$                          | R, 1                       |
| 4 |  |  | $p \wedge q$                 | $\wedge\text{i } 2,3$      |
| 5 |  |  | $q \rightarrow (p \wedge q)$ | $\rightarrow\text{i } 2-4$ |

2. The two assumption lines suggest a result which actually is correct. To find it, figure out how to use the first 4 lines and then correctly finish the proof.

**Exercise 5.25.** Here is a general definition that parallels a concept with the same name which we have already seen with other logics. A sentence  $\varphi$  is *consistent* if  $\not\vdash \neg\varphi$ . That is,  $\neg\varphi$  is *not* provable. If  $\varphi$  is not consistent, it is *inconsistent*.

Prove that

$\varphi$  is consistent if and only if  $\varphi$  is satisfiable

You will need to use either the soundness or completeness of our logic. Please be sure to write down exactly where you used these results.

**Exercise 5.26.** Show that if  $\varphi$  is consistent and  $\psi$  is any sentence, then either  $\varphi \wedge \psi$  is consistent, or else  $\varphi \wedge \neg\psi$  is consistent. Please do this problem without using the completeness of the logic.

[Hint: show the contrapositive.]

**Exercise 5.27.** Example 5.37 shows  $\text{SD}_1$  and  $\text{SD}_2$  explicitly. Let us think now about  $\text{SD}_0$ . In order to figure out what this should be, we must mention an important convention:

|                                                                    |                                                                  |
|--------------------------------------------------------------------|------------------------------------------------------------------|
| $\bigwedge \emptyset = \text{T}$<br>Empty conjunctions count as T. | $\bigvee \emptyset = \text{F}$<br>Empty disjunctions count as F. |
|--------------------------------------------------------------------|------------------------------------------------------------------|

Now figure out what set  $\text{SD}_0$  should be. [Hint: it should be a set of size  $2^0$ .] You should check that Lemmas 5.39–5.43 hold for  $n = 0$ .

**Exercise 5.28.** In this exercise, we introduce some notation that will be used later in the course.

We write  $\oplus(\psi_1, \dots, \psi_n)$  to mean that *exactly one* of  $\psi_1, \dots, \psi_n$  holds. Formally,  $\oplus(\varphi, \psi)$  abbreviates

$$(\varphi \wedge \neg\psi) \vee (\neg\varphi \wedge \psi).$$

So we have  $\models \oplus(p, \neg p)$ , and  $\vdash \oplus(p, \neg p)$  as well. Similarly,  $\oplus(\varphi, \psi, \chi)$  abbreviates

$$(\varphi \wedge \neg\psi \wedge \neg\chi) \vee (\neg\varphi \wedge \psi \wedge \neg\chi) \vee (\neg\varphi \wedge \neg\psi \wedge \chi).$$

In general,  $\oplus(\psi_1, \dots, \psi_n)$  would be a disjunction with  $n$  disjuncts:

$$(\psi_1 \wedge \neg\psi_2 \wedge \dots \wedge \neg\psi_n) \vee (\neg\psi_1 \wedge \psi_2 \wedge \dots \wedge \neg\psi_n) \vee \dots \vee (\neg\psi_1 \wedge \neg\psi_2 \wedge \dots \wedge \psi_n)$$

The  $i$ th disjunct is a conjunction of the  $\psi$ 's with  $\neg$  in front of all the  $\psi$ s except for  $\psi_i$ .

Show that  $\models \oplus(\psi_1, \dots, \psi_n) \rightarrow (\psi_1 \vee \psi_2 \vee \dots \vee \psi_n)$ . [This should be very easy, except that the notation is sticky. If you would rather just do it for  $n = 3$ , this is fine.]

**Exercise 5.29.** Let  $\text{SD}_n$  be the set of state descriptions of order  $n$ . Show that  $\vdash \oplus \text{SD}_n$ . (It's fine to use the Completeness Theorem, provided that you recognize that you are indeed using that result! Doing so makes this very easy.)

**Exercise 5.30.** Here is an exercise in induction. A sentence is in *negation normal form* (nnf) if all of the negation signs are in front of atomic sentences  $p_i$ . For example,  $\neg p_{13} \wedge p_{58}$  is in nnf, but  $\neg(p_3 \vee p_2)$  is not in nnf. Show that for every sentence  $\varphi$ , there is some sentence  $\varphi^*$  in nnf such that  $\vdash \varphi \leftrightarrow \varphi^*$ .

(As an example, if  $\varphi$  is  $\neg(\neg p_1 \wedge p_2)$ , then we can take  $\varphi^*$  to be  $p_1 \vee \neg p_2$ .)

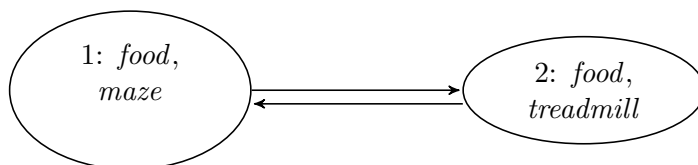
[Hint: you will need to use induction, and it is helpful to prove something which is stronger than what is asked. Show that there are sentences  $\varphi^*$  and  $\varphi^-$  in nnf such that  $\vdash \varphi \leftrightarrow \varphi^*$  and  $\vdash \neg\varphi \leftrightarrow \varphi^-$ . For  $\varphi$  an atomic sentence  $p$ , we can take  $\varphi^* = p$  and  $\varphi^- = \neg p$ . Let us spell out what needs to be done in the induction step for negation. Assume about  $\varphi$  that we have  $\varphi^*$  and  $\varphi^-$ ; we must define nnfs  $(\neg\varphi)^*$  and  $(\neg\varphi)^-$  in terms of them and then verify that  $\vdash \neg\varphi \leftrightarrow (\neg\varphi)^*$  and also  $\vdash \neg\neg\varphi \leftrightarrow (\neg\varphi)^-$ . You also need to do the other steps for the induction which we mentioned in Definition 5.16.]



## Chapter 6

# Meet Ronnie, the Logical Rat

Ronnie is a rat who is very interested in what he sees around him. His owners own a company that manufactures two-chambered rat houses. Each day, they put him in a different house. He runs back and forth all day long. At any given time, Ronnie is very interested in what he finds in the room he currently is in, and also what he sees in the room he is *not* in. For example, here is one of the houses.



In room 1, there is food and a maze (and no treadmill) In room 2, there is food and a treadmill (and no maze).

He's a good reasoner, and so he needs a logical language in which to make assertions. The atomic sentences of this language should correspond to the presence of objects familiar to Ronnie. We'll write these atomic sentences as *water*, *food*, *maze*, etc. However, the exact atomic sentences that we use are not very important, and often we will write them the way we did in Chapter 5, with letters like *p*, *q*, etc.

Ronnie himself does not have access to the room numbers in the various houses. They are written on the outside, and he is stuck indoors. All that he can do is talk (in a particular room) about what is in the current room, what is in the other room, and more complicated variations on these assertions. By “more complicated variations” we mean combinations using the standard sentential connectives  $\neg$ ,  $\wedge$ ,  $\vee$ ,  $\rightarrow$ , and  $\leftrightarrow$ , and crucially a modal operator  $*$  that stands for “in the other room.” We want the language to work as illustrated below:

| English                                                        | $\mathcal{L}(*)$           |
|----------------------------------------------------------------|----------------------------|
| there is water in the room                                     | <i>water</i>               |
| there is food in the other room, and a treadmill in this room  | $(*food) \wedge treadmill$ |
| after switching rooms twice, there would be a maze in the room | $** maze$                  |

We digress to formalize this language in Section 6.1 just below. After that, we return to this more conceptual discussion to explore what kinds of reasoning he could do in this language, and how this reasoning would correspond to various formal assertions. The remainder of this chapter is then an

exploration of some of the formal matters. However, it is always good to keep the informal story in mind in this chapter, and we return at the very end of the chapter to our conceptual points.

## 6.1 The language $\mathcal{L}(*)$

We saw the language  $\text{Sen}$  of propositional logic in Chapter 5. This language is not expressive enough for Ronnie:  $p_1$  means that there is a water dish *here*, and he might want to say that there is a water dish *there*. We add a new operator to  $\text{Sen}$  called  $*$ , which we read as “switch”. Now we get a new language which we call  $\mathcal{L}(*)$ .<sup>1</sup> The syntactic notions of the language may be found in Definition 6.1,

**Definition 6.1: the syntax of  $\mathcal{L}(*)$ , and the statement of proof by induction on sentences**

We define the *switching language*  $\mathcal{L}(*)$  to be the smallest set with the following properties:

1. Each  $p_i \in \text{AtSen}$  belongs to  $\mathcal{L}(*)$ .
2.  $\top$  and  $\text{F}$  belong to  $\mathcal{L}(*)$ .
3. If  $\varphi \in \mathcal{L}(*)$ , then also  $\neg\varphi \in \mathcal{L}(*)$ .
4. If  $\varphi \in \mathcal{L}(*)$  and  $\psi \in \mathcal{L}(*)$ , then also  $\varphi \wedge \psi$ ,  $\varphi \vee \psi$ ,  $\varphi \rightarrow \psi$  and  $\varphi \leftrightarrow \psi$  belong to  $\mathcal{L}(*)$ .
5. If  $\varphi \in \mathcal{L}(*)$ , then  $*\varphi$  belongs to  $\mathcal{L}(*)$ .

To prove a statement \_\_\_\_\_ by induction about all sentences  $\varphi \in \mathcal{L}(*)$ , we use the same steps as in Definition 5.16, with one addition: Assume \_\_\_\_\_ for  $\varphi$ , and prove it for  $*\varphi$ .

The switching language  $\mathcal{L}(*)$  also has complex sentences in it like

$$*(\text{treadmill} \rightarrow *\neg\text{maze}).$$

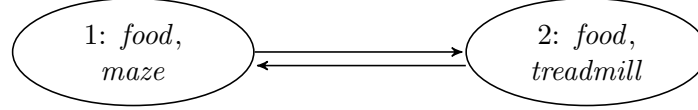
This would say that in the other room from where he currently is, if there is a treadmill, then in the other room (from there), there is no maze.

As you might have guessed, the story of Ronnie’s houses is a *modal setting*. And  $*$  is a *modal operator*. We’ll meet lots of other modal operators in due course. To mention one, Ronnie might want to reason about what he saw *yesterday* in addition to what he sees in the house he’s currently in. So for this, he would want a language with a modal operator for the previous day. But this point gets ahead of what we’ll actually do in this section. We’ll be working with  $*$  alone, doing modal logic in a concrete way. Some of what we do here will not generalize to other settings. Overall, this particular modal setting is one of the simplest of all, and so it should help you to catch on to the subject.

**Models** We said that Ronnie walked around rat houses. We’ll also call those rat houses *models*, because this is the terminology that we’ll use in other ways throughout our study of modal logic.

<sup>1</sup>It might be better to call this  $\text{Sen}(*)$ .

Mathematically, a model for  $\mathcal{L}(\ast)$  is an ordered pair  $M = (R_1, R_2)$  of sets of atomic sentences. We prefer to draw them as graphs, but technically, they're just ordered pairs of sets. For example, the model



is technically the ordered pair

$$(\{food, maze\}, \{food, treadmill\}).$$

The picture and the formal representation have the same content: in room 1, there is food and a maze (and no treadmill); in room 2, there is food and a treadmill (and no maze).

Definition 6.2 defines the semantics of our language. It is very important that we do not speak of a sentence being true or false in one of our models, but rather in *room 1* or *room 2* of one of those models. So we are defining relations  $1 \models \varphi$  and  $2 \models \varphi$  between the two rooms (of a particular house) and sentences. Like all relations, statements of the form  $1 \models \varphi$  can be either true or false (again, this is in a particular house, and  $\varphi$  is a particular sentence.)

In our definition, the main thing to look at is the last clause, the one for the  $\ast$  operator. (All the other clauses in the definition are exactly what we expect from reading the connectives in English.) The clause for  $1 \models \ast \varphi$  and  $2 \models \ast \varphi$  is

$$\begin{aligned} 1 \models \ast \varphi & \text{ iff } 2 \models \varphi \\ 2 \models \ast \varphi & \text{ iff } 1 \models \varphi \end{aligned} \tag{6.1}$$

To understand these, we need to plug in an actual sentence for the sentence variable  $\varphi$ . Let us take  $water \wedge treadmill$ . Then our semantics would say

$$\begin{aligned} 1 \models \ast (water \wedge treadmill) & \text{ iff } 2 \models water \wedge treadmill \\ 2 \models \ast (water \wedge treadmill) & \text{ iff } 1 \models water \wedge treadmill \end{aligned}$$

We'll see examples of this just below.

**Definition 6.2: The semantic notions of  $\mathcal{L}(\ast)$** 

We first define the satisfaction relation between rooms in a model and sentences:

|                                 |     |                                                                                                |                                 |     |                                                                                                |
|---------------------------------|-----|------------------------------------------------------------------------------------------------|---------------------------------|-----|------------------------------------------------------------------------------------------------|
| $1 \models p$                   | iff | if the model says so                                                                           | $2 \models p$                   | iff | if the model says so                                                                           |
| $1 \models \mathbf{T}$          |     | always                                                                                         | $2 \models \mathbf{T}$          |     | always                                                                                         |
| $1 \models \mathbf{F}$          |     | never                                                                                          | $2 \models \mathbf{F}$          |     | never                                                                                          |
| $1 \models \neg\varphi$         | iff | $1 \not\models \varphi$                                                                        | $2 \models \neg\varphi$         | iff | $2 \not\models \varphi$                                                                        |
| $1 \models \varphi \wedge \psi$ | iff | $1 \models \varphi$ and $1 \models \psi$<br>and $\vee, \rightarrow, \leftrightarrow$ similarly | $2 \models \varphi \wedge \psi$ | iff | $2 \models \varphi$ and $1 \models \psi$<br>and $\vee, \rightarrow, \leftrightarrow$ similarly |
| $1 \models \ast\varphi$         | iff | $2 \models \varphi$                                                                            | $2 \models \ast\varphi$         | iff | $1 \models \varphi$                                                                            |

Then we use these relations in the following semantic notions:

$\varphi$  is satisfiable means  $\varphi$  is true in some room of some house

$\Gamma \models_G \varphi$  means for all houses  $M$ ,  
if all sentences in  $\Gamma$  are true in both rooms of  $M$ ,  
then  $\varphi$  is also true in both rooms

$\Gamma \models_L \varphi$  means for all houses  $M$ ,  
if all sentences in  $\Gamma$  are true in a given room  $i$  of  $M$ ,  
then  $\varphi$  is also true in this same room  $i$

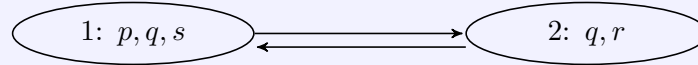
One last point: we can summarize the two lines in (6.1) in one line as

$$i \models \ast\varphi \text{ iff } 3 - i \models \varphi$$

This summarizes what we have seen, since  $3 - 1 = 2$ , and  $3 - 2 = 1$ . Occasionally, the formulation just above saves a few lines in an argument.

**Example 6.3**

Here is a model which we'll call  $M$ .



The model  $M$  itself tells us facts like  $1 \models p$ ,  $2 \not\models p$ ,  $1 \models q$ ,  $2 \models q$ ,  $1 \not\models r$ ,  $2 \models r$ ,  $1 \models s$ , and  $2 \not\models s$ . That is, we see directly from the model which atomic sentences are true in which rooms.

For the more complicated sentences that the switching language contains, we use the other clauses in the definition in Definition 6.2. By way of example, let us show that in  $M$ ,  $1 \models \ast(p \wedge \ast q) \rightarrow \ast r$ . The way to do this is to be more systematic and to evaluate all of the subsentences of the given sentence *at both worlds* of this model  $M$ . The model itself tells us that  $1 \models p$ ,  $2 \not\models p$ ,  $1 \models q$ ,  $2 \models q$ , and  $1 \not\models r$ ,  $2 \models r$ . These last two facts imply that  $1 \models \ast r$  and  $2 \not\models \ast r$ . The earlier facts about  $q$  tell us that  $1 \models \ast q$  and  $2 \models \ast q$ . So  $1 \models p \wedge \ast q$ , while  $2 \not\models p \wedge \ast q$ . Putting these facts together with what we saw about  $\ast r$ , we see that  $1 \models \ast(p \wedge \ast q) \rightarrow \ast r$ , as we said.

It is a good idea to be sure that you see that the definition in Definition 6.2 is actually used. That is, we aren't deciding whether sentences are true or false based on our intuitions and the picture; we have a formal semantics, and this is what we use.

**Remark 6.4**

Here is an important point about our semantics of the binary connectives  $\wedge$ ,  $\vee$ ,  $\rightarrow$ , and  $\leftrightarrow$  in this logic. We consider some particular model, call it  $\mathcal{M}$ . Let us consider an example of the semantics, say  $1 \models (\ast p) \wedge q$ . To evaluate this, we check whether this sentence is true in  $\mathcal{M}$  or not, we check to see if  $1 \models \ast p$  and also if  $1 \models q$ . It is tempting to think that after we see whether  $1 \models \ast p$ , we are in room 2 rather than room 1. So on this view, we should ask about  $1 \models \ast p$  and  $2 \models q$ . Indeed, this reflects the intuitions about moving between the rooms of a rat house with which began this chapter. However natural this is, this is nevertheless not how the semantics actually works. It would be possible to re-formulate the semantics to reflect this more “dynamic” process of evaluation, and we encourage you to think about this.

## 6.2 What kinds of reasoning do we want to formalize using $\mathcal{L}(\ast)$ ?

This chapter opened with a story about Ronnie the rat, and when we did so, we said something about “reasoning.” We were a little “cagey” about this matter of reasoning, and we turned instead to a logical language  $\mathcal{L}(\ast)$ . It is now time to say something more about reasoning, and how it is connected to  $\mathcal{L}(\ast)$ .

There are at least three types of reasoning that we could consider.

*Implicit knowledge of the geometry of rat houses.* Ronnie always visits houses that look the same, houses with two rooms. So some equivalences are going to be *valid* in the sense of being true at all models of all houses. These are implicit knowledge in the sense that they are assertions



about things that a reasoner *should know*. There are two main types of examples.

1. For all sentences  $\varphi$ ,  $\varphi$  and  $**\varphi$  are equivalent. This means that for all rooms of all houses, if  $\varphi$  is true there, then  $**\varphi$  is also true there; and vice-versa.
2. For all sentences  $\varphi$ , if  $\neg*\varphi$  is true at some room, then  $*\neg\varphi$  is also true at that room, and vice-versa.

However, there are other examples of the same sort of implicit knowledge. For example,

- (3) For all sentences  $\varphi$ ,  $\varphi$  and  $****\varphi$  are equivalent.

However, this seems to be a consequence of (1). To see this, let us fix  $\varphi$ . Then (1) applies to  $\varphi$  (since it applies to every sentence), and we infer that  $\varphi$  and  $**\varphi$  are equivalent. By (1) also applies to  $**\varphi$ , (since it applies to every sentence), and we infer that  $**\varphi$  and  $****\varphi$  are equivalent. So now we should be able to reason that indeed  $\varphi$  and  $****\varphi$  are equivalent. The point here is to raise the question of how to incorporate this kind of basic logical reasoning into a logical system. It also raises the question of whether (1) and (2) above constitute *all* of Ronnie's implicit knowledge about the houses he visits.

*Global reasoning from hypotheses.* Suppose that Ronnie learns from experience his owners always produce houses satisfying the following condition: if food is present in a given room, then water is present in the other room. Suppose he also knows that he is currently in a house with food in both rooms. Then he should conclude: water is present in both rooms. We would write this type of global reasoning<sup>2</sup> as

$$food \rightarrow *water, food \models_G water \quad (6.2)$$

Here is a second example. Suppose Ronnie comes to know that he is always in houses with the feature that if both rooms have a treadmill, then neither will have a maze. We would write this as

$$treadmill \models_G \neg maze. \quad (6.3)$$

We call this *global* reasoning to indicate that we have to check an assertion about *both* rooms for the conclusion to follow, and also that the conclusion itself applies to both rooms. The global assertion

$$treadmill \models_G *\neg maze.$$

would have the same force, since on the same assumption (both rooms have a treadmill) we have an equivalent conclusion (from each of the two rooms, if we travel to the other, there will be no maze).

*Local reasoning from hypotheses.* Suppose that Ronnie also learns from experience his owners always produce houses satisfying the following condition: if a given room has food, then the other room has water. Moreover, he is just coming from a room with no water. Even without looking, he should be able to conclude that his current room has no food. We would write this as

$$food \rightarrow *water, *\neg water \models_L \neg food, \quad (6.4)$$

---

<sup>2</sup>Notice that this global reasoning itself uses what we called implicit knowledge of the geometry of rat houses. In general, it is not always clear how to separate the different kinds of knowledge in intuitive examples. Our formalism will be based on precise definitions, so the differences there will be clearer.

or as

$$(food \rightarrow \ast water) \wedge \ast \neg water \models_L \neg food,$$

(The two assertions differ .) This is *local* reasoning in the sense that we only have to look at the room we are in, and check the two hypotheses in (6.4) are true; if it is, we may infer that  $\neg food$  is false in that same room. Contrast this with

$$food \rightarrow \ast water, \ast \neg water \models_G \neg food$$

This would say that if both rooms are such that the presence of food there implies the presence of food in the other room, and if in each room a trip to the other shows no water, then neither room has food. This is a valid inference, but our point here is that it is global rather than local reasoning.

Our inventory of reasoning types thus includes implicit structural reasoning, global reasoning from hypotheses, and local reasoning from hypotheses. Before we turn to formalizations of these, we have some additional remarks.

#### Remark 6.5

First, the implicit structural facts may be taken to be global assertions *without hypotheses*. They may also be taken to be local assertions without hypotheses. Second, it is important to see that *global reasoning may be considered as a special case of local reasoning*. This might seem surprising at first, but it stems from the fact that  $\varphi$  is true at both worlds of a given model if and only if  $\varphi \wedge \ast \varphi$  is true at either world. This means that a global assertion such as (6.3) may be presented in a local form:

$$treadmill \wedge \ast treadmill \models_L \neg maze \wedge \ast \neg maze.$$

The most useful mode of reasoning might well allow both global and local assumptions, and with a local conclusion. Can you give examples of how this would be used? And can you see how using global and local assumptions we could also infer a global conclusion?

### 6.2.1 Formalizing the reasoning styles in terms of different notions of validity

We turn to a formalization of those styles of reasoning.

What we called implicit structural reasoning is going to be formalized implicitly as well in everything we do, so let us set this aside for now.

To formalize global reasoning, we use the definition of  $\Gamma \models_G \varphi$  in Definition 6.2. Let us study some examples with  $\Gamma$  taken to be the empty set  $\emptyset$ . Since every room satisfies every sentence in  $\emptyset$  (since there are no such sentences), The definition of  $\Gamma \models_G \varphi$  in this case boils down to  $\varphi$  is true in both rooms of every model.

**Example 6.6**

Let us check that  $\models_G (**food \wedge ***water) \rightarrow *(water \wedge *food)$ .

Let  $M$  be any model. We'll show that  $1 \Vdash \varphi$ , where  $\varphi$  is the sentence above. Similar work shows that  $2 \Vdash \varphi$ . Assume that  $1 \Vdash **food \wedge ***water$ . We see that  $2 \Vdash *food$ . Let us now unpack the second conjunct. We have  $1 \Vdash ***water$ , and so  $2 \Vdash **water$ . This means that  $1 \Vdash *water$ . So  $2 \Vdash water$ . At this point, we see that  $2 \Vdash water \wedge *food$ . Going back to 1, we have  $1 \Vdash *(water \wedge *food)$ , as desired.

**Example 6.7**

Here is a general case of a fact that we just saw. For all sentences  $\varphi$ , we have  $\models_G **\varphi \leftrightarrow \varphi$ . To see this, fix a model  $M$ . First, let us fix a room  $i$ . We show that in  $M$ ,  $i \Vdash **\varphi \leftrightarrow \varphi$ . We have two steps. First, assume that  $i \Vdash **\varphi$ . Then  $3 - i \Vdash *\varphi$ . And so  $i \Vdash \varphi$ . This shows that  $i \Vdash **\varphi \rightarrow \varphi$ . Second, assume that  $i \Vdash \varphi$ . Then again we have  $3 - i \Vdash *\varphi$ , and so  $i \Vdash **\varphi$ . This shows that  $i \Vdash \varphi \rightarrow **\varphi$ . Putting our two observations together, we have  $i \Vdash \varphi \leftrightarrow **\varphi$ .

**Example 6.8**

Another general fact:  $\models_G \neg *\varphi \leftrightarrow *\neg\varphi$ . Here is the reasoning. Fix a model  $M$ , and also one of its rooms  $i$ . First, assume that  $i \Vdash \neg *\varphi$ . Then  $i \nVdash *\varphi$ , so  $3 - i \nVdash \varphi$ . But concerning room  $3 - i$ , either  $\varphi$  or  $\neg\varphi$  must be true there. So since  $3 - i \nVdash \varphi$ , we must have  $3 - i \Vdash \neg\varphi$ . And thus  $i \Vdash *\neg\varphi$ . This establishes  $i \Vdash \neg *\varphi \rightarrow *\neg\varphi$ . The converse assertion is similar. And this for all rooms in all houses shows that indeed  $\models_G \neg *\varphi \leftrightarrow *\neg\varphi$ .

Recall that our goal in this section is to formalize both the global and local notions of validity. When we come to the local version, we unfortunately have two different options. First, we have  $\Gamma \models_1 \varphi$  defined by

$\Gamma \models_1 \varphi$  means    for all houses  $M$ ,  
                                          if all sentences in  $\Gamma$  are true in room 1 of  $M$ ,  
                                          then  $\varphi$  is also true in room 1

And we have  $\Gamma \models_2 \varphi$  defined the same way, with 1 replaced by 2, of course.

Our next goal is to show that these two notions  $\Gamma \models_1 \varphi$  and  $\Gamma \models_2 \varphi$  coincide. We need a definition, and then a preliminary result.

**Definition 6.9: mirror image models**

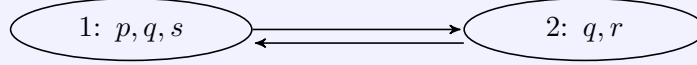
Let  $M$  be a model. Define a new model  $M^*$  by

$$\begin{array}{lll} 1 \Vdash p \text{ in } M^* & \text{iff} & 2 \Vdash p \text{ in } M \\ 2 \Vdash p \text{ in } M^* & \text{iff} & 1 \Vdash p \text{ in } M \end{array}$$

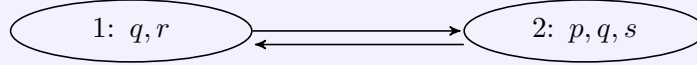
We call  $M^*$  the *mirror image* of  $M$ .

**Example 6.10**

Let  $M$  be the model below.



Then for this  $M$ ,  $M^*$  would be the model

**Lemma 6.11**

For all sentences  $\varphi$ ,

$$\begin{array}{lll} 1 \models \varphi \text{ in } M^* & \text{iff} & 2 \models \varphi \text{ in } M \\ 2 \models \varphi \text{ in } M^* & \text{iff} & 1 \models \varphi \text{ in } M \end{array}$$

*Proof.* We prove both assertions at the same time by induction<sup>3</sup> on  $\varphi$ .

In the case of an atomic sentence  $p_i$ , what we are proving is just the definition of  $M^*$ .

We are going to skip all of the induction steps except for the one for  $\ast\varphi$ . But we encourage you to think about those steps, especially if you are rusty on induction. Assume about a particular sentence  $\varphi$ ,

$$\begin{array}{lll} 1 \models \varphi \text{ in } M^* & \text{iff} & 2 \models \varphi \text{ in } M \\ 2 \models \varphi \text{ in } M^* & \text{iff} & 1 \models \varphi \text{ in } M \end{array}$$

We prove about  $\ast\varphi$  that

$$\begin{array}{lll} 1 \models \ast\varphi \text{ in } M^* & \text{iff} & 2 \models \ast\varphi \text{ in } M \\ 2 \models \ast\varphi \text{ in } M^* & \text{iff} & 1 \models \ast\varphi \text{ in } M \end{array}$$

For this, assume that  $1 \models \ast\varphi \text{ in } M^*$ . Then  $2 \models \varphi \text{ in } M^*$ . By second assumption in this part of the proof,  $1 \models \varphi \text{ in } M$ . And then by our semantics,  $2 \models \ast\varphi \text{ in } M^*$ . This is half of our first goal. The other half is similar. The second goal is proved in much the same way, except that we appeal to our *first* assumption<sup>4</sup>.  $\square$

**Theorem 6.12**

For all sets of sentences  $\Gamma$  and all sentences  $\varphi$ ,  $\Gamma \models_1 \varphi$  iff  $\Gamma \models_2 \varphi$ .

*Proof.* Suppose that  $\Gamma \models_1 \varphi$ . To see that  $\Gamma \models_2 \varphi$ , let  $M$  be any model, and assume that all sentences in  $\Gamma$  are true in room 2 of  $M$ . We show that  $2 \models \varphi \text{ in } M$ .

<sup>3</sup>You might like to review the format of proof by induction for  $\mathcal{L}(\ast)$ . See Definitions 5.16 and 6.1.

<sup>4</sup>The intertwining of assumptions and goals in this proof is the reason that it must be done by a *simultaneous induction*.

Consider  $M^*$ . Since all sentences in  $\Gamma$  are true in room 2 in  $M$ , Lemma 6.11 tells us that those sentences are also true in room 1 of  $M^*$  (not  $M$ !). As a result,  $1 \models \varphi$  in  $M^*$ . Lemma 6.11 applies again, and we see that  $2 \models \varphi$  in  $M$ , as desired.

Similar reasoning shows that  $\Gamma \models_2 \varphi$  implies  $\Gamma \models_1 \varphi$ .  $\square$

### Theorem 6.13

For all sentences  $\varphi$ ,  $\models_1 \varphi$  iff  $\models_2 \varphi$  iff  $\models_G \varphi$ .

*Proof.* We apply Theorem 6.12 with  $\Gamma$  empty to see that  $\models_1 \varphi$  iff  $\models_2 \varphi$ . So we only need to see that  $\models_1 \varphi$  iff  $\models_G \varphi$ .

Assume that  $\models_1 \varphi$ . Then we know that also,  $\models_2 \varphi$ . The conjunction of these is that  $\models_G \varphi$ .

In the other direction, it is clear that  $\models_G \varphi$  implies  $\models_1 \varphi$ .  $\square$

As a result of Theorems 6.12 and 6.13, we make the following definitions.

### Definition 6.14: notions of validity for $\mathcal{L}(*)$

$\models \varphi$  means  $\models_L \varphi$ , and this is the same as  $\models_G \varphi$  by Theorem 6.13.

$\Gamma \models_L \varphi$  means  $\Gamma \models_1 \varphi$ . By Theorem 6.12, this is the same as  $\Gamma \models_2 \varphi$ .

For all of these assertions, if  $\Gamma$  is a finite set  $\{\varphi_1, \dots, \varphi_n\}$ , then instead of writing, e.g.,

$$\{\varphi_1, \dots, \varphi_n\} \models_L \psi$$

we prefer to drop the set braces and write

$$\varphi_1, \dots, \varphi_n \models \psi$$

### Theorem 6.15

For all  $\varphi_1, \dots, \varphi_n$  and  $\psi$ ,

$$\varphi_1, \dots, \varphi_n \models_L \psi \quad \text{iff} \quad \models (\varphi_1 \wedge \dots \wedge \varphi_n) \rightarrow \psi.$$

*Proof.* First, assume that  $\varphi_1, \dots, \varphi_n \models_L \psi$ . To see that  $\models (\varphi_1 \wedge \dots \wedge \varphi_n) \rightarrow \psi$ , consider a model  $M$  and one of its two rooms, say  $i$ . Assume that all of the assumptions  $\varphi_1, \dots, \varphi_n$  are true in room  $i$ . We apply our assumption that  $\varphi_1, \dots, \varphi_n \models_L \psi$  to infer that  $\varphi_1, \dots, \varphi_n \models_i \psi$ . And then it follows that  $i \models \psi$ . Since  $M$  and  $i$  were arbitrary, we see that indeed  $\models (\varphi_1 \wedge \dots \wedge \varphi_n) \rightarrow \psi$ .

In the other direction, assume that  $\models (\varphi_1 \wedge \dots \wedge \varphi_n) \rightarrow \psi$ . This means that  $(\varphi_1 \wedge \dots \wedge \varphi_n) \rightarrow \psi$  is true in both rooms of every model. Now fix a model and one of its two rooms, say  $i$ . Assume that  $i \models \psi_1, \dots, i \models \psi_n$ . Then  $i \models \varphi_1 \wedge \dots \wedge \varphi_n$ . By our assumption,  $i \models \psi$  as well. This for all  $M$  and  $i$  concludes this half of our result.  $\square$

**Theorem 6.16**

For all  $\varphi_1, \dots, \varphi_n$  and  $\psi$ ,

$$\varphi_1, \dots, \varphi_n \models_G \psi \quad \text{iff} \quad \models (\varphi_1 \wedge * \varphi_1 \wedge \dots \wedge \varphi_n \wedge * \varphi_n) \rightarrow \psi.$$

*Proof.* This is similar to the proof of Theorem 6.15, and we leave it for Exercise 6.17.  $\square$

### 6.3 Natural deduction proof system for $\mathcal{L}(*)$

We present a natural deduction style proof system for  $\mathcal{L}(*)$  in Definition 6.18. The basic goal is to show that this system is useful as a model of local reasoning. In the next section, we shall show that it is complete in a precise sense.

**\*-subproofs** We introduce a new feature of proofs called *\*-subproofs*. The idea here is that the outermost level of proofs talk about what is going on in a particular room of a particular house. Then starting a \*-subproof amounts to moving to the *other* room in the house. At some point in the course of doing this reasoning, we might again start a new \*-subproof. This would move us again to the other room (that is, back to where we were when we started). The overall mechanism of \*-subproofs therefore makes our reasoning “local” in the sense that it all is about a particular room rather than about both rooms at the same time.

**Axioms vs. axiom schemes** What we are calling axioms in Definition 6.18 are more properly called *axiom schemes*. For example, we referred to the “Involution Axiom”  $** \varphi \rightarrow \varphi$ . What we mean here is that any of the following sentences may be used on any line in a formal proof:

$$** (p \wedge *q) \rightarrow (p \wedge *q)$$

$$** \perp \rightarrow \perp$$

$$** ((r \vee p) \rightarrow *q) \rightarrow ((r \vee p) \rightarrow *q)$$

And these are just three examples, of course. The main point again is that any sentence of the form  $** \varphi \rightarrow \varphi$  is an axiom in the system of this section.

**Definition 6.17**

We define  $\Gamma \vdash \varphi$  the same way as in Definition 5.31, except that we have additional axioms and rules in the proof system.

**Shortening proofs by appealing to propositional reasoning** Our proof system *includes* the earlier proof system of propositional logic. What we mean is that everything we saw earlier for propositional logic holds again for  $\mathcal{L}(*)$ . What is more, whenever we presented results for **Sen** that were stated in a general way (say using the letter  $\varphi$  as a variable over sentences of **Sen**) will hold again *and this time allowing  $\varphi$  to be a sentence in our larger language  $\mathcal{L}(*)$* . For example, we saw

in Example 5.33 that for all  $\varphi$  in **Sen**,  $\vdash \varphi \vee \neg\varphi$ . At this point, we can see that the same holds for all  $\varphi$  in  $\mathcal{L}(*)$ .

We saw before that for all tautologies  $\varphi$  of **Sen**,  $\vdash \varphi$ . The same thing holds for all sentences in  $\mathcal{L}(*)$  which are *substitution instances* of tautologies. For example, we have seen a tautology  $\pi$  in (5.4). Here is  $\pi$  again:

$$\begin{array}{ccccc} (p_1 \wedge p_2) & \vee & (p_1 \wedge p_3) & \vee & (p_2 \wedge p_3) \\ \vee & (\neg p_1 \wedge \neg p_2) & \vee & (\neg p_1 \wedge \neg p_3) & \vee & (\neg p_2 \wedge \neg p_3) \end{array}$$

Suppose we take this sentence and make a *simultaneous substitution*, for example:

$$p_1 \mapsto *(p_2 \wedge p_3) \quad p_2 \mapsto **p_1 \quad p_3 \mapsto p_3$$

The result of making this substitution<sup>5</sup> in  $\pi$  is

$$\begin{array}{ccccc} (*(p_2 \wedge p_3) \wedge **p_1) & \vee & (*(p_2 \wedge p_3) \wedge p_3) & \vee & (**p_1 \wedge p_3) \\ \vee & (\neg *(p_2 \wedge p_3) \wedge \neg **p_1) & \vee & (\neg *(p_2 \wedge p_3) \wedge \neg p_3) & \vee & (\neg **p_1 \wedge \neg p_3) \end{array} \quad (6.5)$$

---

<sup>5</sup>To explain the notation and idea, we are describing a process of replacing  $p_1$  by  $*(p_2 \wedge p_3)$ ,  $p_2$  by  $**p_1$ , and  $p_3$  by itself. We carry out all three replacements at the same time; if we did them one-by-one, say with “search and replace” on a standard text editor, the result would be different.

**Definition 6.18: Axioms and rules in our natural deduction proof system for  $\mathcal{L}(*)$** 

A *formal proof*  $\Pi$  is defined much as in propositional logic.

There is a new type of subproof, called a *\*-subproof*. See the text for this.

The notion of an *available line* needs to be changed. We say that line  $m$  is *available at*  $n$  if  $m < n$ , line  $m$  does not belong to any closed subproof, finally that if line  $n$  belongs to a \*-subproof  $\Pi'$ , then line  $m$  also belongs to  $\Pi'$ .

We take two new *axioms*, sentences which can be used on any line.

**New Axioms**

All *involution axioms*  $**\varphi \rightarrow \varphi$ .

All *determinacy axioms*  $\neg *\varphi \rightarrow *\neg\varphi$ .

The rules from propositional logic need to be qualified, and there are two additional rules.

**Qualification**

We may not copy any line from outside a given \*-subproof into that \*-subproof using *R* (reiteration). We also may not use rules whose premises are outside of a given \*-subproof in that \*-subproof. The only ways for a line to get into a \*-subproof are *\*e* (see below), *A* (assumption), or some other rule applied inside a the \*-subproof.

**Additional Rules**

- \*i Suppose that line  $n - 1$  is  $\psi$ , and that this line is in a subproof  $\Pi'$  of the main proof. Suppose also that  $\Pi'$  has no open subproofs of its own, and also no assumption lines. Finally, let  $\varphi$  be the last line of  $\Pi'$ . Then  $\Pi'$  may be closed, and line  $n$  may be  $*\varphi$ . The justification is  $*i, m_1 - m_2$ , where  $m_1$  is the first line of  $\Pi'$  and  $m_2$  the last line.
- \*e Suppose that  $m < n$  and line  $m$  is available at  $n$ , and also that line  $n$  belongs to a \*-subproof which was started after line  $m$  and not closed by line  $n$ , and only to one such \*-subproof. If line  $m$  is of the form  $*\varphi$ , then line  $n$  may be  $\varphi$ , justified by  $*e, m$ .



**Example 6.19**

Corresponding to Example 6.6, let us check that  $\vdash (**food \wedge ***water) \rightarrow *(water \wedge *food)$ . We use the following derivation:

|   |  |                                                              |                       |
|---|--|--------------------------------------------------------------|-----------------------|
| 1 |  | $**food \wedge ***water$                                     | assumption            |
| 2 |  | $*   *food$                                                  | $\wedge e, * e, 1$    |
| 3 |  | $**water$                                                    | $\wedge e, * e, 1$    |
| 4 |  | $**water \rightarrow water$                                  | involution axiom      |
| 5 |  | $water$                                                      | $\rightarrow e, 4, 3$ |
| 6 |  | $water \wedge *food$                                         | $\wedge i, 5, 2$      |
| 7 |  | $*(water \wedge *food)$                                      | $*i, 2-6$             |
| 8 |  | $(**food \wedge ***water) \rightarrow *(water \wedge *food)$ | $\rightarrow i, 1-7$  |

The proof begins with an assertion that we assume about one of the two rooms  $i$  in some arbitrary rat house  $M$ . Line 2 begins a  $*$ -subproof, indicating that we shift focus to the other room  $3 - i$ . In that room, the conjuncts in line 1 are true, provided that we drop the outermost  $*$ . One of these statements is  $**water$ . Using an Involution Axiom, we see that  $water$  is indeed true in our current room. We continue to reason about room  $3 - i$  until line 6, and then we pop back to room  $i$ . This shift in view is the reason we add a  $*$  to line 6 when we wrote it in line 7.

**Example 6.20**

Here is an *incorrect derivation*, illustrating why we need to forbid the use of any lines in a  $*$ -subproof that come from outside the  $*$ -subproof, except for lines coming from  $*e$ .

|   |  |                            |                      |
|---|--|----------------------------|----------------------|
| 1 |  | $water$                    | assumption           |
| 2 |  | $*   water$                | R, 1                 |
| 3 |  | $*water$                   | $*i, 2$              |
| 4 |  | $water \rightarrow *water$ | $\rightarrow i, 1-3$ |

If this were correct, it would show that  $\vdash water \rightarrow *water$ . That is, if there's water here, then there's water there. This would render the system unsound. So we have added the stipulation that nothing may be brought into a  $*$ -subproof from the outside, except by using  $*e$ . This means that line 2 is invalid.

**Example 6.21**

Here is another *incorrect derivation*. It explains why our rule of  $\ast e$  insists that there be only one  $\ast$ -subproof opened between a sentence  $\ast\varphi$  and its copied version  $\varphi$ .

|   |  |  |                                          |                      |
|---|--|--|------------------------------------------|----------------------|
| 1 |  |  | $\ast water$                             | assumption           |
| 2 |  |  | $\ast \mid \top$                         | $\top$               |
| 3 |  |  | $\ast \mid water$                        | $\ast e, 1$          |
| 4 |  |  | $\ast water$                             | $\ast i, 3-3$        |
| 5 |  |  | $\ast \ast water$                        | $\ast i, 2-4$        |
| 6 |  |  | $\ast water \rightarrow \ast \ast water$ | $\rightarrow i, 1-5$ |

The mistake is in opening one  $\ast$ -subproof inside another, and then copying two levels inward.

The challenge with this proof system is now to do carry out two tasks. We must make sure that the proof system is complete; this means that we have not left off any important proof principles from our system. And now that we have imposed a restriction on  $\ast$ -subproofs, we still must make sure that this restriction really insures the soundness of the system.

**Example 6.22**

Our system has as an axiom scheme called *determinacy*:  $\neg \ast \varphi \rightarrow \ast \neg \varphi$ . We call this determinacy because it reflects the idea that the “other room” is a determined place<sup>a</sup>. Example 6.23 presents a derivation of the converse.

<sup>a</sup>This might seem cryptic at this point, but later in our study of modal logic, we’ll see settings where there is more than one “other place” to go from a given place. After seeing those other settings, you’ll probably agree that this one is rather special. In any case, there is not much significance to the name *determinacy*.

**Example 6.23: a derivation of the converse of the determinacy axioms**

|    |  |  |                                                      |                       |
|----|--|--|------------------------------------------------------|-----------------------|
| 1  |  |  | $\ast \neg \varphi$                                  | assumption            |
| 2  |  |  | $\ast \varphi$                                       | assumption            |
| 3  |  |  | $\ast \mid \varphi$                                  | $\ast e, 2$           |
| 4  |  |  | $\neg \varphi$                                       | $\ast e, 1$           |
| 5  |  |  | $\varphi \wedge \neg \varphi \rightarrow \ast \perp$ | tautology             |
| 6  |  |  | $\ast \perp$                                         | PR, 3, 4, 5           |
| 7  |  |  | $\ast \ast \perp$                                    | $\ast i, 3-6$         |
| 8  |  |  | $\ast \ast \perp \rightarrow \perp$                  | involution axiom      |
| 9  |  |  | $\perp$                                              | $\rightarrow e, 7, 8$ |
| 10 |  |  | $\neg \ast \varphi$                                  | $\neg i, 3-9$         |
| 11 |  |  | $\ast \neg \varphi \rightarrow \neg \ast \varphi$    | $\rightarrow i, 1-10$ |

**Example 6.24**

We suggested in (6.4) that  $food \rightarrow *water, *\neg water \models_L \neg food$ . Corresponding to this, we have a formal result:  $food \rightarrow *water, *\neg water \vdash \neg food$ . We leave this to Exercise 6.4.

**Example 6.25**

Let  $\varphi$  be the sentence

$$(p \wedge *p) \vee (p \wedge *\neg p) \vee (\neg p \wedge *p) \vee (\neg p \wedge *\neg p).$$

We show that  $\vdash \varphi$  in Example 6.26. In the last line, we used the fact that

$$((p \vee \neg p) \wedge (*p \vee *\neg p)) \rightarrow ((p \wedge *p) \vee (p \wedge *\neg p) \vee (\neg p \wedge *p) \vee (\neg p \wedge *\neg p))$$

is a tautology. (Perhaps the easiest way to see this is to use the distributive laws and a lot of regrouping.) So by our Completeness Theorem for Propositional Logic (Theorem 5.45), there is a formal proof of the line above.

**Example 6.26: a derivation used in Example 6.25 and Lemma 6.30**

|    |                                            |                             |
|----|--------------------------------------------|-----------------------------|
| 1  | $p \vee \neg p$                            | tautology                   |
| 2  | $*p \vee \neg *p$                          | tautology                   |
| 3  | $*p$                                       | assumption                  |
| 4  | $*p \vee *\neg p$                          | $\vee$ i, 3                 |
| 5  | $\neg *p$                                  | assumption                  |
| 6  | $\neg *p \rightarrow *\neg p$              | determinacy axiom           |
| 7  | $*\neg p$                                  | $\rightarrow$ e, 5, 6       |
| 8  | $*p \vee *\neg p$                          | $\vee$ i, 7                 |
| 9  | $*p \vee *\neg p$                          | $\vee$ i, 2, 3–4, 5–8       |
| 10 | $(p \vee \neg p) \wedge (*p \vee *\neg p)$ | $\wedge$ i, 1, 9            |
| 11 | $\varphi$                                  | propositional reasoning, 10 |

**6.3.1 Soundness of the proof system**

The verification that our proof system is sound is quite similar to what we have seen in Section ???. But we must make a few additions and adjustments.

In this discussion, let us write  $\varphi \rightarrow^* \psi$  for  $\varphi \rightarrow * \psi$ . When we have a long series of sentences connected by  $\rightarrow$  and  $\rightarrow^*$ , we understand them as *associated to the right*. For example, we would write  $\varphi_1 \rightarrow \varphi_2 \rightarrow^* \varphi_3 \rightarrow^* \varphi_4$  as a shorthand for

$$(\varphi_1 \rightarrow (\varphi_2 \rightarrow (*(\varphi_3 \rightarrow *\varphi_4))))$$

Let us also use  $\Rightarrow$  as a symbol to stand for either  $\rightarrow$  or  $\rightarrow^\ast$ .

We associate with every line  $i$  in a formal proof an assertion of the form:

$$\models \varphi_1 \Rightarrow \cdots \Rightarrow \varphi_n \Rightarrow \psi \quad (6.6)$$

Here  $\psi$  is the last line, and the sequence  $\varphi_1, \dots, \varphi_n$  is the sequence (in order) of assertions which are open at line  $i$ .

The assertions that correspond to the rules of the proof system which go beyond the propositional part are

(axiom) If  $\psi$  is a determinacy or involution axiom, then for all  $\varphi_1, \dots, \varphi_n$ ,  $\models \varphi_1 \Rightarrow \cdots \Rightarrow \varphi_n \Rightarrow \psi$ .

(\* i) If  $\models \varphi_1 \Rightarrow \cdots \Rightarrow \varphi_n \rightarrow^\ast \psi$ , then  $\models \varphi_1 \Rightarrow \cdots \Rightarrow \varphi_n \rightarrow \ast\psi$ .

(\* e) If  $j \leq n$  and  $\models \varphi_1 \Rightarrow \cdots \Rightarrow \varphi_i \rightarrow \ast\psi$ , then

$$\models \varphi_1 \Rightarrow \cdots \Rightarrow \varphi_i \rightarrow \varphi_{i+1} \rightarrow \cdots \rightarrow \varphi_j \rightarrow \ast(\varphi_{j+1} \rightarrow \cdots \rightarrow \varphi_n \rightarrow \psi).$$

#### Lemma 6.27

Let  $\Pi$  be a formal proof for  $\mathcal{L}(\ast)$ , let

$$\text{seq}(\Pi) = \varphi_1 \Rightarrow \cdots \Rightarrow \varphi_n$$

be the sequence of assumptions in  $\Pi$  (in order) which are not in closed subproofs, and assume that the arrow between  $\varphi_i$  and  $\varphi_{i+1}$  is  $\rightarrow^\ast$  if and only if the number of  $\ast$ -subproofs opened between  $\varphi_i$  and  $\varphi_{i+1}$  is odd. Let  $\psi$  be the last line in  $\Pi$ . Then

$$\models \varphi_1 \Rightarrow \cdots \Rightarrow \varphi_n \Rightarrow \psi,$$

where once again the last  $\Rightarrow$  is  $\rightarrow^\ast$  iff  $\psi$  belongs to a  $\ast$ -subproof.

*Proof.* As with the proof of Lemma ??, this is an induction on the number of lines in a formal proof in the system. We only discuss the step when the last line is justified by (\*e).

Suppose that the last line  $\psi$  of  $\Pi$  is justified by (\*e), so that we have some line  $m < n$  which is available at  $n$ , and the sentence  $\ast\varphi$  is on line  $m$ . Let  $\Delta$  be the formal proof which consists of lines 1– $m$  of  $\Pi$ . Write  $\text{seq}(\Delta)$  as

$$\text{seq}(\Delta) = \chi_1, \dots, \chi_i.$$

Then  $\ast\psi$  is the last line of  $\Delta$ , and by induction hypothesis,

$$\models \chi_1 \Rightarrow \cdots \Rightarrow \chi_i \rightarrow \ast\psi.$$

Since line  $m$  is available at  $n$  in  $\Pi$ , all of the assumptions which were open at line  $m$  are still open at line  $n$ . Thus  $\chi_1 = \varphi_1, \dots, \chi_i = \varphi_i$ . Moreover, the arrows between these sentences are the same in  $\text{seq}(\Delta)$  as in  $\text{seq}(\Pi)$ . So we have

$$\models \varphi_1 \Rightarrow \cdots \Rightarrow \varphi_i \rightarrow \ast\psi. \quad (6.7)$$

Consider the assumption lines in  $\Pi$  that come after line  $m$ . We write these in order as  $\varphi_{i+1}, \dots, \varphi_n$ . There is exactly one  $*$ -subproof which was begun after line  $m$ : there must be at least one, since we apply  $(*e)$  after line  $m$ , and the restriction on this rule  $(*e)$  implies that there is only one which is still open at line  $n$ . Thus at most one of the arrows in  $\text{seq}(\Pi)$  after  $\varphi_i$  can be  $\rightarrow^*$  arrows; the rest must be  $\rightarrow$  arrows. We'll say that the  $*$ -subproof containing  $\psi$  was opened between  $\varphi_j$  and  $\varphi_{j+1}$ . So  $\text{seq}(\Pi)$  is of the form

$$\models \varphi_1 \Rightarrow \dots \Rightarrow \varphi_i \rightarrow \varphi_{i+1} \dots \rightarrow \varphi_j \rightarrow *(\varphi_{j+1} \rightarrow \dots \rightarrow \varphi_n \rightarrow^* \psi)$$

for some  $j \geq i$ ; the important point again is that the arrows after  $\varphi_j$  are  $\rightarrow$  arrows, except for the last arrow. By the assertion corresponding to  $(*e)$  and also (6.7) above, we have

$$\models \varphi_1 \Rightarrow \dots \Rightarrow \varphi_i \rightarrow \dots \rightarrow \varphi_n \rightarrow^* \psi. \quad (6.8)$$

This is the conclusion we are after when the last line of  $\Pi$  is justified by  $(*e)$ . □

to the point

We are not going to present any examples of how this lemma works. You can look back at Example ?? for an example concerning the propositional logic part. The modal part of the system (the  $*i$  and  $*e$  rules) will be generalized in the next chapter, and you can see an example of how this lemma works by going to Example 8.10, changing all  $\Box$  and  $\Diamond$  symbols to  $*$ , and glossing over the second and third lines.

As in Section ??, this last lemma then implies our soundness theorem:

#### Theorem 6.28: Soundness

If  $\Gamma \vdash \varphi$ , then  $\Gamma \models_L \varphi$ .

## 6.4 Completeness of $\mathcal{L}(*)$

We have seen that for  $\mathcal{L}(*)$ , the notions of  $\models_L \varphi$  and  $\models_G \varphi$  coincide. Therefore, we shall write this notion as  $\models \varphi$  without subscript<sup>6</sup>.

In this section, we prove the completeness of our natural deduction proof system. This is the statement that  $\models \varphi$  implies  $\vdash \varphi$ . The work in this section will solidify your understanding of our proof of the completeness theorem for propositional logic. If you understand the parallel work that we did on our proof system for propositional logic in Section 5.6.2, then the work in this section should be fairly easy.

### 6.4.1 State descriptions in $\mathcal{L}(*)$

We adapt the notion of *state descriptions* from propositional logic to switching logic so that versions of our lemmas in Section 5.6.2 hold.

---

<sup>6</sup>However, we urge you to remember that in the presence of non-empty sets  $\Gamma$  of hypotheses, the notions  $\Gamma \models_L \varphi$  and  $\Gamma \models_G \varphi$  are not equivalent.

**Definition 6.29: state descriptions**

A *state description of order  $n$*  is a sentence of the form

$$q_1 \wedge q_2 \wedge \cdots \wedge q_n \wedge \ast(r_1 \wedge r_2 \wedge \cdots \wedge r_n)$$

where each  $q_i$  is either  $p_i$  or  $\neg p_i$ , and also each  $r_i$  is either  $p_i$  or  $\neg p_i$

We again write  $\text{SD}_n$  for the set of state descriptions of order  $n$ .

**Lemma 6.30**

For all  $n$ ,  $\vdash \bigvee \text{SD}_n$ . That is, the disjunction of all state descriptions of order  $n$  is provable in the natural deduction system.

*Proof.* Example 6.25 indicates a derivation when  $n = 1$ . The general case of  $n > 1$  may be argued similarly. If one prefers, this proof could be given an induction argument based on Example 6.25.  $\square$

**Lemma 6.31**

Let  $\varphi$  be any sentence, and let  $n$  be large enough so that  $\text{occ}(\varphi) \subseteq \{p_1, \dots, p_n\}$ . For all  $\alpha \in \text{SD}_n$ , either  $\vdash \alpha \rightarrow \varphi$  or  $\vdash \alpha \rightarrow \neg\varphi$ .

*Proof.* The proof here is similar to that of Lemma 5.41. In fact, the proof of this lemma *includes* the proof of Lemma 5.41. We need to add one more induction step for the sentences  $\ast\varphi$ .

Step 8 We assume about  $\varphi$  that if  $n$  is large enough so that  $\text{occ}(\varphi) \subseteq \{p_1, \dots, p_n\}$ . For all  $\alpha \in \text{SD}_n$ , either  $\vdash \alpha \rightarrow \varphi$  or  $\vdash \alpha \rightarrow \neg\varphi$ . =What we need to prove is that if  $n$  is large enough so that  $\text{occ}(\ast\varphi) \subseteq \{p_1, \dots, p_n\}$ . For all  $\alpha \in \text{SD}_n$ , either  $\vdash \alpha \rightarrow \ast\varphi$  or  $\vdash \alpha \rightarrow \neg\ast\varphi$ .

So, let  $n$  be large enough so that  $\text{occ}(\neg\varphi) \subseteq \{p_1, \dots, p_n\}$ . Let  $\alpha \in \text{SD}_n$ . But the sentence letters that occur in  $\varphi$  are the same as those in  $\neg\varphi$ . That is,  $\text{occ}(\varphi) = \text{occ}(\ast\varphi)$ . Thus

$$\text{occ}(\varphi) = \text{occ}(\ast\varphi) \subseteq \{p_1, \dots, p_n\}$$

This means that our assumption in this step applies. But we don't apply it to  $\alpha$ , but rather to a "mirror image" of it. Write  $\alpha$  as

$$q_1 \wedge q_2 \wedge \cdots \wedge q_n \wedge \ast(r_1 \wedge r_2 \wedge \cdots \wedge r_n),$$

Now let  $\beta$  be the following state description:

$$r_1 \wedge r_2 \wedge \cdots \wedge r_n \wedge \ast(q_1 \wedge q_2 \wedge \cdots \wedge q_n).$$

We have two cases:

- (a)  $\vdash \beta \rightarrow \varphi$
- (b)  $\vdash \beta \rightarrow \neg\varphi$

In case (a), from  $\vdash \beta \rightarrow \varphi$ , we easily get  $\vdash * \beta \rightarrow * \varphi$ . But  $\vdash \alpha \rightarrow * \beta$ , using Exercise 6.5. (Actually, a slight generalization of this exercise is what is being used here.) And so  $\vdash \alpha \rightarrow * \varphi$ .

In case (b), we first see as above that  $\vdash \alpha \rightarrow * \neg \varphi$ . And this time, we use the fact that  $\vdash * \neg \varphi \rightarrow \neg * \varphi$  (see Example 6.22) to see that  $\vdash \alpha \rightarrow \neg * \varphi$ .

This completes the proof.  $\square$

### Lemma 6.32

Each  $\alpha \in \text{SD}_n$  is satisfiable.

*Proof.* Write  $\alpha$  as

$$q_1 \wedge q_2 \wedge \cdots \wedge q_n \wedge *(r_1 \wedge r_2 \wedge \cdots \wedge r_n),$$

where  $q_i$  is either  $p_i$  or  $\neg p_i$ , and  $r_i$  is either  $p_i$  or  $\neg p_i$ .

We make a model  $M$  as follows: Put  $p_i$  in room 1 iff  $q_i$  is  $p_i$ . (And so if  $q_i$  is  $\neg p_i$ , we don't put  $p_i$  in room 1.) Put  $p_i$  in room 2 iff  $r_i$  is  $p_i$ . (And so if  $r_i$  is  $\neg p_i$ , we don't put  $p_i$  in room 2.)

It is not hard to check that this model  $M$  has  $1 \Vdash \alpha$ .  $\square$

### Theorem 6.33

The logic for  $\mathcal{L}(*)$  is complete: if  $\models \varphi$ , then  $\vdash \varphi$ .

*Proof.* The proof is almost the same as that of Theorem 5.45, with a few changes. Appeals to Lemma 5.39 need to be replaced by appeals to Lemma 6.31, and similarly for Lemma 5.41 and Lemma 6.31, and also Lemma 5.43 and Lemma 6.32. We also need to change the proof of the Claim. For this, let us repeat the Claim and revise its proof.

**Claim** For all  $\alpha \in \text{SD}_n$ ,  $\vdash \alpha \rightarrow \varphi$ .

*Proof.* Suppose not. Let  $\alpha \in \text{SD}_n$  be such that  $\not\vdash \alpha \rightarrow \varphi$ . Then by Lemma 6.31,  $\vdash \alpha \rightarrow \neg \varphi$ . By Lemma 6.32, let  $M$  be a model and  $i$  a room such that  $i \Vdash \alpha$  in  $M$ . By assumption that  $\models \varphi$ ,  $i \Vdash \varphi$  in  $M$ . By soundness of the logic,  $\models \alpha \rightarrow \neg \varphi$ . So we also have  $i \Vdash \alpha \rightarrow \neg \varphi$ . But then we must have  $i \Vdash \neg \varphi$ , and this contradicts what we saw above:  $i \Vdash \varphi$ . So our claim is proved.  $\square$

The rest of the proof is the same as that of Theorem 5.45.  $\square$

## 6.5 Hilbert-style proof system for $\mathcal{L}(*)$

At this point, we have seen a natural deduction proof system for  $\mathcal{L}(*)$ , and we have shown it to be complete. As in Chapter 5, we also would like to present a Hilbert-style system. Such a system is easier to present, yet harder to use in deriving particular sentences since it is not designed for that purpose. We present such a system in Definition 6.36.

As you can see, it has *axioms* and *rules*. Going back to the Hilbert-style system for propositional logic, the axiom there would be “all tautologies,” and the rule of inference would be modus ponens.

(There would be no other rules or axioms.) In our system for  $\mathcal{L}(*)$ , we recall our discussion of substitution instances of tautologies from page 146. The tautology in (6.5) would count as an *axiom* in our Hilbert-style system, a sentence which could appear on a line without further ado.

**Remark 6.34**

Note that the determinacy axiom in this system is a biconditional:  $\neg * \varphi \leftrightarrow * \neg \varphi$ . The reason we take this to be a biconditional rather than simply the conditional  $\neg * \varphi \rightarrow * \neg \varphi$  is that I do not know whether the system would be complete with only the conditional assertion. **This is an open problem that I need to clear up.**

As before, we say that  $\Gamma \vdash_{\mathbf{H}} \varphi$  if there is a *formal derivation*. This would be sequence of lines, and each is either an element of  $\Gamma$  (an assumption in the derivation), or a tautology, or an instance of one of the two axioms, or comes from previous lines using one of the two rules of the system.

**Lemma 6.35**

If  $\Gamma \vdash_{\mathbf{H}} \varphi$ , then  $\Gamma \models_G \varphi$ .

*Proof.* By induction on the length of derivations in the Hilbert system. If  $\Pi$  is a derivation of length 1 whose last line is  $\varphi$  and with all assumptions coming from  $\Gamma$ , then either  $\varphi$  itself comes from  $\Gamma$  (in which case  $\Gamma \models_G \varphi$  is easy), or else  $\varphi$  is a tautology (in which case we have the stronger assertion that  $\models_G \varphi$ ). Now assume our result for derivations of length  $n$ , and let  $\Pi'$  have length  $n+1$ . If the last line of  $\Pi$  is justified by modus ponens, the argument is easy. If it is justified by the  $*$ -Rule, then  $\varphi$  is of the form  $*\psi$ . We argue as follows. By induction hypothesis, we have  $\Gamma \models_G \psi$ . It is easy to see semantically that  $\Gamma \models_G *\psi$ . And this means that  $\Gamma \models_G \varphi$ . Omitting some routine steps, this completes our proof.  $\square$

fill in more on  
above!

**Definition 6.36: The Hilbert-style logical system for  $\mathcal{L}(*)$** 

|               |                 |                                                                        |
|---------------|-----------------|------------------------------------------------------------------------|
| <b>Axioms</b> | All tautologies |                                                                        |
|               | (Involution)    | $\varphi \rightarrow **\varphi$                                        |
|               | (Determinacy)   | $*\neg\varphi \leftrightarrow \neg*\varphi$                            |
|               | (Normality)     | $*(\varphi \rightarrow \psi) \rightarrow (*\varphi \rightarrow *\psi)$ |
| <b>Rules</b>  | (Modus Ponens)  | From $\varphi$ and $\varphi \rightarrow \psi$ , infer $\psi$           |
|               | (Switch Rule)   | From $\varphi$ , infer $*\varphi$                                      |

**6.5.1 Using the system**

We present a few examples of how the Hilbert-style system is used.



**Example 6.37**

Example 6.20 presented an incorrect derivation in the natural deduction system. We cannot derive  $water \rightarrow *water$  in the Hilbert-style system, either. However, we do have a derivation of what looks like a similar result:  $water \vdash_H *water$ :

|   |          |            |
|---|----------|------------|
| 1 | $water$  | assumption |
| 2 | $*water$ | * rule, 1  |

It is important to see the difference between two assertions:  $\vdash_H water \rightarrow *water$  and  $water \vdash_H *water$ . The first is the incorrect assertion that we can prove  $water \rightarrow *water$  from no assumptions in the Hilbert-style system. (If we could prove it, then by soundness it would be true that in all rooms of all models, if water was present in that room, it would also be present in the other room of the house.) The second is the correct assertion that we can prove  $*water$  using  $water$  as a global assumption. It would say that if a given model has water in both rooms, then from both rooms it would be true that in the *other* room, water would be found.

7

**Example 6.38**

We saw that  $food \rightarrow *water, food \models_G water$ . Let us check now that

$$food \rightarrow *water, food \vdash_H water.$$

|   |                             |                    |
|---|-----------------------------|--------------------|
| 1 | $food \rightarrow *water$   | assumption         |
| 2 | $food$                      | assumption         |
| 3 | $*water$                    | modus ponens, 1, 2 |
| 4 | $**water$                   | switch rule, 3     |
| 5 | $**water \rightarrow water$ | involution axiom   |
| 6 | $water$                     | modus ponens, 4, 5 |

The important step here is step (4) where we used the switch rule. This rule reflects our use of global reasoning; and the rule would be unsound in a system for local reasoning.

---

<sup>7</sup>the next one could be an exercise

**Example 6.39**

You might like to find a derivation showing that  $\vdash_H (*\varphi \wedge *\psi) \rightarrow *(\varphi \wedge \psi)$ . As a hint, use the propositional logic tautology

$$(p \rightarrow q) \rightarrow ((r \rightarrow p) \rightarrow (r \rightarrow q)).$$

It takes a lot of experience (and/or luck) to solve problems like this. Rest assured that finding such derivations is not a very important skill to have, even for a student of logic.

**Theorem 6.40: Completeness**

$$\vdash_H \varphi \text{ iff } \models \varphi.$$

*sketch.* One way to prove this would be to repeat the steps in the proof of Theorem 6.33 but to work in the Hilbert-style system. Here is a general discussion of what is involved. First, one needs a version of Lemma 6.30, saying that for all  $n$ ,  $\vdash_H \bigvee \text{SD}_n$ . The proof of Lemma 6.30 basically involved only propositional reasoning and the determinacy axiom, and so there is a version of it using  $\vdash_H$ . In the course of Lemma 6.31, we needed the proof system in two places. We needed to know that from  $\vdash \beta \rightarrow \varphi$  we can show  $\vdash *\beta \rightarrow *\varphi$ . In the Hilbert-style system, this is a consequence of the  $*$ -necessitation rule and the normality axiom. We also needed to know facts like

$$\vdash (\varphi \wedge *\psi) \rightarrow *(\psi \wedge *\varphi)$$

(see Exercise 6.5). In the Hilbert-style system, the proof of this would involve the results of Example 6.39 and Exercise 6.18.  $\square$

## 6.6 Strong completeness for local and global consequence

As we prepare to end this chapter, we bring everything together.

**Theorem 6.41**

Concerning the local and global consequence relations and the two deduction systems of this chapter:

1.  $\varphi_1, \dots, \varphi_n \models_L \psi$  iff  $\varphi_1, \dots, \varphi_n \vdash \psi$ .
2.  $\varphi_1, \dots, \varphi_n \models_G \psi$  iff  $\varphi_1, \dots, \varphi_n \vdash_H \psi$ .

In words, the local consequence relation corresponds to derivations with premises in the natural deduction system, and the global consequence relation corresponds to Hilbert-style natural deduction system.

*Proof.* Here are some sketches indicating how we prove the first assertion. The right-to-left direction is Theorem 6.28. For the left-to-right direction of (1), assume that  $\varphi_1, \dots, \varphi_n \models_L \psi$ . Then by Theorem 6.15,  $\models (\varphi_1 \wedge \dots \wedge \varphi_n) \rightarrow \psi$ . So by our Completeness Theorem 6.33,  $\vdash (\varphi_1 \wedge \dots \wedge \varphi_n) \rightarrow \psi$ .

A derivation showing  $(\varphi_1 \wedge \cdots \wedge \varphi_n) \rightarrow \psi$  from no premises can easily be converted into a derivation with assumptions  $\varphi_1, \dots, \varphi_n$  showing  $\psi$ .

We turn to (2). The left-to-right direction of (1) is an induction on derivations in the Hilbert-style system. The only interesting step is for the switch rule. This corresponds to the following semantic fact: if  $\varphi_1, \dots, \varphi_n \models_G \psi$ , then also  $\varphi_1, \dots, \varphi_n \models_G *\psi$ . And this last fact is easy to check.

The right-to-left direction of (2) is perhaps the most interesting point in this theorem. Suppose that  $\varphi_1, \dots, \varphi_n \models_G \psi$ . Then by Theorem 6.16,

$$\models (\varphi_1 \wedge *\varphi_1 \wedge \cdots \wedge \varphi_n \wedge *\varphi_n) \rightarrow \psi. \quad (6.9)$$

Hence the sentence just above is derivable in the Hilbert-style system, by Theorem 6.40. Let  $\Pi$  be a derivation in the system which witnesses (6.9). Here is how to use  $\Pi$  to get a derivation showing  $\varphi_1, \dots, \varphi_n \vdash_H \psi$ . We start with line 1 containing  $\varphi_1$  as an assumption, then line 2 is  $*\varphi_1$  (obtained by the switch rule), then line 3 is  $\varphi_2$  (an assumption), line 4 is  $*\varphi_2$  (switch rule), etc. After getting to  $*\varphi_n$ , we use propositional reasoning to get The conjunction on the left of (6.9). And so using modus ponens, we get  $\psi$  the desired conclusion.  $\square$

### 6.6.1 Summary of our work on $\mathcal{L}(*)$

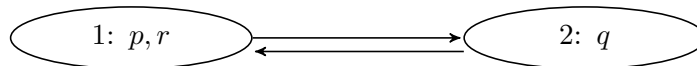
$\mathcal{L}(*)$  is an example of a modal logic because  $*$  is an operation taking sentences to sentences. The language was introduced in order to model reasoning where the “here” location of sentence evaluation might change in a very simple way, alternating between two possible locations. We started with a formal semantics in a particular model. We also saw different modes of reasoning: implicit structural, global, and local. These were presented intuitively, but they were connected to formal definitions of two consequence relations  $\Gamma \models_G \varphi$  and  $\Gamma \models_L \varphi$ . Consideration of the local relation  $\Gamma \models_L \varphi$  led us to a set of semantic results based on the mirror model construction. We then fashioned two proof systems, a natural deduction system and a Hilbert-style system. Each of these turned out to be (weakly) complete in the sense that valid sentences are provable. The systems differed when we considered the relations  $\Gamma \models_G \varphi$  and  $\Gamma \models_L \varphi$ . The first of these corresponds to provability in the Hilbert-style system using hypotheses, and the second to provability in the natural deduction system, again using hypotheses.

As we turn to other modal logics, we’re going to follow the same pattern of discussion. We’ll aim towards modalities of more conceptual interest.

## 6.7 Exercises on $\mathcal{L}(*)$

### 6.7.1 Routine exercises on understanding the semantics and the proof systems

**Exercise 6.1.** Here is a model  $M$  for  $\mathcal{L}(*)$ :



So in room 1,  $p$  and  $r$  are true, and  $q$  is false. In room 2,  $q$  is true, and  $p$  and  $r$  are false.

For each of the following assertions, say whether true or false with a short explanation:

1.  $1 \Vdash p \wedge *q$ .
2.  $2 \Vdash *(p \wedge *q)$
3.  $1 \Vdash *(*p \wedge q)$ .
4.  $2 \Vdash ****q$ .

Here is the solution to (a), to illustrate the amount of detail to aim for:

$1 \Vdash p$  because  $p$  is written in the world 1. Similarly,  $2 \Vdash q$ . Now  $1 \Vdash *q$  because  $2 \Vdash q$ . And since  $1 \Vdash p$  and  $1 \Vdash *q$ , we have  $1 \Vdash p \wedge *q$ .

**Exercise 6.2.** We showed a model  $M$  in problem 6.1 above, and we saw that in  $M$ ,  $1 \Vdash p \wedge *q$ .

1. Draw the “mirror image” model  $M^*$ .
2. Is it true that  $1 \Vdash p \wedge *q$  in  $M^*$ ?
3. Is it true that  $2 \Vdash p \wedge *q$  in  $M^*$ ?
4. What does Lemma 6.11 tell us about this model?

**Exercise 6.3.** Show the following in the natural deduction proof system:

1.  $\vdash * \neg * \varphi \rightarrow ** \neg \varphi$ .
2.  $\vdash maze \rightarrow **** maze$ .
3.  $\vdash \neg * \perp$ .

**Exercise 6.4.** Filling in what we saw in Example 6.24, show that  $food \rightarrow *water, * \neg water \vdash \neg food$ .  
[Hint: use the result of Example 6.22.]

**Exercise 6.5.** Here are some more exercises on the natural deduction proof system.

1.  $\vdash \neg \neg \varphi \rightarrow \varphi$
2.  $\vdash (p \wedge *q) \rightarrow *(q \wedge *p)$ .

**Exercise 6.6.** What is wrong with the following formal derivation in the Hilbert-style system of this chapter?

|   |                        |                                  |
|---|------------------------|----------------------------------|
| 1 | $p_1$                  | assumption                       |
| 2 | $*p_1$                 | Switch Rule, 1                   |
| 3 | $p_1 \rightarrow *p_1$ | $\rightarrow$ -introduction, 1–3 |

### 6.7.2 Exercises on the notions of validity and semantic consequence

**Exercise 6.7.** Show that for *all* models  $M$ ,  $1 \Vdash *(p \rightarrow q) \rightarrow (*p \rightarrow *q)$ . This is a short verification in English. Here is how it starts:

Fix a model  $M$ . We want to show that  $1 \Vdash *(p \rightarrow q) \rightarrow (*p \rightarrow *q)$ . If  $1 \not\Vdash *(p \rightarrow q)$ , we're done. So we can assume that  $1 \Vdash *(p \rightarrow q)$ . We want to show that  $1 \Vdash *p \rightarrow *q$ . As before, if  $1 \not\Vdash *p$ , we are done. So we can assume that  $1 \Vdash *p$ . It's our job to show that  $1 \Vdash *q$ . *You do the rest yourself.*

**Exercise 6.8.** Determine whether each of the following sentences is satisfiable, and also whether it is valid.

1.  $p_1 \rightarrow *p_1$
2.  $p_1 \rightarrow \neg *p_1$
3.  $(*p_1) \wedge (*\neg p_1)$
4.  $(p_1 \wedge *p_2) \vee \neg(p_1 \wedge *p_2)$
5.  $* * * * p_2 \rightarrow *p_2$

[Hint: For the satisfiability part, you might start by evaluating these sentences in both rooms of the model in Example 6.3 above.]

**Exercise 6.9.** Decide which are valid and which are invalid:

1.  $\varphi \rightarrow * * * * \varphi$
2.  $\varphi \rightarrow *\varphi$
3.  $\neg\varphi \rightarrow *\varphi$
4.  $* * \varphi \leftrightarrow \varphi$
5.  $(*\varphi \wedge *\psi) \rightarrow *(\varphi \wedge \psi)$
6.  $*(\varphi \rightarrow \psi) \rightarrow (*\varphi \rightarrow *\psi)$

For the valid ones, give a short proof. For the invalid ones, give a counterexample.

**Exercise 6.10.** One of these is true, one is false. Which is which?

1.  $treadmill \models_L *treadmill$ .
2.  $treadmill \models_G *treadmill$ .

**Exercise 6.11.** Prove that the following are equivalent:

1.  $\varphi_1, \dots, \varphi_n \models_1 \psi$
2.  $\varphi_1, \dots, \varphi_n \models_2 \psi$

**Exercise 6.12.** One of these is true, one is false. Which is which?

1.  $\varphi_1, \dots, \varphi_n \models_L \psi$  iff  $\models (\varphi_1 \wedge \dots \wedge \varphi_n) \rightarrow \psi$ .
2.  $\varphi_1, \dots, \varphi_n \models_G \psi$  iff  $\models (\varphi_1 \wedge \dots \wedge \varphi_n) \rightarrow \psi$ .

**Exercise 6.13.** True or false?  $\varphi_1, \dots, \varphi_n \models_L \psi$  implies  $\varphi_1, \dots, \varphi_n \models_G \psi$ .

**Exercise 6.14.** Recall from Exercise 5.25 that that for propositional logic  $\mathcal{L}$ ,

$$\varphi \text{ is satisfiable iff } \not\models \neg\varphi$$

Show that the same thing is true for  $\mathcal{L}(*):$

$$\varphi \text{ is satisfiable iff } \not\models_G \neg\varphi$$

“Satisfiable” means true in one of the two rooms of some model, and  $\not\models \neg\varphi$  means that  $\models_G \varphi$  is false.

[The reasoning here is almost the same what we saw before, so you can look up your answer to the earlier problem and modify it a little.]

**Exercise 6.15.** Is it true that  $\vdash (*\varphi \rightarrow *\psi) \rightarrow *(\varphi \rightarrow \psi)$ ? Note that this is the converse of the normality axiom that we saw in connection with the Hilbert-style system.

If the sentence is provable, give a derivation. If it is false, give a concrete counterexample (that is, actual sentences instead of the variables  $\varphi$  and  $\psi$ ) along with a reason why  $\not\vdash (*\varphi \rightarrow *\psi) \rightarrow *(\varphi \rightarrow \psi)$ .

[If you are stuck on this, you might like to first think semantically. Is it true or not that  $\models (*\varphi \rightarrow *\psi) \rightarrow *(\varphi \rightarrow \psi)$ ? If you think the answer is yes, then you will want to come up with a derivation in our formal system. If you think the answer is no, then your counterexample plus soundness implies that  $\not\vdash (*\varphi \rightarrow *\psi) \rightarrow *(\varphi \rightarrow \psi)$ .]

**Exercise 6.16.** Using the definitions of  $\models_L$  and  $\models_G$ , tell whether each of the following assertions is true or false, with a reason:

1.  $p \models_L *p$
2.  $p \models_G *p$
3.  $p, *p \models_L **p$
4.  $p, *p \models_L *(p \wedge *p)$

**Exercise 6.17.** Prove that the following are equivalent:

1.  $\varphi_1, \varphi_2, \dots, \varphi_n \models_G \psi$
2.  $\models_L (\varphi_1 \wedge * \varphi_1 \wedge \varphi_2 \wedge * \varphi_2 \wedge \dots \wedge \varphi_n \wedge * \varphi_n) \rightarrow \psi$

[Hint: use Theorem 6.13.]

**Exercise 6.18.** Show in the Hilbert system the converse of what was shown in Example 6.39. That is, give a derivation of  $*(\varphi \wedge \psi) \rightarrow (*\varphi \wedge *\psi)$ . This is not as hard as the derivation in Example 6.39.

### 6.7.3 More difficult exercises

**Exercise 6.19.** Show in the natural deduction system that  $\vdash \varphi \leftrightarrow \neg * \neg * \varphi$ .

**Exercise 6.20.** Is our proof system for  $\mathcal{L}(*)$  *decidable*? Is there an analog of truth tables for  $\mathcal{L}(*)$ ? That is, given a sentence  $\varphi$  in the logic, is there a procedure like the truth-table test for validity?

**Exercise 6.21.** Exercise 5.21 presented the Compactness Theorem for propositional logic.

1. Adapt the method of Exercise 5.21 to show the same result for  $\mathcal{L}(*)$ : if  $\Gamma$  is a set of sentences with the property that each finite subset has a model, then  $\Gamma$  itself has a model.
2. Use the previous part to show the strong completeness theorems for the logical systems of this chapter: For all sets  $\Gamma$  and all sentences  $\psi$ :
  - (a)  $\Gamma \models_L \psi$  iff  $\Gamma \vdash \psi$ .
  - (b)  $\Gamma \models_G \psi$  iff  $\Gamma \vdash_H \psi$ .

This strengthens the result of Theorem 6.41, and you should also use Theorem 6.41 in your work here.

**Exercise 6.22.** (Open problem) Suppose that we replace the two axioms in the natural deduction system with the axioms  $\varphi \leftrightarrow \neg * \neg * \varphi$ . Is the resulting system complete?

**Exercise 6.23.** When we discussed global and local reasoning in Section 6.2, we said that it might be most useful to have a logical system that combined global and local reasoning. Let us define  $\Gamma; \Delta \models \varphi$  to mean that for all models  $M$ , and both rooms  $i$  of  $M$ , if every sentence in  $\Gamma$  is true in both rooms of  $M$ , and if all sentences in  $\Delta$  are true in room  $i$ , then also  $\varphi$  is true in room  $i$ . Modify one of the proof systems in this chapter to propose a proof system of your own, thereby defining a new proof relation  $\Gamma; \Delta \vdash_{my} \varphi$  and prove that your system is sound and complete in the sense that  $\Gamma; \Delta \models \varphi$  iff  $\Gamma; \Delta \vdash_{my} \varphi$ , for all  $\Gamma$ ,  $\Delta$ , and  $\varphi$ .

# Chapter 7

## Relational Semantics

### 7.1 Flavors of structure

Up until now, we have seen logical languages interpreted in various ways. The specific examples have been

- (1) a language about knowledge of cards in Section ??;
- (2) the logic of the interior operation on subsets of  $\mathbf{R}$ , in Section ??;
- (3) propositional logic, interpreted via valuations in Chapter 5;
- (4) the logic  $\mathcal{L}(*)$ , interpreted on very special “rat houses” in Chapter 6.

In this chapter, we unify several of these semantics, the ones that are based on graphs, or something close to graphs.

As it happens, we have a plethora of structural notions that occur on several different “dimensions.”

**Relations and graphs** We begin our study with a discussion of relations and graphs.

#### Definition 7.1: relations

A *relation* on a set  $S$  is a set of ordered pairs from  $S$ .

A *graph* is a structure

$$\mathcal{G} = (G, \rightarrow),$$

such that  $G$  is a set, and  $\rightarrow$  is a relation on  $G$ . The elements of  $G$  are called the *points*, *nodes*, or *vertices* of the graph. The set  $G$  is called the *domain* of the graph  $\mathcal{G}$ . ( $G$  is also called the *underlying set*, or even the *universe* of  $\mathcal{G}$ ).

In this section, we’ll use letters like  $g$  and  $h$ , and also letters like  $x, y, z, \dots$  for the points in a given graph. (These are sometimes called *vertices* or *nodes*), and we write  $x \rightarrow y$  to mean that in the graph  $\mathcal{G}$ ,  $x \rightarrow y$ .

Let us expand on the definition of a graph a bit. By a *structure*, we mean a set together with “extra stuff” in the form of subsets, relations, elements, and the like. We consider the set with the



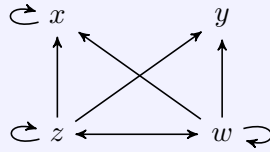
extra stuff to be a single object, as usual in mathematics. We also usually refer to the structure either by the name of its set part, or else we chose a similar name. We do the latter in this chapter using calligraphic letters like  $\mathcal{G}$  for the structures, corresponding to its underlying set  $G$ .

### Example 7.2

As an example of a graph  $\mathcal{G}$ , we take the set  $G = \{x, y, z, w\}$  together with the relation

$$\{(x, x), (z, x), (z, y), (z, z), (z, w), (w, x), (w, y), (w, z), (w, w)\}.$$

It is convenient to picture the graph by taking all of the pairs and drawing an arrow from the first component to the second. We would have the picture below:



In full glory, our graph is a pair

$$\mathcal{G} = (G, \{(x, x), (z, x), (z, y), (z, z), (z, w), (w, x), (w, y), (w, z), (w, w)\}, \{x, z\})$$

**Models** We fix a set  $\text{AtSen}$  of atomic sentences.

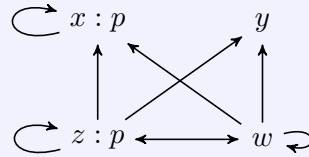
A *model* is a graph which also comes with a *valuation*, a function

$$\text{val} : \mathcal{G} \rightarrow \mathcal{P}(\text{AtSen}).$$

This set  $\mathcal{P}(\text{AtSen})$  is the set of all subsets of  $\text{AtSen}$ . So when we write  $\text{val} : \mathcal{G} \rightarrow \mathcal{P}(\text{AtSen})$ , we are saying that for each point  $g$  of the graph,  $\text{val}(g)$  is an element of  $\mathcal{P}(\text{AtSen})$ , hence a subset of  $\text{AtSen}$ , hence a set of atomic sentences. Since the model has more structure than the graph, we say that the model is *based on* the graph.

**Example 7.3**

As an example of a model  $\mathcal{M}$ , we have the picture below:



This model is based on the graph shown in Example 7.2. It also has the extra information that  $p$  is true exactly at  $x$  and  $z$  (and as a result,  $p$  is false at  $y$  and  $w$ ). If one wants to see this model displayed as a bona fide mathematical structure, it would be

$$\mathcal{M} = (\{x, y, z, w\}, \{(x, x), (z, x), (z, y), (z, z), (z, w), (w, x), (w, y), (w, z), (w, w)\}, (p, \{x, z\}))$$

This  $\mathcal{M}$  is a triple. But we might also re-package  $\mathcal{M}$  a bit and write  $\mathcal{M} = (G, \text{val})$ , where  $\mathcal{G}$  is the graph from Example 7.2. This is convenient when describing a model whose underlying graph is already known.

**Example 7.4**

All of our work in Chapter 6 dealt with models which were based on the graph

$$\text{room 1} \longleftrightarrow \text{room 2}$$

We didn't mention this graph since it was the only one around, and we omitted valuations but instead wrote the content of the valuations inside the rooms.

**7.1.1 Pointed versions, and multi-agent versions**

Returning for a moment to the kinds of pictures we saw back in Chapter 0, there were two aspects of the knowledge graphs that are as yet missing from our structures. The first is the *real world*. We add this to our structure by moving from graphs to *pointed graphs*, and from models to *pointed models*. The name “pointed” comes from areas of mathematics where one talks about *pointed sets*, these are pairs  $(S, s)$ , where  $S$  is a set and  $s \in S$ . So a pointed set is a set together with one of its points, singled out for some purpose or other.

**Definition 7.5: pointed graphs and models**

A *pointed graph* is a pair  $\mathcal{G} = (\mathcal{G}, g)$ , where  $\mathcal{G}$  is a graph and  $g \in \mathcal{G}$ .

A *pointed model* is a pair  $\mathcal{M} = (\mathcal{M}, m)$ , where  $\mathcal{M}$  is a model and  $m \in \mathcal{M}$ .

Note that we are implicitly using some of our conventions about naming here. The ‘ $G$ ’ in the definition of a pointed graph refers to the underlying set of  $\mathcal{G}$ , and similarly for ‘ $M$ ’ and  $\mathcal{M}$ . Also, we usually refer to pointed structures by the name of the underlying un-pointed structure.

In addition to the real world, the graphs of Chapter 0 had more than one type of “arrow”; they had one relation for each agent. In general, we might want to talk about one agent, two, three, or many. So in a specific discussion, we’ll usually have a set  $\mathbf{Ag}$  of *agents* around. Usually we’ll name them with lower-case Roman letters at the beginning of the alphabet.

**Definition 7.6: multi-agent (pointed) graphs and models**

Let  $\mathbf{Ag}$  be a set of agents. A *multi-agent graph over  $\mathcal{A}$*  is a tuple

$$\mathcal{G} = (G, (R_a)_{a \in \mathbf{Ag}}),$$

where  $G$  is a set, and for each  $a \in \mathbf{Ag}$ ,  $R_a$  is a relation on  $G$ .

A *multi-agent model over  $\mathcal{A}$*  is a pair  $\mathcal{M} = (\mathcal{M}, \text{val})$ , where  $\mathcal{G}$  is a multi-agent graph over  $\mathcal{A}$ , and  $\text{val} : M \rightarrow \mathcal{P}(\text{AtSen})$  is a valuation.

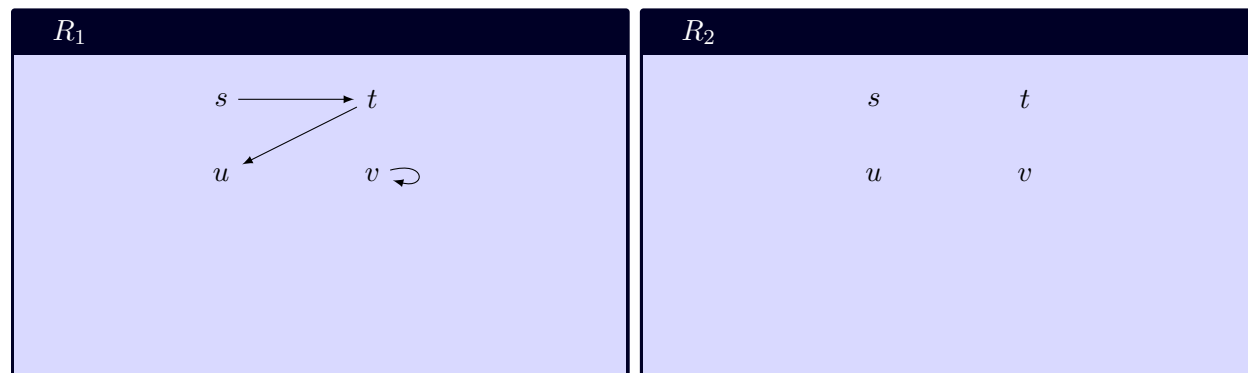
We also speak of pointed multi-agent graphs, and of pointed multi-agent models. These are defined in the obvious ways.

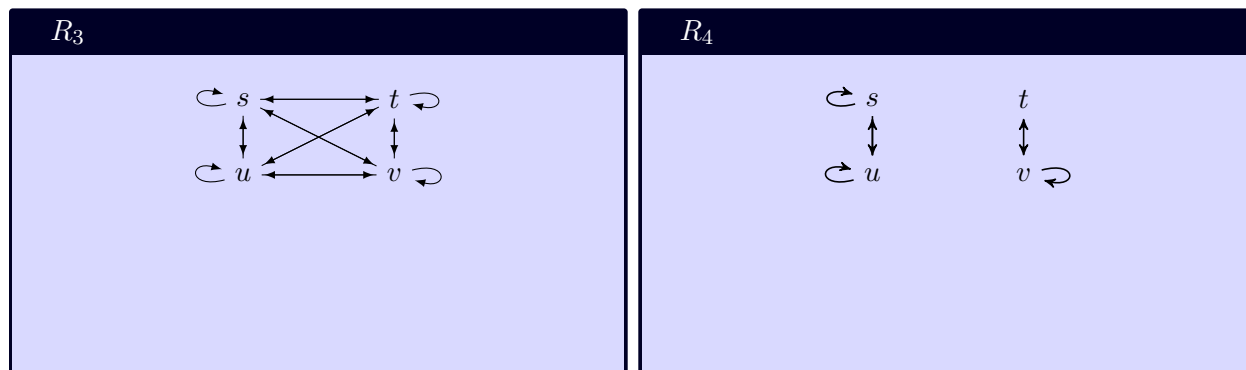
## 7.2 Relations and their properties

In the last section, we introduced several types of structure: graphs, models, multi-agent models, and the pointed versions of all of these. We’ll now take off in a different direction by looking at some definitions that pertain specifically to relations.

We remind you that given a set  $S$ , a *relation on  $S$*  is a set of ordered pairs from  $S$ .

We are going to discuss four examples extensively in this section. They all have  $S = \{s, t, u, v\}$  as their underlying set. They are called  $R_1$ ,  $R_2$ ,  $R_3$ , and  $R_4$ . They are shown below.





As sets, the relations are just what we said they were, sets of pairs. Please remember that the order in which we list the elements of a set do not matter. So even though we say that  $R_1 = \{(s, t), (t, u), (v, v)\}$ , we could equally well write it as  $\{(t, u), (v, v), (s, t)\}$ . We could also repeat elements, since this does not change the nature of a set. It would be confusing to do so, but it would nevertheless be correct to write

$$R_1 = \{(t, u), (t, u), (s, t), (v, v), (v, v), (v, v), (v, v), (t, u)\}.$$

On the other hand, it would be a big mistake to interchange the elements of any ordered pair inside of a relation:

$$R_1 \neq \{(t, s), (t, u), (v, v)\}.$$

The pictures of each relation are obtained from the sets in the following way. If a relation contains  $(x, y)$ , then we put an arrow from  $x$  to  $y$ :  $x \rightarrow y$ . (If a relation has both  $(x, y)$  and  $(y, x)$ , we usually put in a double-headed arrow  $x \leftrightarrow y$  instead of two arrows  $x \rightarrow y$  and  $y \rightarrow x$ . We do this mainly to avoid cluttering the pictures.) These kinds of pictures are called *graphs*, just as we saw in the previous section.

Notice that  $R_2$  is empty. The empty set  $\emptyset$  is a perfectly good relation on any set.

We write  $R_3$  as  $S \times S$ . This is a shorthand notation for a set with 16 pairs, so we should explain how it works. For any two sets  $A$  and  $B$ , the *product set*  $A \times B$  is the set of ordered pairs  $(a, b)$  with  $a$  from  $A$  and  $b$  from  $B$ . In symbols,

$$A \times B = \{(a, b) : a \in A, b \in B\}. \quad (7.1)$$

This is sometimes called the *cartesian product* of the sets  $A$  and  $B$ . For example,

$$\begin{aligned} \{i, j\} \times \{e, f, g\} &= \{(i, e), (i, f), (i, g), (j, e), (j, f), (j, g)\} \\ \{e, f, g\} \times \{i, j\} &= \{(e, i), (f, i), (g, i), (e, j), (f, j), (g, j)\} \end{aligned}$$

Returning to  $R_3$ , we used the notation  $S \times S$  both to shorten our notation and to emphasize that it contains all 16 ordered pairs from  $S$ .

Notice that a relation on  $S$  is the same thing as a subset of  $S \times S$ .

**Notation** We sometimes write, for example,  $s R_4 u$ , instead of the more strictly correct statement  $(s, u) \in R_4$ . The reason is that we think of relations in mathematics like  $<$ . One would never write  $(3, 4) \in <$ , but always writes  $3 < 4$ . We also sometimes will write  $x \rightarrow y$  instead of  $(x, y) \in R$  or  $x R y$ , if the relation  $R$  is understood from the context.

**Summary** The statements below all say the same thing:

$$x R y \quad (x, y) \in R \quad x \rightarrow y \quad x \text{ is related to } y \text{ by } R$$

We decide whether to use one or another of these in a given passage based on the style and content of the passage. For example, when talking about graphs, it is natural to use the arrow notation  $x \rightarrow y$ . If there is a good reason to emphasize that a relation is a set of ordered pairs, it would be sensible to use  $(x, y) \in R$ .

### 7.2.1 Definitions concerning relations

We are going to study various properties of relations, most notably the ones listed in Definition 7.7. Each of the conditions is stated in a shorthand kind of way. For example, the condition of reflexivity at the top of the figure should be understood in more expansive way as: for all sets  $S$  and all relations  $R$  on  $S$ ,  $R$  is *reflexive* if for all  $x \in S$ ,  $x R x$ .

#### Definition 7.7: properties of relations

**reflexive** for all  $x$ ,  $x R x$ .

**symmetric** for all  $x$  and  $y$ , if  $x R y$ , then  $y R x$ .

**anti-symmetric** for all  $x$  and  $y$ , if  $x R y$ , and also  $y R x$ , then  $x = y$ .

**transitive** for all  $x$ ,  $y$ , and  $z$ , if  $x R y$  and  $y R z$ , then also  $x R z$ .

**preorder** reflexive and transitive.

**partial order** reflexive, transitive, and anti-symmetric.

**equivalence relation** reflexive, symmetric, and transitive.

**serial** for all  $x$ , there is some  $y$  such that  $x R y$ .

**Euclidean** for all  $x$ ,  $y$ , and  $z$ , if  $x R y$  and  $x R z$ , then  $y R z$ .

**partial function** for all  $x$ ,  $y$ , and  $z$ , if  $x R y$  and  $x R z$ , then  $y = z$ .

**Example 7.8**

Here is how the relations which we mentioned before fare.

|             | $R_1$ | $R_2$ | $R_3$ | $R_4$ |
|-------------|-------|-------|-------|-------|
| reflexive   | ×     | ×     | ✓     | ×     |
| symmetric   | ×     | ✓     | ✓     | ✓     |
| transitive  | ×     | ✓     | ✓     | ×     |
| equivalence | ×     | ×     | ✓     | ✓     |
| serial      | ×     | ×     | ✓     | ✓     |
| Euclidean   | ×     | ✓     | ✓     | ×     |
| partial fn. | ✓     | ✓     | ×     | ×     |

✓ = yes; × = no

For each of the defining conditions, we'll re-state the definition and then discuss examples, and non-examples.

**7.2.2 Reflexive, symmetric, and transitive**

$R$  is *reflexive* if for all  $x \in S$ , the pair  $(x, x)$  belongs to  $R$ . In words, every point is related to itself. Looking back at our examples concerning  $S = \{s, t, u, v\}$ , we see that  $R_3$  is a reflexive relation on  $S$ , but  $R_1$ ,  $R_2$ , and  $R_4$  are not reflexive.

One fine point on the definition of *reflexive* is that it depends not only on the set of ordered pairs, but also on the underlying set. For example, we took  $R_3$  to be a graph on the set  $S = \{s, t, u, v\}$ . But suppose we enlarge  $S$  to the set  $S' = \{s, t, u, v, w\}$ . Then the same relation  $R_3$  is a relation on the bigger set  $S'$ , because it is a set of ordered pairs that come from  $S'$ . But as a relation on  $S'$ ,  $R_3$  is *not reflexive*: it does not contain  $(w, w)$ .

$R$  is *symmetric* if for all  $s \in S$ , if the pair  $(x, y)$  belongs to  $R$ , then so does  $(y, x)$ . In other words, this would mean that all of the arrows in the picture of a given relation are *two-way* arrows.

We return to our examples concerning  $S = \{s, t, u, v\}$ .  $R_2$ ,  $R_3$ , and  $R_4$  are symmetric but  $R_1$  is not, as you can easily see from the pictures.

But how would we tell symmetry or non-symmetry from the listing of the arrows in a relation (as a set of ordered pairs), rather than from the picture? Let us look at  $R_4$  in a little more detail. The definition of symmetry may be written in logic as  $(\forall x)(\forall y)(x R y \rightarrow y R x)$ . Note that we need to check this even when  $y$  and  $x$  are the same!

| $x$ | $y$ | $x R y$ | $y R x$ | $x R y \rightarrow y R x$ |
|-----|-----|---------|---------|---------------------------|
| $s$ | $s$ | T       | T       | T                         |
| $s$ | $t$ | F       | F       | T                         |
| $s$ | $u$ | T       | T       | T                         |
| $s$ | $v$ | F       | F       | T                         |
| $t$ | $s$ | F       | F       | T                         |
| $t$ | $t$ | T       | T       | T                         |
| $t$ | $u$ | F       | F       | T                         |
| $t$ | $v$ | F       | F       | T                         |

| $x$ | $y$ | $x R y$ | $y R x$ | $x R y \rightarrow y R x$ |
|-----|-----|---------|---------|---------------------------|
| $u$ | $s$ | T       | T       | T                         |
| $u$ | $t$ | F       | F       | T                         |
| $u$ | $u$ | T       | T       | T                         |
| $u$ | $v$ | F       | F       | T                         |
| $v$ | $s$ | F       | F       | T                         |
| $v$ | $t$ | F       | F       | T                         |
| $v$ | $u$ | F       | F       | T                         |
| $v$ | $v$ | T       | T       | T                         |

The fact that we get  $\top$  in all rows means that our relation  $R_4$  really is symmetric. Actually, the same is true of  $R_2$ : Even though there are no arrows in the picture, it counts as symmetric. The reason is that if we fill in the same chart for  $R_2$ , then all of the entries in the column  $x R y$  are  $\text{F}$ , and so all of the entries in the column for the conditional are  $\top$  (!). We say that  $R_2$  is symmetric *vacuously*.

It is usually possible to tell whether a relation has some property by making a table like this<sup>1</sup>. However, the actual verification would be a laborious task.

Returning to our general definitions, a relation  $R$  on a set  $S$  is *transitive* if  $x$  can go to  $y$  and  $y$  to  $z$ , then  $x$  can go to  $z$ . In first-order terms, it is: if  $(\forall x)(\forall y)(\forall z)(x R y \wedge y R z \rightarrow x R z)$ .  $R_1$  is not transitive. To see this, note that  $s \rightarrow t$  and  $t \rightarrow u$ , but  $s \not\rightarrow u$ . (To verify that a relation is does not have a certain property, we only need to find one counterexample.)  $R_2$  is transitive; this holds vacuously, as in our verification of the symmetry of  $R_2$ .  $R_3$  is transitive because all pairs are related.  $R_4$  is not transitive because  $t \rightarrow v$  and  $v \rightarrow t$ , but  $t \not\rightarrow t$ .

### 7.2.3 Preorder, partial order, and equivalence relation

A *preorder* is a relation which is reflexive and transitive. Often one uses a symbol like  $\leq$  for preorders, since this reminds one of the defining properties. But preorders need not be *antisymmetric*. It need not be the case that if  $x \leq y$  and  $y \leq x$ , then  $x = y$ . If this property holds, the preorder is said to be a *partial order*. In our examples,  $R_2$  and  $R_3$  are preorders, but they are not partial orders.  $R_1$  and  $R_4$  are not transitive, so they are automatically not preorders.

#### Example 7.9

Another important class of preorders comes from logic. Consider the set of sentences in propositional logic, and consider the relation  $\leq$  on this set defined by  $\varphi \leq \psi$  iff  $\vdash \varphi \rightarrow \psi$ . This is easily seen to be a preorder. It is not a partial order because  $p \leq p \wedge p$  and  $p \wedge p \leq p$ . (The point is that  $p$  and  $p \wedge p$  are not literally the same.)

$R$  is an *equivalence relation* if  $R$  is reflexive, symmetric, and transitive. In our example relations, only  $R_3$  is an equivalence relation.

The picture of an equivalence relation is that it is a *disjoint set of cliques*. In more detail, a *clique* is a set of points with the property that all points in the set are neighbors of all points (including themselves). A picture is a disjoint set of cliques if it consists of a bunch of cliques with no arrows between them. For example, in  $R_3$ , the entire set  $S$  is a clique. In  $R_4$ , the only clique is  $\{s, u\}$ .

Another way to talk about equivalence relations is in terms of *partitions*.

<sup>1</sup>This is always possible, provided the property in question is expressible in *first-order logic*. All of the conditions studied in this chapter have this feature, but many definitions in graph theory are not first order. For example, we say that a graph is *planar* if it can be drawn on a piece of paper without crossing edges. This condition is important in graph theory, but it is provably not expressible in first-order logic.

**Definition 7.10: partition**

Let  $S$  be any set. A *partition* of  $S$  is a family<sup>a</sup>  $\Pi$  of subsets of  $S$  with the following properties:

1. Every  $X \in \Pi$  is a non-empty subset of  $S$ .
2. If  $X \neq Y$  are different elements of  $\Pi$ , then  $X \cap Y = \emptyset$ .
3. If  $a \in S$ , then there is some  $X \in \Pi$  such that  $a \in X$ .

We frequently call the elements of a partition the *cells* of the partition.

<sup>a</sup>A *family* of sets is just a set of sets. There is no special feature to this terminology.

**Example 7.11**

Let  $S = \{1, 2, 3, 4, 5, 6\}$ . Then the following family of sets is a partition of  $S$ :

$$\{\{1\}, \{2, 3, 6\}, \{4, 5\}\}$$

That is,  $\Pi$  is a set of non-empty sets which cover  $S$  (the union of the sets in  $\Pi$  is all of  $S$ ), and different elements of  $\Pi$  have nothing in common.

Another partition would be  $\{S\}$ . This is a partition with just one cell, the whole set  $S$ .

On the other hand, the following is *not* a partition of  $S$ :

$$\{\{1\}, \{2, 3, 5, 6\}, \{4, 5\}\}$$

The problem is that the elements of  $\Pi$  are not pairwise disjoint:  $\{2, 3, 5, 6\} \cap \{4, 5\} \neq \emptyset$ .

**Proposition 7.12**

Let  $\Pi$  be a partition of  $S$ . Then the following relation  $\equiv_\Pi$  on  $S$  is an equivalence relation:

$$s \equiv_\Pi t \quad \text{iff} \quad s \text{ and } t \text{ are in the same cell of } \Pi.$$

*Proof.* Every  $s$  is in the same cell of  $\Pi$  as  $s$  itself, so  $\equiv_\Pi$  is reflexive.

If  $s$  is in the same cell as  $t$ , then  $t$  is in the same cell as  $s$ . Thus  $\equiv_\Pi$  is symmetric.

Suppose that  $s \equiv_\Pi t$  and  $t \equiv_\Pi u$ . Then  $s$  is in the same cell as  $t$ , and  $t$  in the same cell as  $u$ . So  $s$  is in the same cell as  $u$ . This proves that  $\equiv_\Pi$  is transitive.  $\square$

In the other direction, we start with an equivalence relation  $\equiv$  and we associate to  $\equiv$  a family of sets. For each  $s \in S$ , the set  $\{t : t \equiv s\}$  is called the *equivalence class* of  $s$ . This is often written  $[s]$ . Note that if  $s \equiv s'$ , then  $[s] = [s']$ . (This is an important point!) We need to check that the family of all equivalence is a partition of  $S$ . Each equivalence class is non-empty:  $s \in [s]$ . If  $[s] \cap [t] \neq \emptyset$ , we can see that  $[s] = [t]$  as follows: Let  $x \in [s] \cap [t]$ . Then  $s \equiv x \equiv t$ . So by transitivity,  $s \equiv t$ . Thus  $[s] = [t]$ . Finally, each  $s \in S$  belongs to  $[s]$ . This proves the following result.



**Proposition 7.13**

Every equivalence relation  $\equiv$  on a set  $S$  gives a partition  $\Pi_{\equiv}$ ; the cells of  $\Pi_{\equiv}$  are the equivalence classes  $[s]$ .

**7.2.4 Serial, Euclidean, and partial function**

$R$  is *serial* if  $(\forall x)(\exists y)(x R y)$ . In words, every point has an outgoing edge.

$R$  is *Euclidean* if  $(\forall x)(\forall y)(\forall z)(x R y \wedge x R z \rightarrow y R z)$ . In words, if a point  $x$  has arrows to points  $y$  and  $z$  (one or both of those could be  $x$  itself, and  $y$  could be identical to  $z$ ), then there is an arrow from  $y$  to  $z$ .  $R_1$  is not Euclidean: for a specific counterexample, note that  $t \rightarrow u$  and  $t \rightarrow u$ , but  $u \not\rightarrow u$ .

$R$  is a *partial function* if  $x \rightarrow y$  and  $y \rightarrow z$  implies that  $y \rightarrow z$ . In pictures, every point has at most one outgoing arrow. (“At most one” means zero or one.)

**7.2.5 Relations between sets**

Our definition of a relation on page 163 defined the concept of a relation on a set. We also need a related concept.

**Definition 7.14: relations between sets**

Let  $S$  and  $T$  be relations. A *relation from  $S$  to  $T$*  is a subset  $R \subseteq S \times T$ .

**7.2.6 Multi-agent, pointed, equivalence models**

Now we go back to the discussion of knowledge in Section ???. We want to explain what the pictures that we saw were pictures of, using the terminology that we introduced in this chapter.

Let  $\text{AtSen}$  be a set of atomic sentences, and  $\text{Ag}$  be a set of *agents* (or *players*).

**Definition 7.15: equivalence structures and knowledge structures**

A *knowledge structure over  $\text{Ag}$  and  $\text{AtSen}$*  is a pointed, multi-agent model  $\mathcal{M}$  over  $\text{Ag}$  and  $\text{AtSen}$ .

$\mathcal{M}$  is an *equivalence structure* if each agent’s accessibility relation  $R_a$  is an equivalence relation.

The pictures in Section ??? showed examples of equivalence structures. When dealing with knowledge, one standardly works with such structures. However, we’ll have occasion to also use the more general notion of a knowledge structure.

**7.3 Interpreting modal logic on relational models**

The following is perhaps the standard choice for a basic modal logic.

**Definition 7.16:**  $\mathcal{L}(\Box, \Diamond)$ 

Let  $\text{AtSen}$  be a set of atomic sentences.  $\mathcal{L}$  is the language defined as follows: The elements  $p \in \text{AtSen}$  are sentences, as are the *constants*  $\top$  and  $\bot$ . We construct complex sentences using the boolean connectives  $\neg$ ,  $\wedge$ ,  $\vee$ ,  $\rightarrow$ , and  $\leftrightarrow$ , and also the modal operators  $\Box$  and  $\Diamond$ . These modal operators are read as “box” and “diamond”; they also have other names in various subjects.

Definition 7.16 presents the *syntax* of  $\mathcal{L}(\Box, \Diamond)$ . By the *semantics* of our basic modal language, we mean the definitions that explain how it is to be interpreted.

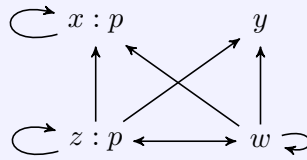
**Definition 7.17:** semantics of modal logic on graphs

|                             |     |                                                               |
|-----------------------------|-----|---------------------------------------------------------------|
| $x \Vdash p$                | iff | the model says so                                             |
| $x \Vdash \neg p$           | iff | $x \not\Vdash p$                                              |
| $x \Vdash \top$             |     | always                                                        |
| $x \Vdash \bot$             |     | never                                                         |
| $x \Vdash \Box \varphi$     | iff | for all $y$ such that $x \rightarrow y$ , $y \Vdash \varphi$  |
| $x \Vdash \Diamond \varphi$ | iff | for some $y$ such that $x \rightarrow y$ , $y \Vdash \varphi$ |

Incidentally, we can interpret  $\mathcal{L}(\Box, \Diamond)$  on graphs even when  $\text{AtSen} = \emptyset$ . That is, even when there are no atomic sentences, there still are sentences like  $(\Diamond \top) \rightarrow \Box(\top \vee \bot)$ . Our semantics allows us to interpret these sentences on graphs. That is, a valuation is not needed for sentences that don’t contain atomic sentences.

**Example 7.18**

We present a number of examples to show how the semantics works, using the model from Example 7.3, repeated below:



Since  $p$  is true at  $x$  and  $z$  but not at  $y$  and  $w$ ,

$$x \Vdash p \quad y \not\Vdash p \quad z \Vdash p \quad w \not\Vdash p$$

Now our semantics of negation tells us

$$x \not\Vdash \neg p \quad y \Vdash \neg p \quad z \not\Vdash \neg p \quad w \Vdash \neg p$$

**Example 7.19**

The sentences  $\top$  and  $\text{F}$  have fixed interpretations

$$\begin{array}{cccc} x \Vdash \top & y \Vdash \top & z \Vdash \top & w \Vdash \top \\ x \nVdash \text{F} & y \nVdash \text{F} & z \nVdash \text{F} & w \nVdash \text{F} \end{array}$$

Actually, these facts have nothing to do with the particular model  $\mathcal{M}$  at hand; they are true for all points in all models.

**Example 7.20**

Next, let's look at the formula  $\Box \text{F}$ . Again, we are concerned with our model  $\mathcal{M}$ . This sentence  $\Box \text{F}$  is only true at an *endpoint*, a point which has not outgoing arrows. [The reason is as follows: if  $x$  is not an endpoint, then we have some  $y$  such that  $x \rightarrow y$ . Then  $x$  does not satisfy  $\Box \text{F}$  because it is not true that all of the successors of  $x$  satisfy  $\text{F}$ ; in particular,  $y$  does not satisfy  $\text{F}$ .] For our model  $\mathcal{M}$ , this means  $y$ . At all other points, the sentence  $\Box \text{F}$  is false. And the *dual* of this sentence is  $\Diamond \top$ . This is only true at points which have some outgoing arrow:  $x$ ,  $z$ , and  $w$ . To summarize:

$$\begin{array}{cccc} x \nVdash \Box \text{F} & y \Vdash \Box \text{F} & z \nVdash \Box \text{F} & w \nVdash \Box \text{F} \\ x \Vdash \Diamond \top & y \nVdash \Diamond \top & z \Vdash \Diamond \top & w \Vdash \Diamond \top \end{array}$$

**7.3.1 The semantic notions  $\models$ ,  $\models_{\mathcal{S}}$ , etc.**

We have already seen the semantic notions  $x \Vdash \varphi$ , and now we need some further definitions of various *consequence relations on classes of relational models*.

**Definition 7.21: more semantic notions**

$\models_{\mathcal{R}} \varphi$  means that for all models  $\mathcal{M}$  whose underlying graph is reflexive, and all points  $x \in M$ ,  $x \Vdash \varphi$ .

$\models_{\mathcal{S}} \varphi$  means the same thing, replacing “reflexive” with “symmetric”.

We similarly define  $\models_{\mathcal{T}} \varphi$  (transitive),  $\models_{\mathcal{Eu}} \varphi$  (Euclidean),  $\models_{\mathcal{R},\mathcal{S}} \varphi$  (reflexive and transitive),  $\models_{\mathcal{eq}} \varphi$  (equivalence relations), etc.

We frequently use  $\mathcal{C}$  as a variable over *classes of models*. (“Classes” here just means collections.) We have in mind cases when  $\mathcal{C}$  is the class of all models based on a reflexive graph, or the class of models based on an equivalence relation, etc. As these examples suggest, usually  $\mathcal{C}$  is the class of all models based on some fixed class of graphs. So when we write, for example  $\models_{\mathcal{C}} \varphi$ , this means any of  $\models_{\mathcal{all}} \varphi$ ,  $\models_{\mathcal{R}} \varphi$ ,  $\models_{\mathcal{S}} \varphi$ , etc. We adopt this notation whenever it shortens the statements of the results without making things too abstract.

A proof of a statement  $\models_{\mathcal{C}} \varphi$  typically starts out

“Let  $\mathcal{M} = (M, \rightarrow, \text{val})$  be a relational model.”

Typically, it will use the defining properties of the class  $\mathcal{C}$ . And it must deal with the formal semantics of the language  $\mathcal{L}(\Box, \Diamond)$ .

To show a statement  $\not\models_{\mathcal{C}} C\varphi$ , we need to find a counterexample. We only need one, but we should be specific. That is, we need to find a model  $\mathcal{M}$  in the class  $\mathcal{C}$ , and a sentence  $\varphi$ , and some point  $x \in G$  such that  $x \not\models \varphi$ .

### Theorem 7.22

The following soundness assertions hold.

1. If  $\varphi$  is a substitution instance of a tautology, then  $\models_{\text{all}} \varphi$ .
2.  $\models_{\text{all}} \Diamond\varphi \leftrightarrow \neg\Box\neg\varphi$ .
3.  $\models_{\text{all}} \Box(\varphi \rightarrow \psi) \rightarrow (\Box\varphi \rightarrow \Box\psi)$ .
4.  $\models_{\text{r}} \Box\varphi \rightarrow \varphi$ .
5.  $\models_{\text{s}} \varphi \rightarrow \Box\Diamond\varphi$ .
6.  $\models_{\text{t}} \Box\varphi \rightarrow \Box\Box\varphi$ .
7.  $\models_{\text{ser}} \Box\varphi \rightarrow \Diamond\varphi$ .
8.  $\models_{\text{Eu}} \Diamond\varphi \rightarrow \Box\Diamond\varphi$ .
9.  $\models_{\text{pf}} \Diamond\varphi \rightarrow \Box\varphi$ .
10. If  $\mathcal{C}$  is any class of models and  $\models_{\mathcal{C}} \varphi \rightarrow \psi$ , and also  $\models_{\mathcal{C}} \varphi$ , then  $\models_{\mathcal{C}} \psi$ .
11. If  $\mathcal{C}$  is any class of models and  $\models_{\mathcal{C}} \varphi$ , then also  $\models_{\mathcal{C}} \Box\varphi$ .

*Proof.* We prove each assertion in turn.

1. Let  $\varphi$  be a substitution instance of a tautology. Let us say that  $\psi$  is a propositional tautology, and  $s : \text{AtSen} \rightarrow \mathcal{L}(\Box, \Diamond)$  is a function which assigns to each atomic sentence  $p_i$  some modal sentence  $s(p_i)$ . Then  $s$  extends to a function  $\bar{s} : \mathcal{L}(\text{AtSen}) \rightarrow \mathcal{L}(\Box, \Diamond)$  in the obvious way, and  $\bar{s}(\psi) = \varphi$ . To see that  $\varphi$  is valid, let  $\mathcal{M}$  be a model and  $x$  one of its points. Define a valuation  $v$  by  $v(p_i) = \text{T}$  iff  $x \models s(p_i)$ . An easy induction<sup>2</sup> on sentences then shows that for all sentences  $\chi$  of propositional logic,  $\bar{v}(\chi) = \text{T}$  iff  $x \models \bar{s}(\chi)$ . Applying this with  $\chi = \psi$ , we see that  $\bar{v}(\psi) = \text{T}$  iff  $x \models \varphi$ . But  $\psi$  is a tautology, and so  $\bar{v}(\psi)$  must be true for all valuations  $v$ . Hence we see that  $x \models \varphi$ , as desired.
2. Let  $\mathcal{M}$  be a model and let  $x \in M$ . Assume first that  $x \models \Diamond\varphi$ . Thus we have some  $y$  such that  $x \rightarrow y$  and  $y \models \varphi$ . If it were true that  $x \models \Box\neg\varphi$ , then this  $y$  would satisfy  $\neg\varphi$ , and it doesn't. So we must have  $x \models \neg\Box\neg\varphi$ . This shows that  $\models_{\text{all}} \Diamond\varphi \rightarrow \neg\Box\neg\varphi$ . The converse assertion is similar.

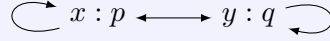
<sup>2</sup>The induction is an easy one if you are familiar with induction. The fact that induction is needed makes this the most difficult part of this entire result.

3. Let  $\mathcal{M}$  be a model and let  $x \in M$ . Assume that  $x \Vdash \Box(\varphi \rightarrow \psi)$ . To show that  $x \Vdash (\Box\varphi \rightarrow \Box\psi)$ , assume that  $x \Vdash \Box\varphi$ . We show that  $x \Vdash \Box\psi$ . For this, let  $x \rightarrow y$ . Then we know two things about  $y$ : first,  $y \Vdash \varphi \rightarrow \psi$ ; and second,  $y \models \varphi$ . So we must have  $y \models \psi$ . This for all  $y$  shows that  $x \Vdash \Box\psi$ . So we are finished.
4. Here is the reasoning. Let  $\mathcal{M} = (M, \rightarrow, \text{val})$  be a reflexive model and let  $x \in M$ . We must show that  $x \Vdash \Box\varphi \rightarrow \varphi$ . We may assume that  $w \Vdash \Box\varphi$ . But since  $\rightarrow$  is reflexive,  $w \rightarrow w$ . And so by the semantics of  $\Box$ ,  $w \Vdash \varphi$ . This completes the proof.
5. Let  $\mathcal{M} = (M, \rightarrow, \text{val})$  be a symmetric model and let  $x \in M$ . We must show that  $x \Vdash \varphi \rightarrow \Box\Diamond\varphi$ . We may assume that  $x \Vdash \varphi$ . In order to show  $x \Vdash \Box\Diamond\varphi$ , suppose that  $x \rightarrow y$ . But since  $\rightarrow$  is symmetric,  $y \rightarrow x$ . Since  $x \Vdash \varphi$ ,  $y \Vdash \Diamond\varphi$ . And since  $y$  is arbitrary,  $x \Vdash \Box\Diamond\varphi$ .
6. Let  $\mathcal{M} = (M, \rightarrow, \text{val})$  be a transitive model and let  $x \in M$ . Assume that  $x \Vdash \Box\varphi$ ; we show  $x \Vdash \Box\Box\varphi$ . Let  $x \rightarrow y$ . To show  $y \Vdash \Box\varphi$ , we let  $y \rightarrow z$  and show  $z \models \varphi$ . We have  $x \rightarrow y$  and  $y \rightarrow z$ ; since  $\rightarrow$  is transitive,  $x \rightarrow z$ . This with  $x \Vdash \Box\varphi$  tells us that  $z \models \varphi$ , as desired.
7.  $\models_{\text{ser}} \Box\varphi \rightarrow \Diamond\varphi$ . Let  $\mathcal{M} = (M, \rightarrow, \text{val})$  be a serial model and let  $x \in M$ . Assume that  $x \Vdash \Box\varphi$ ; we show  $x \Vdash \Diamond\varphi$ . By seriality, let  $x \rightarrow y$ . We have  $y \models \varphi$  because  $x \Vdash \Box\varphi$ . The existence of  $y$  shows that  $x \Vdash \Diamond\varphi$ .
8.  $\models_{\text{Eu}} \Diamond\varphi \rightarrow \Box\Diamond\varphi$ : We leave this to you.
9. Let  $\mathcal{M} = (M, \rightarrow, \text{val})$  be a partial function model and let  $x \in M$ . Assume that  $x \Vdash \Diamond\varphi$ ; we show  $x \Vdash \Box\varphi$ . Let  $x \rightarrow y$  be such that  $y \Vdash \varphi$ . To show  $x \Vdash \Box\varphi$ , let  $x \rightarrow z$ . We want to see that  $z \Vdash \varphi$ . But since  $x \rightarrow y$  and  $x \rightarrow z$ , we have  $y = z$ . Thus indeed  $z \models \varphi$ .
10. If  $\mathcal{C}$  is any class of models and  $\models_{\mathcal{C}} \varphi \rightarrow \psi$ , and also  $\models_{\mathcal{C}} \varphi$ , then  $\models_{\mathcal{C}} \psi$ . Let  $\mathcal{M}$  be a model in  $\mathcal{C}$ , and let  $x \in M$ . Our hypotheses imply that  $x \Vdash \varphi \rightarrow \psi$ , and also  $x \Vdash \varphi$ . Thus  $x \Vdash \psi$ .
11. Let  $\mathcal{M}$  be a model in  $\mathcal{C}$ , and let  $x \in M$ . To see that  $x \Vdash \Box\varphi$ , let  $x \rightarrow y$ . Since  $\varphi$  is true at each point of every model in  $\mathcal{C}$ ,  $y \models \varphi$ . This for all  $y$  shows that  $x \Vdash \Box\varphi$ .

□

**Example 7.23**

We show here that  $\not\models_{\text{eq}} p \rightarrow \Box p$ . Consider the following model:



$x \models p$ , but  $y \not\models p$ . The picture shows that  $x \rightarrow y$ . Therefore  $x \not\models \Box p$ . In other words,  $x \not\models p \rightarrow \Box p$ . Note that  $q$  plays no role here. We could have made our model



with the understanding that  $p$  is false at  $y$ . Further, if we say up-front that the model is to be reflexive and symmetric, it would be simpler to picture it without the loops and double-headed arrows:

$$x : p \text{ ————— } y \quad (7.2)$$

**Example 7.24**

Let us show that  $\not\models_{\text{eq}} p \rightarrow \Box p$ . To do this, we exhibit a model  $\mathcal{M}$  based on an equivalence relation, and a point  $z \in M$  such that  $z \models \Diamond p$ , but  $z \not\models \Box p$ . We take  $\mathcal{M}$  to be the model in (7.2) above, and  $z$  to be  $y$ .

**7.3.2 Duals of the axioms**

You know the *de Morgan laws* from basic logic:

$$\begin{aligned} \neg(\varphi \wedge \psi) &\equiv \neg\varphi \vee \neg\psi \\ \neg(\varphi \vee \psi) &\equiv \neg\varphi \wedge \neg\psi \end{aligned}$$

And also the laws of moving negation across  $\forall$  and  $\exists$ :

$$\begin{aligned} \neg(\forall x)\varphi(x) &\equiv (\exists x)\neg\varphi(x) \\ \neg(\exists x)\varphi(x) &\equiv (\forall x)\neg\varphi(x) \end{aligned}$$

The same *duality phenomenon* can be found with  $\Box$  and  $\Diamond$ :

$$\begin{aligned} \neg\Box\varphi &\equiv \Diamond\neg\varphi \\ \neg\Diamond\varphi &\equiv \Box\neg\varphi \end{aligned}$$

This lets us move negation across  $\Box$  and  $\Diamond$ , flipping them.

For example,

$$\Box\Diamond\Box\neg p \equiv \neg\Diamond\Box\Diamond p.$$

We want to go a step further. Let's look for a moment at the 4 axioms  $\Box\varphi \rightarrow \Box\Box\varphi$ . We always mean this as an axiom scheme, an infinite collection of axioms, one for each sentence  $\varphi$  in

$\mathcal{L}(\Box, \Diamond)$ . Fix a sentence  $\varphi$ . Then one of our axioms is  $\Box\neg\varphi \rightarrow \Box\Box\neg\varphi$ . The contrapositive here is  $\neg\Box\Box\neg\varphi \rightarrow \neg\Box\neg\varphi$ . And this is  $\Diamond\Diamond\varphi \rightarrow \Diamond\varphi$ . Stepping back and considering  $\varphi$  as a variable here, we have constructed the *dual axiom scheme* to the 4 axioms.

### 7.3.3 Semantics on models and multi-agent models

Up until now, we have only been dealing with a language  $\mathcal{L}(\Box, \Diamond)$ . We interpreted this on graphs. When we turn to models, we expand the language to a larger one called  $\mathcal{L}(\Box, \Diamond, \text{AtSen})$ . This language has as atomic sentences the elements of  $\text{AtSen}$ . The semantics is just what we expect:  $x \Vdash p$  in  $\mathcal{M}$  iff  $p \in \text{val}(x)$ .

We also want to expand the language to the multi-agent setting. So the language would be called  $\mathcal{L}(\Box, \Diamond, \text{AtSen}, \text{Ag})$ . This time, instead of the operators  $\Box$  and  $\Diamond$ , we have versions of these indexed by the different agents. For each  $A \in \text{Ag}$ , we have operators  $\Box_a$  and  $\Diamond_a$ . The semantics is

$$\begin{array}{lll} x \Vdash \Box_a \varphi & \text{iff} & \text{for all } y \text{ such that } x \xrightarrow{a} y, y \Vdash \varphi \\ x \Vdash \Diamond_a \varphi & \text{iff} & \text{for some } y \text{ such that } x \xrightarrow{a} y, y \Vdash \varphi \end{array}$$

## 7.4 Bisimulation

We have already seen several situations which called for a *principled* definition of when “two models were basically equivalent.” The topic of *bisimulation* gives us the appropriate definitions.

There are several different, but related, definitions:

bisimulation on a single graph  
 bisimulation between graphs  
 bisimulation between models  
 bisimulation between pointed models  
 bisimulation between multi-agent models  
 bisimulation between pointed multi-agent models

There also are definitions of bisimulation *between two models* and bisimulation *on a model*.

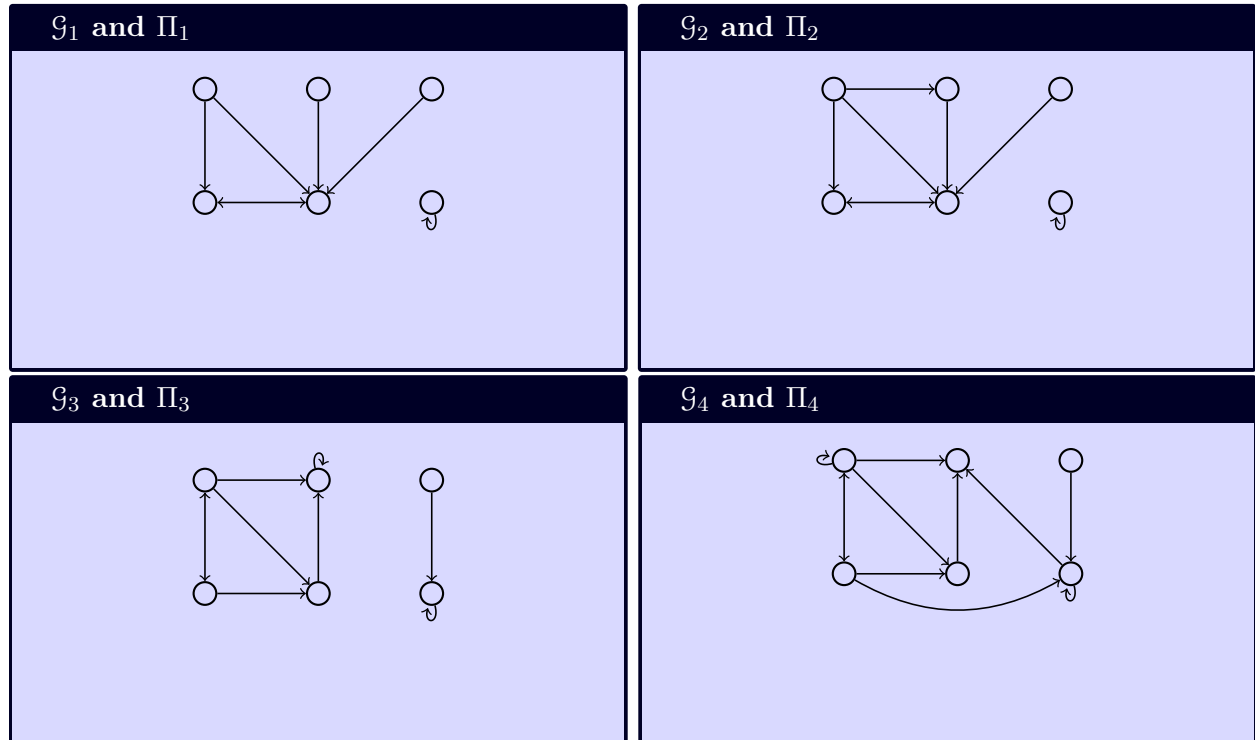
#### Definition 7.25: bisimulation on a graph

Let  $\mathcal{G} = (G, \rightarrow)$  be a graph. A *bisimulation on  $\mathcal{G}$*  is partition  $\Pi$  of the nodes of  $G$  with the following property: if  $x$  and  $y$  are in the same cell of  $\Pi$ , then  $x$  and  $y$  arrow into the same cells of  $\Pi$ . (That is, if one arrows into a given cell, then so does the other.)

As with all partitions, we frequently write  $x \equiv y$  to say that  $x$  and  $y$  are in the same cell of  $\Pi$ .

**Example 7.26**

Three examples and one non-example are shown below.  $\Pi_1$  is a bisimulation on  $\mathcal{G}_1$ . The points in the top cell arrow to the bottom cell (but not to the top), and the points in the bottom cell do not arrow to anything. On the other hand,  $\Pi_2$  is not a bisimulation on  $\mathcal{G}_2$ . The point in the upper-left arrows to the top cell, while the other points in that cell do not arrow into it.  $\Pi_3$  is again a bisimulation on  $\mathcal{G}_3$ .  $\Pi_4$  is a bisimulation on  $\mathcal{G}_4$ . This time, there is nothing really to check: the cells of  $\Pi_4$  are the singleton sets, and every partition of every graph into singleton cells is automatically a bisimulation.

**Theorem 7.27: bisimilar points are modally equivalent**

Let  $\mathcal{G}$  be a graph. Let  $\Pi$  be a bisimulation on a graph  $\mathcal{G}$ . If  $w \equiv x$ , then the worlds  $w$  and  $x$  satisfy all the same sentences:

$$w \models \varphi \quad \text{iff} \quad x \models \varphi.$$

for all sentences  $\varphi$ .

*Proof.* Fix  $\mathcal{G}$  and  $\equiv$ . We show by induction on  $\varphi$  that for all  $w \in M$  and all  $x \in N$ , if  $w \equiv x$ , then

$$w \models \varphi \quad \text{iff} \quad x \models \varphi.$$

The base cases are for  $\top$  and  $\perp$ , and they are easy.



Here is the induction step for  $\wedge$ : Assume our result for  $\varphi$  and for  $\psi$ ; we prove it for  $\varphi \wedge \psi$ . Assume that  $w \equiv x$ . Then

$$\begin{array}{ll}
 & w \models \varphi \wedge \psi \\
 \text{iff} & w \models \varphi \text{ and } w \models \psi \\
 \text{iff} & x \models \varphi \text{ and } x \models \psi \quad (\text{by our inductive assumption}) \\
 \text{iff} & x \models \varphi \wedge \psi
 \end{array}$$

The same steps apply for the other boolean connectives:  $\neg$ ,  $\rightarrow$ , and  $\leftrightarrow$ .

The most interesting step in the induction: assume our result for  $\varphi$ , and prove it for  $\Diamond\varphi$ . We thus assume that *for all pairs*  $w \equiv x$ ,  $w \models \varphi$  iff  $x \models \varphi$ . We prove that *for all pairs*  $v \equiv y$ ,  $v \models \Diamond\varphi$  iff  $y \models \Diamond\varphi$ . We only show one direction: *if*  $v \models \Diamond\varphi$ , then  $y \models \Diamond\varphi$ . (The converse is basically the same.)

So assume  $v \models \Diamond\varphi$ . Let  $z$  be such that  $v \rightarrow z$  and  $z \models \varphi$ .  $z$  belongs to some cell of  $\equiv$ . And  $v$  arrows into that cell. So  $y$  must arrow into that cell as well. Let  $y \rightarrow u$ , and  $u \equiv z$ . They by induction hypothesis,  $u \models \varphi$  iff  $z \models \varphi$ . So  $u \models \varphi$ . And we conclude that  $y \models \Diamond\varphi$ .  $\square$

**Bisimulation between graphs** So far, the concept in this section has been bisimulation *on a particular graph*. We also want the concept of a *bisimulation between two graphs*. Given graphs  $\mathcal{G}$  and  $\mathcal{H}$ , we write  $\mathcal{G} + \mathcal{H}$  for the *disjoint union* of the two graphs. In pictures, this means putting  $\mathcal{G}$  and  $\mathcal{H}$  next to each other with no edges between the two. If we have specific names of the nodes in the two graphs, then when we form the disjoint union we should be careful to “disjointify” the sets of nodes by picking new names, if necessary.

Given graphs  $\mathcal{G}$  and  $\mathcal{H}$ , a *bisimulation between  $\mathcal{G}$  and  $\mathcal{H}$*  is a bisimulation on the disjoint union  $\mathcal{G} + \mathcal{H}$ .

**Bisimulation between pointed graphs** If  $(\mathcal{G}, g)$  and  $(\mathcal{H}, h)$  are pointed graphs, then a *bisimulation between  $(\mathcal{G}, g)$  and  $(\mathcal{H}, h)$*  is a bisimulation  $\equiv$  between  $\mathcal{G}$  and  $\mathcal{H}$  with the extra property that  $g \equiv h$ . In words, it is a partition of the disjoint union which puts the two distinguished points  $g$  and  $h$  into the same cell, and which also has the property that we have seen: nodes in the same cell arrow to the same cells.

**Bisimulation on/between models, and between pointed models** Recall that a model is a graph together with a valuation function  $\text{val}$ .

**Definition 7.28: bisimulation on a model**

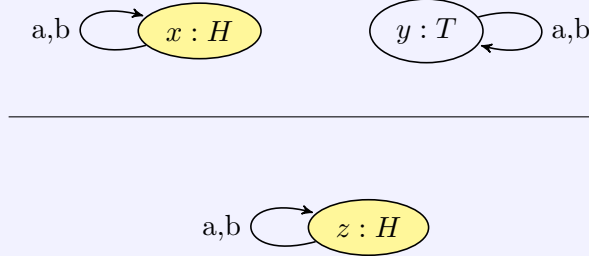
Let  $\mathcal{M} = (M, \rightarrow, \text{val})$  be a model. A *bisimulation on  $\mathcal{M}$*  is a bisimulation on the underlying graph  $(M, \rightarrow)$  which respects atomic sentences. In more detail, it is a partition  $\Pi$  of  $M$  equivalence relation  $\equiv$  on  $M$  with the following property: if  $x$  and  $y$  are in the same cell of  $\Pi$ , then  $\text{val}(x) = \text{val}(y)$ , and  $x$  and  $y$  arrow into the same cells of  $\Pi$ .

From this, we then define bisimulation *between models* and *between pointed models* exactly as before.

**Example 7.29**

Here is an example: from the coin scenarios.

We had two representations of what happens after  $A$  opens the box to see  $H$ :



The coarsest bisimulation on the disjoint union is  $\{\{x, z\}, \{y\}\}$ . Since this relates the two “real worlds,” the pointed models are bisimilar.

**Bisimulations between multi-agent models** Recall that a *multi-agent model*  $\mathcal{M}$  has relations  $R_a$  for each agent  $a$ . We define bisimulation on and between multi-agent graphs models in the same ways as above, except that we require in all cases that points in the same cell arrow *for each agent individually*, into the same cells.

At this point, we want to return to Theorem 7.27. That result was stated for graphs, but the result holds for models as well.

There is a partial converse to Theorem 7.27: the coarsest bisimulation between two finite models is the relation of satisfying the same modal sentences. This also happens when the models are *image finite*: this means that every point has only finitely many successors.

**Theorem 7.30: Hennessey-Milner Theorem**

Let  $\mathcal{M}$  and  $\mathcal{N}$  be *finite* multi-agent models. Let  $m \in M$  and  $n \in N$ . Assume that  $m$  and  $n$  satisfy the same sentences. Then there is a bisimulation  $\equiv$  between  $\mathcal{M}$  and  $\mathcal{N}$  such that  $m \equiv n$ .

*Proof.* Let  $\equiv$  be the relation on the disjoint union of the node sets  $M + N$  defined as follows:

$$x \equiv y \quad \text{iff} \quad \text{for all } \varphi, (x \Vdash \varphi \text{ iff } y \Vdash \varphi)$$

This relation is clearly an equivalence relation. It also preserves atomic sentences: if  $x \equiv y$ , then for atomic  $p$ ,  $x \Vdash p$  iff  $y \Vdash p$ . We check the main property: if  $x \equiv y$ , then  $x$  and  $y$  arrow for all agents into the same cells. Recall that the cells of the partition are the equivalence classes of  $\equiv$ .

So fix  $x$  and  $y$  such that  $x \equiv y$ . Fix an agent  $A$ . Suppose that  $x \xrightarrow{a} u$  so that  $x$  arrows into  $[u]$ . We need to see that  $y$  arrows (for this same agent  $A$ ) into  $[u]$ . We argue by contradiction. Suppose not. Then all of the worlds  $v$  such that  $y \xrightarrow{a} v$  are *not* in  $[u]$ . List these worlds as  $v_1, \dots, v_n$ . (This is where we use the assumption that the model is image finite.) This number  $n$  might be zero. Each  $v_i$  is not related to  $u$  by  $\equiv$ . Thus, for each  $v_i$  there is some sentence  $\psi_i$  such that  $v_i \models \psi_i$  and

$u \models \neg\psi_i$ . Notice now that

$$\begin{aligned} x &\models \Diamond_a((\neg\psi_1) \wedge \cdots \wedge (\neg\psi_n)) \\ y &\models \Box_a(\psi_i \vee \cdots \vee \psi_n) \end{aligned}$$

(If  $n = 0$ , then we want to say  $x \models \Diamond_a \top$ , and  $y \models \Box_a \text{F}$ .) The first of these implies that  $x \not\models \Box_a(\psi_i \vee \cdots \vee \psi_n)$ . But this means that  $x$  and  $y$  do not satisfy the same sentences. So we have our contradiction. This completes the proof.  $\square$

At this point, we need some general definitions concerning partitions.

**Definition 7.31: refinement**

Let  $\Pi$  and  $\Pi'$  be partitions of the same set  $S$ . We say that  $\Pi'$  *refines*  $\Pi$  if every cell of  $\Pi'$  is a subset of a cell of  $\Pi$ . In other words, if  $s \equiv_{\Pi'} t$ , then  $s \equiv_{\Pi} t$ . We write  $\Pi \leq \Pi'$ . We also say that  $\Pi$  is *coarser* than  $\Pi'$ , or that  $\Pi'$  is *finer* than  $\Pi$ .

The coarsest partition of all on  $S$  is  $\{S\}$ . We call this  $\perp$  since it is below all partitions in the order  $\leq$ . The finest partition is  $\{\{s\} : s \in S\}$ . This is the partition of  $S$  into its singleton subsets. For similar reasons, we call this  $\top$ . All partitions  $\Pi$  of this set  $S$  are between  $\perp$  and  $\top$ . That is, for all  $\Pi$ ,  $\perp \leq \Pi \leq \top$ .

**Example 7.32**

Let  $S = \{1, 2, 3, 4\}$ . Then  $\perp$  is  $\{\{1, 2, 3, 4\}\}$ , and  $\top = \{\{1\}, \{2\}, \{3\}, \{4\}\}$ . We also have

$$\perp \leq \{\{1, 2, 4\}, \{3\}\} \leq \{\{1\}, \{3\}, \{2, 4\}\} \leq \top$$

**The partition refinement algorithm for finding the coarsest bisimulation** When discussing bisimulation on a graph, one is usually interested in the *coarsest* bisimulation. The definition of a bisimulation does not immediately give a way to find the coarsest bisimulation. For that matter, it is not even clear that there *is* a unique coarsest bisimulation on a given graph. So more work has to be done to establish these facts. This is the content of the present section.

**Definition 7.33: refine( $\mathcal{G}, \Pi$ )**

Let  $(G, \rightarrow)$  be a graph, and let  $\Pi$  be a partition of the vertices of  $G$ . We get a new partition  $\text{refine}(\mathcal{G}, \Pi)$  in the following way. We put  $g$  and  $h$  in the same cell of  $\text{refine}(\mathcal{G}, \Pi)$  if they arrow into the same cells of  $\Pi$ .

**Algorithm 7.34:** to compute the coarsest bisimulation  $\Pi$  of a given finite graph  $\mathcal{G} = (G, \rightarrow)$ .

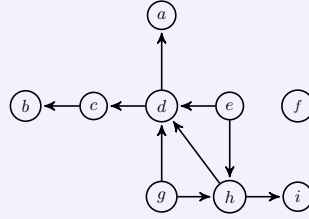
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1:  $n \leftarrow 0$ , and  $\Pi_0 \leftarrow \{G\}$ . ▷  $\Pi_0$  is the coarsest partition of  $G$ 
2: for all  $n$  do
3:    $\Pi_{n+1} \leftarrow \text{refine}(\mathcal{G}, \Pi_n)$ .
4:   if  $\Pi_{n+1} = \Pi_n$  then ▷ fixed point found
5:     output  $\Pi_n$ 
6:   else ▷ refine again
7:     return to line 2
8:   end if
9: end for

```

### Example 7.35

Algorithm 7.34 gives a *partition refinement algorithm* to compute the coarsest bisimulation on a given finite graph. We'll analyze it in the exercises to this chapter. But first, we step through the algorithm on the graph shown below:



Here is how the various steps work out:

$$\begin{aligned}
\Pi_0 &= \{\{a, b, d, c, d, e, f, g, i, h\}\} \\
\Pi_1 &= \{\{a, b, f, i\}, \{c, d, e, g, h\}\} \\
\Pi_2 &= \{\{a, b, f, i\}, \{c\}, \{d, h\}, \{e, g\}\} \\
\Pi_3 &= \{\{a, b, f, i\}, \{c\}, \{d\}, \{e, g\}, \{h\}\} \\
\Pi_4 &= \{\{a, b, f, i\}, \{c\}, \{d\}, \{e, g\}, \{h\}\}
\end{aligned}$$

Since  $\Pi_3 = \Pi_4$ , this partition is the coarsest bisimulation of the graph.

## 7.5 Tree unwindings of pointed models

We end this chapter with a way to take a pointed model  $(\mathcal{M}, x)$  and to convert it to a model which looks like an infinite tree. We'll call this new model  $\mathcal{T}$ . The points of  $\mathcal{T}$  are the finite non-empty sequences  $xx_1x_2 \cdots x_n$  drawn from the set  $M$  that start with  $x$  and follow the arrows of  $\mathcal{M}$ . So the points of  $\mathcal{T}$  are

Given such a sequence, we put an arrow from it to all the sequences that are one term longer. That is, if  $x_n \rightarrow y$  in  $\mathcal{M}$ , then the point above has an arrow to the following point in  $\mathcal{T}$ :

$$x = x_0 \quad x_1 \quad x_2 \quad x_3 \quad \cdots \quad x_n \quad y$$

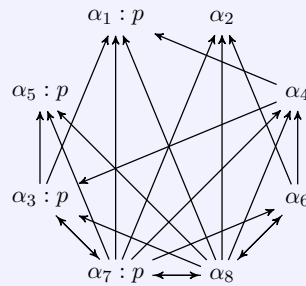


**Remark 7.37**

Here is a query: Suppose that  $\mathcal{M}$  is a model and  $x$  is a point in it, and it just so happens that  $x \models \Box\varphi \rightarrow \varphi$  for all sentences  $\varphi$ . Does it follow that  $x \rightarrow x$ ? If  $x \models \Box\varphi \rightarrow \varphi$  for all  $x \in M$  and all  $\varphi$ , does it follow that  $\mathcal{M}$  is reflexive?

Neither assertion follows. We'll just check that the first one fails by giving an example. Consider a one-point reflexive model  $x \rightarrow x$ , with no atomic sentences true at  $x$ . Call this model  $\mathcal{M}$ . Then  $\mathcal{M}$  is reflexive, and so in  $\mathcal{M}$ ,  $x \models \Box\varphi \rightarrow \varphi$  for all sentences  $\varphi$ . Let  $(\mathcal{T}, x)$  be the tree unwinding of  $(\mathcal{M}, x)$ . As a tree,  $\mathcal{T}$  looks like an infinite line; it is isomorphic to  $0 \rightarrow 1 \rightarrow 2 \rightarrow \dots$ . Trees are not reflexive, and indeed they have no reflexive points at all. As we saw above, there is a bisimulation between  $\mathcal{M}$  and  $\mathcal{T}$ . By Theorem 7.27,  $x$  satisfies all the same sentences as each  $t \in T$ . So the one-point sequence  $x$ , as an element of  $\mathcal{T}$  satisfies all modal sentences of the form  $\Box\varphi \rightarrow \varphi$ .

## 7.6 Exercises on relations and relational semantics

**Example 7.38: a graph for use in Exercise 7.1**


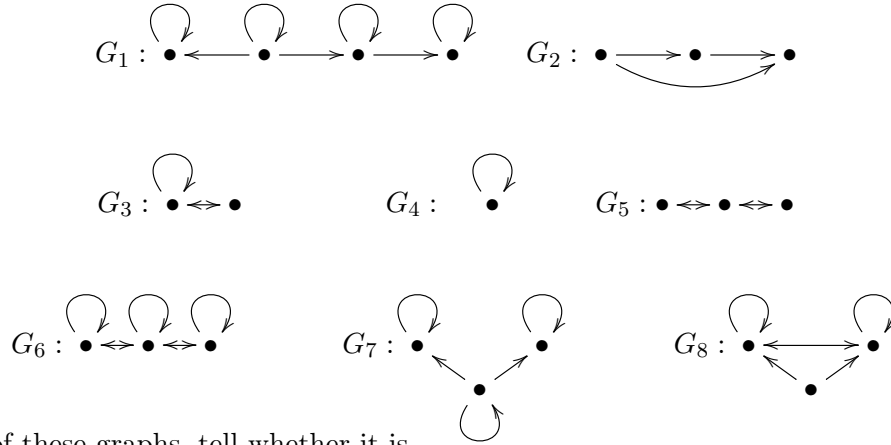
Note that the atomic sentence  $p$  is true at  $\alpha_1, \alpha_3, \alpha_5$ , and  $\alpha_7$ ;  $p$  is false at the other four points.

**Exercise 7.1.** A particular model is shown in Example 7.38. Fill in the chart below, using  $\checkmark$  for

**yes** and  $\times$  for **no**. For example, the  $\checkmark$  means that  $a_8 \Vdash \Diamond p$ , and the  $\times$  means that  $\alpha_3 \not\Vdash \Diamond \neg p$ .

| $\varphi$           | $\alpha_1$ | $\alpha_2$ | $\alpha_3$ | $\alpha_4$ | $\alpha_5$ | $\alpha_6$ | $\alpha_7$ | $\alpha_8$   |
|---------------------|------------|------------|------------|------------|------------|------------|------------|--------------|
| $\top$              |            |            |            |            |            |            |            |              |
| $\text{F}$          |            |            |            |            |            |            |            |              |
| $p$                 |            |            |            |            |            |            |            |              |
| $\neg p$            |            |            |            |            |            |            |            |              |
| $\Box \top$         |            |            |            |            |            |            |            |              |
| $\Diamond \text{F}$ |            |            |            |            |            |            |            |              |
| $\Box \text{F}$     |            |            |            |            |            |            |            |              |
| $\Diamond \top$     |            |            |            |            |            |            |            |              |
| $\Box p$            |            |            |            |            |            |            |            |              |
| $\Diamond \neg p$   |            |            | $\times$   |            |            |            |            |              |
| $\Box \neg p$       |            |            |            |            |            |            |            |              |
| $\Diamond p$        |            |            |            |            |            |            |            | $\checkmark$ |
| $\Diamond \Box p$   |            |            |            |            |            |            |            |              |

**Exercise 7.2.** Here are eight graphs,  $G_1, \dots, G_8$ :

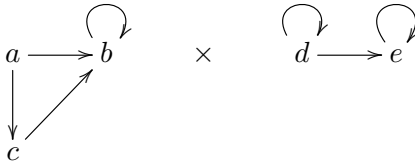


For each of these graphs, tell whether it is

1. reflexive
2. symmetric
3. transitive
4. euclidean
5. an equivalence relation
6. a partial function

Along with your “no” answers, please provide a specific counterexample. (You need not show all the verifications when you answer “yes.”)

**Exercise 7.3.** Compute the product below:



**Exercise 7.4.** For each of the following assertions, tell whether it is true or false. (As always in a problem like this, “true” means *true for all graphs*. So “false” means *false for some graphs*.) Give a short proof of the true ones, and a counterexample for the false ones.

1. The product of two reflexive graphs is reflexive.
2. The product of two euclidean graphs is reflexive.
3. If the product of two graphs is transitive, then both of the given graphs are transitive.
4. If the product of two graphs is reflexive, then both of the given graphs are reflexive.
5. Let  $G$  be a one-point graph with no arrow. Then for all  $H$ ,  $G \times H$  looks like  $H$ .

**Exercise 7.5.** The statements below are about relations on a fixed set  $X$ . For each statement below, say whether it is true or false. If it is true, give a short proof. If it is false, give a counterexample.

1. Every reflexive relation is serial.
2. Every relation which is reflexive and transitive is also symmetric.
3. Every relation which is Euclidean, symmetric and transitive is reflexive.
4. Every relation which is Euclidean and reflexive is symmetric.

**Exercise 7.6.** A modal sentence  $\varphi$  is *satisfiable* if there is some model  $\mathcal{M}$  and some  $x \in M$  such that  $x \Vdash \varphi$ . Show that each of the following sentences is satisfiable.

1.  $(\Diamond \Box p) \wedge (\Box \Diamond p)$ .
2.  $(\Diamond \Box p) \wedge (\neg \Box \Diamond p)$ .



3.  $(\neg\Diamond\Box p) \wedge (\Box\Diamond p)$ .
4.  $(\neg\Diamond\Box p) \wedge (\neg\Box\Diamond p)$ .

**Exercise 7.7.** A modal sentence  $\varphi$  is *satisfiable on a symmetric model* if there is some symmetric model  $\mathcal{M}$  and some  $x \in M$  such that  $x \Vdash \varphi$ . Which of the following sentences is satisfiable on a symmetric model?

1.  $(\Diamond\Box p) \wedge (\Box\Diamond p)$ .
2.  $(\Diamond\Box p) \wedge (\neg\Box\Diamond p)$ .
3.  $(\neg\Diamond\Box p) \wedge (\Box\Diamond p)$ .
4.  $(\neg\Diamond\Box p) \wedge (\neg\Box\Diamond p)$ .

**Exercise 7.8.** Find the duals of the axioms of the standard modal logics, and then fill in the chart below.

| Defining Property of the Relation                                              | Name | Axioms                                            | Dual |
|--------------------------------------------------------------------------------|------|---------------------------------------------------|------|
| reflexive: $x \rightarrow x$                                                   | T    | $\Box\varphi \rightarrow \varphi$                 |      |
| transitive: if $x \rightarrow y \rightarrow z$ , then $x \rightarrow z$        | 4    | $\Box\varphi \rightarrow \Box\Box\varphi$         |      |
| symmetric: if $x \rightarrow y$ , then $y \rightarrow x$                       | B    | $\varphi \rightarrow \Box\Diamond\varphi$         |      |
| serial: there is some $y$ such that $x \rightarrow y$                          | D    | $\Box\varphi \rightarrow \Diamond\varphi$         |      |
| Euclidean: if $x \rightarrow y$ and $x \rightarrow z$ , then $y \rightarrow z$ | 5    | $\Diamond\varphi \rightarrow \Box\Diamond\varphi$ |      |
| partial function: if $x \rightarrow y$ and $x \rightarrow z$ , then $y = z$    |      | $\Diamond\varphi \rightarrow \Box\varphi$         |      |

**Exercise 7.9.** Make up any model you like with the following properties:

1. Your model should be based on a graph with at least six nodes.
2. Using  $p$  and  $q$  for atomic sentences, each should be true at roughly half of the nodes.

Your problem: find the coarsest bisimulation  $B$  on your graph by using the partition refinement algorithm. Then for each of the equivalence classes  $[g]$  of  $B$ , find a modal sentence that is satisfied by the elements of  $[g]$  and not by any other points in the model.

**Exercise 7.10.** Let  $\mathcal{M}$  be any model for modal logic. Let  $\mathcal{N}$  be the same as  $\mathcal{M}$ , except that we add a fresh point called  $*$ . We can any edges we like from  $*$  to the points in  $\mathcal{M}$ , but no edges go back to  $*$ . Further, the valuation of  $*$  can be anything.

Prove that for all  $x \in M$ ,  $x$  satisfies the same modal sentences in  $\mathcal{M}$  as it does in  $\mathcal{N}$ . Do this two ways:

1. First, do it by a direct induction. You will need to say clearly what you are proving, whether points are fixed or not, etc.

2. Second, do it using bisimulation. Find a bisimulation between  $\mathcal{M}$  and  $\mathcal{N}$  and argue the result using the bisimulation which you found.

**Exercise 7.11.** Let  $S$  be any set, and let  $\text{partitions}(S)$  be the set of partitions of  $S$ . We defined a relation  $\leq$  on  $\text{partitions}(S)$  in Definition 7.31. Show that  $\leq$  is a partial order on  $\text{partitions}(S)$ . That is,  $\leq$  is reflexive, transitive, and also anti-symmetric.

**Exercise 7.12.** Let  $\mathcal{G}$  be a graph. Consider the function

$$f : \text{partitions}(G) \rightarrow \text{partitions}(G)$$

given by  $f(\Pi) = \text{refine}(\mathcal{G}, \Pi)$ .

1. Show that  $f$  is *monotone*: if  $\Pi \leq \Pi'$ , then  $f(\Pi) \leq f(\Pi')$ .
2. Give an example where  $\Pi \not\leq f(\Pi)$ , and an example where  $f(\Pi) \not\leq \Pi$ .
3. Show that  $\Pi$  is a bisimulation on  $\mathcal{G}$  iff  $f(\Pi) \leq \Pi$ .

[These are fairly direct verifications.]

**Exercise 7.13.** Let  $(P, \leq)$  be any partial order, and let  $f : P \rightarrow P$  be *monotone*: if  $p \leq q$ , then  $f(p) \leq f(q)$ . Suppose that  $P$  has a least element; call it  $\perp$ . Define  $p_0 = \perp$ ,  $p_1 = f(p_0)$ ,  $p_2 = f(p_1)$ ,  $\dots$

1. Show that  $p_0 \leq p_1 \leq p_2 \leq \dots$ .
2. Suppose that  $P$  is *finite*. Show that there is some  $n$  such that  $p_n = p_{n+1}$ .
3. Let  $p_n$  be as in the last part, so  $p_n = f(p_n)$ . Let  $q$  be any element of  $P$  such that  $f(q) \leq q$ . Show that  $p_n \leq q$ . [Hint: show that  $p_k \leq q$  for all  $k$ , using induction.]

**Exercise 7.14.** Use Exercises 7.11, 7.12, and 7.13 to show that the partition refinement algorithm works. That is, given a finite graph  $\mathcal{G}$ , the algorithm always runs in finite time, and it finds the coarsest bisimulation of  $\mathcal{G}$ .



## Chapter 8

# The Standard Logical Systems of Modal Logic

We now study deductive logical systems that are intended to match the semantic work that we have been doing. This is parallel with what we did with propositional logic in Chapter 5 and the switching logic Chapter 6.

As with our work in previous chapters, the idea is to start with a semantic concept. In our case, the main one is  $\Gamma \models \varphi$  defined by

$$\text{If } U \text{ is a model and } u \in U, \text{ and } u \models \psi \text{ for all } \psi \in \Gamma, \text{ then } u \models \varphi. \quad (8.1)$$

For us,  $\Gamma$  will almost always be a finite set. In that case, its conjunction  $\bigwedge \Gamma$  is a modal sentence. And we may recast (8.1) as:

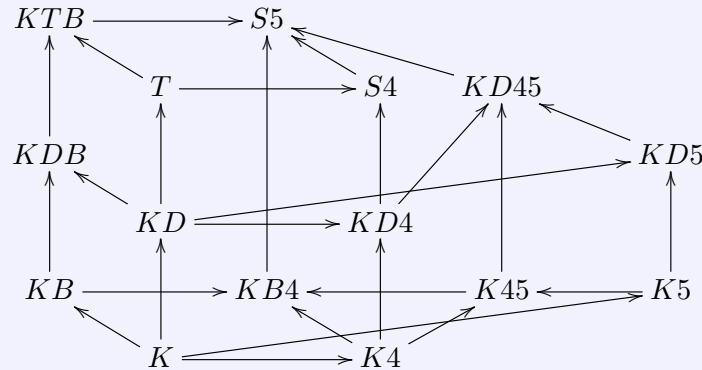
$$(\bigwedge \Gamma) \rightarrow \varphi \text{ holds at all states of all models.}$$

Note that this is a *semantic* definition; it refers to models and states. The whole point of a syntactically defined proof system is to get such a detailed understanding of the semantic notion that we can study and work with it in a purely mechanical way.

**Example 8.1: Classes of models and the sound modal axioms for them**

| Class        | Defining Property of the Relation                                              | Name | Axioms<br>Duals                                                                                    |
|--------------|--------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------|
| <b>Refl</b>  | reflexive: $x \rightarrow x$                                                   | T    | $\Box\varphi \rightarrow \varphi$<br>$\varphi \rightarrow \Diamond\varphi$                         |
| <b>Trans</b> | transitive: if $x \rightarrow y \rightarrow z$ , then $x \rightarrow z$        | 4    | $\Box\varphi \rightarrow \Box\Box\varphi$<br>$\Diamond\Diamond\varphi \rightarrow \Diamond\varphi$ |
| <b>Symm</b>  | symmetric: if $x \rightarrow y$ , then $y \rightarrow x$                       | B    | $\varphi \rightarrow \Box\Diamond\varphi$<br>$\Diamond\Box\varphi \rightarrow \varphi$             |
| <b>Ser</b>   | serial: there is some $y$ such that $x \rightarrow y$                          | D    | $\Box\varphi \rightarrow \Diamond\varphi$<br>$\Box\varphi \rightarrow \Diamond\varphi$             |
| <b>Eucl</b>  | Euclidean: if $x \rightarrow y$ and $x \rightarrow z$ , then $y \rightarrow z$ | 5    | $\Diamond\Box\varphi \rightarrow \Box\varphi$<br>$\Diamond\varphi \rightarrow \Box\Diamond\varphi$ |
| <b>Pre</b>   | Preorder: reflexive<br>and transitive                                          | S4   | KT4                                                                                                |
| <b>Eq</b>    | Equivalence relation: reflexive,<br>symmetric, and transitive                  | S5   | KT45                                                                                               |

We are going to two logical systems of two different kinds: natural deduction systems, and Hilbert-style systems. Each of the logical principles in Example 8.2 will thus give us two systems.

**Example 8.2: Logics built on top of  $K$  by adding  $D$ ,  $T$ ,  $B$ ,  $4$ , and/or  $5$ .**

Common abbreviations:  $T$  is  $KT$ ,  $S4$  is  $KT4$ , and  $S5$  is  $KT5$ .

## 8.1 Natural deduction

You may wish to review the natural deduction proof system for propositional logic that we saw earlier, since everything we do in this section builds on the earlier proof system. We'll have a normality axiom as in Chapter 6, and also an axiom relating  $\Box$  and  $\Diamond$ ; but we'll have nothing like the involution axioms, and usually we won't have the determinacy axioms.

The notion of a  $\star$ -subproof from Chapter 6 is carried over with a new name; we call it a  $\star$ -*subproof*. The basic idea behind  $\star$ -subproofs is that if the overall proof up to that point reflects reasoning about some world  $w$ , such a subproof reflects reasoning about an arbitrary child  $x$  of  $w$ . At any point, one may start a  $\star$ -subproof by drawing a new vertical line. The only way to get a sentence into a  $\star$ -subproof from outside is to use the rule of  $\Box E$ . This rule says that if a sentence  $\Box\varphi$  appears active one level up, then you can write  $\varphi$  inside the  $\star$ -subproof. The justification for this line is  $\Box E$ , and one also gives the line number of the  $\Box\varphi$  line.

We may also make assumptions inside of  $\star$ -subproofs, and we also may use propositional rules such as reiteration (R) and  $(\wedge e)$  inside of  $\star$ -subproofs. A  $\star$ -subproof may itself contain ordinary subproofs and also further  $\star$ -subproofs. We say that a  $\star$ -subproof is *open* at some step if it has a subproof of its own which is not closed at that step. In that case, one may *not* use  $\Box E$  to export sentences from the  $\star$ -subproof. Finally, as soon as we export a sentence from a  $\star$ -subproof, that subproof is *closed*, and we do not return to it.

We obtain logics by taking our system  $K$  and adding all substitution instances of some of the formulas listed in Example 8.2. For example,  $KD4$  is the system  $K$  together with all instances of  $D$  and 4. We also use the following abbreviations:

S4 is KT4

S5 is KT45.

These are the traditional names for these logics. We are mostly going to be interested  $K$ ,  $KT$ ,  $S4$ ,  $S5$  and a few other logics; these are the ones that come up most in applied work. The partial functionality axiom  $D_c$  is not a standard axiom in the literature, and our reason for including it is to make a connection to the work in Chapter 6.

**Simplifying proofs** We simplify formal proofs in a few ways. First, we generally omit all steps of reasoning that have to do with propositional logic, and instead we cite them as PR. Second, we omit mention of the axiom  $\Diamond\varphi \leftrightarrow \neg\Box\neg\varphi$ . So in a sense, we treat  $\Diamond\varphi$  as if it were the same as  $\neg\Box\neg\varphi$ , and  $\Box\varphi$  as if it were the same as  $\neg\Diamond\neg\varphi$ . More generally, in a given logic we not only use the axioms but also their duals, in a sense made precise below. Finally, we often cite previous results and otherwise shorten lengthy derivations whenever possible.

### 8.1.1 Duals of the axioms, used in proofs

We discussed duals of various axioms in Section 7.3.2. It is easy to check that in all of our logical systems, the dual of every axiom is provable. So we simplify our formal derivations by allowing the dual of every axiom to appear as if it were one of the original axioms.

## 8.1.2 Examples

**Example 8.3**

First, a proof of the Normality Axiom  $\vdash \Box(\varphi \rightarrow \psi) \rightarrow (\Box\varphi \rightarrow \Box\psi)$  in  $K$ :

|   |  |                                                                                 |                         |
|---|--|---------------------------------------------------------------------------------|-------------------------|
| 1 |  | $\Box(\varphi \rightarrow \psi)$                                                |                         |
| 2 |  | $\Box\varphi$                                                                   |                         |
| 3 |  | $\star$   $\varphi \rightarrow \psi$                                            | $\Box e, 1$             |
| 4 |  | $\varphi$                                                                       | $\Box e, 2$             |
| 5 |  | $\psi$                                                                          | $\rightarrow e, 3, 4$   |
| 6 |  | $\Box\psi$                                                                      | $\Box i, 5$             |
| 7 |  | $\Box\varphi \rightarrow \Box\psi$                                              | $\rightarrow i, 1, 2-6$ |
| 8 |  | $\Box(\varphi \rightarrow \psi) \rightarrow (\Box\varphi \rightarrow \Box\psi)$ | $\rightarrow i, 1, 2-6$ |

**Example 8.4**

Second, a proof in  $K$  of  $\Box\varphi \wedge \Box\psi \rightarrow \Box(\varphi \wedge \psi)$ . This is an easy exercise, and so we leave it to you.

**Example 8.5**

Third, a proof in  $K$  of  $(\Box\varphi \wedge \Diamond\psi) \rightarrow \Diamond(\varphi \wedge \psi)$ .

|   |  |                                                                                                |             |
|---|--|------------------------------------------------------------------------------------------------|-------------|
| 1 |  | $(\Box\varphi \wedge \Box(\varphi \rightarrow \neg\psi)) \rightarrow \Box\neg\psi$             | Example 8.3 |
| 2 |  | $(\Box\varphi \wedge \neg\Box\neg\psi) \rightarrow \neg\Box\neg(\varphi \rightarrow \neg\psi)$ | PR, 1       |
| 3 |  | $(\varphi \wedge \psi) \rightarrow \neg(\varphi \rightarrow \neg\psi)$                         | tautology   |
| 4 |  | $\Box(\varphi \wedge \psi) \rightarrow \Box(\varphi \rightarrow \neg\psi)$                     | done        |
| 5 |  | $(\Box\varphi \wedge \Diamond\psi) \rightarrow \Diamond(\varphi \wedge \psi)$                  | PR, 2, 4    |

**Example 8.6: necessitation**

If  $\vdash \varphi$  in any of our natural deduction systems, then also  $\vdash \Box\varphi$  in the same system. There is a name for this feature: we say that *the rule of Necessitation is admissible in the system*. To show this, we take a proof of  $\varphi$  in this system, and simply take the whole thing to be a  $\star$ -subproof of a bigger proof. Again, the last line of the  $\star$ -subproof is  $\varphi$ . That outside proof then closes with  $\Box\varphi$ .

**Example 8.7**

Here is an example of necessitation (Example 8.6 at work). Suppose that  $\vdash \varphi \rightarrow \psi$ . We show that  $\vdash \Box\varphi \rightarrow \Box\psi$ :

|   |  |  |                                    |                      |
|---|--|--|------------------------------------|----------------------|
| 1 |  |  | $\Box\varphi$                      | assumption           |
| 2 |  |  | $\star$ $\varphi$                  | $\Box e, 1$          |
| 3 |  |  | $\varphi \rightarrow \psi$         | given                |
| 4 |  |  | $\psi$                             | modus ponens, 2, 3   |
| 5 |  |  | $\Box\psi$                         | $\Box i, 4$          |
| 6 |  |  | $\Box\varphi \rightarrow \Box\psi$ | $\rightarrow i, 2-5$ |

**Example 8.8**

Similarly, if in any system  $\vdash \varphi \rightarrow \psi$ , we argue that  $\vdash \Diamond\varphi \rightarrow \Diamond\psi$ . First, propositional reasoning from  $\vdash \varphi \rightarrow \psi$  shows that  $\vdash \neg\psi \rightarrow \neg\varphi$ . Then the result of Example 8.7 shows that  $\vdash \Box\neg\psi \rightarrow \Box\neg\varphi$ . By propositional reasoning again,  $\vdash \neg\Box\neg\varphi \rightarrow \neg\Box\neg\psi$ . This is  $\vdash \Diamond\varphi \rightarrow \Diamond\psi$ .

**Example 8.9**

We show on the left that all 5 axioms are provable in  $KB4$ . On the right, we show that the 4 axioms are provable in  $KB5$ .

|   |  |                                                           |             |  |   |  |                                                       |                |
|---|--|-----------------------------------------------------------|-------------|--|---|--|-------------------------------------------------------|----------------|
| 1 |  | $\Box\varphi \rightarrow \Box\Box\varphi$                 | 4 axiom     |  | 1 |  | $\Box\varphi \rightarrow \Box\Diamond\Box\varphi$     | B axiom        |
| 2 |  | $\Diamond\Box\varphi \rightarrow \Diamond\Box\Box\varphi$ | Example 8.8 |  | 2 |  | $\Diamond\Box\varphi \rightarrow \Box\varphi$         | 5 axiom (dual) |
| 3 |  | $\Diamond\Box\Box\varphi \rightarrow \Box\varphi$         | B axiom     |  | 3 |  | $\Box\Diamond\Box\varphi \rightarrow \Box\Box\varphi$ | Example 8.7    |
| 4 |  | $\Diamond\Box\varphi \rightarrow \Box\varphi$             | PR, 2, 3    |  | 4 |  | $\Box\varphi \rightarrow \Box\Box\varphi$             | PR, 1-3        |



**Example 8.10**

Here is a derivation in  $K4$  of  $\Box\Diamond p \rightarrow \Box\Box\Diamond p$ .

|    |                                                         |                         |    |                                                                                                        |
|----|---------------------------------------------------------|-------------------------|----|--------------------------------------------------------------------------------------------------------|
| 1  | $\Box\Diamond p$                                        | assumption              | 1  | $\Box\Diamond p \rightarrow \Box\Diamond p$                                                            |
| 2  | $\Box\Diamond p \rightarrow \Box\Box\Diamond p$         | 4-axiom                 | 2  | $\Box\Diamond p \rightarrow (\Box\Diamond p \rightarrow \Box\Box\Diamond p)$                           |
| 3  | $\Box\Box\Diamond p \rightarrow \Box\Box\Box\Diamond p$ | Example 8.7             | 3  | $\Box\Diamond p \rightarrow (\Box\Box\Diamond p \rightarrow \Box\Box\Box\Diamond p)$                   |
| 4  | $\Box\Box\Box\Diamond p$                                | PR, 1–3                 | 4  | $\Box\Diamond p \rightarrow \Box\Box\Box\Diamond p$                                                    |
| 5  | $\star \mid \Diamond p$                                 | $\star$ -rule, 1        | 5  | $\Box\Diamond p \rightarrow^{\Box} \Diamond p$                                                         |
| 6  | $\Box\Box\Diamond p$                                    | $\star$ -rule, 4        | 6  | $\Box\Diamond p \rightarrow^{\Box} \Box\Box\Diamond p$                                                 |
| 7  | $\Box\Diamond\Box\neg p$                                | assumption              | 7  | $\Box\Diamond p \rightarrow^{\Box} \Box\Diamond\Box\neg p \rightarrow \Box\Box\Box\neg p$              |
| 8  | $\star \mid \Box\Diamond p$                             | $\star$ -rule, 6        | 8  | $\Box\Diamond p \rightarrow^{\Box} \Box\Diamond\Box\neg p \rightarrow^{\Box} \Box\Box p$               |
| 9  | $\Diamond\Box\neg p$                                    | $\star$ -rule, 7        | 9  | $\Box\Diamond p \rightarrow^{\Box} \Box\Diamond\Box\neg p \rightarrow^{\Box} \Diamond\Box\neg p$       |
| 10 | $\perp$                                                 | $\wedge$ i, 8, 9        | 10 | $\Box\Diamond p \rightarrow^{\Box} \Box\Diamond\Box\neg p \rightarrow^{\Box} \perp$                    |
| 11 | $\perp \rightarrow \neg p$                              | PR                      | 11 | $\Box\Diamond p \rightarrow^{\Box} \Box\Diamond\Box\neg p \rightarrow^{\Box} \perp \rightarrow \neg p$ |
| 12 | $\neg p$                                                | $\rightarrow$ e, 10, 11 | 12 | $\Box\Diamond p \rightarrow^{\Box} \Box\Diamond\Box\neg p \rightarrow^{\Box} \neg p$                   |
| 13 | $\Box\neg p$                                            | $\star$ -rule, 12       | 13 | $\Box\Diamond p \rightarrow^{\Box} \Box\Diamond\Box\neg p \rightarrow \Box\neg p$                      |
| 14 | $\Diamond p$                                            | R, 5                    | 14 | $\Box\Diamond p \rightarrow^{\Box} \Box\Diamond\Box\neg p \rightarrow \Diamond p$                      |
| 15 | $\perp$                                                 | $\wedge$ i, 13, 14      | 15 | $\Box\Diamond p \rightarrow^{\Box} \Box\Diamond\Box\neg p \rightarrow \perp$                           |
| 16 | $\Diamond\Box\Diamond p$                                | $\neg$ i, etc., 7–15    | 16 | $\Box\Diamond p \rightarrow^{\Box} \Diamond\Box\Diamond p$                                             |
| 17 | $\Box\Box\Box\Diamond p$                                | $\star$ -rule, 16       | 17 | $\Box\Diamond p \rightarrow \Box\Box\Box\Diamond p$                                                    |
| 18 | $\Box\Diamond p \rightarrow \Box\Box\Box\Diamond p$     | $\rightarrow$ i, 1–17   | 18 | $\Box\Diamond p \rightarrow \Box\Box\Box\Diamond p$                                                    |

We have added some assertions on the right that are used in connection with a discussion of the soundness of the proof system in Section 8.2.

## 8.2 Why the proof system works

### Lemma 8.11: Soundness of various logics on classes of models

Let  $\mathcal{C}$  be a class of models, and let  $A$  be a modal scheme which corresponds to  $\mathcal{C}$ . Then the logic  $KA$  is sound for  $\mathcal{C}$ . That is, if  $\Gamma \vdash \varphi$  in  $KA$ , then  $\Gamma \models \varphi$  on models in  $\mathcal{C}$ . Moreover, if  $\mathcal{C}_1, \dots, \mathcal{C}_n$  are classes of models and  $A_1, \dots, A_n$  are corresponding schemes, then  $KA_1 \cdots A_n$  is sound for  $\mathcal{C}_1 \cap \cdots \cap \mathcal{C}_n$ .

*Proof.* The proof is very similar to that of Lemma 8.18, except that we have an extra base case for the instances of the additional axioms.  $\square$

Here are some examples of Lemma 8.11

1. If  $\Gamma \vdash \varphi$  in  $KT$ , then  $\Gamma \models \varphi$  on reflexive models.
2. If  $\Gamma \vdash \varphi$  in  $K4$ , then  $\Gamma \models \varphi$  on transitive models.

3. If  $\Gamma \vdash \varphi$  in  $KB$ , then  $\Gamma \models \varphi$  on symmetric models.
4. If  $\Gamma \vdash \varphi$  in  $KTB$ , then  $\Gamma \models \varphi$  on reflexive, symmetric models

### 8.2.1 Consistency in a logic

We say that a set  $\Gamma$  of sentences in  $\mathcal{L}(\Box, \Diamond)$  is *consistent* in a logic  $L$  if  $\Gamma \not\vdash \mathbf{F}$  in  $L$ . That is, we cannot prove a contradiction from  $\Gamma$  working in the logic  $L$ . If a logic corresponds to a class of models as in Lemma 8.11, then again we get an easy way to prove that a sentence  $\varphi$  is sound in a given logic: find a model of  $\varphi$  in the corresponding class. The same goes for a *set* of sentences, say  $\Gamma$ . To prove that  $\Gamma$  is consistent in a given logic  $L$ , find a model of  $\Gamma$  which has all the properties that are associated with  $L$ .

One has to be careful here, because a sentence might be consistent in one logic but not in another. For example, we claim that  $p \wedge \Box \neg p$  is consistent in  $KB$  but not in  $KT$ . To see that  $p \wedge \Box \neg p$  is consistent in  $KB$ , we consider the following model

$$x : p \longleftrightarrow y$$

Since the model is symmetric, and  $x \Vdash p \wedge \Box \neg p$ , we see that  $p \wedge \Box \neg p$  is consistent in  $KB$ . But we claim that the same sentence is not consistent in  $KT$ . This is because  $p \wedge \Box \neg p$  is equivalent to the negation of an axiom  $\Box \neg p \rightarrow \neg p$  of  $KT$ .

### 8.2.2 Inclusion and equality of logical systems

Example 8.2 shows some arrows between the various logics, and also, one entry seems to be missing. At this point, we want to explain the figure.

Let  $L$  and  $M$  be logical systems such as  $K$ ,  $K4$ ,  $S5$ , etc. We say that  $\varphi$  is a *theorem* of  $L$  if  $\vdash \varphi$  in  $L$ . We write  $L \leq M$  if every theorem of  $L$  is also a theorem of  $M$ . In the setting of this chapter, this boils down to showing that every axiom of  $L$  is a theorem of  $M$ . Example 8.2 shows this  $\leq$  relation by an arrow. For example,  $KB \leq KDB$  as there is an arrow from  $KB$  to  $KDB$ . The reason for this arrow is that every axiom of the first logic is an axiom of the second, so easily every theorem of the first is a theorem of the second.

We say that two logics  $L$  and  $M$  are *equal* if they have the same theorems. This holds iff every axiom of one is provable in the other, and vice-versa. This is the same as having  $L \leq M$  and  $M \leq L$ .

#### Lemma 8.12

$$KD \leq KT.$$

*Proof.* For each  $\varphi$ ,  $\vdash \Box \neg \varphi \rightarrow \neg \varphi$  in  $KT$ . So  $\vdash \varphi \rightarrow \Diamond \varphi$  as well. And since  $\vdash \Box \varphi \rightarrow \varphi$  in  $KT$ , we must have  $\vdash \Box \varphi \rightarrow \Diamond \varphi$ .  $\square$

#### Lemma 8.13

$$KB4 = KB5.$$

*Proof.* See Example 8.13 above. □

#### Theorem 8.14

$$KT5 = KDB5 = KDB4 = KTB4.$$

*Proof.* We first check that the  $B$  axioms are provable in  $KT5$ . Fix some  $\varphi$ . Then one of the 5 axioms is  $\Diamond\Box\neg\varphi \rightarrow \Box\neg\varphi$ . And one of the  $T$  axioms is  $\Box\neg\varphi \rightarrow \neg\varphi$ . So in  $KT5$ ,  $\vdash \Diamond\Box\neg\varphi \rightarrow \neg\varphi$ . Using propositional reasoning,  $\vdash \varphi \rightarrow \Box\Diamond\varphi$ .

As a result of Lemmas 8.12 and 8.12, all axioms of  $KDB5$  are provable in  $KT5$ . To show the two systems are equal, we only need to prove the  $T$  axioms in  $KDB5$ . This is Exercise 8.9.

By Lemma 8.13,  $KDB5 = KDB4$ . Since we also know that the  $T$  axioms hold in  $KDB5$ , we see that  $KDB4 = KTB4$ . □

### 8.2.3 We have not shown the completeness of any logical systems

This chapter has presented a slew of logical systems, and we know that they are sound in the appropriate senses (see Lemma 8.11). But we have not shown the completeness of any of these systems. The work on completeness will appear later, in Chapter 9.

Our soundness fact implies some facts about satisfiability and consistency. For example, if a sentence  $\varphi$  is satisfiable (on any model), then it is consistent in  $K$ . If  $\varphi$  is satisfiable on a transitive model, it is consistent in  $K4$ . If  $\varphi$  is satisfiable on an equivalence relation, then it is consistent in  $S5$ . The converses of these turn out to all be true, and this is tantamount to the completeness of the logics.

#### Remark 8.15

In all our previous logics, there is an *algorithm* to tell whether a given sentence is valid or not. (For propositional logic, the algorithm would ask us to construct the truth table.) For this logic  $\mathcal{L}(K)$ , it is not obvious whether there is such an algorithm or not. After all, we cannot examine all of the different equivalence models in the entire world: there are infinitely many of them.

## 8.3 Hilbert-style systems

We have already seen Hilbert-style (axiomatic) proof systems for propositional logic<sup>1</sup> and the switching logic (see 6.5). We formulate similar logical systems for the logical systems  $K$ ,  $KB$ ,  $KD$ , ...,  $S5$ .

<sup>1</sup>I need to add this discussion or delete this point.

**Definition 8.16: the Hilbert-style logical system  $K$  for  $\mathcal{L}(\Box, \Diamond)$** 

|               |                     |                                                                                 |
|---------------|---------------------|---------------------------------------------------------------------------------|
| <b>Axioms</b> | All tautologies     |                                                                                 |
|               | $(\Diamond - \Box)$ | $\Diamond\varphi \leftrightarrow \neg\Box\neg\varphi$                           |
|               | (Normality)         | $\Box(\varphi \rightarrow \psi) \rightarrow (\Box\varphi \rightarrow \Box\psi)$ |
| <b>Rules</b>  | (Modus Ponens)      | From $\varphi$ and $\varphi \rightarrow \psi$ , infer $\psi$                    |
|               | (Necessitation)     | From $\varphi$ , infer $\Box\varphi$                                            |

For other systems such as  $K4$ ,  $S4$ , etc., we add extra axioms.

The axioms in our systems include substitution instances of tautologies (see page 146), normality

$$\Box(\varphi \rightarrow \psi) \rightarrow (\Box\varphi \rightarrow \Box\psi)$$

and also the definition of  $\Diamond\varphi$  in terms of  $\Box\varphi$ :

$$\Diamond\varphi \leftrightarrow \neg\Box\neg\varphi.$$

Finally, we remind you that as with the natural deduction systems, the Hilbert-style systems include various axioms which we have listed in Example 8.2.

The rules of all of the logics are the same: modus ponens and also necessitation: from  $\varphi$ , infer  $\Box\varphi$ .

As before, proofs are essentially sequences of sentences that are either axioms or come from previous sentences in the sequence according to the rules.

Given a set  $\Gamma$  of modal formulas, we write  $\Gamma \vdash \varphi$  if there are  $\psi_1, \dots, \psi_n$  in  $\Gamma$  so that  $\vdash \psi_1 \wedge \dots \wedge \psi_n \rightarrow \varphi$ . We write  $\Gamma \vdash \varphi$  in this case. Again, usually  $\Gamma$  is a finite set for us, so by taking the conjunction, we usually regard it as a single sentence.

You'll get some practice with deduction in modal logic in the exercises. We show a few facts that will be needed there and in the work of this section. At the same time, we know that this system is not so easy to actually use. It's chief virtue is that it is easy to define and to study theoretically.

**Lemma 8.17**

For all  $\varphi$  and  $\psi$ ,

1.  $\vdash \Box\varphi \wedge \Box\psi \leftrightarrow \Box(\varphi \wedge \psi)$ .
2.  $\vdash \Box\varphi \wedge \Diamond\psi \rightarrow \Diamond(\varphi \wedge \psi)$ .
3. If  $\vdash \varphi \rightarrow \psi$ , then  $\vdash \Diamond\varphi \rightarrow \Diamond\psi$ .

*Proof.* We leave (1) as an exercise.

For (2), note that  $\vdash (\varphi \wedge \neg(\varphi \wedge \psi)) \rightarrow \psi$ , since this is a tautology. So by necessitation,  $\vdash \Box((\varphi \wedge \neg(\varphi \wedge \psi)) \rightarrow \psi)$ . Then using one of our K axioms,  $\vdash \Box(\varphi \wedge \neg(\varphi \wedge \psi)) \rightarrow \Box\psi$ . Using (1),

$$\vdash \Box\varphi \wedge \Box\neg(\varphi \wedge \psi) \rightarrow \Box\neg\psi$$

From this we use propositional logic to get what we want.

Finally, if  $\vdash \varphi \rightarrow \psi$ , then  $\vdash \neg\psi \rightarrow \neg\varphi$ . Hence  $\vdash \Box\neg\psi \rightarrow \Box\neg\varphi$ . Taking the contrapositive gives  $\vdash \Diamond\varphi \rightarrow \Diamond\psi$ .  $\square$

**Soundness** The first result about the system  $K$  is that it is *sound*.

**Lemma 8.18: Soundness of  $K$**

If  $\Gamma \vdash \varphi$ , then  $\Gamma \models \varphi$ .

*Proof.* We prove by induction on the number  $n$  that if  $\Gamma \vdash \varphi$  by a proof  $\Pi$  in  $K$  of height  $n$ , then  $\Gamma \models \varphi$ . **proof hidden**  $\square$

## 8.4 Exercises on the standard modal logics

In the exercises below, we again refer to the sentences  $\alpha_1, \dots, \alpha_8$  from equation (8.2) on page 198.

**Exercise 8.1.** Recall that the  $B$  axioms are the sentences of the form  $\varphi \rightarrow \Box\Diamond\varphi$ . Let's call the  $B'$  axioms the sentences of the form  $(\varphi \wedge \Diamond\psi) \rightarrow \Diamond(\psi \wedge \Diamond\varphi)$ . So we get a logical system  $KB'$  by adding the  $B'$  axioms to  $K$ .

Prove that  $KB' \leq KB$  and  $KB \leq KB'$ . That is, show that the two logical systems can prove each other's axioms, and hence prove all the same sentences.

**Exercise 8.2.** Suppose that a sentence  $\varphi \wedge \Diamond\psi$  is consistent in  $KB$ . Show that  $\psi \wedge \Diamond\varphi$  is also consistent in  $KB$ .

**Exercise 8.3.**

We shall use the following notation at several points later on.

$$\begin{array}{ll}
 \alpha_1 = \Box F \wedge p & \alpha_5 = \Box\neg p \wedge \neg\Diamond p \wedge p \\
 \alpha_2 = \Box F \wedge \neg p & \alpha_6 = \Box\neg p \wedge \neg\Diamond p \wedge \neg p \\
 \alpha_3 = \Box p \wedge \Diamond p \wedge p & \alpha_7 = \Diamond p \wedge \Diamond\neg p \wedge p \\
 \alpha_4 = \Box p \wedge \Diamond p \wedge \neg p & \alpha_8 = \Diamond p \wedge \Diamond\neg p \wedge \neg p
 \end{array} \tag{8.2}$$

Check that  $\alpha_1, \dots, \alpha_8$  are satisfiable and hence consistent in  $K$ .

**Exercise 8.4.** What would happen if we changed the definition of soundness in (8.1) to

If  $U$  is a model and  $u \Vdash \psi$  for all  $u \in U$  and  $\psi \in \Gamma$ ,  
then for all  $u \in U$ ,  $u \Vdash \varphi$ .

Specifically, would part of our proof system become unsound?

**Exercise 8.5.** Show that the  $B$  axiom  $p \rightarrow \Box \Diamond p$  is not provable in  $KD45$ . [The way to start is to find a graph which is serial, transitive, and euclidean, and at the same time is not symmetric. Then one needs to find a point and put a valuation on the graph so that the point does not satisfy the given  $B$ -axiom.]

In the following exercises, we again refer to the sentences  $\alpha_1, \dots, \alpha_8$  from equation (8.2) on page 198.

**Exercise 8.6.** Recall the  $\alpha_i$  sentences from Exercise 8.2 above. Fill in the following chart, putting a  $\checkmark$  in the box if the given sentence is consistent in the given logic, and an  $\times$  if it is not. A few of the entries are given for you.

|      | $\alpha_1$   | $\alpha_2$   | $\alpha_3$   | $\alpha_4$   | $\alpha_5$ | $\alpha_6$ | $\alpha_7$   | $\alpha_8$   |
|------|--------------|--------------|--------------|--------------|------------|------------|--------------|--------------|
| $K$  | $\checkmark$ | $\checkmark$ | $\checkmark$ |              |            |            |              |              |
| $KT$ |              |              | $\checkmark$ | $\times$     |            |            |              |              |
| $KD$ |              |              |              | $\checkmark$ |            |            |              | $\checkmark$ |
| $KB$ |              |              |              | $\checkmark$ |            |            |              |              |
| $K4$ |              |              |              |              |            |            | $\checkmark$ |              |
| $S4$ | $\times$     |              |              |              |            |            |              |              |
| $S5$ | $\times$     |              |              |              |            |            |              |              |

**Exercise 8.7.** Here is another chart, this time about consistency in  $K$  of  $\alpha \wedge \Diamond \beta$ , where  $\alpha$  is the row and  $\beta$  is the column:

|            | $\alpha_1$ | $\alpha_2$ | $\alpha_3$ | $\alpha_4$   | $\alpha_5$   | $\alpha_6$   | $\alpha_7$ | $\alpha_8$ |
|------------|------------|------------|------------|--------------|--------------|--------------|------------|------------|
| $\alpha_1$ |            |            |            |              |              |              |            |            |
| $\alpha_2$ |            |            |            |              |              |              |            |            |
| $\alpha_3$ |            |            |            |              |              |              |            |            |
| $\alpha_4$ |            |            |            |              |              |              |            | $\times$   |
| $\alpha_5$ |            |            |            |              |              |              | $\times$   |            |
| $\alpha_6$ |            |            |            |              |              | $\checkmark$ |            |            |
| $\alpha_7$ |            |            |            |              | $\checkmark$ |              |            |            |
| $\alpha_8$ |            |            |            | $\checkmark$ |              |              |            |            |

For example, the  $\times$  in the last column of the indicated entries mean that  $\alpha_4 \wedge \Diamond \alpha_8$  is not consistent in  $K$ , while the  $\checkmark$  in the bottom row means that  $\alpha_8 \wedge \Diamond \alpha_4$  is consistent in  $K$ .

**Exercise 8.8.** Make a similar chart for  $S4$ . You should not use all eight elements  $\alpha_1, \dots, \alpha_8$  but only those which happen to be consistent in  $S4$ . (You found out exactly which are consistent in  $S4$  in Exercise 8.6.)

**Exercise 8.9.** Show that  $KT \leq KDB5$ .

**Exercise 8.10.** In this section, we considered  $\Box \Diamond \varphi \rightarrow \Box \Diamond \Box \Diamond \varphi$ . As it happens, the converse of this is also valid on transitive models.

1. Show that  $\models_{\mathbf{F}} \Box\Diamond\Box\Diamond\varphi \rightarrow \Box\Diamond\varphi$ .
2. Show that  $\vdash \Box\Diamond\Box\Diamond\varphi \rightarrow \Box\Diamond\varphi$  in  $K4$ .
3. Show that  $\vdash_{\mathbf{H}} \Box\Diamond\Box\Diamond\varphi \rightarrow \Box\Diamond\varphi$  in  $K4$ .

**Exercise 8.11.** This exercise and the next ones hint at the soundness of the natural deduction system. Suppose that

$$\models \varphi_0 \rightarrow \Box(\varphi_1 \rightarrow \Box(\varphi_2 \rightarrow \Box(\varphi_3 \rightarrow \chi_1)))$$

and also that

$$\models \varphi_0 \rightarrow \Box(\varphi_1 \rightarrow \Box(\varphi_2 \rightarrow \Box(\varphi_3 \rightarrow \chi_2))).$$

Show that  $\models \varphi_0 \rightarrow \Box(\varphi_1 \rightarrow \Box(\varphi_2 \rightarrow \Box(\varphi_3 \rightarrow (\chi_1 \wedge \chi_2))))$ . This hints at the justification of  $\wedge I$  in this system.

**Exercise 8.12.** Suppose that

$$\models \varphi_0 \rightarrow \Box(\varphi_1 \rightarrow \Box(\varphi_2 \rightarrow \Box(\varphi_3 \rightarrow \chi_1)))$$

and also that

$$\models \varphi_0 \rightarrow \Box(\varphi_1 \rightarrow \Box(\varphi_2 \rightarrow \Box(\varphi_3 \rightarrow \neg\chi_1))).$$

Show that  $\models \varphi_0 \rightarrow \Box(\varphi_1 \rightarrow \Box(\varphi_2 \rightarrow \Box\neg\varphi_3))$ . This hints at the justification of the rule of  $\neg I$ .

**Exercise 8.13.** Suppose that

$$\models \varphi_0 \rightarrow \Box(\varphi_1 \rightarrow \Box(\varphi_2 \rightarrow \Box\rho)).$$

Show that  $\models \varphi_0 \rightarrow \Box(\varphi_1 \rightarrow \Box(\varphi_2 \rightarrow \Box(\varphi_3 \rightarrow \rho)))$ . This suggests the justification of  $\Box E$ .

## Chapter 9

# Completeness Theorems for Modal Logics

In this chapter, we consider some specific modal logics that come up in applications, and also some specific classes of models, and prove completeness results that relate the logics to the models.

We prove results such as the following:

1.  $\models_{\text{all}} \varphi$  iff  $\vdash \varphi$  in  $K$
2.  $\models_{\text{r,t}} \varphi$  iff  $\vdash \varphi$  in  $S4$
3.  $\models_{\text{eq}} \varphi$  iff  $\vdash \varphi$  in  $S5$

The soundness (right-to-left) directions of these follow from our work in Chapter 7. So the interesting work will be in the other (left-to-right) directions.

### Remark 9.1

Our only for the *weak* completeness results, those of the form  $\models \varphi$  implies  $\vdash \varphi$ . The usual presentation of modal logic gets *strong completeness results* first, and then later reads off the weak completeness results by a method called “filtration.” A strong completeness result is one of the following sort:

For all sets  $\Gamma$  and all  $\varphi$ ,  $\Gamma \models \varphi$  only if  $\Gamma \vdash \varphi$ .

The point is that we have arbitrary sets  $\Gamma$  on the left of the  $\models$  and  $\vdash$  signs. In particular  $\Gamma$  can be *infinite*. The standard textbook treatments of modal logic get strong completeness results first. The weak completeness results are the special cases when  $\Gamma$  is the empty set  $\emptyset$ . The additional results about *finite* models then take additional work. Usually this is done by a technique called *filtration*. Our presentation departs from the usual one since we do not do any of this. We don’t get the strong completeness results at all, but we have a quicker way to get the weak completeness/finite model results.



**Remark 9.2**

Here are the steps that we took for propositional logic.

1. Define the set  $\text{SD}_n$  of *state descriptions of order  $n$* , and show the following:
  - (a) For each  $n$ ,  $\vdash \bigvee \text{SD}_n$ .
  - (b) For each  $n$ , each  $\alpha \in \text{SD}_n$  is satisfiable.
  - (c) For each  $n$  and each  $\varphi$  with  $\text{occ}(\varphi) \subseteq \{p_1, \dots, p_n\}$ , either  $\vdash \alpha \rightarrow \varphi$ , or else  $\vdash \alpha \rightarrow \neg\varphi$ .
2. Then prove completeness in a now-rather-routine way: Assume that  $\models \varphi$ . Let  $n$  be large enough so that  $\text{occ}(\varphi) \subseteq \{p_1, \dots, p_n\}$ . For each  $\alpha \in \text{SD}_n$ , we must have  $\vdash \alpha \rightarrow \varphi$ . (For if not, we have  $\vdash \alpha \rightarrow \neg\varphi$ . So by soundness, we have  $\models \alpha \rightarrow \neg\varphi$ . Now  $\alpha$  is satisfiable, so let  $v$  be a valuation such that  $\bar{v}(\alpha) = \text{T}$ . Then for this  $\bar{v}$ , we also have  $\bar{v}(\neg\varphi) = \text{T}$ . This contradicts  $\models \varphi$ .) And so  $\vdash \bigvee \text{SD}_n \rightarrow \varphi$ . Since we also have  $\vdash \bigvee \text{SD}_n$ , we get  $\vdash \varphi$ .

**Remark 9.3**

Here are the steps that will take to prove the completeness of  $K$  for modal logic, and to get the other results in this chapter.

1. Define the set  $\mathcal{C}_{h,n}$  of *canonical sentences of height  $m$  and order  $n$* , and show the following:
  - (a) For each  $n$ ,  $\vdash \bigvee \mathcal{C}_{h,n}$ . As before, this is tedious, but routine.
  - (b) For each  $n$ , each  $\alpha \in \mathcal{C}_{h,n}$  is satisfiable. This will be much more challenging.
  - (c) For each  $n$  and each  $\varphi$  with  $\text{occ}(\varphi) \subseteq \{p_1, \dots, p_n\}$ , either  $\vdash \alpha \rightarrow \varphi$ , or else  $\vdash \alpha \rightarrow \neg\varphi$ . This will be fairly easy.
2. Then prove completeness in basically the same way as for propositional logic.

We write  $\oplus(\psi_1, \dots, \psi_n)$  to mean that *exactly one* of  $\psi_1, \dots, \psi_n$  holds. We need some results on this, so you should go back and review Exercises 5.28 and 5.29.

**Lemma 9.4**

The following are equivalent in propositional logic:

1.  $\oplus(\varphi_1, \dots, \varphi_k) \wedge \oplus(\psi_1, \dots, \psi_l)$
2.  $\oplus\{\varphi_i \wedge \psi_j : 1 \leq i \leq k, 1 \leq j \leq l\}$

**Lemma 9.5**

The following are equivalent in propositional logic:

1.  $\oplus(\varphi_1, \psi_1) \wedge \cdots \wedge \oplus(\varphi_n, \psi_n)$
2.  $\oplus\{\bigwedge_{i \in S} \varphi_i \wedge \bigwedge_{i \notin S} \psi_i : S \subseteq \{1, \dots, n\}\}$

**Definition 9.6:  $\nabla S$** 

For any set  $S$  of modal sentences, let  $\nabla S$  be defined as follows:

$$\nabla S = \bigwedge_{\psi \in S} \Diamond \psi \wedge \Box \bigvee_{\psi \in S} \psi.$$

The symbol  $\nabla$  is read “nabla.”

In connection with the definition above, here is the convention on empty conjunctions and disjunctions:

**Definition 9.7: empty conjunctions count as  $\top$ , empty disjunctions count as  $\bot$**

$$\bigwedge \emptyset \equiv \top \quad \bigvee \emptyset = \bot$$

**Example 9.8**

Here are some examples of the  $\nabla$ -notation.

$$\nabla(\emptyset) = \bigwedge \emptyset \wedge \Box \bigvee \emptyset \equiv \top \wedge \Box \bot = \Box \bot.$$

$$\nabla(\{\top\}) = \bigwedge \Diamond \top \wedge \Box \bigvee \top = \Diamond \top \wedge \Box \top \equiv \Diamond \top.$$

**Lemma 9.9**

Suppose that  $\vdash \oplus(\psi_1, \dots, \psi_n)$ . For each  $S \subseteq \{\psi_1, \dots, \psi_n\}$ , let

$$\blacktriangledown S = \bigwedge_{\psi_i \in S} \Diamond \psi_i \wedge \bigwedge_{\psi_i \notin S} \neg \Diamond \psi_i.$$

Then in  $K$ ,  $\vdash \oplus\{\blacktriangledown S : S \subseteq \{\psi_1, \dots, \psi_n\}\}$ . Also, in  $K$ ,  $\vdash \oplus\{\nabla S : S \subseteq \{\psi_1, \dots, \psi_n\}\}$ .

*Proof.* We have

$$\vdash \oplus(\Diamond \psi_1 \neg \Diamond \psi_1) \wedge \cdots \wedge \oplus(\Diamond \psi_n \neg \Diamond \psi_n).$$

Thus, the first part follows from Lemma 9.5.

For the second, we show that  $\vdash \blacktriangledown S \leftrightarrow \nabla S$  for all  $S$ . Note that since  $\vdash \oplus(\psi_1, \dots, \psi_n)$ , we also have  $\vdash \bigvee\{\psi_1, \dots, \psi_n\}$  and hence  $\vdash \Box \bigvee\{\psi_1, \dots, \psi_n\}$ . So

$$\vdash \bigwedge_{\psi_i \notin S} \neg \Diamond \psi_i \rightarrow \Box \bigvee_{\psi_i \in S} \psi_i.$$

This implies that  $\vdash \nabla S \rightarrow \nabla S$ . We also prove the converse. Since  $\vdash \oplus(\psi_1, \dots, \psi_n)$ , we have

$$\vdash \bigvee_{\psi_i \in S} \psi_i \rightarrow \bigwedge_{\psi_i \notin S} \neg \psi_i.$$

So

$$\vdash \Box \bigvee_{\psi_i \in S} \psi_i \rightarrow \Box \bigwedge_{\psi_i \notin S} \neg \psi_i.$$

This easily leads to  $\vdash \nabla S \rightarrow \nabla S$ . □

We also need a result on this  $\nabla$  operation.

#### Lemma 9.10

Suppose that  $S$  is a set of satisfiable sentences. Then for all state descriptions  $\sigma$ ,  $\nabla S \wedge \sigma$  is satisfiable.

*Proof.* For each  $\varphi \in S$ , let  $\mathcal{M}_\varphi$  be a model, and  $x_\varphi$  a point in it, such that  $x_\varphi \Vdash \varphi$  in  $\mathcal{M}_\varphi$ . Let  $\mathcal{N}$  be the disjoint union of all the models  $\mathcal{M}_\varphi$ , with a new point  $*$  added. We take as new edges  $* \rightarrow x_\varphi$  for all  $\varphi \in S$ . We also take whatever valuation is needed to insure that  $* \Vdash \sigma$  in  $\mathcal{N}$ . Then in  $\mathcal{N}$ , each  $x_\varphi$  still satisfies  $\varphi$ ; this is an easy argument using bisimulation. (See Exercise 7.10.) And then the semantics of  $\nabla$  implies that in  $*$   $\Vdash \nabla S \wedge \sigma$  in  $\mathcal{N}$ . □

## 9.1 Canonical sentences

### Definition 9.11: height and order

We define the *height* and *order* of an arbitrary sentence  $\varphi$  of modal logic by the following recursions:

|                           |                                 |                            |                                   |
|---------------------------|---------------------------------|----------------------------|-----------------------------------|
| $ht(p_n)$                 | $= 0$                           | $ord(p_n)$                 | $= n$                             |
| $ht(\top)$                | $= 0$                           | $ord(\top)$                | $= 0$                             |
| $ht(\bot)$                | $= 0$                           | $ord(\bot)$                | $= 0$                             |
| $ht(\neg\varphi)$         | $= ht(\varphi)$                 | $ord(\neg\varphi)$         | $= ord(\varphi)$                  |
| $ht(\varphi \wedge \psi)$ | $= \max(ht(\varphi), ht(\psi))$ | $ord(\varphi \wedge \psi)$ | $= \max(ord(\varphi), ord(\psi))$ |
| $ht(\Box\varphi)$         | $= 1 + ht(\varphi)$             | $ord(\Box\varphi)$         | $= ord(\varphi)$                  |

The height measures the maximum nesting depth of boxes, and the order gives the largest subscript on any atomic proposition occurring. We also let

$$\mathcal{L}_{h,n} = \{\varphi : ht(\varphi) \leq h, ord(\varphi) \leq n\}.$$

### Example 9.12

$$\begin{aligned} ht(\Diamond p_3 \wedge \Box \Diamond p_2) &= 2 \\ ord(\Diamond p_3 \wedge \Box \Diamond p_2) &= 3 \end{aligned}$$

**Definition 9.13**

We define the sets  $\mathcal{C}_{h,n}$  of *canonical sentences of height  $h$  and order  $n$*  as follows:  $\mathcal{C}_{0,n}$  is the set  $\mathbf{SD}_n$  of state descriptions of order  $n$ . Given  $\mathcal{C}_{h,n}$ , we let  $\mathcal{C}_{h+1,n}$  be the collection of sentences of the form  $\nabla S \wedge \sigma$ , where  $S \subseteq \mathcal{C}_{h,n}$ , and  $\sigma \in \mathbf{SD}_n$ .

We use Greek letters like  $\alpha$  and  $\beta$  for canonical sentences of some height and order.

**Proposition 9.14**

For each  $h$  and  $n$ ,  $\mathcal{C}_{h,n}$  is a finite subset of  $\mathcal{L}_{h,n}$ .

**Example 9.15: the sentences in  $\mathcal{C}_{1,1}(K)$ , for use throughout this chapter**

$$\begin{array}{ll} \alpha_1 = \nabla \emptyset \wedge p & \alpha_5 = \nabla \{\neg p\} \wedge p \\ \alpha_2 = \nabla \emptyset \wedge \neg p & \alpha_6 = \nabla \{\neg p\} \wedge \neg p \\ \alpha_3 = \nabla \{p\} \wedge p & \alpha_7 = \nabla C_{0,1} \wedge p \\ \alpha_4 = \nabla \{p\} \wedge \neg p & \alpha_8 = \nabla C_{0,1} \wedge \neg p \end{array}$$

**Example 9.16**

$C_{0,1} = \{p_1, \neg p_1\}$ .  $\mathcal{C}_{1,1}$  is a set with eight elements. Because we refer to these elements at various points, it makes sense to adopt names for them. And because we are dealing with  $n = 1$ , we drop the subscript on  $p_1$ . We list these in Example 9.15.

We have used the notation from Definition 9.13 above. For example,  $\nabla \{p\}$  abbreviates  $\Diamond p \wedge \Box p$ , and  $\nabla C_{0,1}$  abbreviates  $\Diamond p \wedge \Diamond \neg p \wedge \Box(p \vee \neg p)$ . (The last conjunct is redundant, so it is better to think of  $\nabla C_{0,1}$  as  $\Diamond p \wedge \Diamond \neg p$ .)

**Lemma 9.17**

For all  $h$  and  $n$ ,  $\vdash \oplus \mathcal{C}_{h,n}$  in  $K$ .

By soundness, we thus have  $\models_{\mathbf{all}} \oplus \mathcal{C}_{h,n}$ . As a result, every pointed model  $(\mathcal{M}, a)$  satisfies a unique element of  $\mathcal{C}_{h,n}$ .

*Proof.* This result is an elaboration of Lemma 9.5. See Exercise 9.6 below.  $\square$

We next present the fact that justifies thinking of the canonical sentences as analogs of state descriptions.

**Lemma 9.18**

Let  $\chi \in \mathcal{L}_{h,n}$  and  $\alpha \in \mathcal{C}_{h,n}$ . Then in  $K$ , either  $\vdash \alpha \rightarrow \chi$  or else  $\vdash \alpha \rightarrow \neg \chi$ .

*Proof.* By induction on  $\chi$ . All of the work is in the induction step for  $\Box$ . So we assume our lemma for  $\chi$  and prove it for  $\Box \chi$ . Let  $h$  and  $n$  be large enough so that  $\Box \chi \in \mathcal{L}_{h,n}$ , and let  $\alpha \in \mathcal{C}_{h,n}$ . We must have  $h > 0$ . Write  $\alpha$  as  $\nabla S \wedge \sigma$ . For each  $\beta \in S$ , the induction hypothesis applies to  $\chi$  and  $\beta$ .

We have two cases. First, assume that for some  $\beta \in S$ ,  $\vdash \beta \rightarrow \neg\chi$ . Then  $\vdash \Diamond\beta \rightarrow \Diamond\neg\chi$ . Since  $\alpha$  is  $\nabla S \wedge \sigma$ , we have  $\vdash \alpha \rightarrow \Diamond\beta$ . So  $\vdash \alpha \rightarrow \Diamond\neg\chi$ . We thus have  $\vdash \alpha \rightarrow \neg\Box\chi$ . The other case is when for each  $\beta \in S$ ,  $\vdash \beta \rightarrow \chi$ . Then  $\vdash \bigvee S \rightarrow \chi$ . So we also have  $\vdash \Box\bigvee S \rightarrow \Box\chi$ . Then  $\vdash \alpha \rightarrow \Box\chi$ . This completes the induction step for  $\Box$ , and hence the proof overall.  $\square$

### Lemma 9.19

For all  $h$  and  $n$ , every  $\alpha \in \mathcal{C}_{h,n}$  is satisfiable: there is a model  $\mathcal{M}$  and some  $x \in M$  such that  $x \Vdash \alpha$  in  $\mathcal{M}$ .

*Proof.* By induction on  $h$ . For  $h = 0$ , this is a fact we know about state descriptions. The induction step is an easy application of Lemma 9.10.  $\square$

## 9.2 Completeness of $K$

In the last section, we introduced the canonical sentences in modal logic and proved their basic properties. All of that work will be applied in this section and the next. We aim in this section to prove the completeness of  $K$ : if  $\models_{\text{all}} \varphi$ , then  $\vdash \varphi$  in  $K$ . The basic idea is to follow the same method that we used in proving completeness for propositional logic and the switching logic  $\mathcal{L}(\ast)$ . The only real difference is that takes more work to show that every canonical sentence is satisfiable. (The parallel results for propositional logic and switching logic were very easy.)

### Theorem 9.20: Completeness

If  $\models_{\text{all}} \varphi$ , then  $\vdash \varphi$  in  $K$ .

*Proof.* Assume that  $\models_{\text{all}} \varphi$ , and let  $h = ht(\psi)$ , and  $n = ord(\varphi)$ .

For all  $\alpha \in \mathcal{C}_{h,n}$ , either  $\vdash \alpha \rightarrow \varphi$  or else  $\vdash \alpha \rightarrow \neg\varphi$ . (Provability here is in  $K$ .) We claim that the second alternative is impossible. For suppose that  $\vdash \alpha \rightarrow \neg\varphi$ . By soundness,  $\models_{\text{all}} \alpha \rightarrow \neg\varphi$ . By Lemma 9.19,  $\alpha$  is satisfiable. But then in the same model,  $\alpha \Vdash \neg\varphi$ . And this contradicts our assumption on  $\varphi$ .

By propositional logic,  $\vdash \bigvee \mathcal{C}_{h,n} \rightarrow \varphi$ . And as we know from Lemma 9.17,  $\vdash \bigvee \mathcal{C}_{h,n}$ . So as desired,  $\vdash \varphi$ .  $\square$

### Corollary 9.21: Decidability

It is decidable whether  $\models_{\text{all}} \psi$  or not.

*Proof.* Given  $\psi$ , compute  $h = ht(\psi)$ ,  $n = ord(\psi)$ , and  $\mathcal{C}_{\leq h,n}$ . If we want to see whether  $\psi$  is true in points of *all* models, we need only look to see if it is true at all points in that one (big, but finite) model. And the evaluation is again a computable matter. (This will not be terribly practical even for small values of  $h$  and  $n$ , as those who work Exercise 9.7 know.)  $\square$

### 9.3 The models $\mathcal{C}_{h,n}(L)$

We have completeness for  $K$ , and we would also like to have it for logics like  $T$ ,  $S4$ , etc. For this, the model of the previous section is not good enough.

**Definition 9.22: the models  $(\mathcal{C}_{h,n}(L), \xrightarrow{c})$**

Fix a logical system  $L$ , such as  $K4$  or  $S5$ . We define the *canonical model of consistent sentences of  $L$  of height  $h$  and order  $n$* :

1. The points of  $\mathcal{C}_{h,n}(L)$  are the elements of  $\mathcal{C}_{h,n}$  which are consistent in  $L$ .
2.  $\alpha \xrightarrow{c} \beta$  iff  $\alpha \wedge \Diamond\beta$  is consistent in  $L$ .
3.  $v(p_i) = \{\alpha : \vdash \alpha \rightarrow p_i\}$ .

In the examples below, we deal with  $n = 1$ . So instead of writing  $p_1$  or  $\neg p_1$ , we simply write  $p$  or  $\neg p$ .

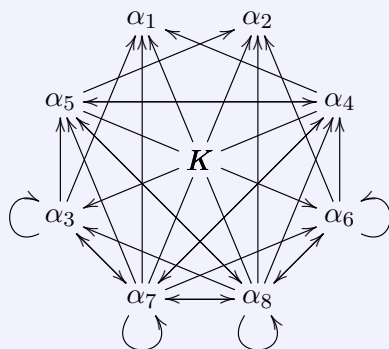
**The pictures on the next page.** Example 9.23 contains  $\mathcal{C}_{1,1}(L)$  for six specific logics  $L$ :  $K$ ,  $K4$ ,  $KB$ ,  $S4$ ,  $KB4$ , and  $S5$ . These models are based on our listing of  $\mathcal{C}_{1,1}$  in Example 9.16.

Here are the details concerning  $\mathcal{C}_{1,1}(K4)$ : Our sentences  $\alpha_1, \dots, \alpha_8$  are all consistent in  $K4$ . But we lose some arrows. Specifically, we lose  $\alpha_5 \rightarrow \alpha_4$ ,  $\alpha_4 \rightarrow \alpha_5$ ,  $\alpha_3 \rightarrow \alpha_7$ , and  $\alpha_6 \rightarrow \alpha_8$ . These facts correspond to various facts about *inconsistency* in  $K4$ ; for example  $\alpha_5 \wedge \Diamond\alpha_4$  is not consistent in  $K4$ , and this is why we lose this arrow.

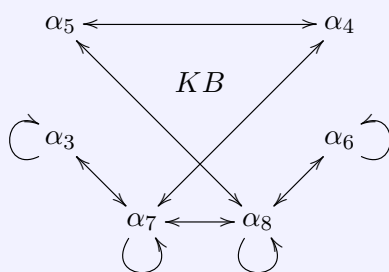
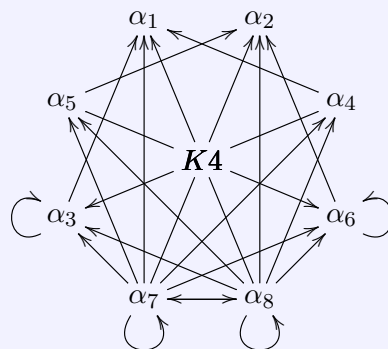
In the models for  $S4$ ,  $KB4$  and  $S5$ , we also lose points. This means that certain of the sentences  $\alpha_1, \dots, \alpha_8$  are not consistent in some of the logics. For example,  $\alpha_3$  is not consistent in  $KB4$ , and  $\alpha_1$  is not consistent in  $S4$ .

**Example 9.23:**  $\mathcal{C}_{1,1}(L)$  for various logics  $L$ 

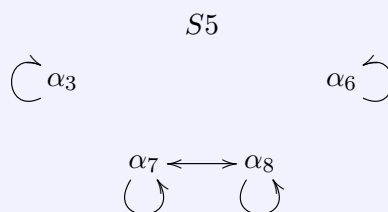
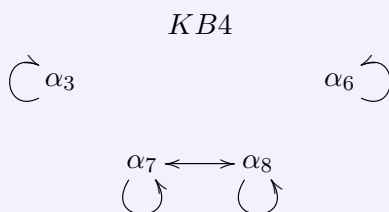
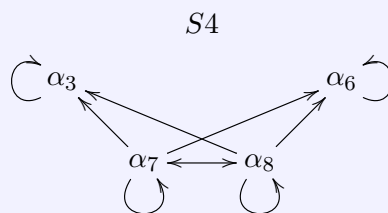
The sentences  $\alpha_1, \dots, \alpha_8$  are from Example 9.15.

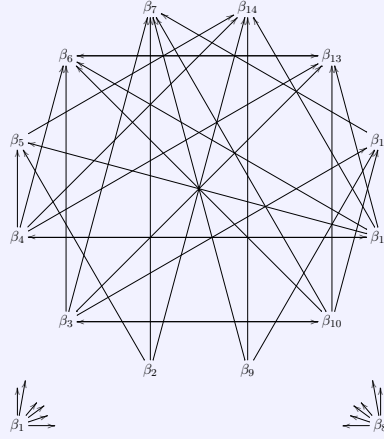


$\alpha_1$        $\alpha_2$



$\alpha_1$        $\alpha_2$



**Example 9.24:**  $\mathcal{C}_{2,1}(S4)$ , omitting self-loops

The sentences in  $\mathcal{C}_{11}(K)$  which happen to be consistent in  $S4$  are  $\alpha_3, \alpha_6, \alpha_7$ , and  $\alpha_8$ . Using these, we list the elements of  $\mathcal{C}_{2,1}$ :

$$\begin{array}{ll}
 \beta_1 = \nabla\{\alpha_3, \alpha_6, \alpha_7, \alpha_8\} \wedge p & \beta_8 = \nabla\{\alpha_3, \alpha_6, \alpha_7, \alpha_8\} \wedge \neg p \\
 \beta_2 = \nabla\{\alpha_3, \alpha_6, \alpha_7\} \wedge p & \beta_9 = \nabla\{\alpha_3, \alpha_6, \alpha_8\} \wedge \neg p \\
 \beta_3 = \nabla\{\alpha_3, \alpha_7, \alpha_8\} \wedge p & \beta_{10} = \nabla\{\alpha_3, \alpha_7, \alpha_8\} \wedge \neg p \\
 \beta_4 = \nabla\{\alpha_6, \alpha_7, \alpha_8\} \wedge p & \beta_{11} = \nabla\{\alpha_6, \alpha_7, \alpha_8\} \wedge \neg p \\
 \beta_5 = \nabla\{\alpha_6, \alpha_7\} \wedge p & \beta_{12} = \nabla\{\alpha_3, \alpha_8\} \wedge \neg p \\
 \beta_6 = \nabla\{\alpha_7, \alpha_8\} \wedge p & \beta_{13} = \nabla\{\alpha_7, \alpha_8\} \wedge \neg p \\
 \beta_7 = \nabla\{\alpha_3\} \wedge p & \beta_{14} = \nabla\{\alpha_6\} \wedge \neg p
 \end{array}$$

These are the elements of  $\mathcal{C}_{2,1}$  consistent in  $S4$ . The structure as always is given by  $\beta_i \hookrightarrow \beta_j$  iff  $\beta_i \wedge \Diamond \beta_j$  is consistent in  $S4$ .

The picture does not show the reflexive arrows on all points, or the arrows from  $\beta_1$  and  $\beta_8$  to all points. It also does not include the information that  $p$  is true exactly at  $\beta_1, \dots, \beta_7$ .

**9.3.1 Properties of  $\mathcal{C}_{h,n}(L)$** 

The models  $\mathcal{C}_{h,n}(L)$  are the vehicles for the main completeness results of this chapter. So we continue now with their study.

**Lemma 9.25**

Let  $\psi \in \mathcal{L}_{h,n}$ . Then

1. In  $K$ ,  $\vdash \psi \leftrightarrow \bigvee\{\alpha \in \mathcal{C}_{h,n} : \text{in } K, \vdash \alpha \rightarrow \psi\}$ .
2. In  $L$ ,  $\vdash \psi \leftrightarrow \bigvee\{\alpha \in \mathcal{C}_{h,n}(L) : \text{in } K, \vdash \alpha \rightarrow \psi\}$ .

*Proof.* Let  $S$  be the set of  $\alpha \in \mathcal{C}_{h,n}$  such that  $\vdash \alpha \rightarrow \psi$ . Then  $\vdash \bigvee S \rightarrow \psi$ . And for  $\alpha \notin S$ ,  $\vdash \alpha \rightarrow \neg\psi$ . So  $\vdash \psi \leftrightarrow \bigvee_{\alpha \in \mathcal{C}_{h,n}} (\alpha \wedge \psi) \leftrightarrow \bigvee_{\alpha \in S} (\alpha \wedge \psi) \leftrightarrow \bigvee_{\alpha \in S} \alpha$ .



(2) follows from (1) and the fact that in  $L$ ,  $\vdash \bigvee \mathcal{C}_{h,n}(L)$ .  $\square$

### Lemma 9.26: Existence Lemma

Let  $\psi \in \mathcal{L}_{h,n}$ , let  $\varphi$  be arbitrary, and suppose that  $\varphi \wedge \Diamond\psi$  is consistent in  $L$ . Then there is some  $\alpha \in \mathcal{C}_{h,n}(L)$  such that  $\varphi \wedge \Diamond\alpha$  is consistent in  $L$ , and  $\vdash \alpha \rightarrow \psi$  in  $K$ .

*Proof.* We write  $\vdash$  for provability in  $L$ . By Lemma 9.25,

$$\vdash \Diamond\psi \leftrightarrow \bigvee \{\Diamond\alpha : \alpha \in \mathcal{C}_{h,n}(L) \text{ and in } K, \vdash \alpha \rightarrow \psi\}.$$

So

$$\vdash \varphi \wedge \Diamond\psi \leftrightarrow \bigvee \{\varphi \wedge \Diamond\alpha : \alpha \in \mathcal{C}_{h,n}(L) \text{ and in } K, \vdash \alpha \rightarrow \psi\}.$$

By our assumption, the finite disjunction on the right just above is consistent in  $L$ . So at least one disjunct must be consistent. This gives some  $\alpha \in \mathcal{C}_{h,n}(L)$  with the desired properties.  $\square$

### Lemma 9.27: Truth Lemma for $\mathcal{C}_{h,n}(L)$

For all  $\alpha \in \mathcal{C}_{h,n}(L)$  and all  $\psi \in \mathcal{L}_{h,n}$ ,  $\alpha \Vdash \psi$  in  $\mathcal{C}_{h,n}(L)$  iff  $\vdash \alpha \rightarrow \psi$  in  $K$ .

*Proof.* By induction on  $\psi$ . The only interesting step is for  $\Diamond\psi$ , assuming the result for  $\psi$ . So assume that  $ht(\Diamond\psi) \leq h$  and  $order \leq n$ . Then  $ht(\psi) < h$ .

The only model in this proof is  $\mathcal{C}_{h,n}(L)$ , and so we shall not mention this again.

Assume that  $\alpha \Vdash \Diamond\psi$ . Let  $\beta \in \mathcal{C}_{h,n}(L)$  be such that  $\alpha \wedge \Diamond\beta$  is consistent in  $L$  and  $\beta \Vdash \psi$ . By induction hypothesis,  $\vdash \beta \rightarrow \psi$ . Thus  $\vdash \Diamond\beta \rightarrow \Diamond\psi$ . We thus see that  $\alpha \wedge \Diamond\psi$  is consistent in  $L$ . As we know, either  $\vdash \alpha \rightarrow \Diamond\psi$  (in  $K$ ), or  $\vdash \alpha \rightarrow \neg\Diamond\psi$  (in  $K$ ). But the latter case cannot happen, because it would contradict the consistency of  $\alpha \wedge \Diamond\psi$ .

In the other direction, suppose that  $\vdash \alpha \rightarrow \Diamond\psi$ . Since  $\alpha$  is consistent in  $L$ , so is  $\alpha \wedge \Diamond\psi$ . So by Existence Lemma 9.26, there is some  $\beta \in \mathcal{C}_{h,n}(L)$  such that  $\alpha \wedge \Diamond\beta$  is consistent in  $L$ , and  $\vdash \beta \rightarrow \psi$  in  $K$ . Thus  $\alpha \xrightarrow{c} \beta$  in the model, and by induction hypothesis,  $\beta \Vdash \psi$ . So  $\alpha \Vdash \Diamond\psi$ .  $\square$

As a result of Lemma 9.27,  $\alpha \Vdash \alpha$  for all  $\alpha \in \mathcal{C}_{h,n}(L)$ .

### Lemma 9.28

The following hold for all  $h$  and  $n$ :

1. If  $KT \leq L$ ,  $\mathcal{C}_{h,n}(L)$  is reflexive.
2. If  $KD \leq L$ ,  $\mathcal{C}_{h,n}(L)$  is serial.
3. If  $KB \leq L$ ,  $\mathcal{C}_{h,n}(L)$  is symmetric.

*Proof.* For (1), let  $\alpha \in \mathcal{C}_{h,n}(L)$ . We must show that  $\alpha \sqsubset \alpha$ ; i.e.,  $\alpha \wedge \Diamond \alpha$  is consistent. If not, then  $\vdash \alpha \rightarrow \Box \neg \alpha$ . But by T,  $\vdash \Box \neg \alpha \rightarrow \neg \alpha$ . So  $\vdash \alpha \rightarrow \neg \alpha$ . This contradicts the consistency of  $\alpha$  in the logic  $L$ .

Next, we prove (2). Let  $\alpha \in \mathcal{C}_{h,n}(L)$ , so  $\alpha \wedge \Diamond \top$  is consistent in any logic which proves the  $D$ -axioms. By the Existence Lemma 9.26, there is some  $\beta \in \mathcal{C}_{h,n}(L)$  such that  $\alpha \wedge \Diamond \beta$  is consistent. Thus  $\alpha \sqsubset \beta$  in  $\mathcal{C}_{h,n}(L)$ .

Finally, we show that  $\mathcal{C}_{h,n}(L)$  is symmetric whenever  $KB \leq L$ . Suppose that  $\alpha \wedge \Diamond \beta$  is consistent, but  $\vdash \beta \rightarrow \Box \neg \alpha$ . Then  $\vdash \Diamond \beta \rightarrow \Diamond \Box \neg \alpha$ . By B,  $\vdash \alpha \rightarrow \Box \Diamond \alpha$ . So  $\Box \Diamond \alpha \wedge \Diamond \Box \neg \alpha$  is consistent (in  $L$ ). This is clearly a contradiction.  $\square$

The results of this section give completeness/decidability results for the logics  $KT$ ,  $KD$ , and  $KB$  on the appropriate classes of structures. The proofs are routine modifications of Theorem 9.20 and Corollary 9.21. We illustrate this point with  $KB$ .

### Theorem 9.29: Completeness

Let  $\psi$  be an arbitrary sentence in  $\mathcal{L}(\Box, \Diamond)$ , the basic language of modal logic. If  $\models_{\mathfrak{S}} \psi$ , then  $\vdash \psi$  in  $KB$ . Moreover, the property of being provable in  $KB$  is decidable.

*Proof.* Let  $h$  and  $n$  be large enough so that  $\psi \in \mathcal{L}_{h,n}$ . The model  $\mathcal{C}_{h,n}(KB)$  is symmetric, and so each element of the model satisfies  $\psi$  in it. By Lemma 9.27, for all  $\alpha \in \mathcal{C}_{h,n}(KB)$ ,  $\vdash \alpha \rightarrow \psi$ . So in  $K$ ,  $\vdash \bigvee \mathcal{C}_{h,n} \rightarrow \psi$ . But in  $KB$ ,  $\vdash \bigvee \mathcal{C}_{h,n}(KB)$ , by Lemma 9.25. Hence in  $KB$ ,  $\vdash \psi$ .

The decidability is again a corollary to the proof above, but we have to do a little more work. Here is what we *would like to say*:  $\psi$  is provable in  $KB$  iff it holds at each world of  $\mathcal{C}_{h,n}(KB)$ , where the numbers  $h$  and  $n$ , and the model  $\mathcal{C}_{h,n}(KB)$  itself, are all constructed from  $\psi$ . The problem is that without knowing what is provable in  $KB$ , we have no real way to construct  $\mathcal{C}_{h,n}(KB)$ . What we have to say is that we know an upper bound  $N$  on the size of that model. (The bound  $N$  would be the size of the set  $\mathcal{C}_{h,n}$ .) And then we know that  $\alpha \in \mathcal{C}_{h,n}$  is consistent in  $KB$  if it has a symmetric model of size  $\leq N$ . So this gives an algorithm to test for consistency in  $KB$ . Using negation, we get an algorithm to test for provability.  $\square$

## 9.4 Completeness of $K4$ and $S4$

At this point, we have the relevant completeness theorem for all logics which extend  $K$  using any of the axioms  $T$ ,  $D$ , and  $B$ . So we have eight theorems in all. We would like to study  $K4$ ,  $KB4$ ,  $S4$ ,  $KD45$ , and  $S5$ .

We should emphasize that the title of our section is not really accurate. What we want to say is that the section has the completeness of  $K4$  for the class of transitive models. And (with small modifications) it has the completeness of  $S4$  for the class of reflexive, transitive models.

### 9.4.1 $K4$

The completeness results for  $K4$  and the analog of Lemma 9.28 for the standard logics extending it are harder to come by.

**Lemma 9.30**

For every  $\alpha \in \mathcal{C}_{h+1,n}$  there is a unique  $\alpha' \in \mathcal{C}_{h,n}$  such that in  $K$ ,  $\vdash \alpha \rightarrow \alpha'$ .

*Proof.* By Lemma 9.17, there is some  $\beta \in \mathcal{C}_{h,n}$  such that in  $\mathcal{C}_{h+1,n}$ ,  $\alpha \Vdash \beta$ . But  $\beta$  must be unique, since we also have  $\models_{\text{all}} \oplus \mathcal{C}_{h,n}$ .  $\square$

**Lemma 9.31**

Suppose that  $K4 \leq L$ . Let  $\alpha \in \mathcal{C}_{h+1,n}(L)$  and  $\beta \in \mathcal{C}_{h,n}$ .

1. Assume that  $\vdash \alpha \rightarrow \Diamond \beta$ . Then  $\beta \in \mathcal{C}_{h,n}(L)$ .
2. If  $\vdash \alpha \rightarrow \Diamond \beta$  and  $\vdash \beta \rightarrow \Diamond \gamma$ , then there is some  $\delta$  such that  $\vdash \alpha \rightarrow \Diamond \delta$  and  $\vdash \delta \rightarrow \gamma$  in  $K$ .

*Proof.* The first assertion boils down to the fact that if  $\alpha$  is consistent in  $L$ , and  $\alpha \rightarrow \Diamond \beta$ , then  $\beta$  must also be consistent in  $L$ . We turn to the second. Assuming that  $\vdash \alpha \rightarrow \Diamond \beta$  and  $\vdash \beta \rightarrow \Diamond \gamma$ , we see that  $\vdash \alpha \rightarrow \Diamond \Diamond \gamma$ . We have  $\vdash \Diamond \Diamond \gamma \rightarrow \Diamond \gamma$ ; this is a 4 axiom. So  $\vdash \alpha \rightarrow \Diamond \gamma$ . Using Lemma 9.26, we must have some  $\delta$  as required.  $\square$

**Lemma 9.32**

Let  $\alpha, \beta \in \mathcal{C}_{h+1,n}(K4)$ . Assume that  $\alpha \hookrightarrow \beta$ . Then  $\vdash \alpha \rightarrow \Diamond \beta'$ , and whenever  $\gamma \in \mathcal{C}_{h,n}$  is such that  $\vdash \beta \rightarrow \Diamond \gamma$ , then also  $\vdash \alpha \rightarrow \Diamond \gamma$ .

*Proof.* First, assume that  $\alpha \wedge \Diamond \beta$  is consistent. Since  $\vdash \beta \rightarrow \beta'$ , we see that  $\alpha \wedge \Diamond \beta'$  is also consistent. Note that  $\Diamond \beta' \in \mathcal{L}_{h+1,n}$ . Therefore  $\vdash \alpha \rightarrow \Diamond \beta'$ , and we have our first point. For the second, suppose that  $\vdash \beta \rightarrow \Diamond \gamma$ . Then again  $\alpha \wedge \Diamond \Diamond \gamma$  is consistent. But  $\vdash \Diamond \Diamond \gamma \rightarrow \Diamond \gamma$ , due to 4. Therefore  $\vdash \alpha \rightarrow \Diamond \gamma$ .  $\square$

**Definition 9.33: the models  $(\mathcal{D}_{h,n}(L), \xrightarrow{d})$** 

We define models

1.  $\mathcal{D}_{0,n}(L)$  is the model  $\mathcal{C}_{0,n}$ .
2. For all  $h$ , the points of  $\mathcal{D}_{h+1,n}(L)$  are the points of  $\mathcal{C}_{h+1,n}(L)$ , and the valuations in the two models are the same.
3. For  $\alpha, \beta \in \mathcal{D}_{h+1,n}(L)$ , whenever  $\vdash \beta \rightarrow \Diamond \gamma$ , then also  $\vdash \alpha \rightarrow \Diamond \gamma$ .

These models  $\mathcal{D}_{h,n}(L)$  are mainly a technical tool used in the study of the earlier models  $\mathcal{C}_{h,n}(L)$ . The point is that these models  $\mathcal{D}_{h,n}(L)$  are easily transitive. (This is the point of introducing them.) But this is not the case for  $\mathcal{C}_{h,n}(L)$ .

It will turn out that for  $L = K4$ , the models are equal:  $\mathcal{D}_{h,n}(K4) = \mathcal{C}_{h,n}(K4)$ , and so we have an indirect way to show that  $\mathcal{C}_{h,n}(K4)$  is transitive. However, we cannot prove that for all logics  $L \geq K4$ ,  $\mathcal{D}_{h,n}(L) = \mathcal{C}_{h,n}(L)$ ; this is false. Thus the construction of the  $\mathcal{D}_{h,n}(L)$  models must itself be modified in order to obtain the completeness of other logics, as we shall see in Section 9.5.

#### Lemma 9.34

Let  $K4 \leq L$ . Then  $\mathcal{C}_{h,n}(L) \subseteq \mathcal{D}_{h,n}(L)$ . And if  $\alpha \mathrel{\mathcal{C}_{h,n}} \beta$ , then  $\alpha \mathrel{\mathcal{D}_{h,n}} \beta$ .

*Proof.* This follows from Lemma 9.32.  $\square$

#### Lemma 9.35

If  $K4 \leq L$ , then  $\mathcal{D}_{h,n}(L)$  is transitive.

*Proof.*  $\mathcal{D}_{0,n}$  is easily transitive. We therefore consider only models  $\mathcal{D}_{h+1,n}(L)$ . Suppose that  $\alpha \mathrel{\mathcal{D}_{h,n}} \beta \mathrel{\mathcal{D}_{h,n}} \gamma$ . We show that  $\alpha \mathrel{\mathcal{D}_{h,n}} \gamma$ . Since  $\beta \mathrel{\mathcal{D}_{h,n}} \gamma$ , we have  $\vdash \beta \rightarrow \Diamond \gamma'$ . It follows from this and the fact that  $\alpha \mathrel{\mathcal{D}_{h,n}} \beta$  that  $\vdash \alpha \rightarrow \Diamond \gamma'$ . This is half of what we want. For the other half, suppose that  $\vdash \gamma \rightarrow \Diamond \delta$ . Then  $\vdash \beta \rightarrow \Diamond \delta$ , and so also  $\vdash \alpha \rightarrow \Diamond \delta$ .  $\square$

#### Lemma 9.36: Truth Lemma for $\mathcal{D}_{h,n}(L)$

Let  $K4 \leq L$ . Fix  $h$  and  $n$ . For all  $\alpha \in \mathcal{D}_{h,n}(L)$  and all  $\psi \in \mathcal{L}_{h,n}$ ,  $\alpha \Vdash \psi$  in  $\mathcal{D}_{h,n}(L)$  iff  $\vdash \alpha \rightarrow \psi$  in  $K$ .

*Proof.* By induction on  $\psi$ . We only do the step for  $\Diamond \psi$ . Assume that  $ht(\Diamond \psi) \leq h$ .

Assume that  $\alpha \Vdash \Diamond \psi$  in  $\mathcal{D}_{h,n}(L)$ . Let  $\beta \in \mathcal{D}_{h,n}(L)$  be such that  $\alpha \mathrel{\mathcal{D}_{h,n}} \beta$  and  $\beta \Vdash \psi$ . By induction hypothesis,  $\vdash \beta \rightarrow \psi$  in  $K$ . Note now that  $ht(\psi) \leq h - 1$ . And by the definition of  $\mathcal{D}_{h,n}(L)$ ,  $\beta' \in \mathcal{D}_{h-1,n}$ . We claim that in  $K$ ,  $\vdash \beta' \rightarrow \psi$ . (For if not, then we have  $\vdash \beta' \rightarrow \neg \psi$ . And since  $\vdash \beta \rightarrow \beta'$ , we also have  $\vdash \beta \rightarrow \neg \psi$ . This contradicts the consistency of  $\beta$ .) And since  $\vdash \alpha \rightarrow \Diamond \beta'$ , we have  $\vdash \alpha \rightarrow \Diamond \psi$ .

In the other direction, suppose that  $\vdash \alpha \rightarrow \Diamond \psi$ . Then  $\alpha \wedge \Diamond \psi$  is consistent in  $K$ . By Existence Lemma 9.26, let  $\beta \in \mathcal{C}_{h,n}(L)$  be such that  $\alpha \mathrel{\mathcal{C}_{h,n}} \beta$  and  $\vdash \beta \rightarrow \psi$  in  $K$ . Then  $\beta \in \mathcal{D}_{h,n}(L)$ , and by Lemma 9.34,  $\alpha \mathrel{\mathcal{D}_{h,n}} \beta$ . By induction hypothesis,  $\beta \Vdash \psi$ . So we have  $\alpha \Vdash \Diamond \psi$ , as desired.  $\square$

#### Lemma 9.37

$\mathcal{C}_{h,n}(K4) = \mathcal{D}_{h,n}(K4)$ .

*Proof.* By definition,  $\mathcal{D}_{h,n}(L) \subseteq \mathcal{C}_{h,n}(L)$ . Lemma 9.34 shows that  $\mathcal{C}_{h,n}(L) \subseteq \mathcal{D}_{h,n}(L)$  for all  $L \geq K4$ .

Continuing, Lemma 9.32 shows that  $\mathrel{\mathcal{C}_{h,n}}$  is a subrelation of  $\mathrel{\mathcal{D}_{h,n}}$ . In the other direction, suppose that  $\alpha \mathrel{\mathcal{D}_{h,n}} \beta$ . Using Lemma 9.36, we see that for  $\mathcal{D}_{h,n}(K4)$ ,  $\alpha \Vdash \alpha \wedge \Diamond \beta$ . Since the model  $\mathcal{D}_{h,n}(K4)$

is transitive (by Lemma 9.35), we see that  $\alpha \wedge \Diamond\beta$  is satisfiable on a transitive model and hence is consistent in  $K4$ . Thus  $\alpha \xrightarrow{c} \beta$ .

The models  $\mathcal{C}_{h,n}(K4)$  and  $\mathcal{D}_{h,n}(K4)$  agree on atomic sentences. □

### Theorem 9.38

Every  $\alpha \in \mathcal{C}_{h,n}(K4)$  is satisfied on some finite transitive model. Thus  $K4$  is complete for transitive models, and also decidable.

This concludes our work on the completeness of  $K4$  for transitive models. A simple observation extends the work here to prove the completeness of  $S4$  for reflexive and transitive models.

## 9.5 Completeness of $S5$

We conclude this chapter with the completeness of  $K45$  for the class of models which are transitive and Euclidean. With small modifications that we indicate in Exercise 9.14, we also obtain the completeness of  $S5$  for the class of models which are equivalence relations.

### Definition 9.39: the models $(\mathcal{E}_{h,n}(L), \xrightarrow{c})$

We define models  $(\mathcal{E}_{h,n}(L), \xrightarrow{c})$  by:

1.  $\mathcal{E}_{h,n}(L)$  has the same points and valuation as  $\mathcal{D}_{h,n}(L)$  from Definition 9.33.
2. For  $\alpha, \beta \in E_{h+1,n}(L)$ ,  $\alpha \xrightarrow{c} \beta$  iff  $\alpha \xrightarrow{d} \beta$ , and also: whenever  $\gamma \in \mathcal{C}_{h,n}$  is such that  $\vdash \alpha \rightarrow \Diamond\gamma$ , then also  $\vdash \beta \rightarrow \Diamond\gamma$ .

### Lemma 9.40

Let  $K45 \leq L$ . If  $\alpha \xrightarrow{c} \beta$ , then  $\alpha \xrightarrow{c} \beta$ .

*Proof.* Let  $\alpha \xrightarrow{c} \beta$ . By Lemma 9.32.  $\alpha \xrightarrow{d} \beta$ . Suppose that  $\gamma \in \mathcal{C}_{h,n}$  and  $\vdash \alpha \rightarrow \Diamond\gamma$ ; we show that  $\vdash \beta \rightarrow \Diamond\gamma$ . If not, then  $\vdash \beta \rightarrow \Box\neg\gamma$ . So  $\vdash \Diamond\beta \rightarrow \Diamond\Box\neg\gamma$ . But  $\alpha \wedge \Diamond\beta$  is consistent, and so is  $\alpha \wedge \Diamond\Box\neg\gamma$ . Using an axiom of  $K5$ ,  $\vdash \Diamond\Box\neg\gamma \rightarrow \Box\neg\gamma$ . so  $\alpha \wedge \Box\neg\gamma$  is consistent. By Lemma 9.18,  $\vdash \alpha \rightarrow \Box\neg\gamma$ . But this contradicts the consistency of  $\alpha$ . □

### Lemma 9.41

$\mathcal{E}_{h,n}(L)$  is transitive and Euclidean.

*Proof.*  $\mathcal{E}_{0,n}(L)$  is easily Euclidean. We therefore consider only the models  $\mathcal{E}_{h+1,n}(L)$ . For the transitivity, suppose that  $\alpha \xrightarrow{c} \beta \xrightarrow{c} \gamma$ ; we show that  $\alpha \xrightarrow{c} \gamma$ . By Lemma 9.35,  $\alpha \xrightarrow{d} \gamma$ . Further, suppose that  $\vdash \alpha \rightarrow \Diamond\delta$ , where  $\delta \in \mathcal{C}_{h,n}(L)$ . Then also  $\vdash \beta \rightarrow \Diamond\delta$ , and in addition we have  $\vdash \gamma \rightarrow \Diamond\delta$ .

We turn to the Euclidean property. Suppose that  $\alpha \vDash \beta$  and  $\alpha \vDash \gamma$ . We show that  $\beta \vDash \gamma$ . First, since  $\vdash \alpha \rightarrow \Diamond \gamma'$  and  $\vdash \alpha \vDash \beta$ , we have  $\vdash \beta \rightarrow \Diamond \gamma'$ . Suppose that  $\vdash \beta \rightarrow \Diamond \delta$ . Then  $\vdash \alpha \rightarrow \Diamond \delta$ , since  $\alpha \vDash \beta$ . And then since  $\alpha \vDash \gamma$ , we have  $\vdash \gamma \rightarrow \Diamond \delta$ , as desired.  $\square$

#### Lemma 9.42

For  $L = K45$ ,  $\mathcal{C}_{h,n}(L) = \mathcal{E}_{h,n}(L)$ .

*Proof.* We follow the proof of Lemma 9.37. We need Lemma 9.40. We also need a result like Lemma 9.36, and the proof would be nearly the same as that of Lemma 9.36 itself. We also use Lemma 9.41.  $\square$

#### Theorem 9.43

Every  $\alpha \in \mathcal{C}_{h,n}(K45)$  is satisfied on some finite transitive, Euclidean model. Thus  $K45$  is complete for transitive, Euclidean models, and also decidable.

## 9.6 Exercises on completeness theorems

**Exercise 9.1.** What does Lemma 9.9 say concretely when  $n = 1$ ?

**Exercise 9.2.** Let  $\mathcal{M}$  be any relational model. Fix a number  $n$ . For every  $a \in M$  and every number  $h \geq 0$ , we define the sentence  $\varphi_a^{h,n}$ . The definition is by recursion on  $h$  (simultaneously for all  $a \in A$ ) as follows:  $\varphi_a^{0,n}$  is the unique canonical sentence of height 0 and order  $n$  satisfied by  $a$ . (It is the conjunction of all atomic sentences satisfied by  $a$  and all negations of atomic sentences not satisfied by  $a$ .) Given  $\varphi_b^{h,n}$  for all  $b \in A$ , we define

$$\varphi_a^{h+1,n} = \nabla \{ \varphi_b^{h,n} : a \rightarrow b \} \wedge \varphi_a^{0,n}.$$

1. Prove that for all models  $\mathcal{M}$ , all points  $a \in M$ , and all numbers  $h$  and  $n$ ,  $a \Vdash \varphi_a^{h,n}$ .
2. Prove that each  $\varphi_a^{h,n}$  is a canonical sentence of height  $h$  and order  $n$ .
3. Prove every canonical sentence of height  $h$  and order  $n$  is of the form  $\varphi_a^{h,n}$  for some model  $\mathcal{M}$  and some  $a \in M$ .

**Exercise 9.3.** This exercise outlines the modification of Theorem 9.20 needed for  $KD$ .

1. Check that every state description of order  $n$  is consistent in  $KD$ . [Hint: find a model and appeal to soundness.]
2. Check that the model  $\mathcal{C}_{\leq h,n}(KD)$  is almost serial: the only points with no outgoing arrows are the elements of  $\mathcal{C}_{0,n}$ .

3. Find a simple modification of the arrow relation of  $\mathcal{C}_{\leq h,n}$  that defines  $\mathcal{C}_{\leq h,n}(KD)$  as a serial model.
4. Check Lemma 9.18 in your model. (This is the hardest part of the whole exercise. It is basically a careful re-reading of Lemma 9.18.)
5. Check that we have completeness for  $KD$  on serial models.

**Exercise 9.4.** Draw a picture of  $\mathcal{C}_{1,1}(KT)$ .

**Exercise 9.5.** Are  $\mathcal{C}_{1,1}(KT)$ ,  $\mathcal{C}_{1,1}(K4)$ ,  $\mathcal{C}_{1,1}(KB)$ , and  $\mathcal{C}_{1,1}(KB4)$  submodels of  $\mathcal{C}_{1,1}(K)$ ? Why or why not?

**Exercise 9.6.** Prove Lemma 9.17. Use induction on  $h$ . In the case of  $h = 0$ , the result follows from Lemma 9.5; we take  $\varphi_i = p_i$  and  $\psi_i = \neg p_i$ . The induction step uses Lemma 9.5.

**Exercise 9.7.** How many elements does  $\mathcal{C}_{2,2}$  have?

**Exercise 9.8.**  $\mathcal{C}_{h,n}(L)$  is *strongly extensional*: the only bisimulation on it is the partition into singleton sets.

**Exercise 9.9.** We return to Examples 9.15 and 9.24. In those examples, we had some explicit definitions of the points  $\alpha_3, \alpha_6, \alpha_7, \alpha_8$  of  $\mathcal{C}_{1,1}(S4)$  and the points  $\beta_1, \dots, \beta_{14}$  of  $\mathcal{C}_{2,1}(S4)$ . In this notation, tell what  $\beta'_i$  is for all  $i$ . For example, is  $\beta'_1 = \alpha_6$ ? If not, which  $\alpha_j$  is it?

**Exercise 9.10.** Prove that  $\mathcal{D}_{h,n}(S4) = \mathcal{C}_{h,n}(S4)$  and conclude that  $S4$  is complete for reflexive transitive models, and also decidable.

**Exercise 9.11.** State and prove a similar result for  $KD4$ .

**Exercise 9.12.** Check that  $\mathcal{D}_{h,n}(KB4) \neq \mathcal{C}_{h,n}(KB4)$ .

**Exercise 9.13.** This exercise shows how to modify the work of this section to get the completeness result for  $KB4$ . Define a model  $\mathcal{D}'_{h,n}(L)$  to be the same as  $\mathcal{D}_{h,n}(L)$ , except that its relation  $\alpha \xrightarrow{dd} \beta$  should hold iff both  $\alpha \xrightarrow{d} \beta$  and  $\beta \xrightarrow{d} \alpha$ . State the analogs of Lemmas 9.34–9.37, and discuss the points where the proofs must be modified for the new models. Finally, deduce the desired results for  $KB4$ .

**Exercise 9.14.** Section 9.5 proved the completeness of  $K45$  for models which are transitive and Euclidean. Use the following steps to extend these results to get the completeness of  $S5 = KT45$  for equivalence relations.

1. Recall that a relation which is reflexive, transitive, and Euclidean is symmetric.
2. In Lemma 9.41, show that if  $KT \leq L$ , then  $\mathcal{E}_{h,n}(L)$  is reflexive.

3. State and prove a version of Lemma 9.36 for  $S5$ .
4. Use previous parts to show that for  $L = KT45$ ,  $\mathcal{C}_{h,n}(L) = \mathcal{E}_{h,n}(L)$ . [Follow the steps in Lemma 9.42.]
5. Deduce that  $S5$  is complete for equivalence relations.





# Chapter 10

## Dynamic Epistemic Logic

### 10.1 Common knowledge

One of the most important *group-level* knowledge phenomena is *common knowledge*. A sentence  $S$  is *common knowledge* for a group  $\mathcal{G}$  of people if

$$S \text{ is true, and everyone in the group } \mathcal{G} \text{ knows } (10.1). \quad (10.1)$$

It follows that

- ★ Everyone in  $\mathcal{G}$  knows  $S$ .
- ★ Everyone in  $\mathcal{G}$  knows that everyone in  $\mathcal{G}$  knows  $S$ .
- ★ Everyone in  $\mathcal{G}$  knows that everyone in  $\mathcal{G}$  knows that everyone in  $\mathcal{G}$  knows  $S$ .
- ★ etc., forever.

We hasten to add that this is an *informal* definition; very soon we'll put forward a *proposal* on how this might be modeled with modal logic.

Before that, here is an example of the intuitive idea. We have already seen the most basic “coin scenario.”

$A$  and  $B$  enter a large room containing a remote-control mechanical coin flipper. One presses a button, and the coin spins through the air, landing in a small box on a table. The box closes.  $A$  and  $B$  are much too far to see the coin.

In reality, the coin shows heads.

The group  $\mathcal{G}$  here is the set  $\{A, B\}$ .

It should be *common knowledge* for this group that neither knows the state of the coin. For that matter, it should be common knowledge that it is common knowledge that nobody knows.

**Another example of the intuitive idea** From our introduction:

We have a deck with three cards  $\heartsuit$ ,  $\clubsuit$ , and  $\diamondsuit$ . We also have three players:  $B$ ,  $C$ , and  $D$ . We deal the cards out, one to each player. The deal was ( $B : \diamondsuit, C : \clubsuit, D : \heartsuit$ ). Then,  $C$  and  $D$  show each other their cards while  $B$  looks on.

Let us consider the group  $\mathcal{G} = \{C, D\}$ . It should be *common knowledge* for this group that  $C$  has  $\clubsuit$ ,  $D$  has  $\heartsuit$ , and that  $B$  doesn't know any of this. But if we considered the larger group  $\{B, C, D\}$ , then it is *not common knowledge* that  $C$  has  $\clubsuit$ .

**Agents and groups of agents** A *group*  $\mathcal{G}$  of agents is a non-empty subset of  $\mathbf{Ag}$ . As with all set notation, the order in which elements of a set are listed doesn't count, and neither do repeats.

**Proposal:  $E_{\mathcal{G}}\varphi$  and  $CK_{\mathcal{G}}\varphi$**  We are building on epistemic logic as we studied it earlier this semester. So we have a set  $\mathbf{Ag}$  of agents, and a set  $\mathbf{AtSen}$  of atomic sentences, and we make new sentences using the boolean connectives  $\neg$ ,  $\wedge$ ,  $\vee$ ,  $\rightarrow$ , and  $\leftrightarrow$ ; the modal operators  $K_a$  for each agent  $A$ ; and (for the first time), new operators  $E_{\mathcal{G}}$  and  $CK_{\mathcal{G}}$  for each group.

We read  $E_{\mathcal{G}}\varphi$  as “everyone in  $\mathcal{G}$  knows that  $\varphi$ .” We read  $CK_{\mathcal{G}}\varphi$  as “it is common knowledge for the group  $\mathcal{G}$  that  $\varphi$ .”

If we have a “small” group of agents, such as  $\{a, b\}$ , then often we drop the set braces from the notation. We might write  $CK_{a,b}\varphi$  instead of  $CK_{\{a,b\}}\varphi$ . In fact, the set braces on this look a little pedantic.

**Semantics of  $E_{\mathcal{G}}\varphi$**  Here is our the formal definition of the *semantics* of  $E_{\mathcal{G}}\varphi$ . We start with a multi-agent model  $\mathcal{M} = (W, (R_a)_{a \in \mathbf{Ag}}, val)$ . For a world  $x \in W$ ,

$$x \Vdash E_{\mathcal{G}}\varphi \quad \text{iff} \quad \text{for all } a \in \mathcal{G}, \text{ and all worlds } y \text{ such that } x \xrightarrow{a} y, y \Vdash \varphi$$

**The logic of  $E_{\mathcal{G}}\varphi$**  The main sound logical principle for the  $E_{\mathcal{G}}$  operators is:

$$\models_{\mathbf{all}} E_{\mathcal{G}}\varphi \leftrightarrow \bigwedge_{a \in \mathcal{G}} K_a \varphi.$$

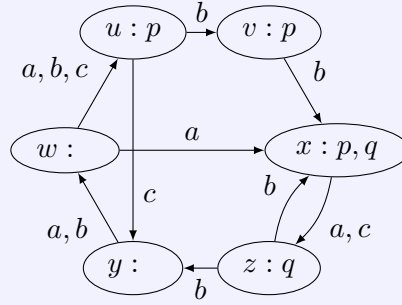
For example

$$\models_{\mathbf{all}} E_{\{b,c\}}\varphi \leftrightarrow (K_b \varphi \wedge K_c \varphi)$$

For a one-element set such as  $\{b\}$ ,  $E_{\{b\}}\varphi$  would be literally the same as  $K_b\varphi$ .

**Semantics of  $CK_{\mathcal{G}}\varphi$**  Given a multi-agent model  $\mathcal{M} = (W, (R_a)_{a \in \mathbf{Ag}}, val)$  and a group  $\mathcal{G} \subseteq \mathbf{Ag}$ :  
For a world  $x \in W$ ,

$$x \Vdash CK_{\mathcal{G}}\varphi \quad \text{iff} \quad \text{for all } y \text{ such that } x \xrightarrow{\mathcal{G}^*} y, y \Vdash \varphi$$

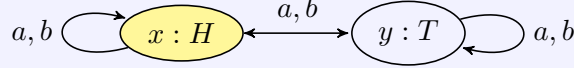
**Example 10.1**

$$\begin{array}{lll}
 u \models CK_b p & x \models CK_{a,c} q & z \models CK_b (q \rightarrow p) \\
 u \not\models CK_b p & w \not\models CK_a p & z \not\models CK_{a,b,c} (q \rightarrow p)
 \end{array}$$

The statements which are common knowledge typically formalize “hard facts” which we bring to various social situations.

**Example 10.2**

Recall that our  $\oplus$  notation means *exclusive or*.



Then

$$x \models CK_{a,b} ((H \oplus T) \wedge \neg K_a H).$$

**The logic of  $CK_{\mathcal{G}}\varphi$**  Here are some sound principles.

The main sound logical principles for the  $CK_{\mathcal{G}}$  operators are: the *Mix Axiom*

$$CK_{\mathcal{G}}\varphi \rightarrow \varphi \wedge E_{\mathcal{G}}CK_{\mathcal{G}}\varphi$$

The name comes from the fact that the axiom deals with the interactions of the two operators  $CK_{\mathcal{G}}$  and  $E_{\mathcal{G}}$ .

**Lemma 10.3**

The Mix Axiom is sound.

$$\models_{\text{all}} CK_{\mathcal{G}}\varphi \rightarrow \varphi \wedge E_{\mathcal{G}}CK_{\mathcal{G}}\varphi$$

*Proof.* Let  $M$  be a model, and assume that  $x \in M$  satisfies  $CK_{\mathcal{G}}\varphi$ . Then  $x \models \varphi$ . And we claim that if  $x \xrightarrow{a} y$  for some  $a \in \mathcal{G}$ , then  $y \models CK_{\mathcal{G}}\varphi$ . For this, let  $z$  be such that  $y \xrightarrow{a^*} z$ . Then  $x \xrightarrow{a} y \xrightarrow{a^*} z$ , so  $x \xrightarrow{a^*} z$  as well. Therefore,  $z \models \varphi$ .  $\square$

Also, we have the *Induction Rule*:

$$\text{from } \chi \rightarrow \psi \wedge K_a \chi \text{ for all } a \in \mathcal{G}, \text{ infer } \chi \rightarrow CK_{\mathcal{G}}\psi.$$

**Lemma 10.4**

The Induction Rule is sound. Assume that  $\models_{\text{all}} \chi \rightarrow \psi \wedge E_{\mathcal{G}}\chi$ . Then  $\models_{\text{all}} \chi \rightarrow CK_{\mathcal{G}}\psi$ .

*Proof.* Let  $\mathcal{M}$  be a model. Assume that every point  $x \in M$  satisfies  $\chi \rightarrow \psi \wedge E_{\mathcal{G}}\chi$ . We show about all points  $x \in M$  that  $x \models \chi \rightarrow CK_{\mathcal{G}}\psi$ .

So let us fix some  $x$  such that  $x \models \chi$ . We show that all points reachable from  $x$  for the group  $\mathcal{G}$  satisfy  $\psi \wedge \chi \wedge E_{\mathcal{G}}\chi$ . This is true for  $x$  itself. Suppose that  $y \models \psi \wedge \chi \wedge E_{\mathcal{G}}\chi$ , and also that  $y \xrightarrow{a} z$  for some  $a \in \mathcal{G}$ . Since  $y \models E_{\mathcal{G}}\chi$ ,  $z \models \chi$ . And thus we also have  $z \models \psi \wedge E_{\mathcal{G}}\chi$ . This completes the induction. It now follows that all points reachable from  $x$  for the group  $\mathcal{G}$  satisfy  $\psi$ .  $\square$

Using this logic, one can prove the important properties of common knowledge. For example, the *transitivity* ( $K4$ ) *property*:

$$CK_{\mathcal{G}}\varphi \rightarrow CK_{\mathcal{G}}CK_{\mathcal{G}}\varphi.$$

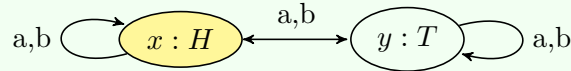
If  $\varphi$  is common knowledge in a group, then the fact of its being common knowledge is itself common knowledge in the group.

(To formally prove, take  $\chi$  to be  $CK_{\mathcal{G}}\varphi$ , and use Mix.)

**Soundness and Completeness** The logical system for  $S5^*$  is sound and complete. We aren't going to prove this, but I can send you some pointers if you are interested. The logic is also *decidable*. That is, a computer program could be written that would say yes or no (correctly) to questions about whether a given sentence is or is not provable in the logic.

**Proposition 10.5**

Consider the model  $\mathcal{M}_1$  from the first coin scenario:



Let  $\chi_h$  be the following sentence:

$$H \wedge CK_{a,b}((H \oplus T) \wedge \neg K_a H \wedge \neg K_a T \wedge \neg K_b H \wedge \neg K_b T)$$

Let  $\mathcal{N}$  be any model, and  $w \in \mathcal{N}$  any world such that  $w \models \chi$ . Then  $(\mathcal{M}_1, x) \equiv (\mathcal{N}, w)$ . That is, there is a bisimulation between  $\mathcal{M}_1$  and  $\mathcal{N}$  such that  $x \equiv w$ .

*Proof.* Let  $\chi_t$  be the same sentence as  $\chi_h$ , except that we use  $T$  as the first conjunct instead of  $H$ . We use the following partition of the disjoint union  $\mathcal{M}_1 + \mathcal{N}$ : (a) put  $x$  in the same cell as all  $w \in \mathcal{N}$  such that  $w \models \chi_h$ ; (b) put  $y$  in the same cell as all  $w \in \mathcal{N}$  such that  $w \models \chi_t$ ; (c) put all other  $w \in \mathcal{N}$  in their own cell.

To check that this partition is a bisimulation, we must check the following facts: (1) if  $w \models \chi_h$ , then  $w$  arrows for each agent to some  $u$  such that  $u \models \chi_h$ , (2)  $w$  arrows for each agent to some  $u$  such that  $u \models \chi_t$ ; (3) every world  $u$  that  $w$  arrows to satisfies either  $\chi_h$  or  $\chi_t$ .

All of these follow from the Mix Axiom and from basic facts about our model  $\mathcal{M}_1$ . To simplify our notation, let us write  $\chi_h$  as  $H \wedge CK_{a,b}\psi$ . Then by Mix,

$$\models_{\text{all}} CK_{a,b}\psi \rightarrow E_{a,b}CK_{a,b}\psi,$$

and also

$$\models_{\text{all}} CK_{a,b}\psi \rightarrow (H \oplus T).$$

These imply (1)–(3) above. □

## 10.2 Public announcements

A *public announcement* is the simplest kind of epistemic action, and this is why we’re studying it first. But in a sense, this chapter is just a warm-up to the work we’ll do later.

Let’s return to the *muddy children scenario*. This was really a series of different models, connected by *announcements* on the part of either the father, or the children themselves.

We currently can talk about models that correspond to different steps in the scenario, but we have no “first class” concept of an announcement. Now is the time to rectify this.

Our current language  $\mathcal{L}$  is defined as follows: We begin with a set **AtSen** of atomic sentences, and also a set **Ag** of agents. The elements of **AtSen** are sentences, and the sentences are closed under the boolean connectives and also the knowledge operators  $K_a$  for agents  $a$ , and finally under common knowledge operators:  $\mathcal{G}$  is a group of agents and  $\varphi$  a sentence, then  $E_{\mathcal{G}}\varphi$  and  $CK_{\mathcal{G}}\varphi$  are sentences.

Now we want to add a new clause to the definition of the language:

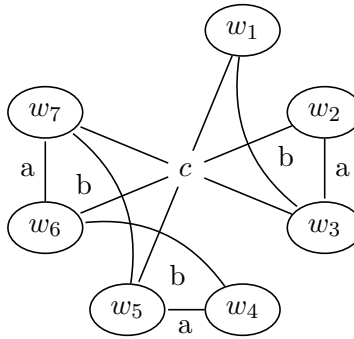
If  $\varphi$  and  $\psi$  are sentences so are  $\langle\varphi\rangle\psi$  and  $[\!\varphi\!]\psi$ .

Read  $[\!\varphi\!]\psi$  as “if  $\varphi$  happened to be true, then announcing it publicly results in  $\psi$ .” Read  $\langle\varphi\rangle\psi$  as “ $\varphi$  is true, and after announcing it publicly,  $\psi$  is true”

### Definition 10.6: restricting models

Let  $\mathcal{M}$  be any model, and  $\varphi$  any sentence. We define a new model called  $\mathcal{M} \upharpoonright \varphi$  called *the restriction of  $\mathcal{M}$  to  $\varphi$*  by taking as worlds the worlds of  $\mathcal{M}$  where  $\varphi$  is true, and then using the relations and valuation from  $\mathcal{M}$  in the obvious way.

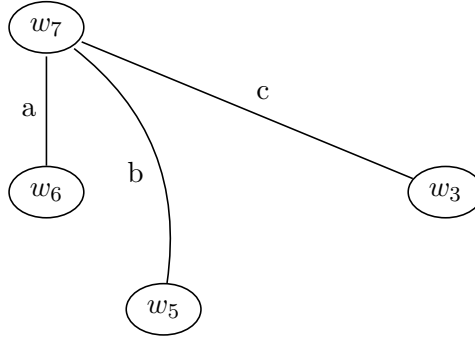
For example, recall  $\mathcal{M}$  from above. Then  $\mathcal{M} \upharpoonright (D_A \vee D_B \vee D_C)$  is show below:



Another example. Let's call the previous model  $\mathcal{N}$ . Let's let  $\varphi$  be the sentence

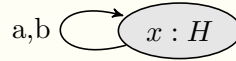
$$\begin{aligned} & \neg K_A D_A \wedge \neg K_A \neg D_A \\ \wedge & \neg K_B D_B \wedge \neg K_B \neg D_B \\ \wedge & \neg K_C D_C \wedge \neg K_C \neg D_C \end{aligned}$$

Then  $\mathcal{N} \upharpoonright \varphi$  is shown below:



#### Remark 10.7

But what if the announcement is false? Suppose we have the basic coin scenario, but after the people look. It is modeled as a one-world model:



Suppose that someone shouts out “*it's tails!*” It's not clear what the model should be afterwards, because the announcement says more about the speaker than about the real situation!

In what follows in this chapter, we only model announcements which are *true in the real world of a pointed model*. But later in the course we'll have more sophisticated models which allow us to talk about belief, deception, etc.

### 10.2.1 Adding public announcements: semantics

#### Definition 10.8

Let model  $\mathcal{M}$  be a model.

$$\begin{array}{ll} w \Vdash \langle \varphi \rangle \psi & \text{iff } w \Vdash \varphi \text{ in } \mathcal{M}, \\ & \text{and in } \mathcal{M} \upharpoonright \varphi, w \Vdash \psi \\ w \Vdash [\varphi] \psi & \text{iff } w \Vdash \varphi \text{ in } \mathcal{M} \text{ implies that} \\ & \text{in } \mathcal{M} \upharpoonright \varphi, w \Vdash \psi \end{array}$$

We can now go back to our original  $\mathcal{M}$  for the muddy children and check our desired facts (and others):

$$\begin{aligned} w_1 &\models [!(D_A \vee D_B \vee D_C)]K_A D_A. \\ w_1 &\not\models [!(D_A \vee D_B \vee D_C)]K_B D_B. \\ w_0 &\models [!(D_A \vee D_B \vee D_C)]\varphi \text{ for all } \varphi. \end{aligned}$$

**duality of  $\langle !\varphi \rangle \psi$  and  $[!\varphi] \psi$**  Note that

The “diamond form”  $\langle !\varphi \rangle \psi$  is usually more natural in thinking about actual models and scenarios. But  $[!\varphi] \psi$  is more natural, and more standard, when it comes to writing axioms in a logic. This is a parallel to what we did before, with  $\Box$  as basic and  $\Diamond$  as an abbreviation.

### 10.2.2 The Logic of Public Announcements

#### Example 10.9: Public announcement logic (PAL)

$$\begin{array}{ll} [\text{Pub}\varphi]p \leftrightarrow (\varphi \rightarrow p) & \text{atomic permanence} \\ [\text{Pub}\varphi]\neg\psi \leftrightarrow (\varphi \rightarrow \neg[\text{Pub}\varphi]\psi) & \text{partial functionality} \\ [\text{Pub}\varphi]K_a \psi \leftrightarrow (\varphi \rightarrow K_a [\text{Pub}\varphi]\psi) & \text{announcement-knowledge} \end{array}$$

$$\begin{array}{l} \text{From } \chi \rightarrow [\text{Pub}\varphi]E_g(\chi \wedge \varphi) \\ \text{infer } \chi \rightarrow [\text{Pub}\varphi]CK_g\psi \end{array} \quad (\text{the Announcement Rule})$$

The sentence  $p$  above must be an atomic sentence.

## 10.3 Epistemic actions and the update product

**The basic goal** We have seen a model of *public announcement of some sentence  $\varphi$* : given a model, we just delete the worlds where  $\varphi$  is false. But what about more complicated *epistemic actions*:

- public announcements to groups
- completely private announcements to groups, where the outsiders don’t suspect that anything is going on.
- private announcements to groups, where the outsiders do suspect that something is going on.

How can we model those?

**Products of sets** Given two sets  $A$  and  $B$ , we form the *product set*  $A \times B$  by taking all ordered pairs  $(a, b)$ , where  $a \in A$  and  $b \in B$ . For example,

$$\{1, 2, 3\} \times \{1, 2\} = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2)\}.$$

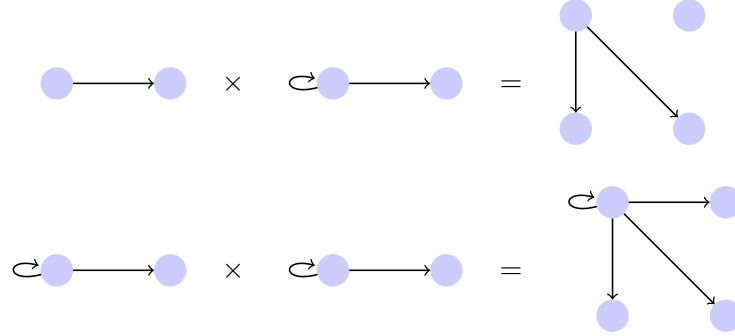


**Products of graphs** Given two graphs  $G = (G, \rightarrow_1)$  and  $H = (H, \rightarrow_2)$ , we form a new graph  $G \times H$  by taking the product set  $G \times H$  as the set of nodes, and the edge relation in  $G \times H$  is given by

$$(g, h) \rightarrow (g', h') \text{ iff } g \rightarrow_1 g' \text{ and } h \rightarrow_2 h'$$

The new graph  $G \times H$  is called the *product of  $G$  and  $H$* .

### Examples of products of graphs



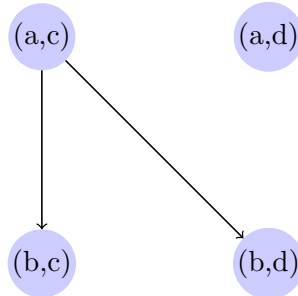
**The first example in detail** It makes sense to introduce names for the nodes.



In terms of our definitions,  $G$  is  $\{a, b\}$  with  $a \rightarrow_1 b$ .  $H$  is  $\{c, d\}$  with  $c \rightarrow_2 c$  and  $c \rightarrow_2 d$ . The node set of  $G \times H$  is  $\{(a, c), (a, d), (b, c), (b, d)\}$ .

|                             |     |                             |    |                             |    |                             |    |
|-----------------------------|-----|-----------------------------|----|-----------------------------|----|-----------------------------|----|
| $(a, c) \rightarrow (a, c)$ | no  | $(a, d) \rightarrow (a, c)$ | no | $(b, c) \rightarrow (a, c)$ | no | $(b, d) \rightarrow (a, c)$ | no |
| $(a, c) \rightarrow (a, d)$ | no  | $(a, d) \rightarrow (a, d)$ | no | $(b, c) \rightarrow (a, d)$ | no | $(b, d) \rightarrow (a, d)$ | no |
| $(a, c) \rightarrow (b, c)$ | yes | $(a, d) \rightarrow (b, c)$ | no | $(b, c) \rightarrow (b, c)$ | no | $(b, d) \rightarrow (b, c)$ | no |
| $(a, c) \rightarrow (b, d)$ | yes | $(a, d) \rightarrow (b, d)$ | no | $(b, c) \rightarrow (b, d)$ | no | $(b, d) \rightarrow (b, d)$ | no |

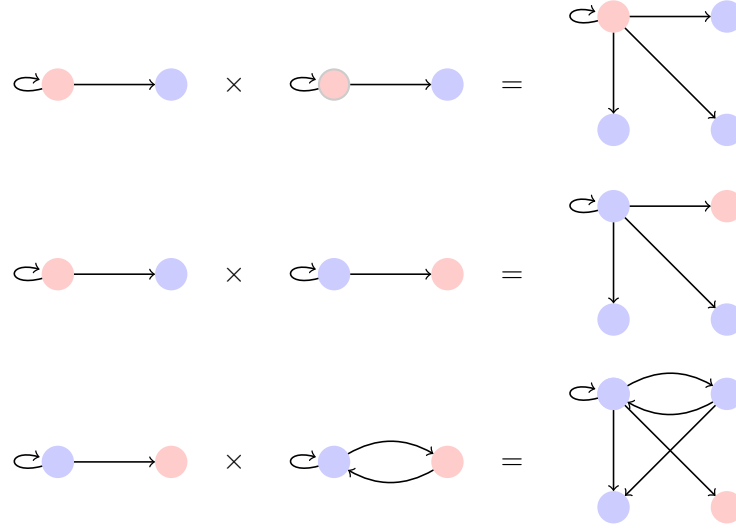
We therefore draw the product as



In a picture, we might as well forget the names of all nodes. That is, the identities of the objects which serve as our nodes are not important, but we do need names for the nodes to calculate the product graph.

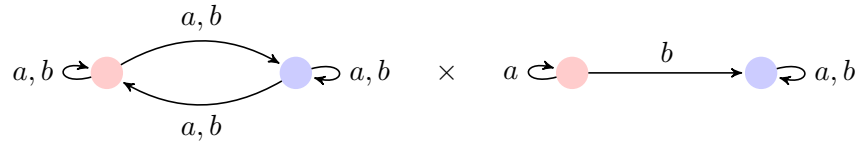
**Products of pointed graphs** A *pointed graph* is a graph with a distinguished node. Sometimes this is called the *start node*, or (in a modal logic setting) the *real world*. We would a pointed graph as  $(G, \rightarrow, g)$ , displaying the distinguished node. We have already defined the product of two graphs  $G = (G, \rightarrow_1, g)$  and  $H = (H, \rightarrow_2, h)$ . We next define the *product of pointed graphs*  $G = (G, \rightarrow_1, g)$  and  $H = (H, \rightarrow_2, h)$  by using the set  $G \times H$  as the nodes, the  $\rightarrow$  relation just as before, and taking the pair  $(g, h)$  to be the distinguished point.

### Products of pointed graphs

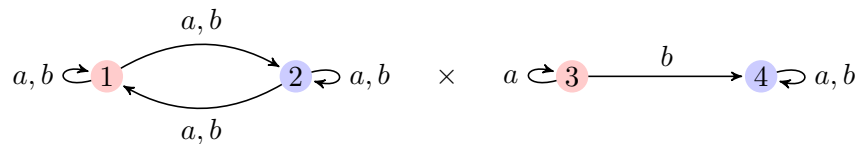


**Products of multi-agent graphs** We always start with a set  $\text{Ag}$  of agents. A *multi-agent graph* has one set of nodes with edge relations, one relation  $R_a$  for each agent  $a$ . (We also could call this relation  $\rightarrow^a$ .) Given two multi-agent graphs  $G = (G, (R_a)_{a \in \text{Ag}})$  and  $H = (H, (S_a)_{a \in \text{Ag}})$ , we form the *product multi-agent graph*  $G \times H$  by using the different agent relations separately, then combining the results. And if we have pointed, multi-agent graphs, ...

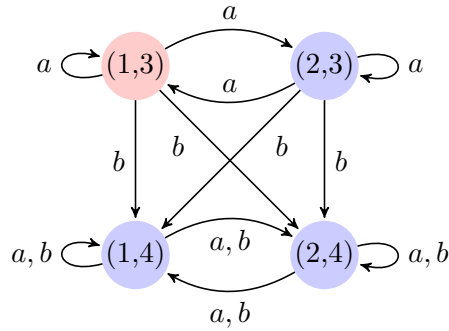
**Example of a product of pointed multi-agent graphs** We calculate



It makes sense to introduce names for the nodes; in fact, I am not sure how one could do this without naming the nodes.



We get



**Products of models?** A *model* is a graph with a valuation function  $val$ , telling which atomic sentences are true at the nodes. There is no product operation on models, because it is not clear how to combine two valuations into one.

#### Definition 10.10

Let  $\mathbf{AtSen}$  be a set of atomic sentences, and  $\mathbf{Ag}$  be a set of *agents*. Let  $\mathcal{L}$  be a modal language as we have studied them.

An *action model* is a structure

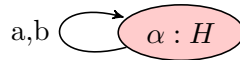
$$\Sigma = (\Sigma, (R_a)_{a \in \mathbf{Ag}}, pre)$$

such that

- ★  $\Sigma$  is a set whose elements are called *actions*
- ★ for each agent  $a$ ,  $R_a$  is a relation on  $\Sigma$ , called  $a$ 's *accessibility relation*
- ★  $pre$  is a function,  $pre : \Sigma \rightarrow \mathcal{L}$ , that gives for each action  $\alpha$  a sentence  $pre(\alpha) \in \mathcal{L}$  called *the precondition of  $\alpha$*

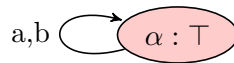
We also speak of *pointed* action models.

#### Examples of action models



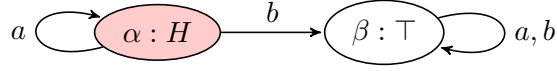
We have one action,  $\alpha$ , and  $pre(\alpha) = H$ .

Another action model

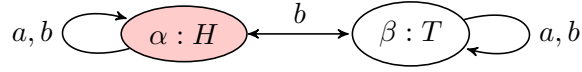


This time,  $pre(\alpha) = \top$ . Please do not confuse  $\top$  (a tautology) with  $T$  (tails).

## Two more examples of an action models



This will represent a *private announcement of heads to a*



This will represent *a looking at the coin in the presence of b* (who does not look). Notice that the precondition of  $\beta$  is  $\top$  in the first action model, and  $T$  in the second.

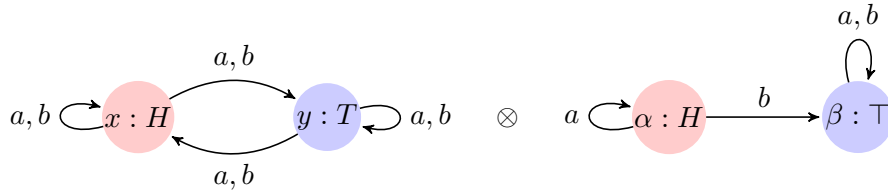
**Definition 10.11**

Given a multi-agent model  $\mathcal{M}$  and an action model  $\Sigma$ , we define the *update product*  $\mathcal{M} \otimes \Sigma$  to be the following model:

- ★ The node set of  $\mathcal{M} \otimes \Sigma$  is a certain subset of the product set  $W \times \Sigma$ , namely

$$\{(w, \alpha) : w \models \text{Pre}(\alpha)\}.$$

- ★  $(w, \alpha) \xrightarrow{a} (x, \beta)$  iff  $w \xrightarrow{a} x$  and  $\alpha \xrightarrow{a} \beta$ .
- ★  $\text{val}(w, \alpha) = \text{val}_{\mathcal{M}}(w)$ .

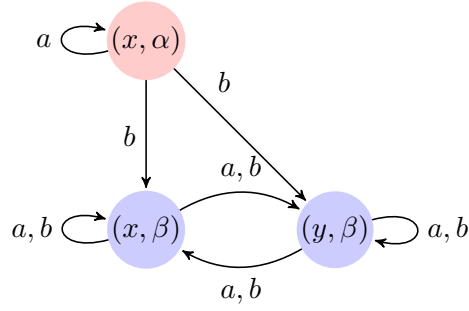
**Example: Amina “cheats” and views the coin**

The first step is to compute the set of nodes in  $W \otimes \Sigma$ .

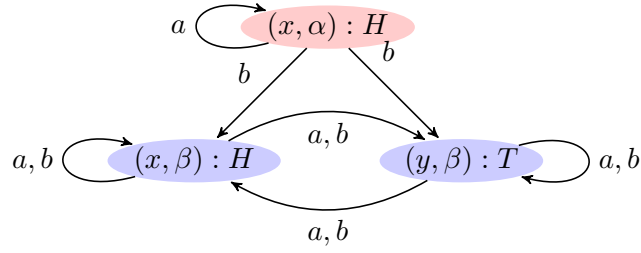
| node $z$ | action $\sigma$ | $\text{Pre}(\sigma)$ | $z \models \text{Pre}(\sigma)$ ? |
|----------|-----------------|----------------------|----------------------------------|
| $x$      | $\alpha$        | $H$                  | yes                              |
| $x$      | $\beta$         | $\top$               | yes                              |
| $y$      | $\alpha$        | $H$                  | no                               |
| $y$      | $\beta$         | $\top$               | yes                              |

Please make sure you understand what all of this means. Our node set is thus  $\{(x, \alpha), (x, \beta), (y, \beta)\}$ . The arrows for  $a$  and  $b$  work as in the product. The valuation is “inherited from the first compo-

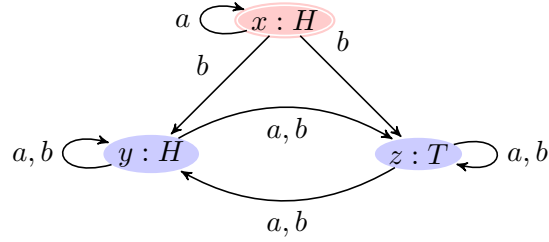
nent". We have



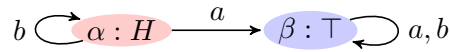
We get



Let's simplify the picture by renaming the nodes:



Let's derive the double cheating model



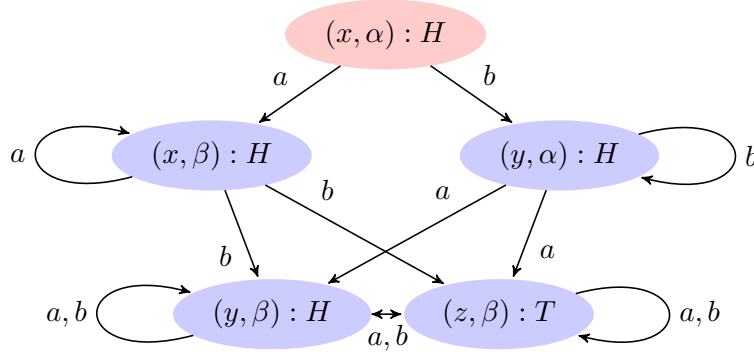
The first step is to compute the set of nodes in  $W \otimes \Sigma$ .

| node $z$ | action $\sigma$ | $\text{Pre}(\sigma)$ | $z \Vdash \text{Pre}(\sigma)?$ |
|----------|-----------------|----------------------|--------------------------------|
| $x$      | $\alpha$        | $H$                  | yes                            |
| $x$      | $\beta$         | $\top$               | yes                            |
| $y$      | $\alpha$        | $H$                  | yes                            |
| $y$      | $\beta$         | $\top$               | yes                            |
| $z$      | $\alpha$        | $H$                  | no                             |
| $z$      | $\beta$         | $\top$               | yes                            |

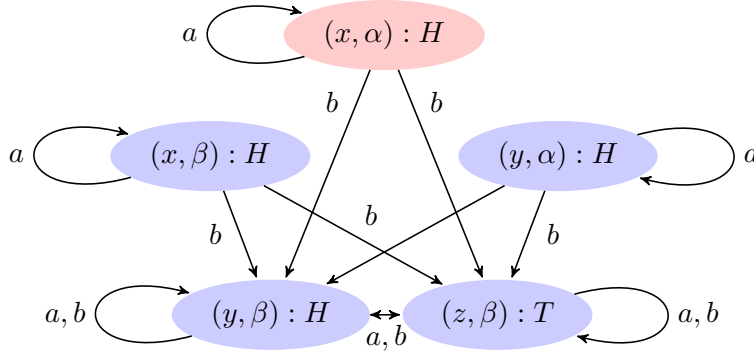
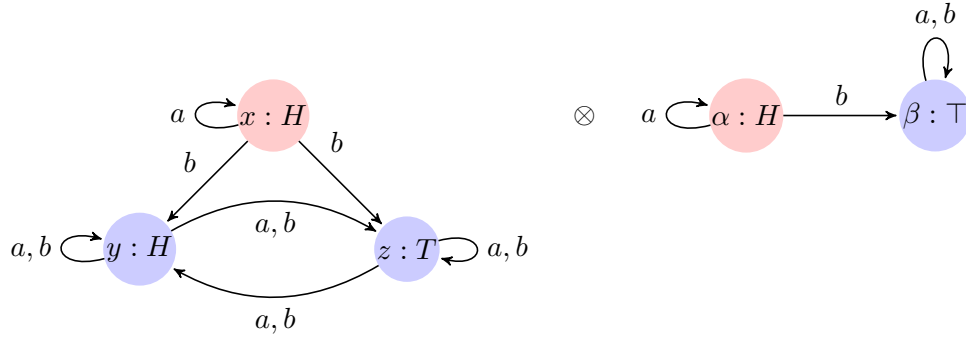
We thus have the node set

$$\{(x, \alpha), (x, \beta), (y, \alpha), (y, \beta), (z, \alpha)\}$$

**The double cheating example concluded** The arrows and the valuation are then calculated as before.



What if Amina cheats twice?



**Public announcements again** The action model corresponding to a public announcement of  $\varphi$  is

$$a, b, \dots \hookrightarrow \alpha : \varphi$$

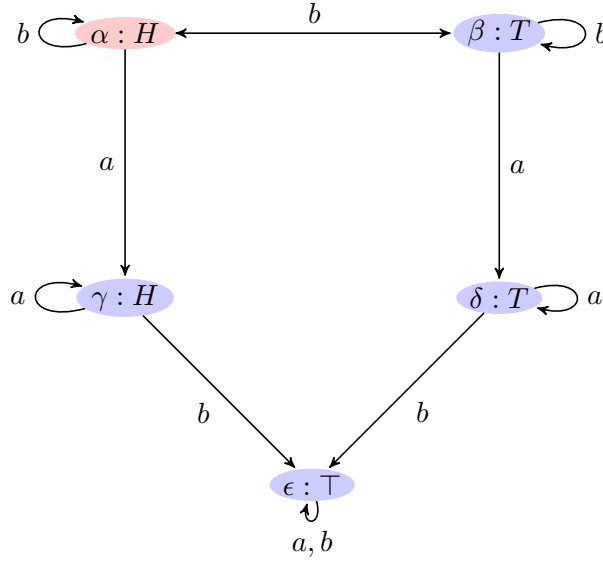
(All agents are on the loop.) We call this action model  $\Sigma_{\text{Pub}} \varphi$ .

The point is that for all models  $\mathcal{M}$ ,

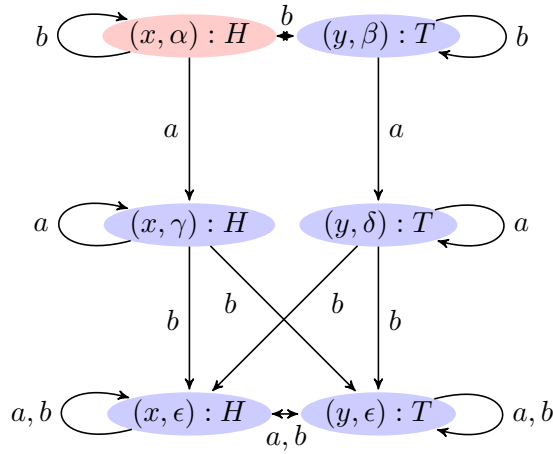
$$\mathcal{M} \otimes \Sigma_{\text{Pub}} \varphi \cong \mathcal{M} \upharpoonright \varphi.$$

And so the update product operation generalizes what we did earlier, by allowing more complex types of actions, as we have seen.

**So you thought you were cheating?** After the first coin scenario,  $A$  secretly opens the box herself.  $A$  believes that  $B$  did not suspect that anything happened at all. But in reality,  $B$  did see  $A$  open the box (but not the coin itself).



**So you thought you were cheating?** The product is then



The point is that one can now ask if this model *matches our intuitions*.

**Justifying action models** This gives us a way to check if action models are “correct.” We check if using them preserves our intuitions. That is, if we believe a model  $\mathcal{M}$  to match our intuitions about a scenario *before* an action, then we should believe  $\mathcal{M} \otimes \Sigma$  after the action. If we don’t, then  $\Sigma$  is not a good model of what is happening.

**Adding logic** We saw before the logic of public announcements. It added sentences  $[\text{Pub}\varphi]\psi$  to epistemic logic. We could also now add more complicated sentences which correspond to more complicated actions.

For example

$$\langle \text{Pri}_a \varphi \rangle \psi$$

This would say that  $\varphi$  is true, and announcing it to  $a$  in a completely private way results in  $\psi$  being true. What should the semantics of this sentence  $\langle \text{Pri}_a \varphi \rangle \psi$  be? And what should the main logical laws be?

## 10.4 Exercises on dynamic epistemic logic

**Exercise 10.1.** (The knowledge flip-flop) Suppose we have four players: Amina, Bao, Chandra, and Dianne. They have a deck with two indistinguishable  $\spadesuit$  cards, one  $\diamond$  and one  $\clubsuit$ . The cards are dealt, and the players look at their own cards.

We assume that the following are common knowledge: the distribution of cards in the deck, the fact that each player knows which card was dealt to them, and that they do not initially know any other player's card.

The deal is

$$w = (A\spadesuit, B\spadesuit, C\diamond, D\clubsuit).$$

So this is the “real world”, and we call this world  $w$ .

We take a modal language with atomic sentences

$$A\spadesuit, A\diamond, A\clubsuit, B\spadesuit, B\diamond, B\clubsuit, C\spadesuit, C\diamond, C\clubsuit, D\spadesuit, D\diamond, D\clubsuit,$$

We have agents  $a, b, c$ , and  $d$ , and so we have knowledge sentences involving those agents. We shall be concerned with the sentence  $\varphi$ , given as

$$\neg K_b A\spadesuit \wedge \neg K_b A\diamond \wedge \neg K_b A\clubsuit.$$

It says that Bao doesn't know Amina's card.

1. Show that  $w \models K_c \varphi$ .
2. Then Amina announces, “I do not have  $\diamond$ .” Show that now  $w \models \neg K_c \varphi$ .
3. After this, Dianne announces, “I do not have  $\spadesuit$ .” Show that now  $w \models K_c \varphi$ .

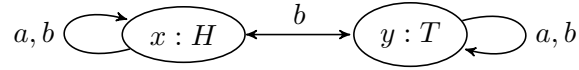
You *can* solve this problem by constructing the initial model explicitly. It has twelve worlds. Some people will no doubt prefer to do this, since it is very concrete and also satisfying. But you don't *have* to do it this way. You could argue about the model, mentioning various worlds and accessibility relations, without writing the whole thing out in full.<sup>1</sup>

---

<sup>1</sup>This problem is due to my former Ph.D. student Joshua Sack. It is the simplest known example of knowledge flip-flopping back and forth during a dialog. Sack's dissertation was on combining epistemic logic (as we are studying it in this course) with temporal logic, so that he could say “ $A$  knows  $\psi$  now but didn't know it one action ago.”



**Exercise 10.2.** This problem again concerns the model



Tell whether the following are true or false, with a short explanation:

1.  $x \models \langle !H \rangle CK_{a,b} H$ .
2.  $x \models \langle !T \rangle CK_{a,b} H$ .
3.  $x \models [!T] CK_{a,b} H$ .
4.  $x \models \langle !K_b H \rangle K_a K_b H$ .

**Exercise 10.3.** Prove that

$$\models_{all} \langle \text{Pub}\varphi \rangle K_a \psi \leftrightarrow (\varphi \wedge K_a (\varphi \rightarrow \langle \text{Pub}\varphi \rangle \psi))$$

**Exercise 10.4.** Recall that throughout these notes, letters like  $p$  denote *atomic sentences*. Show that  $\models_{all} [!p] CK_{\mathcal{A}} p$ . Here  $\mathcal{A}$  is the set of all agents. Intuitively,  $[!p] CK_{\mathcal{A}} p$  says, “if  $p$  is true, then announcing it truthfully to everyone results in  $p$  becoming common knowledge.”